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Felipe Braga Ribas

EXPLORANDO OS OBJETOS TRANSNETUNIANOS PELO MÉTODO DE OCULTAÇÕES ESTELARES:
predição, observação, Quaoar e os primeiros resultados.

EXPLORATION DES OBJETS TRANS-NEPTUNIENS PAR LA METHODE DES OCCULTATIONS STELLAIRES:
prédictions, observations, Quaoar et les premiers résultats.

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Tese de Doutorado apresentada ao Programa de Pós-graduação em Astronomia do Observatório Nacional/MCTI em regime de co-tutela com o Observatoire de Paris-Meudon / LESIA, como requisito parcial à obtenção do título de Doutor em Astronomia e Astrofísica.

Orientadores: Prof. Dr. Roberto Vieira Martins (ON/BR);
Prof. Dr. Bruno Sicardy (Obspm/FR).

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Thèse dirigée par:
Prof. Dr. Bruno Sicardy (Obspm/FR);
Prof. Dr. Roberto Vieira Martins (ON/BR).

Soutenue le 26 Février 2013

devant le jury composé par:

President: Dr. Antonella Barucci / Observatoire de Paris-Meudon - LESIA / FR
Rapporteur: Dr. Jean-Marc Petit, Observatoire de Besançon / FR
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Doctorat
Spécialité
Astronomie et Astrophysique
École Doctorale
d'Astronomie & Astrophysique d'Île-de-France

RAPPORT DE SOUTENANCE DE THÈSE

Thèse soutenue le mardi 26 Février 2013 à 14h00

par M. Felipe BRAGA-RIBAS

Sujet de la thèse : Etude d'occultations stellaires par de objets transneptuniens

Président du Jury : Rodney GOMES, Chercheur Titulaire/ON/au MCTI au Brésil

Composition du jury :
Mme Antonella BARUCCI, Astronome au LESIA à l'Observatoire de Paris
M. Sylvio FERRAZ-MELLO, Professeur émérite à l'Instituto Astronomico e Geofísico au Brésil
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RAPPORT DE SOUTENANCE:

voir le vers de cette feuille.

MENTION (HONORABLE AU TRÈS HONORABLE) : TRÈS HONORABLE

LE PRESIDENT ET LES MEMBRES DU JURY (SIGNATURES) :

Mme Antonella BARUCCI M. Sylvio FERRAZ-MELLO M. Rodney GOMES

M. Jean-Marc PETIT M. Bruno SICARDY M. Roberto VIEIRA MARTINS

Le jury, dont la liste est donné au recto du présent document, a assisté ce jour (26 février 2013) à la soutenance de thèse de M. Felipe Braga-Ribas, sur le sujet suivant: "Exploration des objets trans-neptuniens par la méthode des occultations stellaires: prédictions, observations, Quaoar et les premiers résultats".

Le jury a apprécié la présentation orale du candidat, qui a répondu de manière tout à fait satisfaisante aux questions du jury.

Le manuscrit est également apprarue comme bien structuré et très complet.

Le jury a souligné la quantité et la qualité du travail effectué. Le candidat a mi en place tout ma chaîne qui permet de prédire les événements, organiser les campagnes d'observation, et finalement analyser les donnés et en déduire les propriétés des objets étudié: taille, forme, densité, albédo, et contraints sur la présence d'atmosphère éventuelles.

Par toutes ces raisons, le jury est très heureux de décerner le titre de docteur à Felipe Braga-Ribas dnas la spécialité Astronome et Astrophysique, aussi bien an titre de l'Observatoire de Paris que de l'Observatório Nacional de Rio de Janeiro, et avec mention très honorable.

Fait à Rio de Janeiro
26 février 2013

Braga-Ribas, Felipe

EXPLORANDO OS OBJETOS TRANSNETUNIANOS PELO MÉTODO DE OCULTAÇOES ESTELARES: Predição, Observação, Quaoar e os primeiros resultados / Felipe Braga Ribas-

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Resumo

Os objetos transnetunianos (TNOs) são corpos primitivos do Sistema Solar que orbitam além de Netuno. Mais de 1200 TNOs já foram descobertos desde de 1992, e nesta população encontram-se planetas anões como Plutão, Makemake e Éris. Sua dinâmica orbital, a sua composição interna e superficial, e sua distribuição de tamanho e massa contêm informações sobre a formação e evolução de todo o Sistema Solar.

Observações fotométricas, espectroscópicas, e térmicas (Infravermelho distante), fornecem informações como seu período de rotação, composição superficial e estimativas de tamanho e albedo.

No entanto, dada suas distâncias (maiores que 30 unidades astronômicas) e seus tamanhos, atualmente é impossível medir suas dimensões a partir de imagens diretas, bem como detectar diretamente suas atmosferas, já que seus diâmetros angulares são inferiores a 50 milissegundos de grau (mas).

O método de ocultações estelares consiste em observar a passagem do TNO em frente a uma estrela. A duração precisa do fenômeno, observada de diferentes sítios, permite deduzir o tamanho e a forma destes objetos com precisões na ordem do quilômetro. Então, valores precisos do albedo podem ser calculados e, para os corpos que têm sua massa determinada a partir da órbita de seu satélite, valores precisos de suas densidades podem também ser calculados. Além disso, possíveis atmosferas atenuam a luz da estrela instantes antes da ocultação, o que permite detectar atmosferas com pressões tão pequenas quanto o nanobar.

Este trabalho descreve (1) o método de predição de ocultações aplicado a mais de cinquenta TNOs, com precisão na ordem de 30 mas, (2) a organização de campanhas mundiais envolvendo dezenas de observatórios profissionais e amadores, (3) a observação, análise e interpretação de ocultações promovidas por Quaoar e outros grandes objetos transnetunianos.

Desde 2009, quatorze ocultações por TNOs já foram observadas: doze pelo grupo

Paris/Rio e duas pela equipe do MIT. Além da discussão detalhada sobre as ocultações por Quaoar, um resumo dos resultados sobre o tamanho, forma, albedo, densidade e atmosfera, obtidos através das ocultações estelares por objetos transnetunianos também é apresentado. Em particular, são abordados os eventos envolvendo Éris, Makemake, 2003 AZ_{84}, Varuna e Sedna, todas inéditas.

Palavras-chave: Ocultação estelar, TNO, KBO, 50000 Quaoar, Objeto Maclaurin.

Résumé

Les objets trans-neptuniens (OTNs) sont des corps primitifs du système solaire qui orbitent au-delà de la planète géante Neptune. Plus de 1200 objets ont été découverts depuis 1992, et on compte dans cette population les planètes naines Pluton, Makemake et Eris. Ce sont des corps primitifs, et à ce titre, leur dynamique orbitale, leur composition interne et surfacique et leur distribution en taille nous renseignent sur la formation et l'évolution du système solaire dans son ensemble.

De nombreuses observations photométriques, spectroscopiques et thermiques (IR lointain) ont fourni des renseignements sur leurs périodes de rotation, leurs compositions de surface et des estimations de leurs tailles et de leurs albédos.

Cependant, au vu de leurs distances (plus de 30 unités astronomiques) et de leur taille, il est actuellement impossible de résoudre leur diamètre angulaire (en général bien inférieur à 50 miliseconds d'arc (mas) et de détecter directement leur atmosphère.

La méthode d'occultation stellaire consiste à observer le passage d'OTNs devant une étoile. Les durées précises du phénomène, observé en différents endroits, permettent ensuite de retrouver la taille et la forme des objets avec des précisions kilométriques. Ainsi, des valeurs précises de l'albédo peuvent être déduites, et pour les corps dont la masse est connue via le mouvement de satellites, des valeurs précises de la masse volumique peuvent également être calculées. Par ailleurs, des atmosphères éventuelles provoquent des baisses de signal juste avant et après l'occultation, permettant des détections à des niveaux de pression de quelques nanobars.

Cette thèse décrit (1) les méthodes de prédiction des occultations par une quarantaine d'OTNs, avec des précisions de l'ordre de 30 mas, (2) l'organisation de campagnes mondiales faisant intervenir des dizaines de sites professionnels et amateurs, (3) l'observation, l'analyse et l'interprétation de plusieurs occultations par de gros OTNs.

Depuis 2009, onze occultations par des OTNs ont été observées : neuf par le consortium dont fait partie le groupe du LESIA, et deux par une équipe du MIT. Dans cette thèse

sont décrites en particulier les occultations par Varuna, Eris, 2003 AZ_{84}, Makemake et Quaoar, qui sont toutes des premières.

Nous donnons des contraintes sur les tailles, albédos, forme de ces corps, ainsi que des limites de leur atmosphère jusqu'à des niveaux de l'ordre de quelques nanobars.

Mots clefs: occultation stellaire, OTN, KBO, 50000 Quaoar, objet de Maclaurin.

Abstract

Transneptunian objects (TNOs) are fossils of the Solar System beyond Neptune. Over 1200 TNOs have been found since 1992, and among them we find dwarf-planets such as Pluto, Makemake and Eris. The dynamics, internal and surface composition, size and mass distribution may have information about the early ages and evolution of the Solar System.

Photometry, spectroscopic, thermal (far infra-red) give information about their rotational period, surface composition, size and albedo estimations.

Nevertheless, due to their distance (greater than 30 Astronomical Units) and their sizes, we do not have the necessary technology to directly measure their size and detect possible atmospheres, as their angular sizes are smaller than a 50 miliarc seconds (mas).

The stellar occultation technique consists of observing the transit of a TNO in front of a star. The precise duration of the event, observed from several different sites, allows to deduce their size and shape with kilometre accuracy. So, precise determination of their albedos can be derived, and for those bodies that have their mass deduced from its satellite orbit, precise determination of their densities is allowed. Besides, tenuous atmospheres at the nanobar level, can be detected as they promote an attenuation of the star light right before and after the occultation.

In this work it is described (1) the prediction method applied to more then fifty TNOs, with accuracy of about 30 mas, (2) the world-wide campaigns with tens of professional and amateurs observatories, (3) the observation, analysis, and interpretation of the observed occultation by Quaoar and other large transneptunian objects.

Since 2009, fourteen occultations by TNOs were already observed, twelve by the Paris/Rio consortium.In addition to the detailed discussion about the Quaoar events, a compilation of all the other results comprising as size, shape, albedo, density and atmosphere, obtained from TNO stellar occultations is presented. In particular occultations by Eris, Makemake, 2003 AZ_{84}, Varuna and Sedna, all of which are new.

We present constraints over size, shape, albedo of these bodies, and also upper limits on possible atmospheres up to the nanobar level.

Keywords: Stellar occultation, TNO, KBO, 50000 Quaoar, Maclaurin object.

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Definições, acrônimos e constantes

ua $\Longleftrightarrow$ unidade astronômica = 1.49598×10^8 km

G $\Longleftrightarrow$ Constante Gravitacional = 0,865 cm^3 $g^{-1}h^{-2}$; $6{,}674 \times 10^{-8}$ cm^3 $g^{-1}s^{-2}$

σ_{SB} $\Longleftrightarrow$ Constante de Stefan-Boltzmann = $5.670{\times}10^{-8}$ W m^{-2} K^{-4}

IAU $\Longleftrightarrow$ União Astronômica Internacional, do inglês: *International Astronomical Union.*

AR, DEC $\Longleftrightarrow$ Ascenção Reta, Declinação.

TU $\Longleftrightarrow$ Tempo Universal.

LST $\Longleftrightarrow$ Tempo Solar Local, do inglês: *Local Solar Time.*

MSTG $\Longleftrightarrow$ Tempo Sideral Médio em Greenwich , do inglês: *Mean Sideral Time in Greenwich.*

PA $\Longleftrightarrow$ Ângulo de posição, do inglês: *Position Angle.*

C/A $\Longleftrightarrow$ Maior aproximação aparente entre dois corpos, do inglês: *Closest Approach.*

GPS $\Longleftrightarrow$ Sistema de Posicionamento Global, do inglês: *Global Positioning System.*

NTP $\Longleftrightarrow$ Protocolo de Tempo de Rede, do inglês: *Network Time Protocol.*

PRAIA $\Longleftrightarrow$ Plataforma para Redução Automática de Imagens Astronômicas (Assafin et al. 2011).

FITS $\Longleftrightarrow$ Sistema de transporte de imagem flexíveis, do inglês: *Flexible Image Transport System.*

EM-CCD $\Longleftrightarrow$ Dispositivo de cargas acopladas - Multiplicador de Elétrons, do inglês: *Electron-multiplying - Charge Coupled Device.*

ITV $\Longleftrightarrow$ Incrustador de Tempo em Vídeo.

DDT $\Longleftrightarrow$ Tempo de Diretor, do inglês: *Director's Discretionary Time.*

HST $\Longleftrightarrow$ Telescópio Espacial Hubble, do inglês: *Hubble Space Telescope.*

Capítulo 1

Introdução

O estudo dos objetos transnetunianos (TNOs) está intimamente ligado ao estudo da evolução do Sistema Solar. A evolução dinâmica pode ter deixado assinaturas na disposição atual das órbitas (anexo A), densidade e distribuição de massa dos TNOs. Os elementos primordiais da nuvem protoplanetária também podem ter sido preservados por estes objetos, e entender sua abundância, distribuição e processos químicos aos quais estes corpos foram submetidos é fundamental para entender as condições iniciais de formação do conjunto. Daí a necessidade de conhecer as propriedades físicas, como tamanho, densidade, albedo e composição destes objetos. Uma ampla e detalhada visão do que conhecemos sobre os objetos transnetunianos é dada no livro *The Solar System Beyond Neptune* (Barucci et al. 2008).

Para obter informações sobre estas propriedades físicas é preciso lançar mão de diversas áreas de estudo como: evolução dinâmica, espectroscopia de superfície, fotometria, medidas térmicas, determinação de massa, etc. Um dos parâmetros físicos mais fundamentais é o tamanho de cada objeto. No entanto, como esses pequenos corpos estão a mais de 30 `ua` (unidades astronômicas) do Sol, não há telescópio em solo que consiga medir estes objetos através de imagem direta.

Utilizando a técnica de ocultações estelares, que consiste em observar o TNO enquanto este passa na frente de uma estrela para um dado observador, procura-se determinar os tamanhos e formas destes corpos com precisões da ordem do quilômetro e, por consequência, seus albedos e densidades são obtidos com grande precisão. Isto tem implicação direta sobre sua composição superficial e estrutura interna. Além disso, hoje em dia, a ocultação estelar é a única técnica que permite detectar atmosferas com pressão superficial na ordem do nanobar em torno destes corpos, o que tem implicações sobre os processos térmicos aos quais os objetos são submetidos.

Antes de observar uma ocultação estelar por um grande objeto, é preciso realizar a predição do evento (Capítulo 2). Para isto compara-se suas órbitas à posição de estre-

las de catálogos. No entanto, suas órbitas apresentam imprecisões até dez vezes maiores que o necessário para o cálculo destas ocultações. Como uma das causas destes erros, tem-se a recente descoberta destes objetos em relação com seus longos períodos orbitais (>200 anos), de forma que somente uma pequena porcentagem de sua órbita é conhecida. Seu fraco brilho, em consequência de sua distância ao Sol, também dificulta sua observação. Para contornar este problema, realizam-se medidas astrométricas dos objetos para determinar qual é a diferença em relação à posição dada por sua efeméride. Esta diferença é aplicada como um desvio constante, para então se calcular os eventos de ocultação.

Além disso, o lento deslocamento dos TNOs no céu faz com que eventos de ocultação sejam raros, obrigando a recorrer a catálogos estelares mais densos, com estrelas mais fracas. Isto traz outro problema, já que a precisão da posição astrométrica destas estrelas também é inferior à necessária para uma boa predição de ocultação. Para contornar estas dificuldades, fez-se um programa junto ao telescópio de 2,2m com do ESO/MPG[1], equipado com a câmera WFI (Wied Field Imager), no qual se observou os caminhos que dez entre os maiores TNOs percorrerão até 2015. Utilizando técnicas astrométricas adequadas, criou-se catálogos estelares até magnitude R= 18, com precisão melhor que 40 milissegundos de grau (mas) (anexos C e D). Com isto, predições de ocultações estelares por TNOs confiáveis foram obtidas (Assafin et al. (2010) & Assafin et al. (2012)).

Centenas de eventos são previstos para cada ano, e como é impossível observar todos, uma seleção é feita seguindo critérios como magnitude da estrela ocultada e localização da sombra, conforme a rede de colaboradores. Dias antes de um evento, quando o TNO e a estrela candidata podem ser observados numa mesma imagem, tomam-se medidas astrométricas relativas. Com isto evita-se erros sistemáticos de catálogos. Assim, precisões da ordem de 20 mas são atingidas, e o local por onde a sombra de ocultação passará sobre a Terra é definido.

Campanhas internacionais são realizadas, envolvendo dezenas de pesquisadores e astrônomos amadores. Uma página internet para cada evento é publicada[2] com informações aos colaboradores.

Neste trabalho, além metodologia de predição exposta acima, discute-se em detalhes todo o procedimento de redução dos dados provenientes das ocultações de Quaoar em 2011 e 2012 (Capítulo 4). A ocultação ocorrida em 04 de maio de 2011 contou com detecção em 5 sítios diferentes. Cada sítio obtém uma curva de luz que é modelada por uma ocultação de um corpo sem atmosfera, convolvida com a difração e tamanho da estrela e o tempo de integração da observação. Assim, obtém-se os instantes de início e fim do evento, o que caracteriza uma medida do perfil de Quaoar, chamada de corda. As cordas são utilizadas

[1]European Southern Observatory/Max-Planck-Gelsellschaft

[2]`http://www.lesia.obspm.fr/perso/braga-ribas/campaigns`

para se ajustar uma elipse que reflete o tamanho, forma, posição e orientação do corpo.

Pela disposição das cordas, Quaoar parece ter uma forma bastante irregular. Neste caso apresentam-se duas soluções. A primeira caracteriza-se por ter uma cratera que, como é discutido, seria demasiadamente profunda. A outra solução apresenta uma montanha, fisicamente improvável, considerando o tamanho e composição de Quaoar.

Explora-se então uma terceira solução, em que se considera a possibilidade de haver problemas no registro do tempo absoluto em alguns observatórios. Devido ao seu tamanho, rotação e composição superficial, espera-se que Quaoar esteja em equilíbrio hidrostático. Por isso, ele deve apresentar a forma de um objeto Maclaurin. Para se obter tal figura, as cordas são deslocadas no tempo, obtendo-se assim a superfície equivalente, que é comparada à curva de equilíbrio de um objeto Maclaurin. Obtém-se assim todos os parâmetros físicos pertinentes de Quaoar (anexo H, (Braga-Ribas et al. 2013)).

Considerando todas as soluções, as seguintes faixas de resultados são obtidas: raio equivalente 504 – 566 km, achatamento 0,0722 – 0,33, albedo geométrico no visível 0,102 – 0,139 e densidade 1,40 – 2,46 g cm^{-3}. Ainda coloca-se um limite superior de 21 nanobar para a presença de uma atmosfera global de metano em Quaoar, mas não é possível descartar a presença de atmosferas locais.

A metodologia aplicada à Quaoar também foi utilizada em outras ocultações por TNOs observadas pelo grupo, nominalmente Éris (anexo F, (Sicardy et al. 2011)), Makemake (anexo G, (Ortiz et al. 2012)) e 2003 AZ_{84} (anexo E, (Braga-Ribas et al. (2011) e Braga-Ribas et al. (2012)). No capítulo 5, um sumário de todos os resultados já obtidos a partir de ocultações estelares por TNOs é apresentado.

Capítulo 2

Predição de Ocultações

Predizer ocultações estelares por pequenos corpos do Sistema Solar consiste essencialmente em comparar, sobre a esfera celeste, a posição do objeto num dado momento oriunda de sua efeméride, com a posição de estrelas fornecidas por um catálogo. Se as posições forem coincidentes, uma ocultação irá ocorrer. É evidente que diversos detalhes, e por vezes dificuldades, estão envolvidos neste processo, e todos serão apresentados nas seções que seguem. Nas publicações Assafin et al. (2010) e Assafin et al. (2012) os procedimentos descritos a seguir também são discutidos.

Para eventos observáveis a partir da Terra, compara-se a posição de estrelas de um catálogo com a efeméride geocêntrica de um dado objeto, calculada a cada minuto, por exemplo. Se a distância estrela/objeto for menor que um dado valor, uma possível ocultação é encontrada. Tal valor é definido como um raio de busca, que considera o tamanho angular da Terra e do próprio objeto. A uma distância de 30 ua, a Terra tem raio de 293 masİncluindo-se o tamanho do próprio objeto, não maior que 50 mas para os TNOs, estes valores definem o raio mínimo de busca de uma ocultação sobre nosso planeta ($d_{min} = 318$ mas).

No que segue, serão consideradas posições aparentes no plano tangente à esfera celeste com o observador no centro da Terra. A partir de duas posições arbitrárias do corpo nos instantes t_1 e t_2, a uma distância D, anterior e posterior ao encontro com a estrela S (ver Figura 2.1), pode-se determinar a distância geocêntrica aparente no plano do céu $(a - b)$ entre a estrela e o objeto. A distância mínima entre os dois objetos (C/A), definida por d, pode ser calculada com a equação 2.1. O instante de maior aproximação t_0 é definido pela equação 2.2, onde assume-se que $t_2 > t_1$. A velocidade v relativa entre o objeto e a estrela pode ser calculada com a equação 2.3, onde velocidades positivas significam movimento aparente prógrado (ascensão reta crescente). Ainda podemos definir o ângulo de posição (PA) como sendo o ângulo entre o norte (zero grau) e a posição de maior aproximação C/A, contado no sentido horário. Para saber as condições de observabilidade (dia/noite)

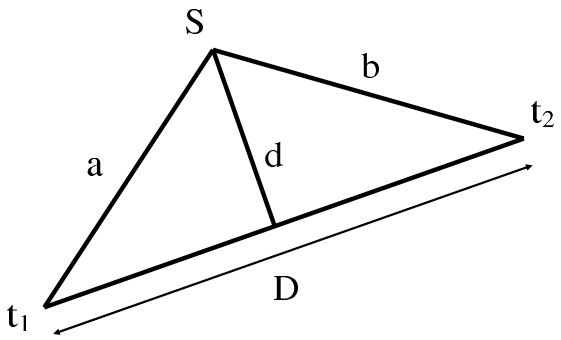


Figura 2.1 Figura esquemática para o cálculo dos parâmetros de uma ocultação estelar. Com duas posições do corpo, anterior e posterior ao encontro, a uma distância D, em instantes arbitrários t_1 e t_2, pode-se determinar a sua distância geocêntrica aparente (a e b) à estrela S. A distância mínima d entre a estrela e o objeto é conhecida como *Closest Approach* (C/A). Com estes valores é possível calcular o instante de maior aproximação t_0, a velocidade com que sombra da ocultação se desloca sobre a Terra v, o ângulo de posição (PA) e o tempo solar local (LST).

ao longo da faixa de ocultação, pode-se calcular o tempo solar local aproximado (LST), usando o t_0, a ascensão reta (AR) e o Tempo Sideral Médio em Greenwich (do inglês MSTG) como na equação 2.4. A diferença entre a AR e MSTG constitui a longitude do ponto sub-planetário. Valores positivos correspondem a longitudes leste e negativo a oeste.

$$d = \sqrt{a^2 - \left(\frac{a^2 - b^2 + D^2}{2D}\right)^2} \tag{2.1}$$

$$t_0 = t_1 + (t_2 - t_1)\sqrt{\frac{a^2 - d^2}{D^2}} \tag{2.2}$$

$$v = \frac{A\sin(D)}{t_2 - t_1} \tag{2.3}$$

$$LST = t_0 + AR - MSTG = t_0 + long \tag{2.4}$$

Para destacar fenômenos lentos, normaliza-se as magnitudes para velocidade típica dos TNOs durante a oposição v=20 km/s. Com isto, obtém-se a magnitude M* da estrela que será menor que a magnitude real se a velocidade for menor que 20 km/s, ou maior que a magnitude real se a velocidade for maior que 20 km/s (Equação 2.5). Eventos lentos podem permitir exposições mais longas, facilitando a observação.

$$M^* = M + 2,5 \log_{10} \left(\frac{v}{20\ km\ s^{-1}} \right) \tag{2.5}$$

À exceção de Plutão, que apresenta ∼105 milissegundos de grau (mas) no plano do céu, os TNOs têm tamanhos angulares menor que 50 mas, portanto acurácia da mesma ordem é necessária para se fazer predição de ocultações estelares confiáveis. Infelizmente, nem a posição do corpo, nem da estrela são conhecidos com tal exatidão. A origem destes erros e as técnicas para minimiza-los serão discutidas a seguir.

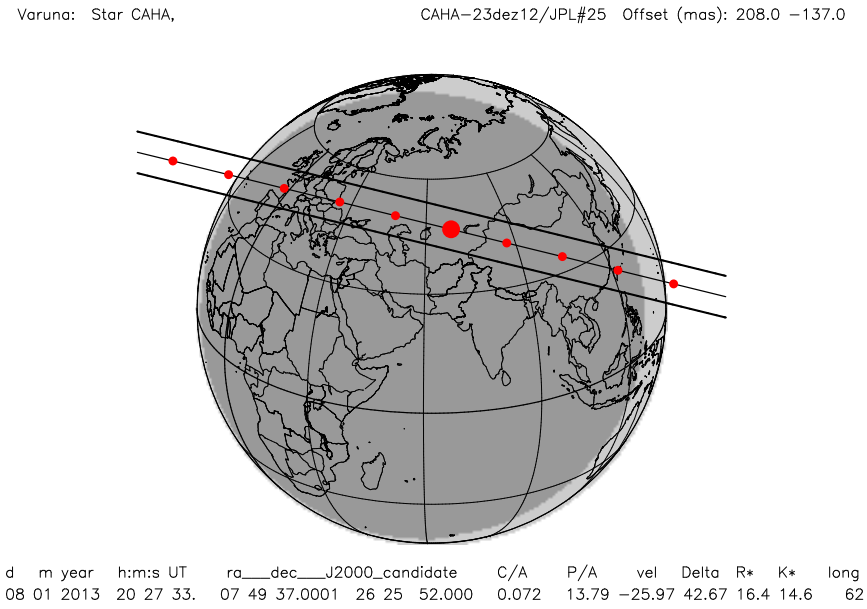


Figura 2.2 Mapa de predição de ocultação: As barras paralelas que cruzam a Terra delimitam o tamanho da sombra projetada pelo objeto ao se deslocar sobre o planeta. A parte cinza escura representa a noite no momento do C/A, a cinza clara o crepúsculo civil com o Sol a uma altura <-18 graus. O ponto vermelho grande representa o ponto de maior aproximação (C/A), os pontos vermelhos estão separados de minuto em minuto. Na legenda acima, tem-se o nome do objeto e o *offset* aplicado à efeméride (ver seção 2.2). No rodapé são apresentadas as informações, na sequencia: data e horário do C/A, coordenadas geocêntricas J2000 da estrela, C/A em segundos de arco, ângulo de posição em grau, velocidade da sombra em km/s, a distância do objeto ao observador em unidades astronômicas, as magnitudes R* e K* (ver Eq. 2.5) e a longitude do ponto sub-planetário em graus.

2.1 Catálogos estelares

Encontram-se disponíveis em bases de dados, como o VizieR (Ochsenbein et al. 2000), dezenas de catálogo estelares que fornecem posições astrométricas para estrelas de todo o céu. Destacam-se catálogos com estrelas brilhantes: HIPPARCOS (Perryman et al. 1997) e TYCHO-2 (Høg et al. 2000); ou catálogos mais densos em número de estrelas: 2MASS (Skrutskie et al. 2006) e USNO B1.0 (Monet et al. 2003); ou ainda aqueles atrometricamente mais precisos: UCAC2 (Zacharias et al. 2004a), recentemente atualizado para UCAC4 (Zacharias et al. 2013). Estes catálogos apresentam precisões astrométricas e quantidades de estrelas muito diferentes (ver Tabela 2.1), fatores que são muito importantes quando se considera a busca por estrelas candidatas à serem ocultadas.

Catálogo	N^o estrelas	mov. próprio	Mag. limite	precisão astrométrica
HIPPARCOS	118218	sim	9	1 mas
TYCHO-2	2.5×10^6	sim	12	32 mas
2MASS	$4{,}71 \times 10^8$	não	16*	100 mas
USNO B1.0	$1{,}05 \times 10^9$	sim	21	200 mas
UCAC4	$1{,}13 \times 10^8$	sim	16	40 mas

Tabela 2.1 Alguns dos principais catálogos astrométricos, com a quantidade de estrelas catalogadas, se fornecem movimento próprio ou não, qual o limite em magnitude V e a precisão astrométrica média. *O catálogo 2MASS refere-se à magnitudes no infravermelho e o limite dado é para a banda J (1.25 μm).

Uma vez que um evento é identificado, novas medidas astrométricas da estrela são necessárias para melhor determinar sua posição, já que erros da própria medida do catálogo e na determinação do seu movimento próprio são facilmente superiores à precisão exigida para uma boa predição de ocultação por TNO (>50 mas).

Catálogos poucos densos, somados ao pequeno tamanho angular e a baixa velocidade de deslocamento no céu dos TNOs, levarão à poucas ou nenhuma ocultação por ano. Catálogos densos mas com baixa precisão astrométrica, levarão à muitos falsos eventos, o que implica desperdício de esforço observacional, e de dinheiro de financiamentos para viagens e equipamentos.

Para superar este problema fizemos, durante um período de colaboração entre o Observatório Nacional e o ESO (*European Southern Observatory*), pedidos de tempo do telescópio de 2,2m com do ESO/MPG[1], equipado com a câmera WFI (Wied Field Imager). Esta câmera, composta por um mosaico de oito CCDs, tem um campo final no plano do céu de 30' × 30' com escala de placa de 0",238. Com este equipamento, observamos o caminho que o sistema de Plutão, Éris, Haumea, Makemake, Quaoar, Varuna, Sedna, Ixion, Orcus, 2003 AZ_{84} e 2002 TX_{300} farão no céu de 2008 a 2015 (Figura 2.3). Estes

[1]European Southern Observatory/Max-Planck-Gelsellschaft

objetos foram selecionados entre os maiores e cientificamente mais relevantes na época do pedido.

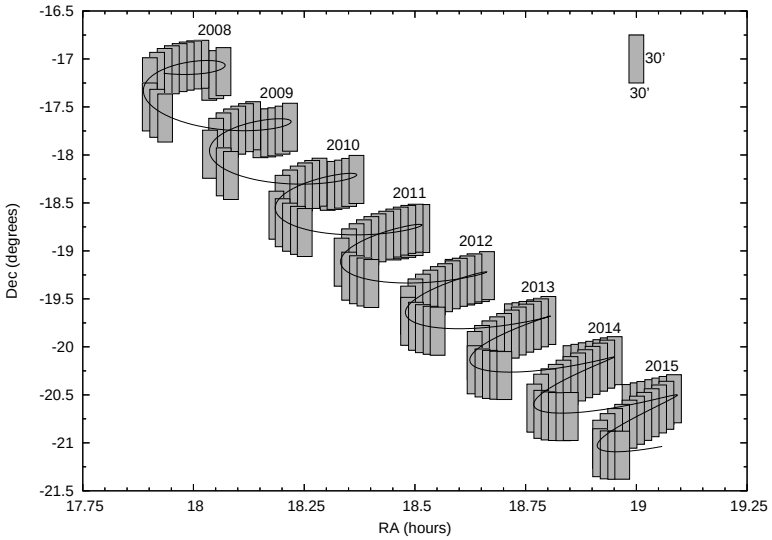


Figura 2.3 Caminho de Plutão entre 2008 e 2015 (AR, DEC) como exemplo de caminho observado com o telescópio do ESO 2,2 m para 11 objetos. Os retângulos representam o campo de 30' × 30' da camera WFI, efetivamente observados. A linha contínua representa o movimento de Plutão no céu ao longo do período.

Estas observações se estenderam de 2007 a 2009 (programas: 077.C-0283, 075.C-0154, 079.A-9202, 079.C-0354) com mais de 7,5 × 10^6 estrelas identificadas, com magnitudes R entre 11 e 21. As observações foram feitas em filtro R de banda larga (λ = 651,7 nm, $\Delta\lambda$ = 162,2 nm). A posição de cada estrela foi determinada tendo o catálogo UCAC2 como referência, e apresentam erros de 30 mas para estrelas entre magnitudes R 11 e 18. Os movimentos próprios foram utilizados diretamente do catálogo UCAC2, quando disponível, ou calculados tendo as posições dadas pelos catálogo 2MASS ou USNO B1, nesta ordem de preferência, como posições de primeira época. Mais detalhes sobre estas observações podem ser encontrados nas publicações de Assafin et al. (2010) & Assafin et al. (2012).

Isto nos permitiu gerar catálogos estelares para cada objeto com posições de alta precisão astrométrica (<30 mas) e bastante densos. Estes catálogos serão referidos ao longo do texto como catálogos WFI. Por consequência as predições de ocultações geradas à partir deles são mais confiáveis, o que permite que concentremos nossos esforços sobre eventos com maior potencial de serem efetivamente observados.

Desde que o Brasil se tornou membro[2] do ESO, pedimos e fomos contemplados com

[2]A adesão oficial ainda não foi sancionada! Mas tempo de telescópio ja são concedidos aos brasileiros.

mais tempo no telescópio Max-Planc 2.2 m para observarmos o caminho de 2012 a 2014, de mais 39 TNOs e centauros (programas: 088.C-0434, 089.C-0356, 090.C-0118 e 091.C-0454). Estas observações já foram realizadas e reduzidas, e as predições de ocultações por estes objetos já estão disponíveis (Camargo et al. 2013). No decorrer do texto os programas de predição de ocultação publicados em Assafin et al. (2010) & Assafin et al. (2012) e (Camargo et al. 2013) serão referenciados como caminhos.

2.2 Órbitas

Desde a descoberta do primeiro objeto transnetuniano (15760) 1992 QB_1 (Jewitt & Luu 1993) (excluindo Plutão), pouco mais de 1260 corpos deste tipo já foram descobertos, sendo a maioria entre os anos 1999 a 2006. A recente descoberta, ligada ao período orbital maior do que 200 anos, faz com que arcos individuais não maiores do que 10% das órbitas sejam conhecidos. Além disso, devido ao seu tamanho e distância do Sol, suas magnitudes V aparente são superiores a 18 dificultando sua observação, salvo poucas exceções.

Por estes fatores, as órbitas dos TNOs apresentam erros que podem chegar a 500 mas, ou seja, mais de dez vezes superiores à precisão necessária para uma boa predição de ocultação. Para melhor compreensão do que isto representa, uma incerteza de 100 mas sobre um objeto a 45 ua, projetados sobre a Terra, representa uma incerteza de 3300 km sobre a posição da sombra.

O caso mais conhecido é a deriva de Plutão em relação à sua efeméride desde 2005 (Assafin et al. 2010). Quando um TNO oculta uma estrela, do ajuste das curvas de luz de ocultação observadas é possível obter a posição do TNO relativa à estrela com precisão abaixo de 1 mas. Assim, à medida que ocultações estelares por Plutão foram observadas com sucesso, foi possível verificar que, a partir de 2005, sua posição real se desvia numa taxa de aproximadamente 30 mas/ano em declinação (DEC) da posição prevista pela efeméride DE413 (disponível pelo sistema HORIZONS/NASA: (Giorgini et al. 1997)). Hoje em dia, a diferença entre a efeméride e a posição real em declinação já chega a quase 200 mas. Com observações de 2005 a 2012 a deriva linear em declinação é muito clara (Figura 2.4). Esta deriva pode estar ligada à efeito de refração cromática não corrigidas nas medidas astrométricas. Em ascensão reta (AR) observa-se uma oscilação conforme a distância do evento à oposição. Uma hipóteses é que um erro na distância heliocêntrica poderia provocar tal efeito.

Com o exemplo acima, fica claro que, ao se fazer predições de ocultações estelares, é preciso considerar o erro da efeméride do objeto. Infelizmente, apenas Plutão já foi observado em diversos eventos de ocultação que permitem verificar tal deriva. Para os demais objetos, a estratégia usada para contornar o problema do erro de suas efemérides,

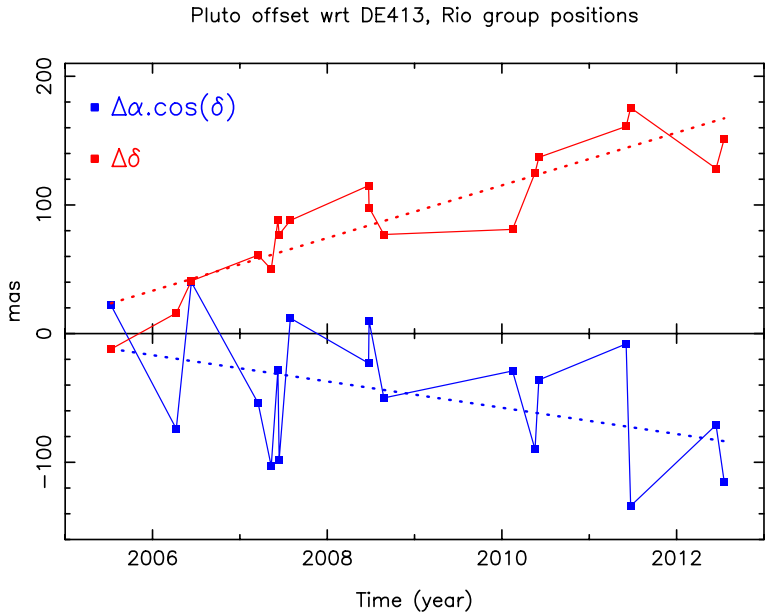


Figura 2.4 Deriva linear da posição observada de Plutão em relação à efeméride DE413. Em declinação a deriva é de aproximadamente 30 mas/ano. Em ascensão reta o efeito é menos pronunciado (Assafin et al. 2010).

consiste em observa-los para medir suas posições astrométricas e então determinar a diferença entre esta posição medida e a prevista pela efeméride (do inglês, *offset*). Aplica-se o *offset* medido à sua efeméride para então obter as predições de ocultações.

Os *offset* utilizados em nossas predições vêm principalmente de observações feitas no Observatório do Pico dos Dias (OPD) que pertence ao Laboratório Nacional de Astrofísica (LNA) e no telescópio Max-Planc 2,2 m do ESO.

Para os novos 39 TNOs e centauros para os quais observamos seus caminhos até 2014, obtivemos mais tempo no telescópio Max-Planc 2,2 m para determinar seus *offset* (Camargo et al. 2013) (programas: 090.C-0118, 091.C-0454). Parte destes objetos já foi observada no segundo semestre de 2012 e o restante será observado no primeiro semestre de 2013.

Capítulo 3

Seleção, Atualizações e Campanhas

A predição de ocultações é apenas o passo inicial para se atingir o objetivo principal do trabalho, medir o tamanho e forma, e deles derivar diversos parâmetros físicos dos TNOs. Todas as predições dos programas dos caminhos são disponibilizadas na internet, acessíveis pela página de Bruno Sicardy: `http://www.lesia.obspm.fr/perso/bruno-sicardy/`. Obviamente não é possível observar todos os eventos previstos, uma vez que cada observação demanda um grande esforço logístico na mobilização de pessoal, preparação de observadores, telescópios, instrumentos, que reflete também no uso de grandes montantes de recursos financeiros.

3.1 Seleção

Das centenas de mapas de predição de ocultações gerados, é preciso selecionar quais os eventos mais favoráveis à serem efetivamente observados. Alguns parâmetros disponíveis nos mapas podem ser utilizados, mas há ainda fatores como maior interesse científico, raridade do evento, rede de colaboradores e o clima.

Dos 10 objetos (mais Plutão) para os quais observamos seu caminho no céu até 2015, alguns tem como fundo estelar o plano da galáxia, e, portanto, há maior probabilidade de ocultarem uma estrela. No entanto, alguns objetos, mesmo estando em regiões mais desprovidas de estrelas que significa uma baixa taxa de ocultações por ano, foram selecionados por seu interesse científico, à exemplo de Éris, Haumea, Makemake e Sedna. Isto significa que seus eventos serão raros mas bastante importantes. Portanto, mesmo que as condições não sejam ideais, um grande esforço observacional será feito na busca da detecção do evento.

A quantidade de colaboradores e a facilidade de acesso à telescópios distribuídos pelo globo é bastante heterogênea (Figura 3.1). Por exemplo, eventos que passem sobre o Chile, Europa e Estados Unidos, têm mais chace de serem observados do que eventos que

cruzem a Africa Central ou a Rússia. Então eventos que exijam telescópios maiores e com maior incerteza só serão observados se passarem por uma região favorável. No entanto, se o evento for muito raro ou envolver uma estrela brilhante (mag. V $<$ 12), aventuramos a utilização de telescópios em quase qualquer parte do globo.

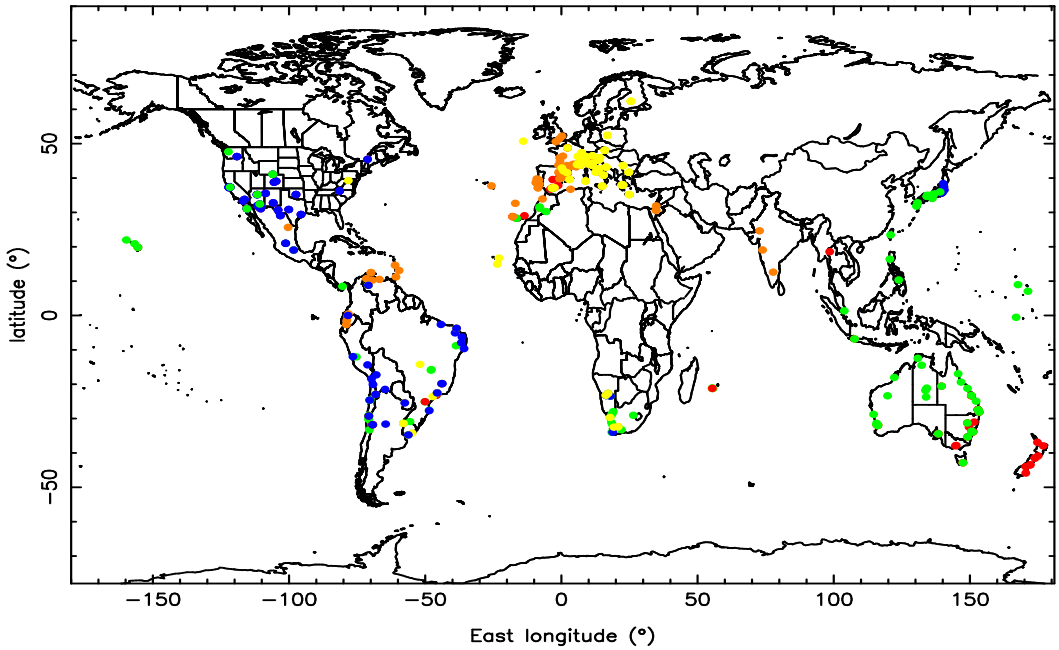


Figura 3.1 Mapa com todos os sítios que colaboram com nosso grupo regularmente ou já colaboraram em uma ou mais campanhas observacionais de ocultações estelares. A densidade de colaboradores pelo globo é bastante heterogênea e composta por profissionais e amadores. As diferentes cores representam diferentes campanhas, mas em geral um mesmo sítio já colaborou em diversas campanhas.

Nos mapas de predição (Fig. 2.2) são fornecidas as magnitudes normalizadas R* e K* (ver equação 2.5), estes dois valores nos dão uma ideia dos tamanhos de telescópios que podem ser utilizados numa possível campanha observacional. Quanto mais brilhante a estrela, maior o número de telescópios que podem ser utilizados, uma vez que telescópios da ordem de 0,25 m de diâmetro de espelho são muito mais numerosos que telescópios de 0,5 m ou maiores. A diferença entre as magnitudes R* e K* também podem nos dizer se uma estrela é bastante avermelhada. Isto costuma ser uma vantagem para muitos CCDs, já que são mais sensíveis na parte vermelha do espectro.

Eventos mais lentos permitem maiores tempos de exposição. Por exemplo, uma ocultação por um objeto de 900 km a uma velocidade típica de 20 km/s pode durar até 45 segundos, mas há eventos que podem durar até mais de 150 segundos. Assim, CCDs que sejam menos sensíveis e que tenham o tempo de descarga da imagem maiores, que são mais comuns em telescópios pequenos, podem ser utilizados para a observação.

Outro fator importante é a possibilidade de atualizações da predição. Como será

discutido a seguir, esta é uma etapa fundamental da predição de ocultações. Para isto verifica-se a elongação do Sol. Se ela for pequena e a leste, significa que não será possível observar o corpo e a estrela a ser ocultada, nos meses que precedem o evento. Então, caso não exista boa determinação do *offset* e/ou da posição da estrela, não será possível lançar grandes campanhas observacionais.

É muito comum utilizar-se a queda de magnitude durante a ocultação como um parâmetro de seleção. A queda em magnitude está relacionada à precisão fotométrica necessária na observação, isto é, à razão sinal sobre ruído (S/R). Quanto menor a queda maior a precisão fotométrica necessária, e, por consequência, maior o tempo de exposição necessário para um dado equipamento. No entanto, no caso contrário, quanto maior a queda mais fácil é a observação do evento, já que um S/R baixo é suficiente para sua detecção. Como quase todos os TNOs têm magnitudes R superiores a 18, e as estrelas acessíveis à maioria dos telescópios têm mag. R < 18, a queda de magnitude é, em geral, superior a um. Por exemplo, para um baixo S/R$= 5$, uma queda de uma magnitude já seria facilmente detectável. Portanto, este não é um parâmetro muito relevante para a seleção de eventos envolvendo os TNOs. O mesmo não é verdade quando a ocultação é por Plutão, já que este tem magnitude R$\sim$14,5.

Para o cálculo da queda de magnitude Δ_{mag}, utiliza-se a magnitude do objeto ocultador m_p e da estrela ocultada m_s relacionados na equação 3.1. Uma relação aproximada entre a precisão em magnitude σ_{mag} e o sinal sobre ruído é dada pela equação 3.2. Não há dúvida que maiores valores de S/R trazem resultados mais precisos. Por exemplo, para detecção de atmosferas muito tênues, alta frequência de imagens e alto S/R são fundamentais. Entretanto, para a determinação do tamanho e forma dos objetos, precisamos do maior número de detecções positivas em sítios diferentes. Então, não é incomum haver uma grande heterogeneidade de S/R entre os dados utilizados para a determinação do perfil de um objeto.

$$\Delta_{mag} = m_p - m_s + (2,5 \cdot (\log_{10}(10^{(m_s - m_p)\cdot 0,4} + 1))) \tag{3.1}$$

$$\sigma_{mag} \approx 1,0857 \cdot \left(\frac{S}{R}\right)^{-1} \tag{3.2}$$

A seleção dos eventos mais favoráveis é feita ao final de cada ano para o ano subsequente, e disponibilizada em um site de internet com acesso restrito aos colaboradores para o pedido de tempo em telescópios (`http://devel2.linea.gov.br/~braga.ribas/selec`). Desde 2010 mais de 450 eventos já foram selecionados, dos quais cerca de 12% foram efetivamente observados e uma parcela menor ainda foi detectada.

3.2 Atualizações e Campanhas

3.2.1 Atualizações

Como já foi dito, a atualização das predições é uma parte muito importante do trabalho pré-ocultação. Má determinação de movimento próprio da estrela e do *offset* do objeto são os fatores que justificam este procedimento.

Quando uma predição é feita com muitos anos de antecedência é preciso aplicar o movimento próprio para determinar a posição da estrela. Para o programa dos caminhos, o movimento próprio aplicado é oriundo do catálogo UCAC2, ou foi calculado tendo os catálogos 2MASS ou USNO B1 como primeira época. Estes movimentos próprios contêm imprecisões que, propagadas à data da ocultação, podem ser superiores à precisão exigida para uma boa predição de ocultação, isto é, o diâmetro angular do objeto (~ 30 mas). A melhor maneira de superar este problema é realizar observações astrométricas da estrela próximo à época do evento.

Como já discutido, as efemérides costumam apresentar *offset* e até mesmo deriva ao decorrer do tempo em relação à posição real do objeto, que podem chegar a centenas de mas. Então, da mesma forma que as estrelas candidatas, os objetos ocultadores também precisam de observações astrométricas próximo à data do evento. As posições medidas são comparadas à efeméride para se obter o *offset*, que será propagado linearmente para a data da ocultação.

Considerando uma velocidade típica de um minuto de grau por dia, cerca de uma semana antes do evento é possível observar o TNO e a estrela candidata numa mesma imagem obtida com telescópios que tenham campo da ordem de 15' $\times$ 15' no plano do céu. É neste momento que normalmente se consegue as predições mais precisas, já que obtém-se astrometria relativa entre os objetos, isto é, erros sistemáticos de catálogo são cancelados. Nestas condições pode-se chegar à posições relativas com erro menor que 30 mas, ou seja, aproximadamente o diâmetro aparente de um TNO.

Vários telescópios, como o 1 m do Pic du Midi (Obs. Paris / IMCCE), 1,2 m Calar Alto (IAA), 1,5 Sierra Nevada, 1,6 m e 0,6 m Observatório do Pico dos Dias (Laboratório Nacional de Astrofísica), são utilizados por colaboradores nesta fase, e as medições astrométricas feitas com a "Plataforma para Redução Automática de Imagens Astronômicas" (PRAIA) por nossa equipe (Assafin et al. 2011) e também por colaboradores como Dr. Raoul Behrend do Observatório de Zurique / Suiça.

3.2.2 Campanhas

Uma vez com as novas medidas, novos mapas de predição são gerados (Fig. 2.2). Se as condições da nova predição forem favoráveis, lança-se um alerta a possíveis observadores. Contactamos observatórios profissionais e amadores pedindo para observarem o evento. Em algumas oportunidades, câmeras especiais são levadas aos telescópios que não tenham equipamento adequado.

Como muitos eventos só podem ser confirmados com alguns dias de antecedência, nem sempre é possível obter telescópios profissionais de grande porte. Uma alternativa é solicitar uma quantidade de tempo de alvo de oportunidade (ToO) durante o período normal de pedidos e usá-lo quando um evento é confirmado. Assim o fizemos para os telescópios VLT 8 m (Very Large Telescope) e NTT 3.5 m (New Technology Telescope) do ESO, recebendo 3 horas/semestre em cada instrumento para os semestres 2009a, 2009b, 2010a, 2011a, 2012b. O mesmo vem sendo feito desde 2011 com telescópios de Calar Alto 1,2m, 0,77 em La Hita, 1,5 m Sierra Nevada, 3,6 m TNG (Telescopio Nazionale Galileo) em La Palma Ilhas Canárias. No caso de não termos tempo assegurado em telescópios profissionais em localização favorável, fazemos pedido de tempo de diretor (DDT, *Director's Discretionary Time*).

Para cada campanha uma página internet é aberta (veja figura 3.2 como exemplo), contendo informações importantes para a observação, cartas de campo para a localização do alvo e os mapas de predição. Desde 2010 mais de 40 campanhas já foram realizadas e as informações disponibilizadas nos sítios: `http://www.lesia.obspm.fr/perso/braga-ribas/list` para os eventos de 2011, `http://www.lesia.obspm.fr/perso/braga-ribas/campaigns` para os eventos de 2012 e `http://dev.linea.gov.br/~braga.ribas/campaigns` para 2013.

3.3 Observação

A ocultação estelar é um evento fotométrico e temporal, isto é, mede-se a variação da quantidade de luz recebida da estrela ao longo do tempo, o que chamamos de curva de luz de ocultação, a exemplo da figura 5.4. Quanto maior a frequência de imagens, melhor a precisão, em quilômetros, sobre o tamanho do objeto. Normalmente não se utiliza filtro, uma vez que precisa-se da maior quantidade de fótons possíveis num tempo curto de exposição. Em circunstâncias de Lua próxima ou poluição luminosa intensa, o emprego de filtro vermelho (R ou I) pode melhorar o contraste entre a estrela ocultada e o ruído do céu. Ou ainda, no caso da existência de brumas na atmosfera do objeto ocultador, a forma da curva de luz pode ser diferenciada caso a observação seja feita com filtros

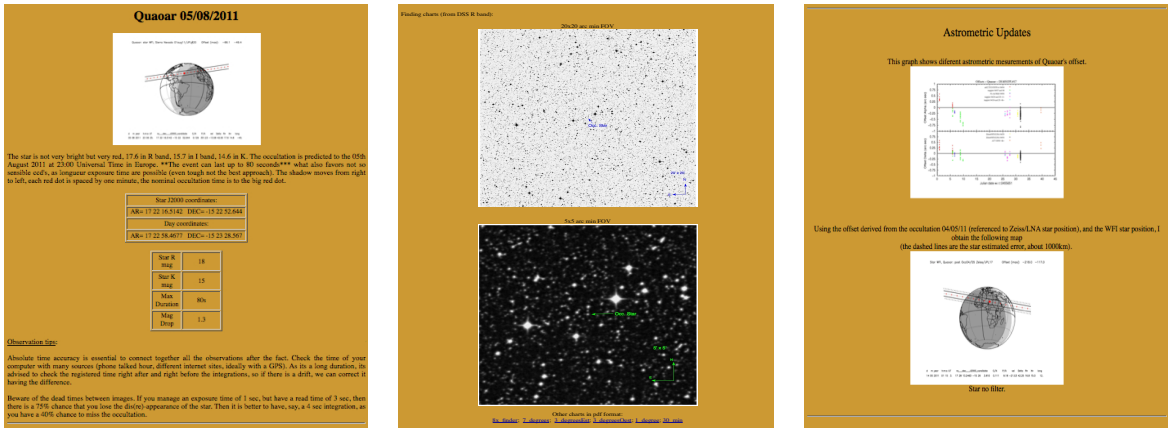


Figura 3.2 Exemplo de página internet de uma campanha para a observação de uma ocultação. Varias informações são dadas aos colaboradores como: o mapa de predição, as coordenadas da estrela J2000 e "do dia", magnitudes R e K da estrela, queda de magnitude prevista, motivação, dicas de observação, cartas de campo para busca do alvo e referências e mapas de predição atualizados. (Acessado em 26/01/2013: `http://www.lesia.obspm.fr/perso/braga-ribas/list/2011/2011_08_05_Quaoar.html`)

diferentes (Sicardy et al. 2006b). Até hoje, somente Plutão e Tritão[1] são conhecidos por terem atmosfera no Cinturão de Kuiper.

O objetivo é determinar o tamanho do objeto com precisão de alguns quilômetros, utilizando para isto observações de sítios diferentes. Então, o registro do tempo absoluto por cada telescópio é essencial. Para tanto, utiliza-se antenas GPS (Global Positioning System) que fornecem o tempo absoluto com precisão melhor que o milésimo de segundo. Caso uma antena GPS não esteja disponível, outras fontes de tempo absoluto são empregadas, como serviços NTP (Network Time Protocol), internet ou rádio.

Em boas condições de observação, isto é, céu escuro e limpo, e uma câmera CCD sensível e rápida, estrelas de magnitude R<18 podem ser observadas por telescópios a partir de 0,5 m de abertura; estrelas de magnitude R<16,5 são acessíveis a telescópios de 0,3 m, estrelas de mag. R<15 podem ser registradas por espelhos de 0,25 m, pequenos telescópios de 0,15 m podem detectar estrelas de mag. R<12 e, a partir de mag. R<11, os eventos podem ser observados visualmente com um telescópio de ao menos 0,2 m. No programa caminhos (Seção 2.1), entre 2009 e 2015, há apenas 6 ocultações previstas para todo o globo que envolvam estrelas de magnitudes entre 11 e 13, para magnitudes entre 13 e 15 são 34 eventos, entre magnitudes 15 e 17 são 174 e acima de magnitude 17 são milhares (Figura 3.4).

[1]Tritão, satélite de Netuno, foi provavelmente um TNO capturado por Netuno no período da migração planetária (McKinnon & Leith (1995) e Nogueira et al. (2011)).

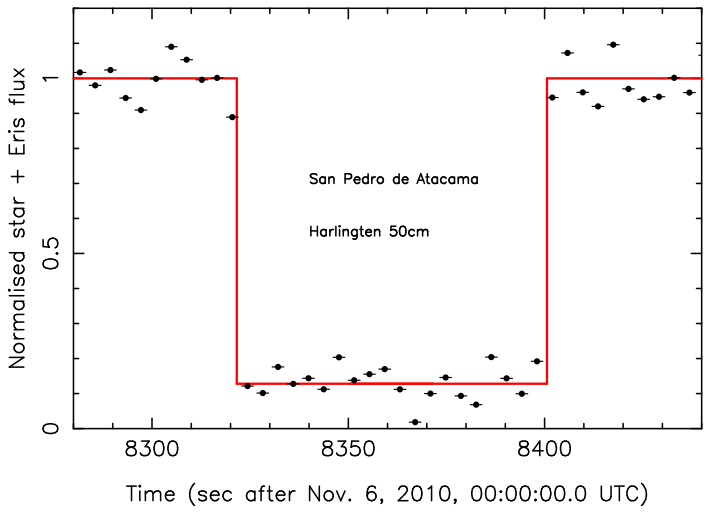


Figura 3.3 Exemplo de curva de luz obtida da ocultação por Éris observada em 06 de novembro de 2010 (Sicardy et al. 2011). Os pontos pretos são o fluxo medido em cada imagem, os segmentos de traços sobre eles representam o tempo de integração de cada imagem, o traço vermelho é a modelagem geométrica da ocultação.

3.3.1 Câmeras

Como as ocultações são eventos rápidos que necessitam de imagens sequenciais, câmeras especiais são necessárias. Infelizmente as câmeras mais comumente encontradas nos observatórios não são adequadas por diversos fatores. O principal deles é a existência de um intervalo entre cada integração. Este intervalo, conhecido como tempo-morto, é o tempo que a câmera CCD precisa para descarregar a imagem para a memória e salvá-la em disco. Isto representa uma real perda de informação, já que durante este intervalo não temos nenhuma informação sobre a possível variação do fluxo da estrela. Se, por exemplo, a ocultação se iniciar durante o tempo-morto, haverá uma incerteza sobre o início do evento igual ao tempo-morto. Se, por exemplo, o tempo-morto for de 2 segundos num evento a 20 km/s, a incerteza será de até 40 km sobre o limbo do objeto.

Algumas ações podem diminuir o tempo-morto, como, por exemplo, binar o CCD ou selecionar apenas uma parte da matriz, ou ainda selecionar um modo de leitura rápido. Mas, como ele é inevitável em algumas câmeras, procura-se fazer o tempo de exposição ao menos duas vezes mais longo que o tempo-morto, assim ele representará não mais que 33% do tempo total do ciclo (integração + tempo-morto), diminuindo a probabilidade do evento iniciar ou terminar durante tal intervalo.

Uma câmera que tem sido bastante utilizada, por ter sido concebida para as necessidades observacionais de ocultações estelares, é a Raptor/247 Merlin. Foi desenvolvida pela empresa *Raptor photonics* em conjunto com pesquisadores do *Observatoire de Paris-Meudon* Com um sensor EM-CCD (*Electron Multiplying - Charging Coupled Device*) de

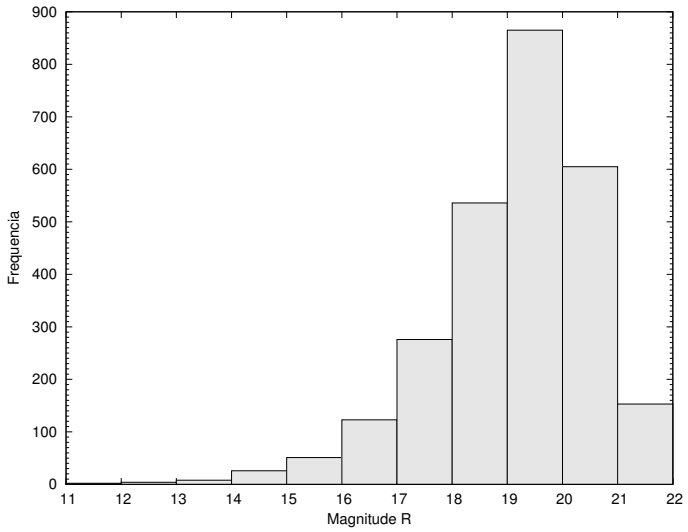


Figura 3.4 Histograma mostrando a quantidade de eventos previstos pelo programa caminhos por magnitude R entre os anos 2009 e 2015. Há apenas 2 eventos envolvendo estrelas de magnitude 11 a 12 e 4 envolvendo estrelas de mag. 12 a 13. Como o catálogo é completo até magnitude 19, há uma queda em quantidade de eventos a partir de magnitude 20.

alta sensibilidade e ruído de leitura menor que 1 elétron, opera em *frame transfer*, o que permite ter tempo-morto desprezível. Com dimensões menores que 9 cm e menos de 400 g pode ser facilmente transportada e instalada em qualquer tipo de telescópio. O programa de aquisição de imagem também foi otimizado para a observação astronômica. Por exemplo, registra no cabeçalho de cada imagem o exato instante em que a imagem foi tomada e permite visualizar a imagem em tempo real.



Figura 3.5 O conjunto da câmera Raptor/Merlin à direita e alguns acessórios, como a antena GPS à esquerda, fundamental para o registro dos instantes corretos

Outra alternativa é a utilização de câmeras de vídeo de alta sensibilidade. Alguns modelos, como a Watec - 120N+, permitem o controle do tempo de integração pelo usuário. Elas precisam de um sistema paralelo que escreve o horário proveniente de uma antena GPS diretamente sobre a imagem (ITV, Incrustador de Tempo em Vídeo). As imagens precisam ser digitalizadas para pós-processamento e fotometria. O pesquisador Fréderic Vachier do *Observatoire de Paris / IMCCE* desenvolveu, o programa "Aquisição e Tratamentos de Ocultações Estelares"(ATOS, do francês *Acquisition et Traitement des Occultations Stellaires*), no qual colaborei, para implementação do ajuste das curvas de luz para obtenção dos instantes de início e fim de evento.

3.4 Redução

3.4.1 Fotometria

A calibração de imagens obtidas para ocultação é feita de maneira tradicional, retirado-se *bias* e *corrente de escuro* e dividindo pelo $flat - field$. O fluxo da estrela + objeto é medido com fotometria de abertura. A abertura deve ser otimizada para o momento da ocultação, já que minutos antes/depois os objetos podem não estar totalmente fundidos no plano do céu, havendo perda de fluxo medido. Devido às variações de transparência do céu, utiliza-se estrelas próximas como referência, obtendo fotometria diferencial de abertura. Todas as curvas de luz, obtidas nos diferentes sítios, são posteriormente normalizadas ao nível do fluxo da estrela + objeto nos períodos anterior e posterior à ocultação, aplicando-se um polinômio de primeiro ou segundo grau aos dados.

Devido à diferente resposta espectral dos sensores ou ao emprego de filtros, a queda de magnitude será diferente para cada equipamento. Para calcular a razão entre os fluxos dos dois objetos e então a queda de magnitude para aquele instrumento, imagens anteriores e/ou posteriores ao evento são obtidas, de tal maneira que o ocultador e a estrela ocultada possam ter seus fluxos medidos separadamente. Estas imagens devem ser tomadas preferencialmente à mesma altura do evento, já que o índice de cor dos objetos pode ser diferente. Neste caso, a absorção cromática diferencial pode dar origem a razões de fluxo diferentes para alturas diferentes. Este procedimento também é essencial para os casos de objetos com atmosfera. Devido à refração na atmosfera do corpo, recebe-se luz residual da estrela durante todo o evento, e este fluxo nos trará informação sobre a pressão da atmosfera e gradiente de temperatura próximos à superfície do objeto (Sicardy et al. 2006b). Então é essencial conhecer precisamente a razão de fluxo entre o corpo e a estrela.

3.4.2 Instantes da ocultação

O instante médio de cada imagem é lido diretamente do cabeçalho da imagem juntamente com o procedimento de fotometria. No caso de observação por vídeo, o tempo é lido diretamente da imagem. Um procedimento fundamental para o registro de uma ocultação é referenciar o horário registrado à uma fonte segura de tempo, preferencialmente via GPS. Nem sempre este é o caso, e por vezes programas que buscam o horário via NTP são utilizados.

Alguns programas de aquisição, como o comumente utilizado *MaxImDL*, truncam o horário registrado, guardando apenas a parte inteira do segundo (Sicardy et al. 2011). Recuperar a fração de segundo é possível, e para isto ajusta-se uma reta aos instantes de tempo $(i, t_{h,i})$, sendo i o número da imagem e $t_{h,i}$ o instante escrito no cabeçalho. Os resíduos deste ajuste mostram um padrão de serra com dispersão entre -0,5 e +0,5 segundos, mostrando que há truncagem e, mostrando também, o período em que o ciclo de aquisição é regular (Figura 3.6). Deste ajuste obtém-se o ciclo de aquisição t_c, o qual é utilizado para obter o instante de cada imagem t_{fit}, agora com a fração do segundo, e um instante inicial t_0 (Equação 3.3). Meio segundo é adicionado aos instantes para corrigir o deslocamento imposto pela truncagem. Com este procedimento os instantes de cada imagem tem uma precisão interna que dependerá da raiz do número de images utilizas e será inferior ao segundos.

$$t_{fit} = t_0 + (t_c \cdot i) + 0,5 \tag{3.3}$$

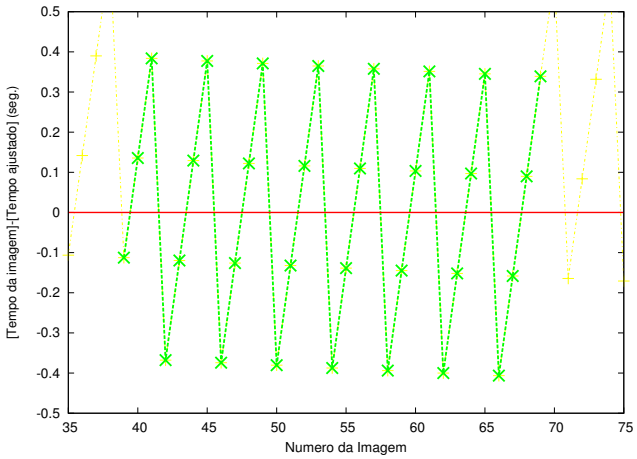


Figura 3.6 A fração de segundo pode ser recuperada quando o programa de aquisição trunca, no segundo, o instante escrito no cabeçalho. A linha tracejada verde é um ajuste linear aos pontos de onde se obtém o tempo de ciclo. Para este ajuste utilizou-se os pontos no entorno do evento, durante um período em que a aquisição foi regular. Os pontos amarelos são os instantes derivados para o restante das imagem a partir do ajuste.

Os instantes de início e fim da ocultação são obtidos das curvas de luz, ajustando aos dados um modelo geométrico de um corpo sem atmosfera ocultando uma estrela pontual (Figura 3.7). O modelo é então convolvido com: o padrão de difração de Fresnel de uma fonte pontual à distância do objeto; o diâmetro da estrela; a banda em comprimento de onda utilizado; e o tempo de integração (Widemann et al. 2009). Os programas utilizados para obter os resultados apresentados adiante, foram escritos por Bruno Sicardy.

A difração de Fresnel é calculada para uma fonte pontual à distância do objeto ocultador como descrito em (Roques et al. 1987). O tamanho da principal franja de difração pode ser calculado com a equação de escala de Fresnel $L_f = \sqrt{\lambda D/2}$, onde λ é o comprimento de onda da observação e D a distância ao objeto. O diâmetro angular da estrela no plano do céu (Θ_V e Θ_B), pode ser estimado como descrito por van Belle (1999) utilizando as magnitudes aparente da estrela nas bandas B ou V e K (Equações 3.4).

$$\begin{aligned} \Theta_V &= 10^{0,669+0,223\cdot(V-K)} \\ \\ \Theta_B &= 10^{0,648+0,220\cdot(B-K)} \end{aligned} \tag{3.4}$$

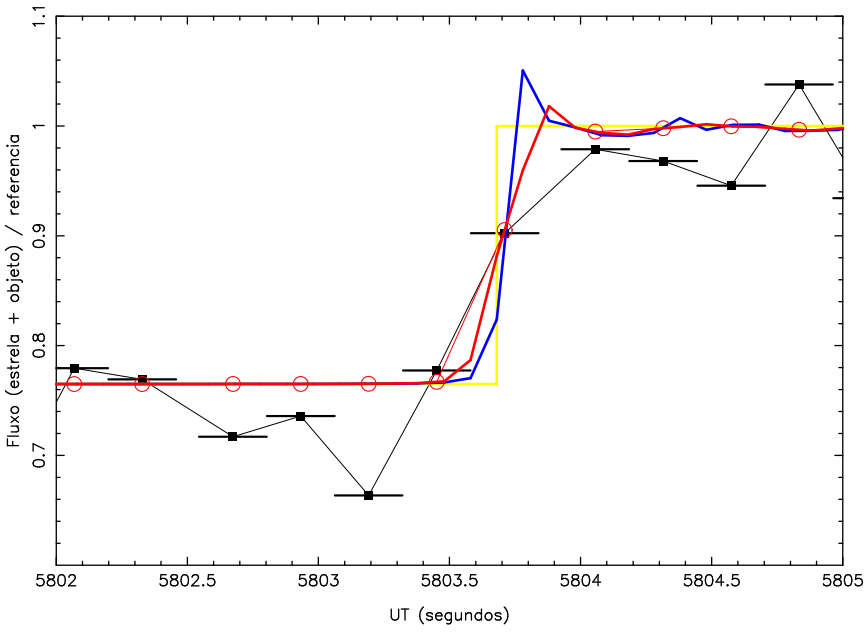


Figura 3.7 Determinação do instante de ocultação. Dados da ocultação por Makemake em 24 de abril de 2011 (Ortiz et al. 2012). O modelo geométrico (linha amarela) é convolvido à difração de Fresnel, ao tamanho da estrela, à banda espectral da observação (linha azul) e ao tempo de integração (segmentos pretos). A curva sintética resultante (linha e círculos vermelhos) é comparada ao dados (pontos pretos) calculando-se o χ^2_{min}.

O modelo é comparado aos dados em uma série de instantes i para N pontos no entorno de um instante dado (início ou fim do evento). O valor de χ^2 é obtido comparando o fluxo observado e calculado ($\Phi_{\mathsf{i,obs}}$ e $\Phi_{\mathsf{i,cal}}$, respectivamente) sobre o quadrado da dispersão dos dados (σ_i^2, equação 3.5). O valor mínimo χ^2_{min} é tomado como o instante do início/fim da ocultação. As barras de erro a 1σ são calculadas pela variação de χ^2_{min} a $\chi^2_{min} + 1$, e para 3σ até $\chi^2_{min} + 9$.

$$\chi^2 = \sum_{1}^{\mathsf{N}} (\Phi_{\mathsf{i,obs}} - \Phi_{\mathsf{i,cal}})^2 / \sigma_{\mathsf{i}}^2 \tag{3.5}$$

Capítulo 4

Propriedades físicas de Quaoar a partir de ocultações estelares

A partir da detecção positiva de uma ocultação, obtém-se o que chamamos de "corda". Cada curva de luz é de fato a medida de uma latitude do objeto, e a duração da corda dependerá da posição relativa observador - ocultador - estrela e do tamanho do corpo. Com a observação simultânea de vários locais, obtém-se o perfil instantâneo do objeto. As cordas são combinadas corrigindo sua posição do efeito paralático da posição de cada sítio (Figura 4.1).

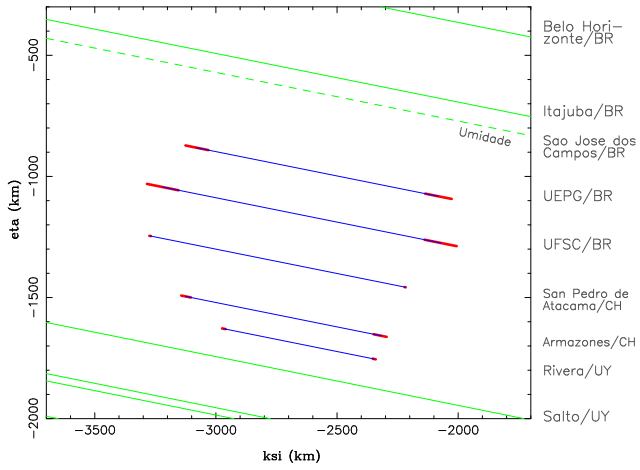


Figura 4.1 Cordas observadas na ocultação de Quaoar de 04 de maio de 2011. Cada observatório obtém uma medida do perfil do objeto numa dada latitude (linhas azuis), e ao recombina-las obtém-se a forma do corpo. À direita os nomes dos observatórios. Os segmentos vermelhos correspondem à barra de erro da determinação do instante de início/fim do evento. As linhas verdes correspondem à observatório que obtiveram dados mas não detectaram o evento. Em São José dos Campos (linha tracejada), alta umidade impediu a observação do evento.

Uma vez as cordas recombinadas, considera-se uma elipse como o caso mais geral para descrever o perfil do objeto. Esta elipse será ajustada às cordas para então obter-se os parâmetros: tamanho, elipticidade, centro e ângulo de posição do corpo.

Outro parâmetro que pode-se calcular à partir da solução do ajuste de uma elipse é o raio equivalente (R_{equiv}) dado pela equação 4.1. Ele corresponde ao raio de um círculo com mesma área superficial de uma elipse de semi-eixo maior a e semi-eixo menor c. Outros métodos de determinação de tamanhos de objetos celestes costumam apresentar o raio equivalente como resultado. Alguns destes métodos, como o utilizado pelos telescópios espaciais *Spitzer* e *Herschel* (Stansberry et al. (2008) e Santos-Sanz et al. (2012)), baseiam-se sobre o fluxo observado do objeto, portanto não podem medir sua elipticidade. Apresentam então o raio que um objeto circular teria conforme o fluxo medido, o raio equivalente.

$$R_{equiv} = \sqrt{a \cdot c} \tag{4.1}$$

Para discussão da metodologia de obtenção dos parâmetros físicos do objeto e as conclusões deles derivadas, apresta-se os resultados obtidos a partir da análise da ocultação estelar por Quaoar em 04 de maio de 2011, 17 de fevereiro e 15 de outubro de 2012. Estes resultados são apresentados e discutidos no artigo (Braga-Ribas et al. 2013).

4.1 Quaoar

Quaoar foi descoberto em junho de 2002 em imagens do *Palomar Observatory 1,2 m Oschin Telescope* (Williams et al. 2002) (Figura 4.2), recebendo a denominação 2002 LM_{60}. Após a confirmação de sua órbita, a União Astronômica Internacional (UAI) o designou o número 50000 em comemoração pela descoberta de um dos maiores objetos da região. Após, objetos maiores como Éris, Makemake e Haumea foram encontrados, mas suas numerações foram dadas seguindo a ordem normal de descoberta.

O semieixo-maior de Quaoar é de 43.51 `ua`, com excentricidade de 0,035, inclinação de 7,9° e período orbital é de 287 anos. Quaoar é classificado dinamicamente como um objeto quente clássico do cinturão de Kuiper, também chamados de *cubewanos*. Em 2007, descobriu-se um satélite em torno de Quaoar (Brown & Suer 2007). Denominado *Weywot*, ele orbita à uma distância média de 14,500 km em aproximadamente 12 dias. A determinação de sua órbita nos permite conhecer a massa de Quaoar que é utilizada para calcular sua densidade.

À época da descoberta, o diâmetro de Quaoar foi estimado em 1900 km, assumindo um albedo de 0,04. Com imagens do *Hubble Space Telescope* (HST), estimou-se seu albedo em $p = 0,1$, então seu tamanho foi calculado em 1260 ± 190 km (Brown & Trujillo 2004).

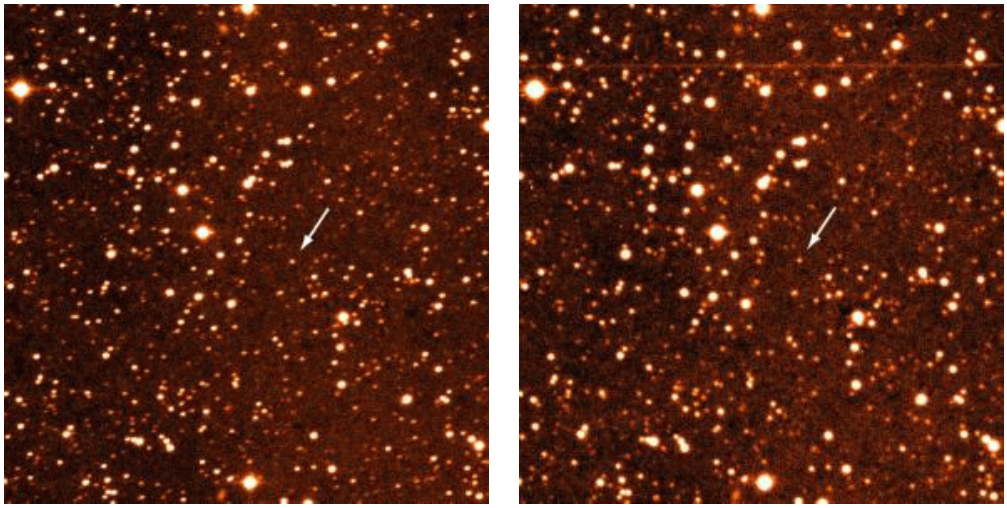


Figura 4.2 Dois instantes das imagens obtidas pelo *Palomar Observatory 1.2 m Oschin Telescope* que permitiram a descoberta de Quaoar (Brown & Trujillo 2004).

Mais tarde, com medidas do *Spitzer Infra-Red Telescope*, seu tamanho foi estimado em 844^{+207}_{-189} km (Stansberry et al. 2008). Com imagens diretas do HST, Fraser & Brown (2010) mediram um diâmetro de 890 ± 70 km.

Do estudo espectroscópico sabe-se que Quaoar possui gelos de água na sua fase amorfa e cristalizada, gelo de metano, e possivelmente gelo de etano e nitrogênio molecular (Jewitt & Luu (2004), Schaller & Brown (2007b), Guilbert et al. (2009) e Dalle Ore et al. (2009)). Isto significa que Quaoar reteve voláteis em sua superfície e pode, portanto, também ter retido uma atmosfera.

4.2 Predição

A descoberta da estrela candidata a ser ocultada foi feita pelo programa caminhos. A estrela é a UCAC2 26257103, com magnitudes B=17,1[†], V= 15.9[§], R= 15,8[‡], I= 14,9[†], J=13,6[♭], H= 12,9[♭], K=12,8[♭] [1]. A estrela e Quaoar foram monitorados desde algumas semanas antes do evento até poucos dias antes, para obtenção de posições astrométricas relativas na época do evento. Os seguintes telescópios contribuíram com observações pré-evento: Observatoire du Pic du Midi T-1m, TRAPPIST T-0,6m, ASH T-0,45m, Observatório do Pico dos Dias (OPD) Zeiss T-0,6m. As observações foram feitas com filtros R, I e sem filtro. Com estas observações, obteve-se astrometria relativa, uma vez que Quaoar e a estrela estavam nas mesmas imagens. O erro da predição ficou limitado pela dispersão da medida, e não por erros sistemáticos, o que corresponde menos de 30

[1] [†](Monet et al. 2003), [§](Zacharias et al. 2004a), [‡](Assafin et al. 2012), [♭](Skrutskie et al. 2006).

mas. Na figura 4.3 são mostrados alguns dos mapas de atualização gerados dias antes do evento.

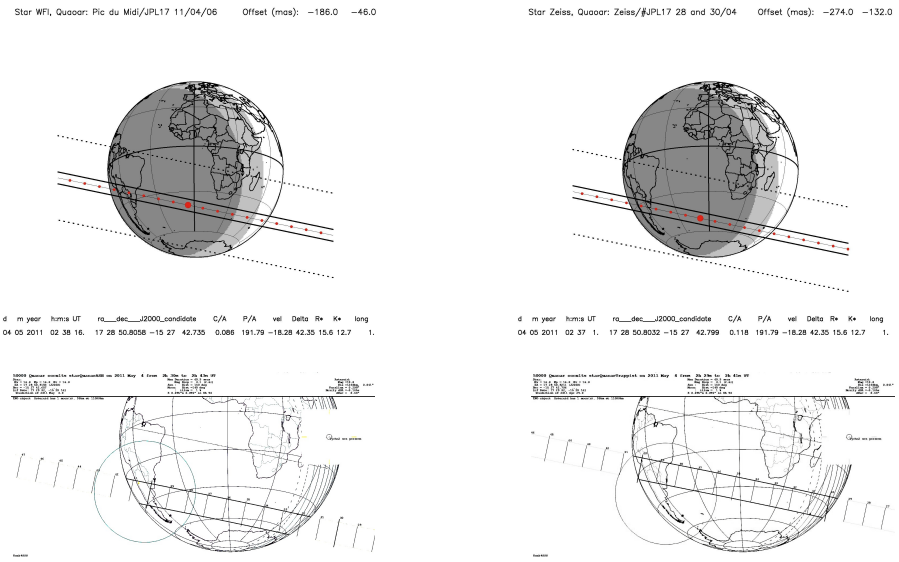


Figura 4.3 Alguns dos mapas obtidos com o resultado de observações dias antes à ocultação. Com astrometria relativa, o erro é dado pela dispersão da medida, o que corresponde a menos de 30 mas. Partindo do canto superior esquerdo em sentido horário (observatório - data): Pic du Midi - 06/abril; OPD/Zeiss - 28 a 30/abril; ASH - 26/abril; TRAPPIST - 24 a 28/abril.

Como posição final J2000 da estrela à época do evento obteve-se:

$$\begin{cases} \alpha = +17^{\mathrm{h}}\ 28^{\mathrm{min}}\ 50.801^{\mathrm{s}} \pm 0.007'' \\ \\ \delta = -15^{\circ}\ 27'\ 42.770'' \pm 0.014'' \end{cases} \tag{4.2}$$

4.3 Observação

Com as atualizações favoráveis a uma ocultação ocorrendo no sul da América do Sul, diversos observatórios naquela região foram contactados para realizarem a observação. Uma página na internet foi criada com informações e dicas aos observadores, como os mapas de predição, cartas de campo para busca do alvo, coordenadas, magnitudes da estrela, duração, queda de magnitude esperada, etc (`http://www.lesia.obspm.fr/perso/braga-ribas/list/2011/2011_05_04_Quaoar.html`).

Uma grande diversidade de instrumentos foi utilizada, de telescópios profissionais de até 2 m a telescópios amadores de 0.254 m de diâmetro, formando uma rede de 16 telescópios do centro do Brasil ao centro do Chile, passando pelo sul do Brasil, norte do Chile,

centro da Argentina e Uruguai. Desses, 13 puderam monitorar o evento, e seis detectaram a ocultação (Figura 4.4). Dois dos telescópios que detectaram a ocultação se encontram no mesmo observatório, portanto o evento foi detectado de cinco sítios diferentes. Dois outros sítios estavam próximos o bastante para delimitar o tamanho do corpo, mesmo não tendo detectado o evento, foram eles Salto/UY e OPD/BR. A tabela 4.1 traz informações sobre as características dos instrumentos, suas coordenadas, resultado e observadores.

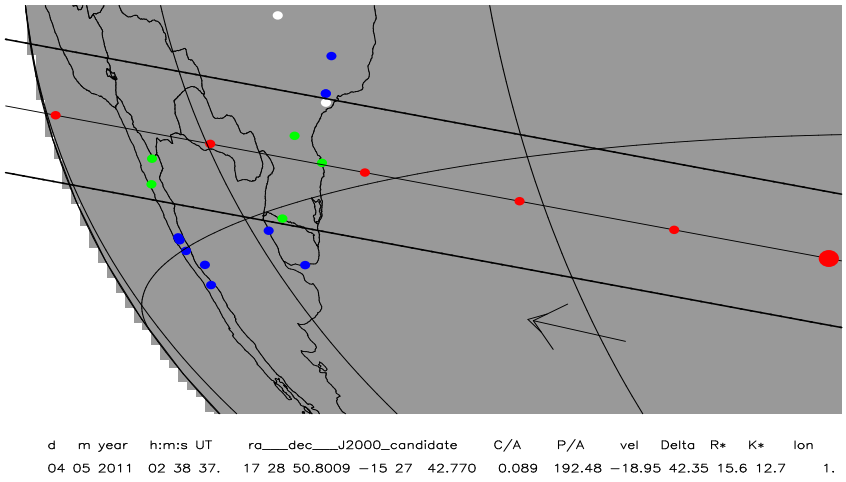


Figura 4.4 A trajetória da sombra de Quaoar observada na ocultação de 04 de maio de 2011. Os ponto vermelho grande à direita corresponde ao instante de C/A, os vermelhos pequenos estão em intervalos de um minuto. A sombra se desloca da direita para a esquerda. Os pontos verdes correspondem ao sítios que detectaram o evento, os azuis os sítios que obtiveram dados mas não detectaram o evento. Os pontos brancos correspondem aos sítios que não puderam observar por questões climáticas. Todos estão listados na Tabela 4.1.

4.4 Fotometria

A qualidade das imagens obtidas foi bastante variada, como pode-se ver na figura 4.5. Todas as curvas de luz foram obtidas através de fotometria de abertura diferencial, obtidas com o programa PRAIA. As aberturas foram otimizadas para obter-se o melhor S/R durante o evento. Posteriormente todas as curvas de luz foram normalizadas ao nível do fluxo da estrela mais Quaoar, isto é, nos momentos anteriores e posteriores ao evento.

O instante médio de cada imagem foi obtido diretamente do cabeçalho de cada imagem. Dos seis telescópios que detectaram positivamente o evento, dois registraram somente a parte inteira do segundo. O centésimo do segundo foi recuperado como descrito na seção 3.4.2. Quase todos os telescópios ajustaram seus horários com um servidor NTP. Apenas

Tabela 4.1 Características dos instrumentos, coordenadas, resultado, S/R e observadores.

Observatório	Longitude Latitude Altitude	Telescópio Abertura	Exposição Tempo-morto (segundos)	Resultado comentário	Observador
Brasília / BRA	47° 54' 41" O 15° 53' 29" S 1,072 m	0.254 m	8.0 2.0	 nuvens	P. Cacella
Belo Horizonte CEAMIG-REA / BRA	43° 59' 51" O 19° 49' 49" S 825 m	0.30 m	6.0 3.0	Negativo S/R=21	C. Jacques E. Pimentel B. Giacchini
Pico dos Dias - Brasópolis / BRA	45° 43' 58" O 22° 32' 08" S 1,864 m	Carl Zeiss 0.60 m	0.95 1.05	Negativo S/R=85 nuvens finas	M. Assafin L. Almeida
Pico dos Dias - Brasópolis / BRA	45° 43' 58" O 22° 32' 08" S 1,864 m	Boller & Chivens 0.60 m	- -	Problemas instrum.	V. Magalhães C. Montaña C. Rodrigues
São José dos Campos - INPE / BRA	45° 51' 44" O 23° 12' 33" S 617m	0.275 m	7.5 2	umidade	A. Milone
Ponta Grossa - UEPG / BRA	50° 05' 56" O 25° 05' 22" S 909 m	0.40 m	14 4.25	Positivo S/R=20	M. Emilio C. Neves
Florianópolis - UFSC / BRA	48° 31' 21" O 27° 36' 12" S 20 m	0.254 m	10 2.75	Positivo S/R=12	W. Schoenell A. Amorim
San Pedro de Atacama / CHL	68° 10' 48" O 22° 57' 12" S 2,397 m	Harlingten 0.5 m	3.0 1.1	Positivo S/R=52	operado remotamente by N. Morales
San Pedro de Atacama / CHL	68° 10' 48" O 22° 57' 12" S 2,397 m	ASH2 0.4 m	9.0 7.1	Positivo S/R=22	operado remotamente by N. Morales
Armazones / CHL	70° 11' 46" O 24° 35' 52" S 2,706 m	OCA 0.84 m	7.0 2.4	Positivo S/R=42	E. Unda-Sanzana J.P. Colque
Rivera / URY	55° 36' 00" O 30° 51' 48" S 200 m	0.305 m	4.0 2.3	Positivo S/R=26	S. Roland L. Almenares A. Bergengrue
Salto - OLASU / URY	57° 58' 43" O 31° 23' 33" S 37 m	0.305 m	4.0 2.7	Negativo S/R=18	S. Bruzzone E. Alvarez
La Silla / CHL	70° 44' 22" O 29° 15' 17" S 2,315 m	TRAPPIST 0.60 m	6.0 1.55	Negativo S/R=50	operado remotamente by E. Jehin
Los Molinos OALM / URY	56° 11' 25" O 34° 45' 19" S 80 m	0.35 m	4.0 1.0	Negativo S/R= 10	R. Salvo M. Martinez J. Capeche
CASLEO / ARG	69° 17' 45" O 31° 47' 56" S 2,492 m	J. Sahade 2.15 m	6.0 4.5	Negativo S/R= 250	R. Gil-Hutton
Santa Martina / CHL	70° 32' 04" O 33° 16' 09" S 1,450 m	0.40 m	1.0 0.0	Negativo S/R= 4	R. Leiva L. Vanzi I. Toledo

Rivera acertou manualmente o horário do computador a partir da hora dada por um GPS de mão, de forma que erros em tempo absoluto maiores que um segundo não são esperados.

4.5 Instantes da ocultação

Os instantes de ingresso e egresso da estrela (início e fim da ocultação) em cada telescópio foram determinados como pela modelagem apresentada em 3.4.2 e (Widemann et al. 2009). Na figura 4.5, sobrepostos aos dados, são apresentados o modelo geométrico e a curva de luz sintética após a convolução com a difração de Fresnel, o diâmetro da estrela e o tempo de exposição.

A escala de Fresnel à distância de Quaoar no momento da ocultação ($D = 42,35$ ua) para a banda típica de observação ($\lambda = 0,65\ \mu m$) é de 1,4 km. O diâmetro angular da estrela, calculado a partir das magnitudes B= 17,1 e K= 12,8, projetado à distância de Quaoar é de 0,46 km. O menor tempo de integração utilizado pelos telescópios que detectaram o evento foi 3 segundos, o que equivale à 55 km perpendicular ao limbo de Quaoar durante a ocultação. Portanto, as variações de fluxo nas curvas de luz são dominadas pelo tempo de integração e pelo ruído da medida e não pela difração de Fresnel nem pelo diâmetro da estrela.

4.6 Geometria

Como discutido, uma vez os instantes de ingresso e egresso determinados, obtém-se as cordas do objeto (Tabela 4.2), isto é, tem-se medidas do perfil do corpo no instante da ocultação em diferentes latitudes. Como a ocultação por Quaoar em 4 de maio de 2011 foi observada de cinco sítios diferentes, obteve-se cinco cordas (Fig. 4.1), que são utilizadas para reconstruir sua forma e tamanho.

Considerando a elipse como o caso mais geral para perfil do objeto no plano do céu, tem-se cinco parâmetros livres à serem ajustados às extremidades das cordas: o centro do objeto relativo à posição da estrela no plano do céu (f, g), o semi-eixo maior da elipse a', o achatamento ϵ' e a orientação da elipse no plano do céu PA (contado à partir do Norte celeste em direção ao Leste). O centro de Quaoar para cada sítio, f_c e g_c, é calculado usando a efeméride geocêntrica JPL#21 (Giorgini et al. 1997), onde f_c corresponde à posição relativa em ascensão reta, positiva se a estrela está a leste do objeto, e g_c é a posição em declinação, positiva se a estrela esta ao norte do centro do objeto. Estes valores também podem ser interpretados como o *offset* de Quaoar em relação à efeméride.

Um objeto no regime de baixo momento angular apresenta a forma esferoide Maclaurin

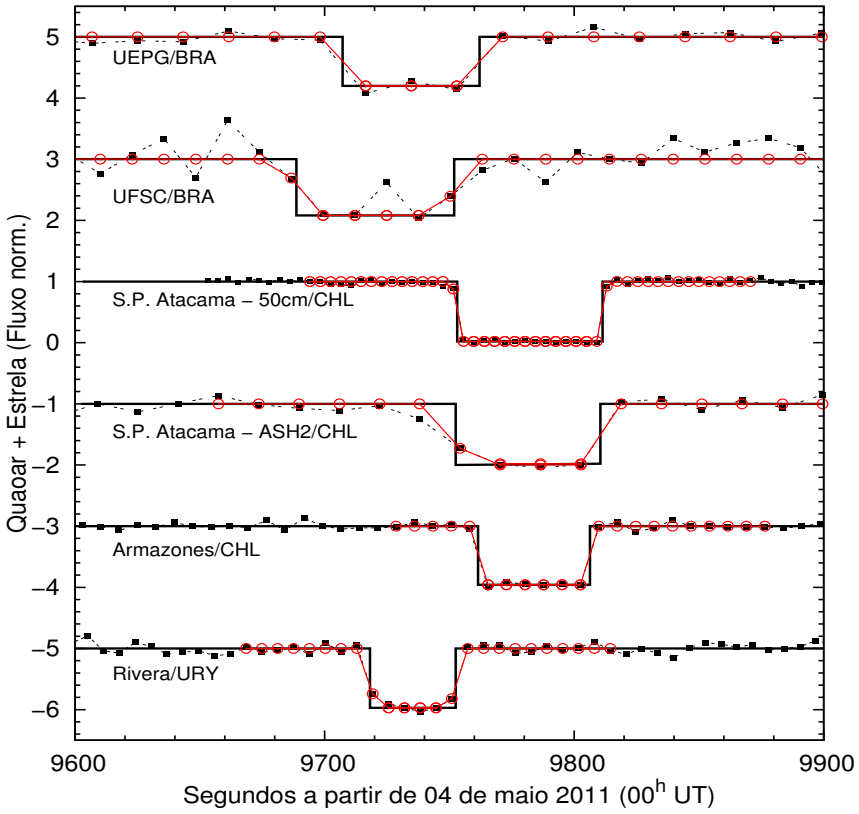


Figura 4.5 Curvas de luz que detectaram a ocultação por Quaoar em 04 de maio de 2011. Os pontos quadrados correspondem ao fluxo medido em cada imagem, os círculos e linhas vermelhas à curva de luz sintética após a convolução, e as linhas pretas contínuas ao modelo geométrico. Todas as curvas foram normalizadas à 1 na parte fora de evento. No gráfico acima elas foram deslocadas de quantidades inteiras para melhor visualização, com a exceção de S.P. Atacama - 50cm.

(Chandrasekhar 1987) (seção 4.8), e seus eixos são definidos como $a = b > c$ sendo a e c os raios equatorial e polar, respectivamente. O seu achatamento é dado por $\epsilon = (a-c)/a$, que, para pequenos valores está relacionado ao achatamento aparente por $\epsilon' \sim \epsilon \cdot sin^2(\zeta)$, sendo ζ o ângulo entre o eixo de rotação c e o ângulo de visada do observador ($\zeta = 0$ correspondendo ao polo na direção de visada). O achatamento aparente é dado por $\epsilon' = (a' - b')/a'$, sendo os semi-eixos $a' > b'$. A solução circular também é contemplada, uma vez que é um caso específico da elipse, quando o achatamento é nulo ($\epsilon = 0$).

De cada corda tem-se dois pontos que são usados para determinar o perfil de Quaoar, um ponto no ingresso e outro no egresso. Cada ponto tem seu erro correspondente ($\sigma_{i,r}$) que é dado pela incerteza em tempo sobre a determinação de tais instantes, multiplicada pela velocidade perpendicular ao limbo do TNO. Para se ajustar a elipse utiliza-se os N=10 pontos para ajustar M=5 parâmetros, buscando o χ^2 mínimo da equação 4.3. Os valores $f_{i,obs}$ e $g_{i,obs}$ são as coordenadas das extremidades das cordas, e os valores $f_{i,calc}$ e

$g_{i,calc}$ são as coordenadas das extremidades correspondentes da elipse ajustada. O valor do χ^2 por grau de liberdade (χ^2_{pdf}) é calculado como $\chi^2_{\mathsf{pdf}} = \chi^2_{\mathsf{min}}/(N-M)$ e também pode ser usado como um critério de qualidade do ajuste.

$$\chi^2 = \sum_{1}^{\mathsf{N}} \left[\left(f_{\mathsf{i,obs}} - f_{\mathsf{c,calc}}\right)^2 + \left(g_{\mathsf{i,obs}} - g_{\mathsf{c,calc}}\right)^2 \right] / \sigma^2_{\mathsf{i,r}}, \tag{4.3}$$

O ajuste é feito pelo método *Downhill Simplex* com a rotina *Amoeba* (Press et al. 1990), no qual parâmetros iniciais são fornecidos, e os instantes derivados das curvas de luz (Tabela 4.2), considerados como os pontos a serem ajustados. Por uma série de passos, o programa procura minimizar uma função, que neste caso é o χ^2. Para explorar todo o

Tabela 4.2 Instantes de início/fim da ocultação dos sítios que detectaram o evento e circunstâncias de observação.

Observatório	Instrumento Escala de Placa (")	Telescópio Abertura Filtro	Exposição Tempo-Morto S/R	Imersão ttempo (UT) h:mm:ss±ss	Emersão tempo (UT) h:mm:ss±ss
Pico dos Dias - Brasópolis / BRA	Andor/Ixon EMCCD 1,11	Carl Zeiss 0,60 m não	0,95 s 1,05 s S/R=85	Inicio obs.: 2:26:10	Fim obs.: 2:58:20 Negativo
Ponta Grossa - UEPG / BRA	SBIG STL-6303E 1,74	RCX 400 0,40 m V	14 s 4,25 s S/R=20	2:41:47,2±2,7	2:42:42,1±3,1
Florianópolis - UFSC/ BRA	SBIG ST-7 1,48	CM-1100 0,254 m não	10 s 2,75 s S/R=12	2:41:28,7±2,8	2:42:31,9±2,9
San Pedro de Atacama / CHL	Apogee U42 1,61	Harlingten 0,5 m não	3,0 s 1,1 s S/R=52 Ortiz	2:42:33,1±0,13	2:43:31,4±0,15
San Pedro de Atacama / CHL	SBIG STL11000 1,21	ASH2 0,4 m não	9,0 s 7,1 s S/R=22 Ortiz	2:42:32,5±1,0	2:43:30,30±4,3
Armazones / CHL	SBIG STL6303E 0,57	OCA 0,84 m não	7,0 s 2,4 s S/R=42	2:42:41,5±2,2	2:43:26,3±2,0
Rivera / URY	SBIG ST-7 2,57	LX200 0,305 m não	4,0 s 2,3 s S/R=26	2:41:58,2±0,4	2:42:32,5±0,4
Salto - OLASU / URY	QSI 516wsg 1,22	LX200 0,305 m não	4,0 s 2,7 s S/R=18	Inicio obs.: 2:20:29	Fim obs.: 2:54:04 Negativo

espaço de fase, o ajuste foi feito com valores pré-definidos de PA e ϵ'. Variou-se PA entre 0 e 180° com passos de 3° e, para cada PA explorou-se ϵ' de 0 a 0,5 com passos de 0,004. Portanto, os parâmetros a serem ajustados são o raio equatorial a' e o centro da elipse f_c e g_c. Para cada conjunto de parâmetros, a solução com seu correspondente χ^2_{min} é salvo. O conjunto de parâmetros com o menor χ^2_{pdf} entre todas as soluções é tido como o melhor ajuste, e o erro da solução a 1σ é obtido tomando-se os valores de $\chi^2_{min} + 1$.

Com todos estes ajustes cria-se um mapa 2D em cores, no qual o χ^2 de cada conjunto de parâmetros é apresentado em função do achatamento e raio equivalente. As cores são codificadas, sendo os pontos pretos a região de 1σ, e os amarelos todas as soluções acima de 3σ. Os tamanhos dos pontos são inversamente proporcionais ao valor do seu χ^2. Estes mapas apresentam uma forma em V. Isto acontece porque as maiores cordas impõem que a' seja maior que um valor L, próximo ao valor da corda mais longa. O alinhamento destas cordas é próximo ao alinhamento do eixo-maior da elipse, o que impõe que b' seja menor que L. Então o raio equivalente será limitado por $L\sqrt{1-\epsilon'} < R_{equiv} < L/\sqrt{1-\epsilon'}$. Estes limites correspondem às bordas à esquerda à e direita, respectivamente, da forma em V, e a base do V corresponde à $\epsilon' = 0$ e $\mathrm{R}_{equiv} = \mathrm{L}$, sendo que L depende do modelo ajustado.

A melhor solução obtida ao se ajustar as cordas observadas, apresenta um $\chi^2_{min} = 9,1$, quando um valor próximo a 1 seria esperado. Isto significa que o modelo não é satisfatório. Como pode-se ver na figura 4.6 a elipse não passa por várias extremidades das cordas. Da figura 4.1 também pode-se perceber que o perfil dado pelas cordas realmente não corresponde à uma elipse, o que pode significar que Quaoar não apresenta a forma de um objeto em equilíbrio hidrostático.

Espera-se que um objeto com cerca de 1000 km de diâmetro, como Quaoar, esteja em equilíbrio hidrostático, isto é, que tenha a forma de um objeto Maclaurin ou Jacobi (Seção 4.8). O diâmetro crítico para isto pode ser determinado assumindo-se a composição (gelo ou rocha) e sua temperatura (Tancredi & Favre 2008). Para corpos formados principalmente por gelo, o diâmetro crítico varia entre $D \sim 200 - 900\ km$; para corpos rochosos, a variação se dá entre $D \sim 500 - 1200\ km$. Na seção 4.8 é feita uma discussão detalhada deste tema.

Como o histórico da formação e colisão, bem como a estrutura interna de Quaoar não são conhecidos, não pode-se descartar soluções que não correspondam às formas de equilíbrio. Sendo o objetivo maior do trabalho, melhorar o conhecimento sobre o tamanho e forma de Quaoar e utiliza-los para derivar albedo e densidade, assume-se casos extremos, explorando assim todas as possíveis formas para o corpo.

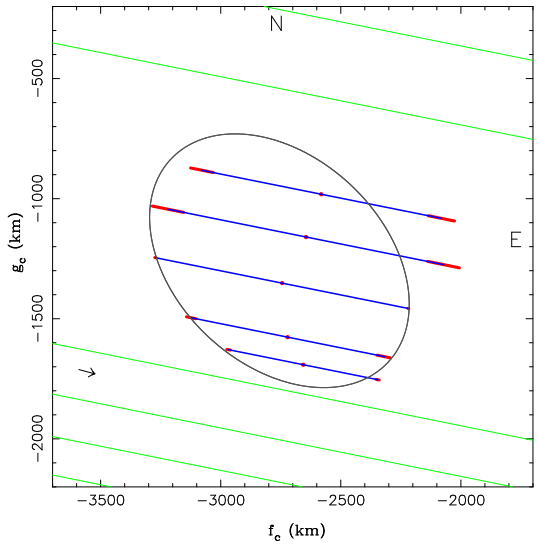


Figura 4.6 **Solução nominal:** O ajuste sobre todos os pontos não é satisfatório. Nem todas as extremidades das cordas são contempladas pela elipse ajustada.

4.7 Soluções extremas

Para investigar soluções extremas, elimina-se pontos específicos artificialmente. Para isto, considera-se que Quaoar pode ter grandes características sobre sua superfície, o que o difere de uma elipse. Isto é feito ao se associar à eles erros muito grandes, e, portanto tais pontos passam a ter peso muito pequeno sobre o ajuste (Eq. 4.3). Os pontos escolhidos para serem "eliminados" são aqueles que têm maior peso sobre a forma da elipse previamente ajustada. Seguindo este procedimento, dois casos são apresentados a seguir. Um terceiro caso também é estudado, onde considera-se que os instantes registrados estão em desacordo com o Tempo Universal Coordenado e que Quaoar encontra-se em equilíbrio hidrostático. Os parâmetros derivados, bem como suas barras de erro são apresentados na tabela 4.3.

4.7.1 Solução: Cratera

A primeira solução apresentada foi obtida eliminando-se o ponto do egresso em San Pedro de Atacama, considera-se então que tem-se N=9 pontos para serem ajustados. Esta é a solução mais alongada dentre as apresentadas e tem uma grande "falha" no seu bordo leste, por isto é chamada de *Cratera*. Ela tem um achatamento de 0,29, ângulo de posição a 21° , raio equivalente de 557 km e $\chi^2_{pdf,min} = 3,8$ (Tabela 4.3). Na figura 4.7 pode-se ver a elipse correspondente ao melhor ajuste, bem como um mapa de χ^2_{min} com os valores correspondes para cada solução testada. No mapa de χ^2_{min} pode-se visualizar a região

em que os ajustes apresentam menor valor de χ^2_{min}, cores escuras, que correspondem aos melhores ajustes.

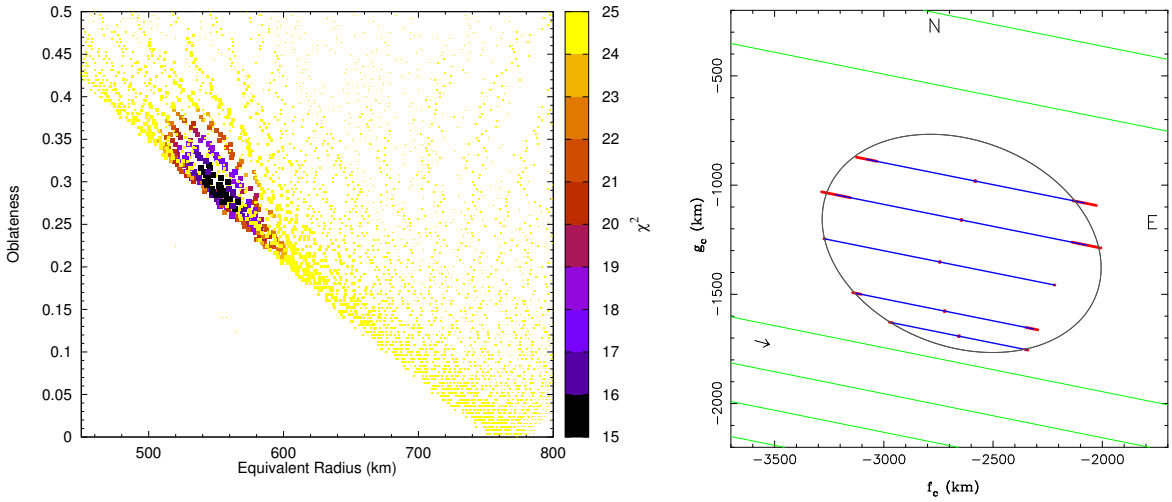


Figura 4.7 **Solução Cratera:** O gráfico da esquerda corresponde ao mapa de χ^2_{min} para cada solução testada em função do achatamento e do raio equivalente. O ponto do egresso de San Pedro de Atacama foi desconsiderado para esta solução. A região preta equivale à valores de $\chi^2_{min}+1$, isto é, 1σ. A palheta de cores foi ajustada de forma que todas as cores, que não amarelo, correspondem à valores dentro da região de 3σ. O tamanho dos pontos também é inversamente proporcional ao valor do seu χ^2_{min}. Pode-se ver que as melhores soluções se restringem à uma pequena gama de valores. No gráfico à direita tem-se a elipse correspondente ao melhor ajuste. As linhas azuis são as cordas observadas, os segmentos vermelhos os erros associados à cada ponto e as linhas verdes cordas negativas.

Observando a distribuição de tamanhos das crateras no Sistema Solar, e com ensaios de colisões em laboratório Leliwa-Kopystyński et al. (2008) determinaram uma relação para calcular o diâmetro máximo de uma cratera sobre um corpo de gelo, ou de rocha, sem que ele seja quebrado pelo impacto de origem. Seja R o raio do corpo, o diâmetro máximo da cratera será dado por $D_{\text{c,gelo}} \sim 1,2R$ se o objeto for predominantemente formado por gelo, ou $D_{\text{c,rocha}} \sim 1,6R$ se ele for formado por silicatos.

Considerando o $R_{equiv} = 557\ km$ obtido nesta solução, tem-se $D_{\text{c,gelo}} \sim 668$ km e $D_{\text{c,rocha}} \sim 891$ km. Do painel direito da figura 4.7, pode-se estimar um diâmetro de $\sim 430\ km$ para a suposta cratera. Este valor é compatível com os diâmetros máximos calculados.

A profundidade máxima de uma cratera sobre um corpo de gelo ou silicato também é investigado em Leliwa-Kopystyński et al. (2008). Sendo o corpo formado predominantemente por H_2O, CO_2 ou silicatos, as profundidades máximas variam de 25%, 31% e 26%, respectivamente, em relação ao diâmetro da cratera. Estas proporções correspondem, respectivamente, à 108, 133 e 111 km para a cratera de 430 km. Na figura 4.8, tem-se a

diferença entre as extremidades das cordas e o limbo da elipse ajustada. Para San Pedro de Atacama esta diferença é de $\sim -200\ km$, ou 46% do diâmetro estimado para a cratera.

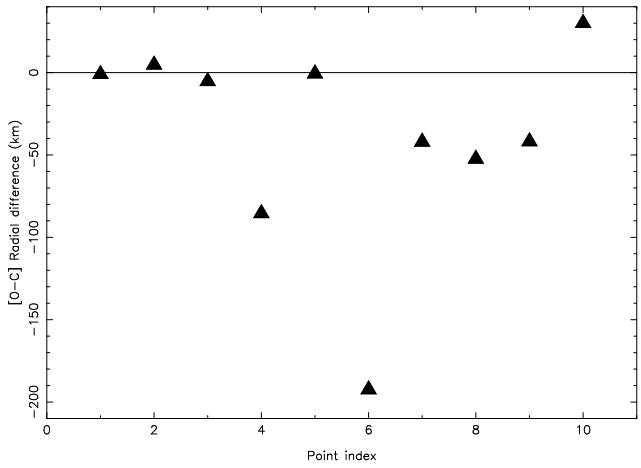


Figura 4.8 **Solução Cratera:** Diferença entre as extremidades das cordas e o limbo da elipse ajustada. Do sul para o norte, Rivera é representada pelos índices 1 e 2, Armazones 3 e 4, San Pedro de Atacama 5 e 6, Florianópolis 7 e 8, e Ponta Grossa 9 e 10, onde índices ímpares correspondem aos pontos a oeste e pares a leste. Tal diferença para o ponto do egresso em San Pedro de Atacama (índice 4) é de $\sim 200\ km$.

Considerando os pontos discutidos acima, pode-se concluir que, ou a solução é fisicamente impossível, ou o modelo de Leliwa-Kopystyński et al. (2008) não é complexo o suficiente, subestimando os valores máximos das características das crateras. Uma possível explicação para esta última hipótese seria uma inesperada forte coesão interna em Quaoar.

4.7.2 Solução: Montanha

A presente solução foi obtida eliminando-se o ponto do egresso da corda de Rivera, a mais ao sul a detectar o evento. A solução é menos alongada, e apresenta uma importante proeminência no seu bordo a sudeste, por isto é chamada de *Montanha*. No painel direito da figura 4.9 pode-se ver a elipse que corresponde ao melhor ajuste. Ela apresenta um achatamento de 0.138, ângulo de posição de 7,5°, raio equivalente de 525 km e $\chi^2_{pdf,min} = 3,9$ (Tabela 4.3). No painel esquerdo da figura 4.9 também é apresentado o mapa de χ^2_{min} (ver seção 4.7.1 para maiores explicações).

As diferenças entre as extremidades das cordas e o limbo da elipse ajustada para esta solução são mostradas na figura 4.10. Para o ponto do egresso em Rivera, a diferença é de $\sim 165\ km$ o que equivale à 31% do raio equivalente.

Em (Johnson & McGetchin 1973) pode-se verificar na Figura 1 (reproduzida aqui na Figura 4.11) que, em um corpo formado predominantemente por gelo, com 525 km de raio,

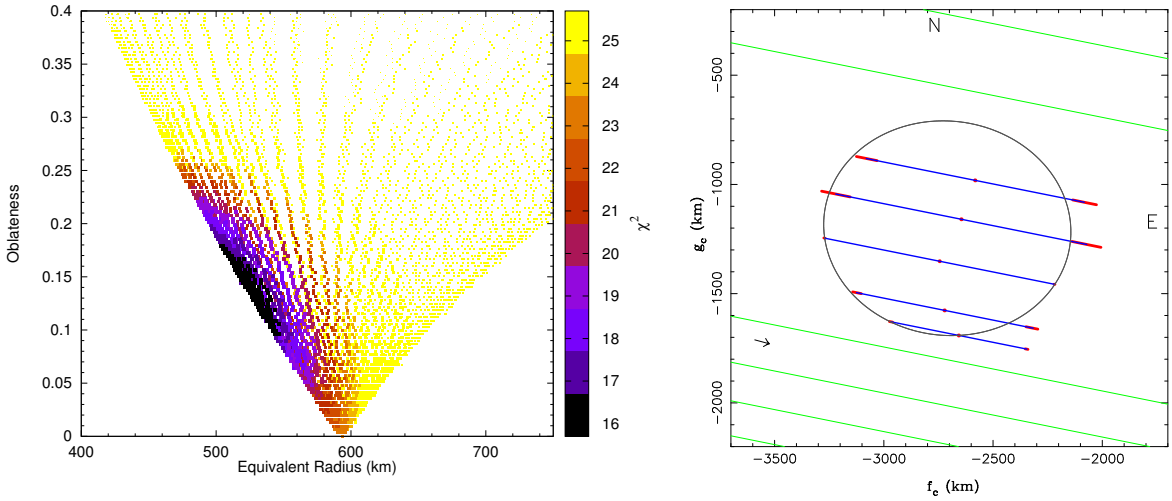


Figura 4.9 **Solução Montanha:** O gráfico da esquerda corresponde ao mapa de χ^2_{min} para cada solução testada em função do achatamento e do raio equivalente. O ponto do egresso de Rivera, corda mais ao sul, foi desconsiderado para esta solução. A região preta equivale à valores a 1σ (veja cabeçalho da figura 4.7 para mais detalhes). No gráfico à direita tem-se a elipse correspondente ao melhor ajuste. A sudeste, a elipse não se ajusta às cordas de Rivera e Armazones, dando lugar à uma grande proeminência.

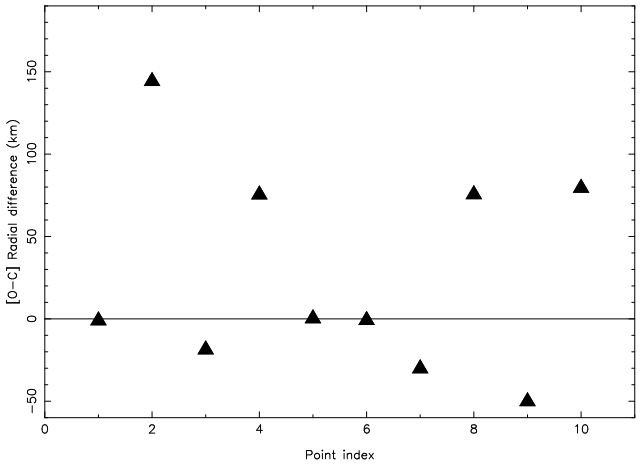


Figura 4.10 **Solução Montanha:** Diferença entre as extremidades das cordas e o limbo da elipse ajustada. Para o egresso em Rivera (índice 2), esta diferença é de $\sim 144\ km$. Para o egresso em Armazones, Florianópolis e Ponta Grossa (índices 4, 8 e 10 respectivamente), a diferença é de aproximadamente 80 km.

a maior proeminência suportada sobre a superfície é de $\sim 20\ km$. Em um corpo formado por silicatos com o mesmo raio, a maior amplitude seria de $\sim 150\ km$. No entanto, como será mostrado, esta solução corresponde à uma densidade próxima a 2 g/cm^3, o que implica em uma fração importante de gelo, portanto grandes proeminências não seriam suportadas em seu relevo. Isto pode significar que a montanha mostrada nesta solução

seja fisicamente improvável.

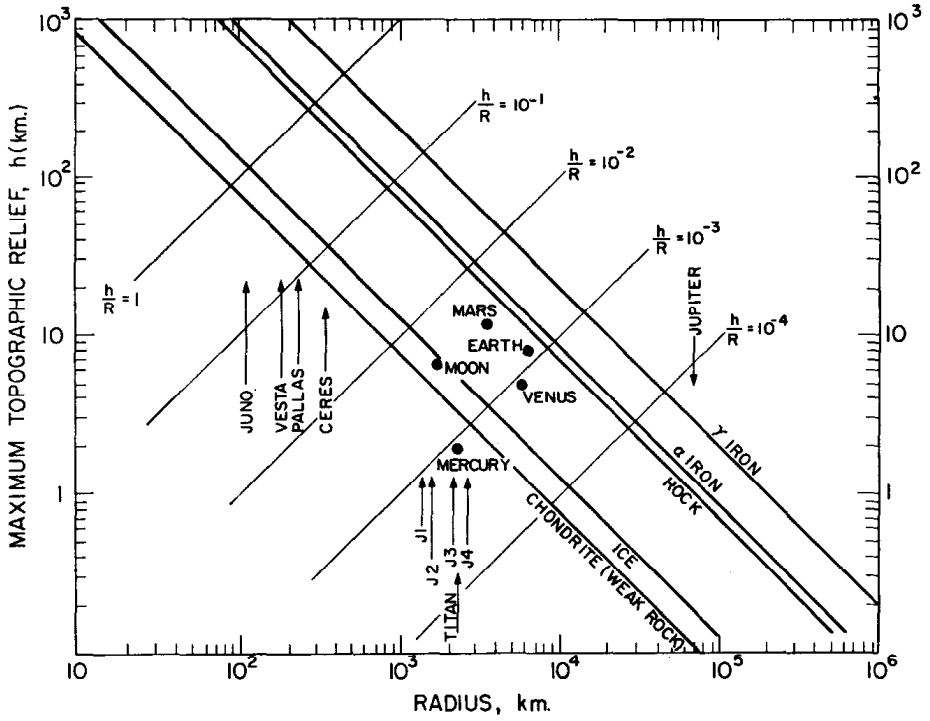


Figura 4.11 Máxima carga estática topográfica que pode ser suportada pela força dos materiais em função do raio planetário. Proeminências observadas na Terra, Mercúrio, Vênus, Marte e Lua são mostradas pelos pontos sólidos. Os satélites galileanos e os maiores satélites também são indicados. Valores da rugosidade h/R são apresentados como parâmetro, onde h é a escala da proeminência e R o raio do planeta (Johnson & McGetchin 1973).

4.8 Objeto Maclaurin com instantes deslocados

Como dito e será explicado em detalhe a seguir, espera-se que Quaoar tenha a forma de um objeto Maclaurin. As soluções acima não condizem com esta expectativa. Então, assume-se que o horário absoluto registrado por alguns sítios apresentem deslocamentos em relação ao tempo universal, para então obter a solução correspondente à um objeto Maclaurin.

Não há nenhum indício do que poderia causar tais deslocamentos, já que todos os sítios estavam conectados à um servidor de tempo via internet (NTP). A única exceção foi Rivera, que foi uma estação móvel, que utilizou um GPS de mão e manualmente acertou o horário do computador que era usado para registrar as imagens.

Antes de deslocar as cordas e seguir para o ajuste Maclaurin, é preciso entender porque espera-se que Quaoar apresente tal figura e como pode-se utilizá-la para restringir os

resultados.

Figuras de equilíbrio

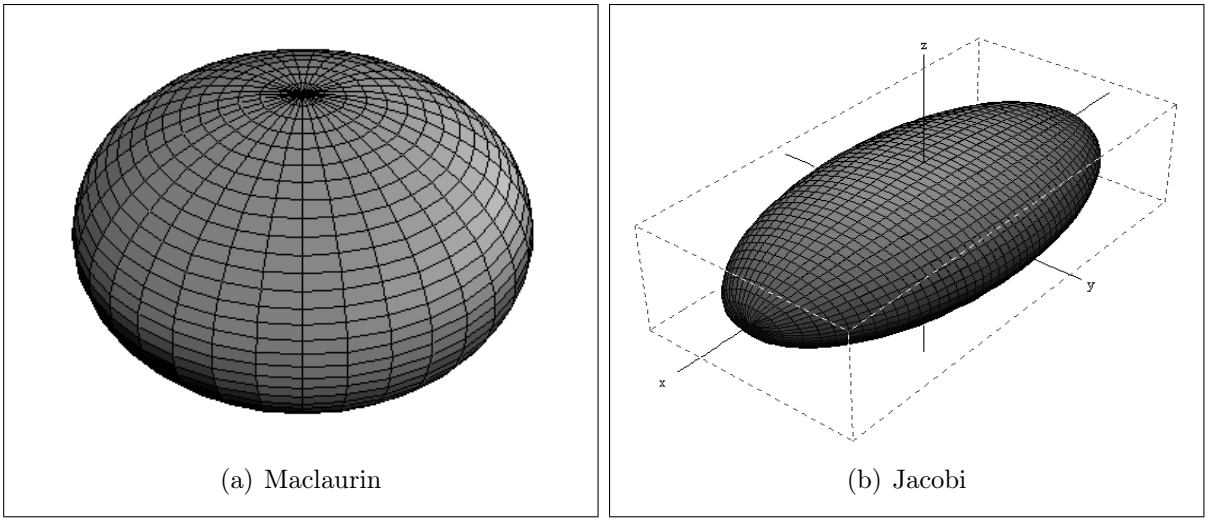
Um corpo fluido isolado em rotação irá assumir formas esferoidais dependendo do seu momento angular. Sendo a equação de um esferoide dada pela função 4.4, pode-se definir três casos para a forma do objeto em revolução (Chandrasekhar 1987):

1. Se não estiver em rotação sua forma será esférica, com todos os eixos cartesianos partilhando da mesma dimensão ($a = b = c$).

2. Se seu momento angular for baixo, assumirá a forma de um esferoide oblato com apenas dois eixos com o mesmo tamanho ($a = b > c$), conhecido como esferoide **Maclaurin** (Fig. 4.12(a)).

3. Se seu momento angular for alto, irá assumir a forma de um elipsoide de **Jacobi**, onde ($a > b > c$), isto é, seus três eixos tem dimensões diferentes (Fig. 4.12(b)).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \tag{4.4}$$



(a) Maclaurin (b) Jacobi

Figura 4.12 À esquerda um esferoide Maclaurin, definido por $a = b > c$. À direita um elipsoide Jacobi, definido quando $a > b > c$.
Figuras adaptadas de (a) http://en.wikipedia.org/wiki/File:OblateSpheroid.png acessada em 16/01/2013; (b) http://commons.wikimedia.org/wiki/File:Ellipsoide.png acessada em 16/01/2013.

Objetos do Sistema Solar em que a própria gravidade supera a força de corpo rígido, assumem as formas supracitadas. Isto ocorre a partir de uma dada dimensão que depende da resistência do material S, da densidade ρ e do diâmetro D do objeto, entre outros fatores. Para diferentes faixas de ρ *e* S, Tancredi & Favre (2008) calcularam que para

corpos com densidade entre $\rho \sim 1 - 2$ g/cm^3 e dureza de material $S \sim 1 - 10$ MPa, tipicamente formados por gelo, o diâmetro crítico varia entre $D \sim 200 - 900$ km; corpos rochosos com densidades entre $\rho \sim 3 - 4$ g/cm^3 e $S \sim 30 - 100$ MPa, a variação é entre $D \sim 500 - 1200$ km. Quando o diâmetro crítico é atingido dize-se que o objeto está em equilíbrio hidrostático e assumirá uma das três formas acima.

É possível definir a zona de transição entre um objeto Maclaurin e Jacobi. Em Chandrasekhar (1987) foram definidos dois parâmetros adimensionais (Equação 4.5) : (Γ) associado ao momento angular L, à constante gravitacional G, à massa M do corpo e ao seu raio R; e (Ω) associado à velocidade angular ω e à densidade ρ. Para momentos angulares altos, o corpo atinge o estado de transição quando $\Omega_t > 0,347$ e $(c/a)_t = 0,583$, passando da forma Maclaurin para um elipsoide Jacobi (Figura 4.13(a)). Se o momento for ainda mais alto, o corpo atingirá um valor crítico quando ele se torna instável, à ponto de sofrer disrupção. Isto é atingido quando a razão entre os maiores eixos é $(a/b)_{max} = 0,432$, o que corresponde à $\Gamma_{max} = 0,390$ (Tancredi & Favre 2008).

$$\Gamma = \frac{L}{\mathsf{GM^3R}^{1/2}}, \qquad \Omega = \frac{\omega^2}{\pi \mathsf{G}\rho} \tag{4.5}$$

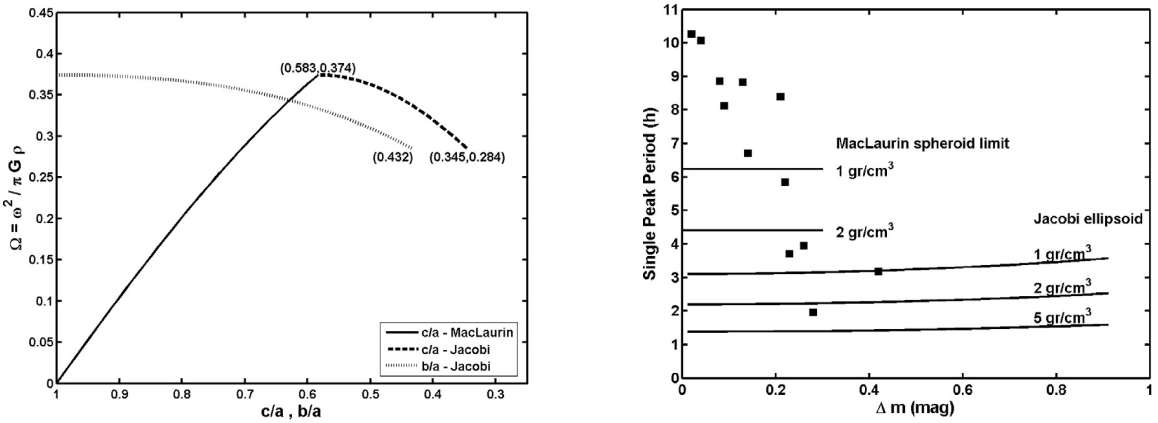
Pode-se relacionar estes valores de transição à valores do achatamento pela relação $(a/c) = 1 - \epsilon = \cos(\psi)$. Portanto, o valor de transição entre um objeto Maclaurin e Jacobi em função do achatamento é $\epsilon_t = 0,417$, e o achatamento máximo de um objeto Jacobi é $\epsilon_{max} = 0,655$. Lembrando que, o que obtém-se das cordas de ocultação é uma projeção do achatamento real ϵ em função do ângulo ζ definido entre o eixo de rotação c e a linha de visada, dada por $\epsilon' \sim \epsilon \cdot \sin^2(\zeta)$.

Um objeto Maclaurin não apresentará curva de luz de rotação se tiver uma superfície uniforme. Já um objeto Jacobi terá uma curva de luz com dois picos de mesma amplitude que podem ser calculados pela equação 4.6, onde $\beta = (b/a)$ e $\gamma = (c/a)$. A presença de diferenças locais de albedo, crateras e facetas irão causar desvios à estas curvas (Trilling & Bernstein 2006). No caso do objeto ser visto pelo equador ($\zeta = 90°$) a amplitude será máxima, e a equação 4.6 se simplifica na relação $\Delta m_{max} = 2,5 \log(\beta)$.

$$\Delta m = -1,25 \cdot \log_{10}\left[\frac{1 + \gamma^2 \tan^2\zeta}{1 + (\gamma/\beta)^2 \tan^2\zeta}\right] \tag{4.6}$$

Retornando aos pontos de transição e instabilidade, é possível determinar uma relação entre o período de rotação $\mathrm{P} = 2\pi/\omega$, a densidade, e os valores de Ω_t e Ω_{max} (Equação 4.7). Na figura 4.13(b), Tancredi & Favre (2008) relacionaram o período de rotação conhecido de alguns TNOs à variação de magnitude observada. Sobre este gráfico traçaram algumas linhas de transição entre forma Maclaurin e Jacobi e de instabilidade, para determinadas densidades.

$$P^2 = \frac{4\pi}{\Omega \mathsf{G} \rho} \tag{4.7}$$



(a) Ω em função de c/a e b/a (b) Período em função da variação de magnitude

Figura 4.13 **(a)** O parâmetro adimensional Ω, associado à velocidade angular, em relação à razão entre os eixos (c/a) e (b/a). **(b)** Limites de transição entre Maclaurin e Jacobi e disrupção, para densidades $\rho = 1$, 2 e 5 g/cm^3. Os quadrados pretos correspondem ao período de rotação de único pico em relação à amplitude da variação de magnitude de alguns TNOs (Tancredi & Favre 2008).

De maneira mais geral, um objeto Maclaurin deve satisfazer à equação 4.8 (Plummer (1919) e Sicardy et al. (2011)). Para pequenos achatamentos, a equação 4.8 pode ser aproximada por $\omega^2 a^3/\mathrm{GM} \sim 4\epsilon/5$, levando ao período de rotação P $\sim 2,33/\sqrt{\epsilon}$ horas.

$$\frac{\omega^2 a^3}{G\mathrm{M}} = \frac{3\left\{2\psi\left[2+\cos(2\psi)\right]-3\sin(2\psi)\right\}}{4\sin^3(\psi)} \tag{4.8}$$

Relacionando o volume de um objeto oblato à sua densidade, é possível substituir $\mathrm{M} = (4\pi a^3 \cos(\psi)\rho)/3$ na equação 4.8 para obter a equação 4.9, que fornece a densidade em função do achatamento para uma dada velocidade angular. Então, uma vez que o período de rotação de qualquer objeto é conhecido, pode-se ter a curva de equilíbrio hidrostático em função de seu achatamento, caso ele seja um corpo Maclaurin (Figura 4.14).

$$\rho = \frac{\omega^2}{G\pi} \frac{\sin^3(\psi)}{\cos(\psi)\left\{2\psi\left[2+\cos(2\psi)\right]-3\sin(2\psi)\right\}} \tag{4.9}$$

No caso mais geral para um objeto Jacobi, a velocidade angular, β e γ, as razões entre os eixos do elipsoide são relacionadas pelas integrais 4.10, onde $\Delta(\beta,\gamma,u)$=$[(1+u)(\beta^2+u)(\gamma^2+u)]^{1/2}$ (Lacerda & Jewitt (2007) e Sicardy et al. (2011)).

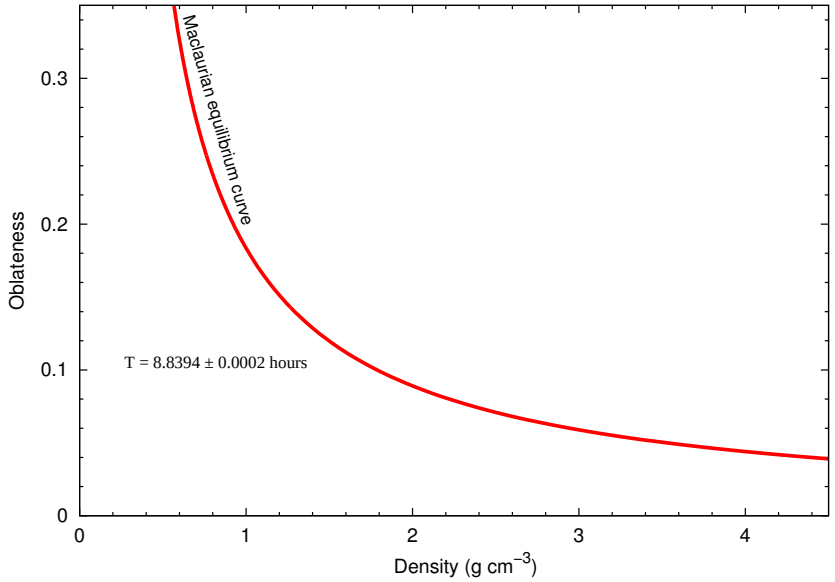


Figura 4.14 Curva de equilíbrio hidrostático de um objeto Maclaurin. A figura é uma descrição da equação 4.9, relacionando o achatamento à densidade de um objeto com período de rotação de 8.8934 horas.

$$\beta^2 \int_0^\infty \frac{du}{(1+u)(\beta^2+u)\Delta(\beta,\gamma,u)} = \gamma^2 \int_0^\infty \frac{du}{(\gamma^2+u)\Delta(\beta,\gamma,u)} \tag{4.10}$$

$$\Upsilon = \frac{\omega^2 a^3}{GM} = \frac{3}{2} \int_0^\infty \frac{udu}{(1+u)(\beta^2+u)\Delta(\beta,\gamma,u)} \tag{4.11}$$

A partir de um dado valor de β obtém-se γ, e calculando a integral da equação 4.11, pode-se determinar a densidade (Equação 4.12). Então, novamente, uma vez que o período de rotação de qualquer objeto é conhecido, pode-se ter a curva de equilíbrio hidrostático em função de seu achatamento, para o objeto Jacobi.

$$\rho = \frac{3\ \omega^2}{4\pi \mathsf{G} \beta\gamma\Upsilon} \tag{4.12}$$

Verifica-se as condições acima para o caso específico de Quaoar. Ele apresenta curva de luz de rotação de único pico com amplitude $\Delta\ m = 0,112 \pm 0,011$ e período P $= 8,8394 \pm 0,0002$ horas (Figura 4.15). A possibilidade de se tratar de uma curva de pico duplo com P $= 17,6788$ horas não é descartada, mas pouco provável já que a densidade de um objeto Jacobi com tal período de rotação seria proibitivamente baixa (Ortiz et al. 2003).

Considerando o período de rotação de pico único e uma densidade típica entre os TNOs de 2 g/cm^3 pode-se calcular Ω para Quaoar. Usando a equação 4.7, obtém-se

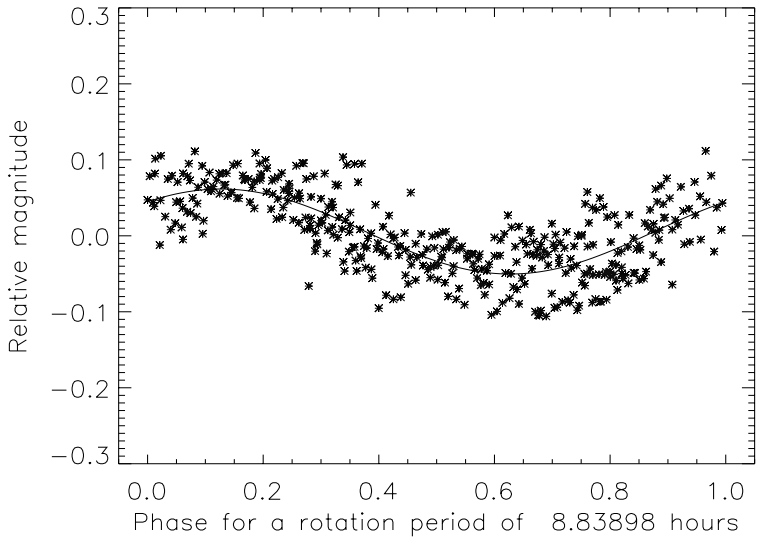


Figura 4.15 Curva de luz de rotação de Quaoar com P $= 8,8394 \pm 0,0002$ horas e $\Delta m = 0,112 \pm 0,011$ (Ortiz et al. 2003).

Ω_Q = 0,093, portanto encontra-se dentro da faixa de estabilidade de um corpo Maclaurin (Figura 4.13(a)).

A partir das soluções já apresentadas, calcula-se as densidades e considerando também do período de rotação de pico único, determina-se Ω para Quaoar $\Omega_Q = 0,08 - 0,13$. Portanto ele se encontra dentro da faixa de estabilidade de um corpo Maclaurin (Figura 4.13(a)). Da solução *cratera* tem-se o maior achatamento possível para Quaoar $\epsilon'_{max} = 0,324$. Comparando ao valor de achatamento de transição $\epsilon'_{max} < \epsilon_t = 0,417$, também vê-se que Quaoar está na faixa de equilíbrio Maclaurin. Ainda, na figura 4.13(b) pode-se verificar que Quaoar está bem acima das linhas de transição traçadas no gráfico.

Deslocando os instantes

Após as considerações acima, investiga-se o deslocamento em tempo absoluto sobre as cordas, necessário para obter a figura de um objeto Maclaurin.

Para qualquer elipsoide, os instantes centrais de todas as cordas que medem seu perfil devem estar alinhados em uma linha reta. Então, desloca-se os instantes absolutos das cordas observadas de maneira a satisfazerem esta condição. Uma vez alinhados, se for possível ajustar sobre estas cordas uma elipse, significa que também será possível obter o elipsoide Maclaurin para Quaoar. De fato, infinitos elipsoides podem ser obtidos com diferentes deslocamentos das cordas, mas somente um será Maclaurin, como será discutido a seguir.

Sendo o raio equivalente de uma elipse definido como o raio de um círculo que possui

a mesma área superficial que a dada elipse (Eq. 4.1), alinha-se as cordas de maneira a se obter um círculo. Para isto, a linha que une as cordas será perpendicular à direção das cordas. A área superficial deste círculo será conservada por qualquer elipsoide que contemple as cordas observadas, inclusive o elipsoide Maclaurin.

Deslocando as cordas de forma a ficarem alinhadas na direção perpendicular à direção das cordas, obtém-se um círculo com raio de R=555 ± 2,5 km (figura 4.16). Este raio nos fornece o raio equivalente do elipsoide que se procura, no caso o Maclaurin.

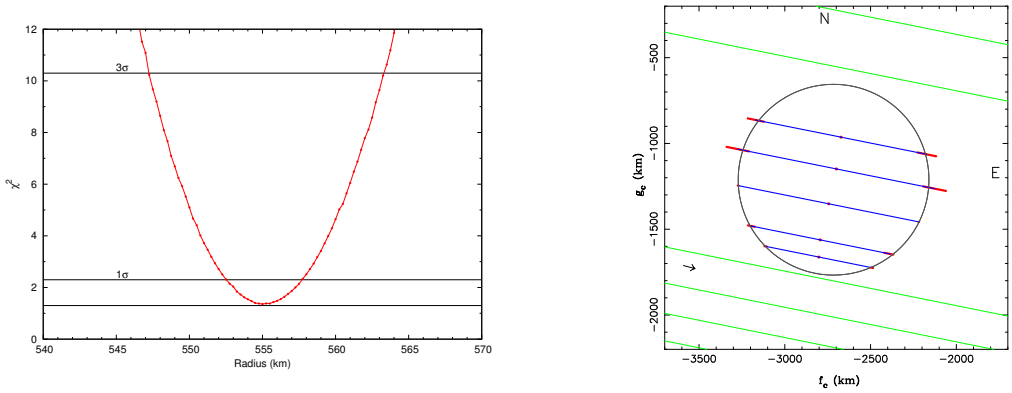


Figura 4.16 Após deslocar as cordas de maneira que os seus centros fique alinhados em uma reta perpendicular à direção das cordas, ajusta-se um círculo à elas. Deste círculo determina-se o raio equivalente $R_{equiv} = 555 \pm 2,5$. No gráfico à esquerda, pode-se ver a curva de χ^2 versus diversos raios testados como solução. Os níveis de 1σ e 3σ são indicados. À direita, o ajuste circular sobre as cordas deslocadas.

Elipsoides equivalentes

Obtida a superfície, investiga-se quais elipses possuem áreas equivalentes. Sendo a o semi-eixo maior, o semi-eixo menor aparente b' será dado pela equação 4.13, em que c é o semi-eixo menor real e ζ é o ângulo entre o eixo de c e o ponto de visada do observador (sedo $\zeta = 0°$ quando o polo esta voltado para o observador).

$$b' = a\sqrt{\cos^2(\zeta) + (c/a)^2 \sin^2(\zeta)}, \tag{4.13}$$

Sendo a área dada por $S = \pi a b' = \pi R^2_{equiv}$, e o achatamento por $\epsilon = 1-(c/a)$, define-se uma relação (Equação 4.14 entre um dado achatamento e seu semi-eixo maior para uma superfície (ou R_{equiv}), para um dado ângulo ζ).

$$\epsilon = 1 - \frac{\sqrt{(S/\pi a^2)^2 - \cos^2(\zeta)}}{\sin(\zeta)} \tag{4.14}$$

Como $0 < (1-\epsilon)^2 < 1$, isto impõe uma limitação a valores de a (Equação 4.15)

$$R_{\text{equiv}} < a < \frac{R_{\text{equiv}}}{\sqrt{|\cos(\zeta)|}} \tag{4.15}$$

Finalmente, para cada semi-eixo maior, pode-se calcular uma densidade pela equação 4.16, onde M é a massa do corpo obtida à partir da órbita de Weywot (Vachier et al. (2012) e (Fraser et al. 2013)).

$$\rho = \frac{3M}{4\pi a^3} \cdot \frac{1}{1-\epsilon} \tag{4.16}$$

Assim, as equações 4.14 e 4.16 definem infinitas elipses, limitadas pela relação 4.15, que satisfazem as cordas de observação (com seus tempos deslocados de acordo), e que possuem o mesmo $\mathrm{R}_{equiv} = 555 \pm 2,5$ obtido do ajuste circular. Estas elipses, descritas por ϵ e ρ, são apresentadas na figura 4.18, na qual as curvas para os extremos $\zeta = 90°$ a $\zeta = 0°$ definem uma região em V em que a solução real deve se encontrar. Esta região pode ser comparada à curva de equilíbrio de um objeto Maclaurin.

Ainda é possível definir o ângulo de posição PA à partir do achatamento. Partindo de uma geometria onde as cordas são horizontais, primeiro considera-se as cordas deslocadas para ajustar um circulo de raio R_{equiv}. Deslocando-se cada corda de maneira que os seus centros continuem alinhados numa linha inclinada, transforma-se o círculo numa elipse de semi-eixo maior a, semi-eixo menor b' e ângulo de posição β (Figura 4.17).

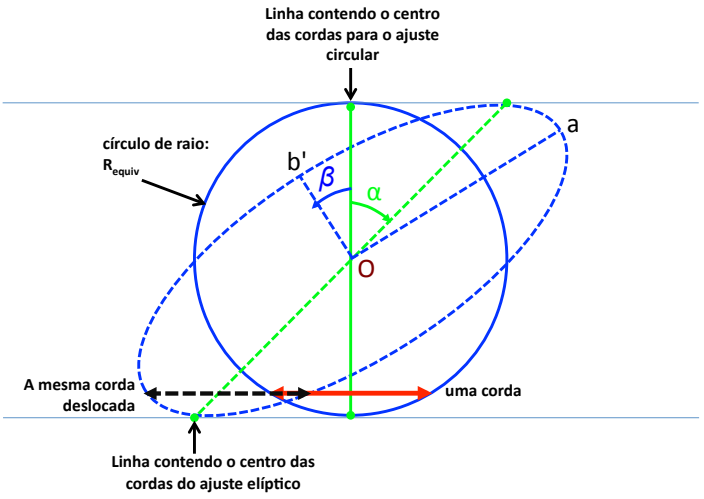


Figura 4.17 Figura esquemática para descrever os ângulos α e β utilizados para determinar o ângulo de posição PA a partir de um ϵ dado, com a condição de que a superfície equivalente seja conservada.

Colocando a origem das coordenadas no centro do círculo, tem-se a equação do cículo $X^2 + Y^2 = R^2_{\text{equiv}}$. Deslocando o ponto (X, Y) horizontalmente, até o ponto $x = X +$

$Y\tan(\alpha)$ e $y = Y$ da equação do círculo, obtém-se a equação 4.17, onde α é o ângulo o qual a linha que une o centro das cordas faz com a vertical.

$$\frac{x^2\cos^2(\alpha)}{R^2_{\rm equiv}} + \frac{y^2}{R^2_{\rm equiv}} - xy\frac{\sin(2\alpha)}{R^2_{\rm equiv}} = 1. \tag{4.17}$$

A equação da elipse com semi-eixos a e b' e ângulo de posição β é dada pela equação 4.18.

$$x^2\left[\frac{\cos^2(\beta)}{a^2} + \frac{\sin^2(\beta)}{b'^2}\right] + y^2\left[\frac{\sin^2(\beta)}{a^2} + \frac{\cos^2(\beta)}{b'^2}\right] + xy\frac{a^2-b'^2}{a^2b'^2}\sin(2\beta) = 1 \tag{4.18}$$

Relacionando os termos das equações 4.17 e 4.18, tem-se a tabela de equações 4.19:

$$\begin{cases} \dfrac{\cos^2(\alpha)}{R^2_{\rm equiv}} = & \dfrac{\cos^2(\beta)}{a^2} + \dfrac{\sin^2(\beta)}{b'^2} \\ \dfrac{1}{R^2_{\rm equiv}} = & \dfrac{\sin^2(\beta)}{a^2} + \dfrac{\cos^2(\beta)}{b'^2} \\ -\dfrac{\sin(2\alpha)}{R^2_{\rm equiv}} = & \dfrac{a^2-b'^2}{a^2b'^2}\sin(2P_0) \end{cases} \tag{4.19}$$

Da tabela 4.19 é possível obter as relações entre α e β (Equação 4.20). Deve-se notar que, para α variando de $-90° \to 0^- \to 0^+ \to +90°$, β varia de $0 \to +45° \to -45° \to 0$, com uma descontinuidade em $\alpha = 0$. Deve-se notar também que os dois ângulos (α e β) têm sinais opostos e define-se α positivo. Finalmente, para obter PA ainda é preciso considerar o ângulo das cordas em relação a horizontal PA$= \beta + \phi$.

$$\tan^2\beta = \frac{b'}{a} = (1-\epsilon) \quad \text{e} \quad \tan(2\beta) = -\frac{2}{\tan(\alpha)} \tag{4.20}$$

Solução Maclaurin

Na figura 4.18, a intersecção entre a a região em V e a curva de equilíbrio Maclaurin fornece a solução Maclaurin para Quaoar. Assim, define-se o achatamento verdadeiro de Quaoar $\epsilon = 0,0897 \pm 0,006$ e a densidade $\rho = 1,99 \pm 0,14 \mathrm{g/cm^3}$. Considerando o erro sobre a massa de Quaoar M$= 1,40 \pm 0,21 \times 10^{21}$ kg, as barras de erro são alargadas para $\epsilon = 0,0897^{+0,026}_{-0,016}$ e a densidade $\rho = 1,99 \pm 0,42$ g/cm^3.

Considerando o achatamento encontrado $\epsilon = 0,0897$, calcula-se o ângulo de posição. Sendo α positivo, $\beta = -44,1 = -44,1 + 180 = 135,9°$. Considerando a inclinação das cordas de 12,5° em relação ao eixo Leste–Oeste, o ângulo de posição será PA$= \beta + 12,5 =$

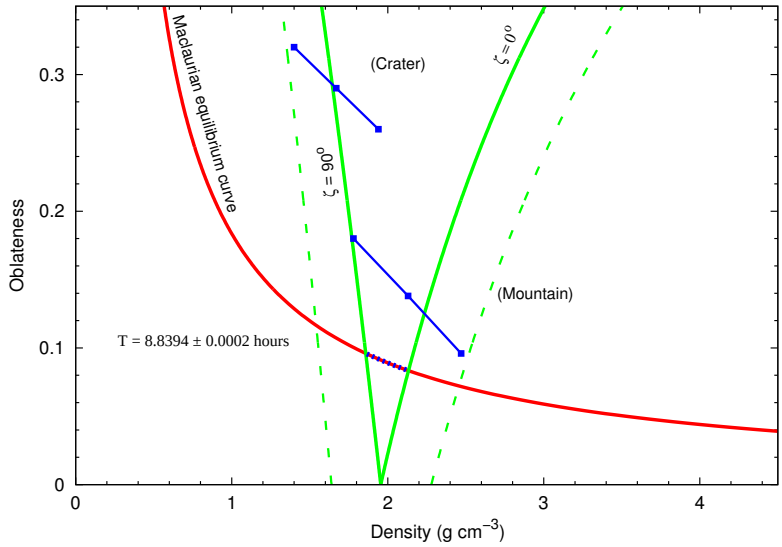


Figura 4.18 Curva de equilíbrio hidrostático para objeto Maclaurin (linha vermelha, Figura 4.14) comparada à curvas de elipses com $R_{equiv} = 555 \pm 2,5$, com ζ de 0 a 90° (linhas verdes). Considerando o erro sobre a determinação da massa ($M = 1,40 \pm 0,21 \times 10^{21}$ kg) a região em V se estende às linhas verdes tracejadas. Os segmentos em azul correspondem aos valores calculados para as soluções *cratera* e *montanha* com suas barras de erro a 1σ, obtidos mais adiante (tabela 4.5). A intersecção entre a região em V e a curva de equilíbrio, tracejada em azul, define a faixa de solução Maclaurin para Quaoar.

148, 4° em relação ao eixo Norte–Sul.

A figura 4.19 apresenta a elipse para a solução Maclaurin. Para seu ajuste, os únicos parâmetros livres são f_c e g_c, o centro da elipse, já que todos os outros parâmetros são deduzidos a partir de R_{equiv} e da massa de Quaoar. O ajuste apresenta um $\chi^2_{min} = 1,33$. Para estimar a qualidade do ajuste, deve-se considerar ainda que quatro cordas foram deslocadas e ajustadas para se obter a superfície equivalente. Então, o modelo Maclaurin exige o ajuste de M=7 parâmetros, quatro cordas deslocadas, R_{equiv} e f_c e g_c, então $\chi^2_{pdf} = \chi^2/(10-7) = 0,44$, indicando que o ajuste é melhor do que o esperado pelas barras de erro.

Este ajuste é obtido com a suposição de que os instantes das cordas podem ser deslocados, mesmo sem nenhum motivo aparente para isto. No entanto, nota-se que os deslocamentos necessários sobre as cordas de Ponta Grossa e Florianópolis estão dentro da faixa de 1σ dos instantes de início/fim de evento obtidos. Já para Armazones o deslocamento é de cerca de $2{,}5\sigma$ acima do erro dos instantes de evento.

Na tabela 4.3 os parâmetros das elipses obtidas para o ajuste das diferentes soluções são apresentados. Os valores de χ^2 e χ^2_{pdf} podem ser usados como medidas de qualidade

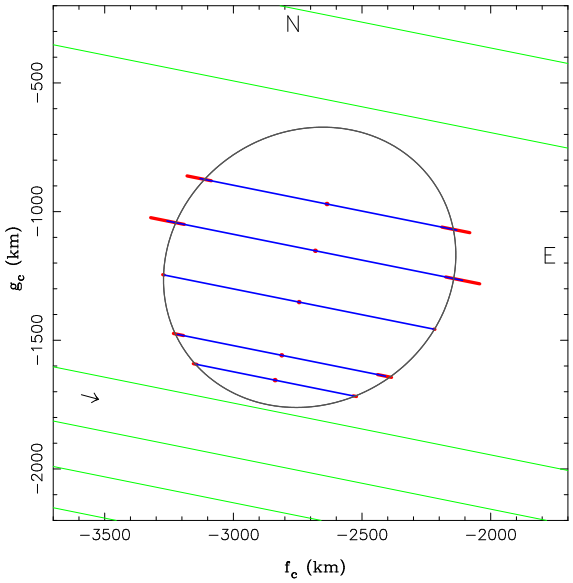


Figura 4.19 A elipse correspondente à solução Maclaurin. A partir do raio equivalente $R_{equiv} = 555 \pm 2,5$ determina-se o achatamento de $\epsilon = 0,0897^{+0,026}_{-0,016}$, a densidade $\rho = 1,99 \pm 0,42 g/cm^3$ e o ângulo de posição PA$= 148,4 \pm 0,9°$. Para esta solução, as cordas tiveram que ser deslocadas da seguinte maneira: San Pedro de Atacama: fixada; Rivera: -9,1 s; Armazones: -4,6 s; Florianópolis: -2,4 s; e Ponta Grossa: -3,8 s.

dos ajustes.

Tabela 4.3 Os resultados obtidos nas diferentes soluções são apresentados aqui. O raio equatorial ($R_{\text{equat.}}$), que corresponde ao semi-eixo maior a' da elipse ajustada, também é apresentado. As barras de erro a 1σ podem ser visualizadas nos mapas χ^2_{min}, sendo a região em preto do painel esquerdo das figuras 4.7 e 4.9. A determinação das barras de erro da solução Maclaurin foi discutida ao longo do texto.

Soluções:	Cratera	Montanha	Maclaurin
$R_{\text{equat.}}{}^{\dagger}$ (km)	656 ± 12	568 ± 12	569^{+24}_{-17}
Achatamento†	$0,29 \pm 0.04$	$0,14 \pm 0,05$	$0,0897^{+0,0268}_{-0,0175}$
f_c (km)	-2647 ± 19	-2712 ± 16	-2710 ± 4
g_c (km	-1253 ± 20	-1202 ± 32	-1216 ± 5
PA (°)	21 ± 3	$7,5 \pm 11$	147 ± 2
χ^2_{min}	15,1	15,7	1,33
χ^2_{pdf}	1,68	1,74	0,44

†Para as soluções cratera e montanha, o semi-eixo maior e o achatamento são aparentes, já para solução Maclaurin, são os valores reais.

Na tabela 4.4 a posição astrométrica derivada do ajuste da ocultação é dada. Ainda são apresentados o instante e distância de C/A e os *offset* de Quaoar em relação à efeméride JPL#21/DE405.

Tabela 4.4 Do ajuste da ocultação pode-se obter: a posição relativa entre a estrela e Quaoar; o instante de maior aproximação em relação ao geocentro, dado em h:mm:ss,s±ss, de 04 de maio de 2011; e o *offset* correspondente médio das soluções obtidas, em relação à efeméride JPL#21/DE405, dado em mas, onde 1mas equivale à 30,714 km à distância de 42,35 ua, o erro da posição da estrela é considerado sobre o *offset*. A posição da estrela é dada na tabela4.2.

Data	04 de maio 2011
Instante do C/A	02:38:37,32±2,6 s
Distancia do C/A	37±1,7 mas Sul
offset AR	-86,4±9 mas
offset DEC	-63±15 mas

4.9 Albedo

O albedo geométrico de um objeto é definido como a razão do seu fluxo com ângulo de fase nulo ao fluxo de um disco plano Lambertiano com a mesma área refletora. Um disco de Lambert é aquele que espalha a luz de maneira perfeitamente difusa, ou seja, que reflete a luz de maneira isotrópica. Ele é importante para se derivar possíveis propriedades da superfície do objeto, como tamanhos das partículas. Éris foi observado em uma ocultação estelar em 2010 (Sicardy et al. 2011) e seu albedo geométrico determinado na banda do visível $p_V = 0,96^{+0,09}_{-0,04}$. Segundo este resultado, o seu albedo pode ser superior a um. Isto se explica pelo fato de que as superfícies dos TNOs não são Lambertianas, e muitos apresentam reflexão preferencial na direção do observador quando o ângulo de fase é pequeno. Este efeito é conhecido como efeito de oposição (Schaefer et al. 2009).

Pode-se calcular o albedo geométrico de Quaoar a partir dos resultados das diferentes soluções acima apresentadas, e sua magnitude absoluta $\mathsf{H_Q}$ encontrada na literatura. A magnitude absoluta corresponde à magnitude que o objeto teria se estivesse à 1 ua do Sol e observado à 1 ua da Terra exatamente na oposição ($\alpha = 0$). Na banda do visível, a magnitude absoluta de Quaoar é $\mathsf{H_{Q,V}}$= 2,82 ± 0,06.

A relação entre a refletividade I/F, o albedo geométrico p e a magnitude absoluta é dada pela equação 4.21, onde $\mathsf{ua_{km}}$= 1,49598 $\times 10^8$ km e $H_\odot$ corresponde à magnitude aparente do Sol à 1 ua. A refletividade $(I/F)(\alpha)$ é a razão entre o brilho de Quaoar I observado com o ângulo de fase α, e o brilho F de uma superfície de Lambert. A função de fase $\Phi(\alpha)$ é a correção à ser aplicada ao albedo, sendo $\Phi(0) = 1$, que depende de propriedades da superfície. Normalmente ela é empiricamente determinada, observando o objeto em uma grande gama de ângulos de fase. Devido à descoberta relativamente recente, associada ao longo período orbital, ainda não é possível conhecer corretamente este parâmetro para os TNOs.

$$(I/F)(\alpha) = \Phi(\alpha) \cdot p = \Phi(\alpha) \cdot (\mathsf{ua_{km}}/R_{\mathsf{equiv}})^2 \cdot 10^{0.4(\mathsf{H_\odot - H_Q})} \tag{4.21}$$

Usando $H_{\odot,\mathsf{V}} = -26,74$ e os valores obtidos nas soluções acima calcula-se os possíveis valores para o albedo de Quaoar apresentados na tabela 4.5. Se considerarmos toda a gama de raios equivalentes obtidos, de 504 a 566 km, a gama de albedos geométricos no visível para Quaoar é $p_{\mathsf{v,Q}} = 0,102$ *a* $0,139$.

4.10 Densidade

A densidade global é um dos parâmetros físicos mais fundamentais de um objeto. Ela é a primeira estimativa da composição geral do corpo. De maneira simplista, se a densidade for baixa (~ 1 $\mathsf{g/cm^3}$) ele é constituído de gelo de água e/ou outros gelos; se for alta (~ 3 *a* 4 $\mathsf{g/cm^3}$), é constituído principalmente por rochas. É também o primeiro indicativo de sua estrutura interna. Se a densidade global for similar à densidade dos componentes da superfície, é um indicativo que seu interior é homogêneo; se for mais baixa, indica alta porosidade; se for mais alta pode indicar a presença de uma camada porosa (formada, por exemplo, por gelo ou regolito) e interior mais denso (Carry 2012).

De fato as propriedades físicas como a densidade contêm informações sobre os processos térmicos e químicos do disco proto-planetário, ou de onde e quando, neste disco, estes objetos foram formados. Por exemplo, a razão entre quantidade de rocha e gelo $a \sim 70/30$ é associada à um processo de acreção no Sistema Solar exterior (McKinnon et al. 2008).

Com a descoberta de Weywot e a determinação de sua órbita, tem-se o seu período de revolução (P). A partir de terceira Lei de Kepler pode-se determinar a massa do sistema (Equação 4.22), onde G é a constante gravitacional universal, e a_{orb} é o semi-eixo maior da órbita.

$$M = \frac{4\pi^2 a_{orb}^3}{G\mathsf{P^2}} \tag{4.22}$$

Diferentes soluções orbitais foram calculadas (Fraser & Brown (2010) e Vachier et al. (2012)) e delas a massa determinada ($\mathrm{M}_Q = 1,6 \pm 0,3 \times 10^{21}$ kg). Com observações adicionais, realizadas em 2011, Fraser et al. (2013) determinaram uma órbita mais precisa e portanto uma massa mais precisa: $\mathrm{M}_Q = 1{,}40 \pm 0{,}21 \times 10^{21}$ kg. Com esta massa e o volume de um elipsoide de revolução à partir dos raios equatorial e polar obtidos nas diferentes soluções apresentadas, calcula-se a densidade de Quaoar (Eq. 4.23). Os resultados para cada solução são apresentados na tabela 4.5, considerando toda a gama das soluções com $R_{equat} = 555$ a 668 km e $\epsilon' = 0,073$ a $0{,}33$, a faixa de densidades derivadas para Quaoar é $\rho_Q = 1,40$ a $2{,}46$ $\mathsf{g/cm^3}$.

$$\rho = \frac{3M}{4\pi a^2 c} \tag{4.23}$$

Tabela 4.5 Usando as diferentes soluções calculou-se o albedo e a densidade de Quaoar. Para calcular as densidades utilizou-se o volume de um esferoide oblato (ver seção 4.8), a partir dos raios equatorial e polar oriundos das soluções. As barras de erro correspondem às faixas de valores para cada solução, combinadas com os erros de $H_{\mathsf{Q,V}}$ e da massa M disponíveis na literatura (DeMeo et al. (2009) e Fraser et al. (2013), respectivamente). A última linha é a combinação de todas as faixas de resultados apresentadas acima.

Solução	$R_{\mathsf{equiv.}}$ (km)	Achatamento	Albedo (banda V)	Densidade ($\mathsf{g/cm^3}$)
Cratera	552 ± 14	0,29±0,04	0,110 ± 0,002	1,67 ± 0,27
Montanha	524 ± 20	0,14±0,05	0,123 ± 0,016	2,12 ± 0,34
Maclaurin	555 ± 2,5	$0{,}0897^{+0{,}0268}_{-0{,}0175}$	0,109 ± 0,007	1,99 ± 0,42
Faixas	504 – 566	0,73 – 0,33	0,102 – 0,139	1,40 – 2,46

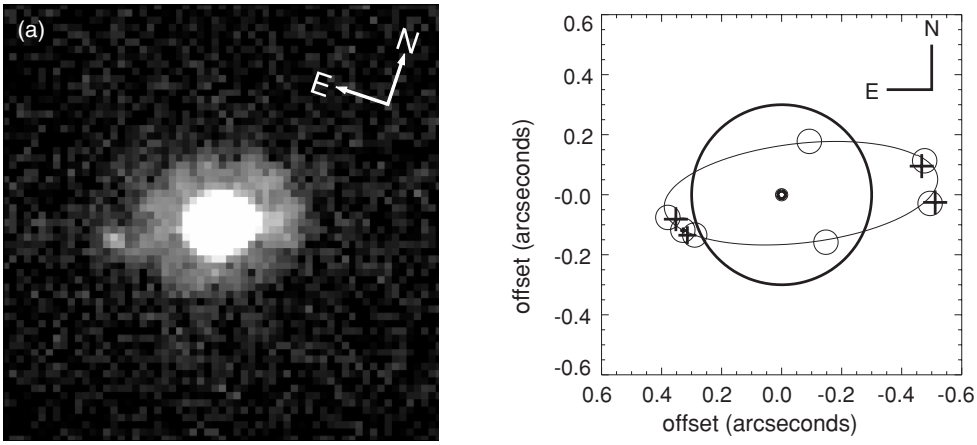
A solução Maclaurin parece ser a melhor, já que as condições sobre as propriedades físicas de Quaoar levam à conclusão de que ele deve estar em equilíbrio hidrostático. Caso Quaoar não esteja em equilíbrio hidrostático, duas hipóteses podem ser levantadas: (1) O processo que leva os objetos do Sistema Solar ao equilíbrio ainda não é bem compreendido; (2) Um mecanismo recente atuou sobre Quaoar tirando-o do equilíbrio. Por exemplo, se ele sofreu um grande impacto na sua história recente e ainda não retornou para o estado de equilíbrio hidrostático. Essa hipótese não é sustentada pelo albedo calculado, que é característico de uma superfície antiga.

4.11 Ocultação por Weywot

Como já foi dito, Quaoar tem uma lua, (50000/1) Weywot, descoberta em 2006 (Brown & Suer 2007). O satélite tem cerca de 5 magnitudes a menos que Quaoar (Figura 4.20(a)), então, assumindo que tenham o mesmo albedo, a razão entre seus tamanhos é de 12:1. Usando o tamanho de Quaoar, determinado do ajuste da ocultação, Weywot teria um diâmetro equivalente de ∼90 km e massa de 7,1 × 10^{17} kg, se a densidade for igual à de Quaoar (1,93 ± 0,5 g/cm^3).

A órbita de Weywot tem semi-eixo maior de 14000 km, excentricidade 0,13, inclinação em relação ao plano J2000 de -38° e período orbital de 12,4 dias (Vachier et al. (2012)). No plano do céu, o plano orbital projetado de Weywot é dado entre 5° e 15° . Pode-se comparar este plano com o ângulo de posição obtido na solução Maclaurin (PA = 148,4°), que se encontra a cerca de 130° do plano orbital de Weywot. Ou seja, esta solução descarta

a possibilidade de Weywot ter uma órbita equatorial.



(a) Imagem de Quaoar e Weywot (b) Orbita Weywot

Figura 4.20 **(a)** Imagem de Quaoar e Weywot que permitiu a descoberta da lua. **(b)** Ajuste de órbita, onde as cruzes representam observações diretas do satélite e o círculo central grande a região onde a lua estaria escondida pelo brilho de Quaoar. (Fraser & Brown 2010).

De acordo com as efemérides para Weywot (Vachier et al. 2012), disponibilizadas [2] pelo IMCCE (do francês, *Institut de Mécanique Céleste et de Calcul des Éphémérides*), a lua ocultaria a mesma estrela cerca de onze minutos depois para sítios ∼800 km ao norte de Quaoar. A ocultação poderia durar até 5 segundos.

O Observatório do Pico dos Dias (OPD) estava a cerca de 700 km ao norte da sombra e CEAMIG a cerca de 1000 km ao norte (Tabela 4.1), mas nenhum evento secundário foi detectado nas curvas de luz. No entanto, a estratégia de observação não foi otimizada para detectar o satélite e sim o corpo principal. As observações no CEAMIG foram feitas com 6 segundos de exposição com um tempo-morto de 3 segundos. Então, se a ocultação de Weywot, com 5 segundos de duração, tivesse ocorrido durante uma integração, uma queda de 2,8 magnitudes teria sido observada. Se tivesse ocorrido logo no início de um tempo-morto, uma queda de 1,1 magnitudes na imagem seguinte seria esperada. Casos intermediários apresentariam quedas de magnitude intermediárias, mas nenhuma queda desta ordem foi observada. No OPD o tempo de exposição foi de 0,95 segundos com tempo-morto de 1,05 segundos. Então, considerando uma ocultação de 5 segundos, detectaria-se queda de 3 magnitudes em pelo menos 2 imagens subsequentes, mas nenhuma queda foi observada próximo ao momento esperado.

[2] `http://www.imcce.fr/fr/ephemerides/formulaire/form_ephepos.php`

4.12 Atmosfera

O estudo espectroscópico dos objetos transnetunianos mostra que eles retiveram voláteis sobre suas superfícies. Gelo de água, em fase cristalina e amorfa, foi detectado em muitos objetos. Elementos como metano, amônia, nitrogênio molecular e até metanol e etano em alguns dentre os maiores objetos (Barucci et al. (2005) e Guilbert et al. (2009)). Em contraste, entre os objetos menores apenas gelo de água ou mesmo apenas um espectro plano são observados. De maneira simplista, os TNOs parecem se dividir em dois grupos: os maiores e ricos em voláteis, e o menores e pobres em voláteis (Schaller & Brown 2007).

A presença de voláteis é um dos pré-requisitos para a presença de uma atmosfera. É preciso também uma fonte de energia para transformar os voláteis em gás, e que a taxa de perda deste gás, por mecanismo Jeans ou hidrodinâmico, não seja tão grande a ponto de já terem perdido todo o gás na idade do Sistema Solar (Stern & Trafton 2008). Os pequenos TNOs não têm força gravitacional suficiente para terem retido uma atmosfera até a época atual, mas alguns objetos ainda podem possuir uma.

Três processos físicos podem dar origem à atmosferas nos TNOs: sublimação, que depende da temperatura de equilíbrio do corpo e dos elementos presentes; perda de gás do seu interior, devido a presença de gêiseres por exemplo; e impactos que podem ter energia suficiente para sublimar uma quantidade suficiente de gelos para originar uma atmosfera. Ainda, a temperatura no ponto subsolar pode ser suficientemente alta para gerar uma fina atmosfera local, que se recondensaria em outras regiões do objeto, mudando as características da superfície, como pode ser o caso de Makemake (Ortiz et al. 2012). Devido à excentricidade de suas órbitas, importantes mudanças sazonais são esperadas, como já se observou em Tritão e Plutão (Sicardy et al. 2003), e até mesmo recondensação completa da atmosfera, como deve ser o caso de Éris (Sicardy et al. 2011).

Até hoje, só se conhecem dois pequenos corpos do Sistema Solar externo que exibem atmosferas sustentadas pela gravidade do corpo, que apresentam estrutura térmica e evolução: Tritão e Plutão. Tritão é provavelmente um TNO que foi capturado por Netuno durante a fase de migração planetária (McKinnon & Leith (1995) e Nogueira et al. (2011)). Possui uma temperatura de equilíbrio (Tabela 4.6) de 32 K, suficiente para sublimar parcialmente N_2, CH_4 e CO. Sua atmosfera tem pressão de 2,3 μbar a 1400 km de altitude (seu raio é de 1352 km) (Elliot et al. 1998). Plutão tem temperatura de equilíbrio de 40 K, pressão de 4,5 μbar a 1215 km do centro (raio de 1180 $\pm$ 22 km) (Lellouch et al. 2009).

Para investigar a estabilidade dos voláteis nos TNOs ao longo da idade do Sistema Solar, Schaller & Brown (2007) desenvolveram um modelo simples de perda de atmosfera pelo mecanismo Jeans, analisando os gelos de N_2, CH_4 e CO. Para Quaoar, levando em consideração as propriedades físicas, como diâmetro, temperatura média e distância média ao Sol, eles concluíram que N_2 é muito leve para ter ficado retido nesta escala de tempo,

mas CH_4 e CO estariam próximos à curva de estabilidade (Figura 4.21). É importante frisar que o escape Jeans fornece apenas um limite inferior para a taxa de escape. Outros mecanismos de escape, se atuantes, serão mais eficientes em retirar os gases do objeto.

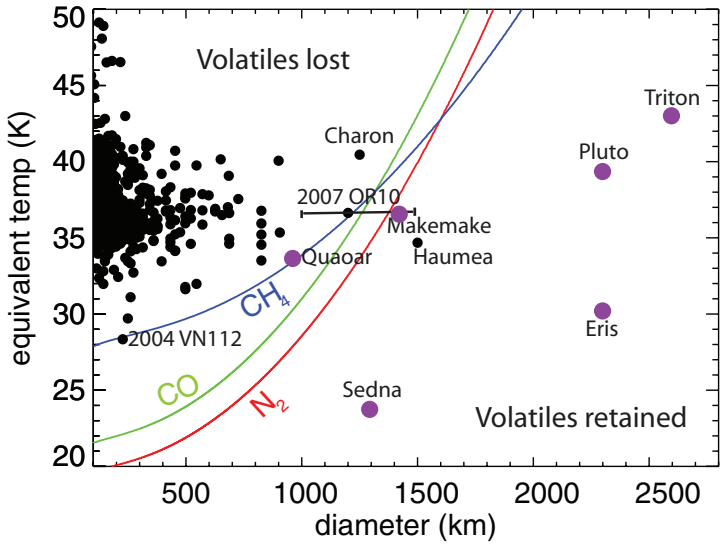


Figura 4.21 Retenção de voláteis pelos TNOs ao longo da idade do Sistema Solar. Objetos à esquerda das curvas de N_2, CO e CH_4 são muito pequenos e/ou quentes para terem retido voláteis, os à direita os mantiveram. Todos os objetos marcados por pontos roxos tiveram CH_4 detectado em suas superfícies (Brown et al. (2011), atualizado de Schaller & Brown (2007)). Quaoar encontra-se numa zona de transição.

A presença de gelo de água na sua fase cristalina em diversos TNOs, inclusive Quaoar, implica que suas superfícies estiveram expostas à temperaturas maiores que ∼100 K numa época relativamente recente, $< 10^7$ anos (Cooper et al. (2003) e Jewitt & Luu (2004)). Alguns voláteis, ao atingirem esta temperatura, seriam rapidamente perdidos, podendo dar origem ao escape hidrodinâmico. A zona de transição entre uma atmosfera que perde lentamente seus gases pelo mecanismo Jeans e uma atmosfera que é rapidamente perdida pelo mecanismo hidrodinâmico é investigada (Levi & Podolak 2009). Na figura 4.22 é apresentado um gráfico entre a densidade do corpo e seu tamanho, com curvas de temperatura máxima que poderiam atingir antes de submeter o gás de metano ao mecanismo hidrodinâmico. Alguns TNOs são mostrados, Quaoar encontra-se numa zona de transição. Para o N_2 e CO, apenas o escape Jeans teria ocorrido.

No entanto, os regimes de escape dependem mais da razão de mistura entre os gases do que de suas pressões de vapor individuais, mesmo que um regime hidrodinâmico possa aumentar a perda de CH_4 parcialmente. Além disso, a firme detecção de gelo de metano na superfície indica que estes efeitos não foram suficientemente eficazes a ponto de terem retirado todo o metano de Quaoar.

A massa de Quaoar e o raio equivalente derivado (R_{equiv} = 555 km) implicam numa

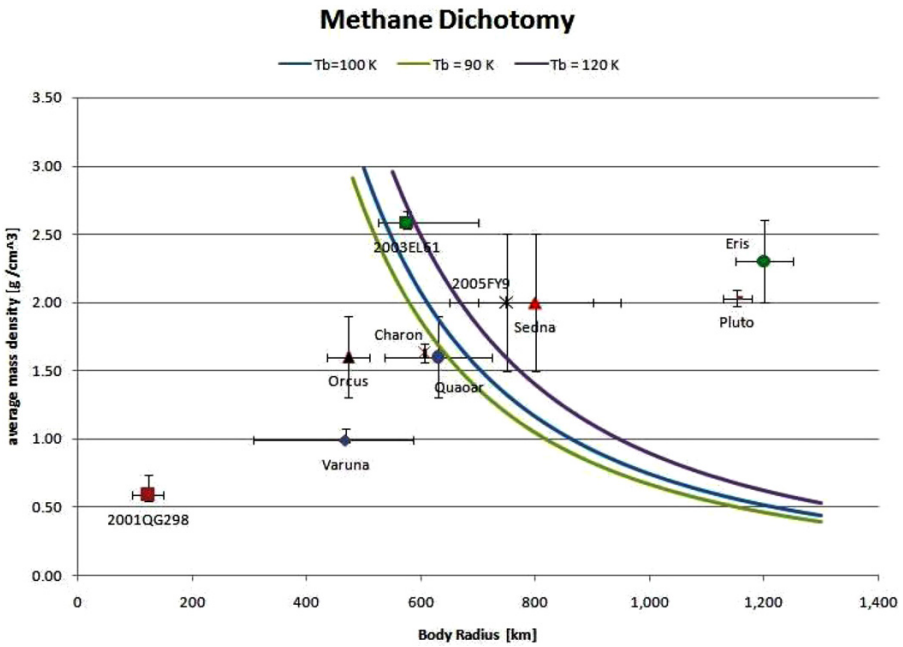


Figura 4.22 Transição entre escape Jeans ou Hidrodinâmico para uma suposta atmosfera de metano. As curvas correspondem à temperatura máxima que o corpo poderia se submeter ao mecanismo hidrodinâmico. Para os objetos à esquerda, o metano é perdido pelo mecanismo hidrodinâmico; para os à direita, apenas o escape Jeans atua. Quaoar encontra-se numa zona de transição. (Levi & Podolak 2009).

gravidade superficial de $\sim 0,4\ m/s^2$. Este valor é intermediário aos valores calculados para Plutão e Caronte, 0,64 e 0,30 m/s^2 respectivamente (Tholen et al. 2008).

Muitos voláteis já foram detectados sobre a superfície de Quaoar (Figura 4.23): ∼36% de gelo de água em fase cristalina e amorfa (Jewitt & Luu 2004), ∼13% de gelo de metano e possivelmente ∼4% etano (Schaller & Brown (2007b) e Guilbert et al. (2009)), e talvez ∼20% de nitrogênio molecular (Dalle Ore et al. 2009). Segundo Schaller & Brown (2007b) seu espectro é de um objeto geologicamente morto e frio, perdendo seus voláteis.

Então, como Quaoar se encontra em zonas de transição entre ter perdido ou não sua atmosfera pelos mecanismo Jeans ou hidrodinâmico, e possui voláteis em sua superfície, é possível que possua uma atmosfera, assim como Plutão. Como a ocultação estelar é a técnica mais eficaz para se detectar (ou colocar limite superior à presença de) tênues atmosferas à pressões tão baixas quanto o nanobar, investiga-se sua assinatura nas curvas de luz de ocultação obtidas em 2011.

A estrutura térmica da atmosfera de Quaoar obviamente não é conhecida, mas pode-se esperar que seja composta principalmente por metano ou dióxido de nitrogênio (se presente na superfície). A parte baixa da atmosfera deve estar em equilíbrio térmico com a superfície, então calcula-se a temperatura de equilíbrio da superfície correspondente à

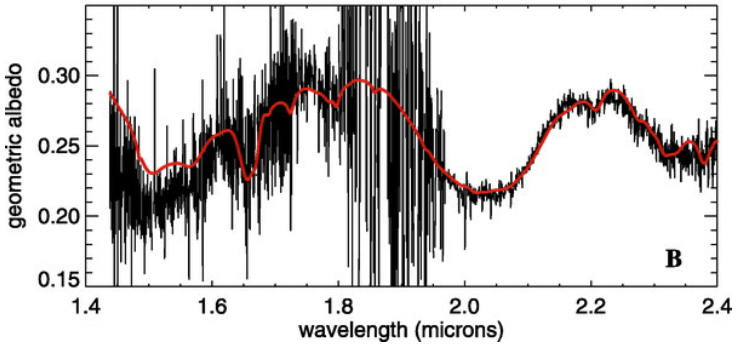


Figura 4.23 O espectro de Quaoar mostra a presença de gelos de H_2O em fase amorfa e cristalizada, de CH_4 e possivelmente de CO e N_2 (Brown & Suer 2007).

um corpo sem atmosfera usando as equações da tabela 4.6.

A temperatura no ponto sub-solar de uma superfície lambertiana com emissividade 1 é T_0, onde $F_\odot(1\ \mathtt{ua})$= 1367 W m^{-2} é o fluxo solar integrado em todos os comprimentos de onda a 1 ua, e σ_{SB}= 5,670×10^{-8} W m^{-2} K^{-4} é a constante de Stefan-Boltzmann. No momento da ocultação de 2011, Quaoar se encontrava a $r = 42{,}35$ ua, então $T_0 = 60{,}5$ K. A temperatura de Quaoar no ponto sub-solar $T_{\mathsf{sub}\odot}$ depende do albedo ($p_v = 0,120 \pm 0,018$) e da integral de fase q que é a razão entre o albedo de Bond A e o albedo geométrico p. Para Quaoar $q \sim 0,52$, então $T_{\mathsf{sub}\odot} = 59,5$ Kelvins.

A temperatura de equilíbrio da superfície depende da velocidade de rotação do objeto. As equações dadas na tabela 4.6 correspondem: à esquerda, a temperatura sub-solar de equilíbrio de um objeto com rotação lenta T_{ss}, e sua temperatura global T_{ssg}; à direita, a temperatura de um corpo com rotação rápida T_{equ}, e sua temperatura global T_{eqg}. A emissividade ε, que é a eficiência do corpo em emitir globalmente o fluxo térmico, e o fator relacionado à eficiência do corpo para emitir radiação na direção do observador η, também são considerados (Stansberry et al. 2008). Para os TNOs mede-se um valor típico de $\varepsilon = 0,9$. Para Quaoar calcula-se $\eta = 1,7$ (Fornasier et al. (2013), comunicação privada).

Para Quaoar calculam-se as temperaturas $T_{\mathsf{ss}} = 54$ K e $T_{\mathsf{equ}} = 40$ K. No entanto, com um período de rotação de 8,8394 h, Quaoar coloca-se com rotação intermediária, e por isto assume-se como temperatura superficial de Quaoar T$_{sup} = 42$ K, que é tomada como temperatura de sua atmosfera inferior.

A temperatura da parte superior de sua atmosfera pode ser calculada pelo "modelo-termostato" do metano (Yelle & Lunine 1989). Nele o aquecimento pela absorção de radiação no infra-vermelho é balanceada pela emissão na banda do metano a 7,7μm. À distância de Quaoar, o fluxo solar é quase duas vezes menor que em Plutão, a 31 ua. Para Plutão, o modelo prevê que a atmosfera superior atinja 106 K à partir de 10 km acima

Tabela 4.6 Temperaturas de equilíbrio de um corpo sem atmosfera: primeira linha: T_0 temperatura no ponto sub-solar de uma superfície lambertiana com emissividade 1; segunda linha: $T_{\mathsf{sub}\odot}$ temperatura no ponto sub-solar de um dado objeto; terceira linha: T_{ss} objeto com rotação lenta, T_{equ} objeto com rotação rápida; quarta linha: respectivas temperaturas globais T_{ssg} e T_{eqg}.

$T_0 = [F_\odot/(r^2\sigma_{\mathsf{SB}})]^{1/4}$	
$T_{\mathsf{sub}\odot} = T_0 \cdot [(1 - p_{\mathsf{V}}q)]^{1/4}$	
Rotação lenta	Rotação rápida
$T_{\mathsf{ss}} = T_0 \cdot [(1 - p_{\mathsf{V}}q)/(\varepsilon\eta)]^{1/4}$	$T_{\mathsf{equ}} = T_0 \cdot [(1 - p_{\mathsf{V}}q)/(\pi\varepsilon\eta)]^{1/4}$
$T_{\mathsf{ssg}} = T_0 \cdot [(1 - p_{\mathsf{V}}q)/(2\varepsilon\eta)]^{1/4}$	$T_{\mathsf{eqg}} = T_0 \cdot [(1 - p_{\mathsf{V}}q)/(4\varepsilon\eta)]^{1/4}$

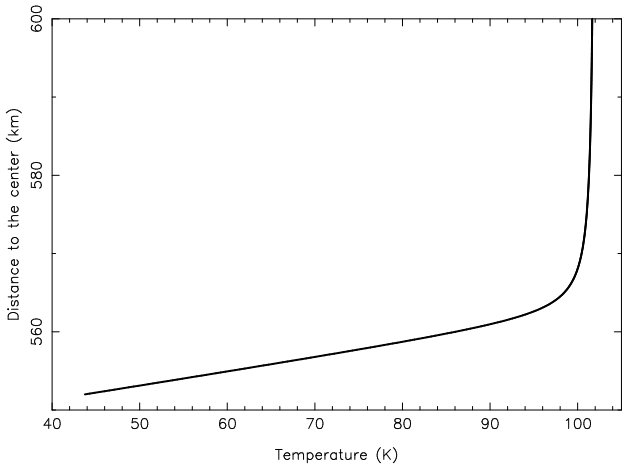


Figura 4.24 Estrutura térmica da atmosfera de Quaoar, como gradiente de 5,7 k/km e um perfil isotérmico com temperatura de 102 K que se estende até a estratosfera.

da superfície (Lellouch et al. 2009). À partir destas considerações, a atmosfera de Quaoar é modelada com temperatura de 102 K, à partir de 10 km de altitude e com 42 K na troposfera (parte inferior). Isto resulta no perfil de temperatura mostrado na figura 4.24, onde o gradiente de temperatura, entre a superfície e 10 km de altitude, é de 5,7 K/km, seguido por uma parte isotérmica a 102 K até a estratosfera.

A pressão de vapor de voláteis $\mathsf{p}_{\mathsf{vap}}$, a uma dada temperatura, pode ser calculada (Figura 4.25). Usando relações termodinâmicas propostas por Fray & Schmitt (2009), calculou-se a pressão de vapor para CO, CH_4 e N_2 de 30 a 60 K (Tabela 4.7). Pode-se então verificar qual seria a pressão superficial da atmosfera (teórica) para diferentes elementos na superfície de Quaoar à $T_{sup} = 42$ K: $N_2 = 176$ μbar; CO = 31 μbar; CH_4 = 32 nanobar.

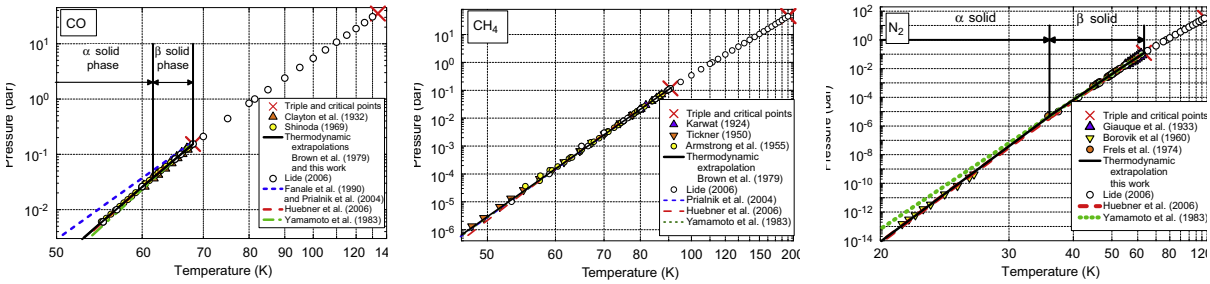


Figura 4.25 Temperatura versus pressão de volatilização e saturação por diferentes modelos para CO, CH_4 e N_2. Para o CO a fase alfa esta compreendida de 14 a 61,55 K e a fase beta de 61,55 a 61,8 K. Para o N_2 a fase alfa esta entre 10 a 35,61 K e a beta de 35,61 a 63,14 K (Fray & Schmitt 2009).

Tabela 4.7 Pressão de vapor para entre 30 a 60 K para CO, CH_4 e N_2, calculados à partir de relações termodinâmicas (tabela 5 de Fray & Schmitt (2009)). Para 42 K, a temperatura superficial de Quaoar, a pressões de vapor para CO, CH_4 e N_2 são 176 μbar, 31 μbar e CH_4 = 32 nanobar respectivamente.

T	CO	CH_4	N_2
(K)	p_{vap} (μbar)		
30	3,6E-03	4,3E-07	3,896E-02
32	2,63E-02	4,97E-06	0,2644
34	0,1518	4,32E-05	1,4288
36	0,7208	2,96E-04	6,4172
38	2,9008	1,66E-03	21,8446
40	10,1402	7,89E-03	65,5998
42	31,4007	3,227E-02	176,9673
44	87,5542	0,1162	435,1745
46	222,8098	0,3748	987,2374
48	523,4024	1,0964	2086,879
50	1145,8708	2,9436	4145,052
52	2357,0208	7,3245	7791,259
54	4587,0654	17,0332	13943,44
56	8496,5449	37,2864	23883,00
58	15056,529	77,3097	39330,62
60	25640,482	152,6466	62518,28

A curva de luz provocada pela presença de gases no entorno de um objeto pode ser modelada. Ao passar pela atmosfera do corpo, a luz da estrela sofrerá refração que dependerá do seu elemento constituinte, sua temperatura e pressão. A curva de luz observada ainda dependerá da distância do observador ao objeto D e da distância y à parte central da ocultação (Figura 4.26).

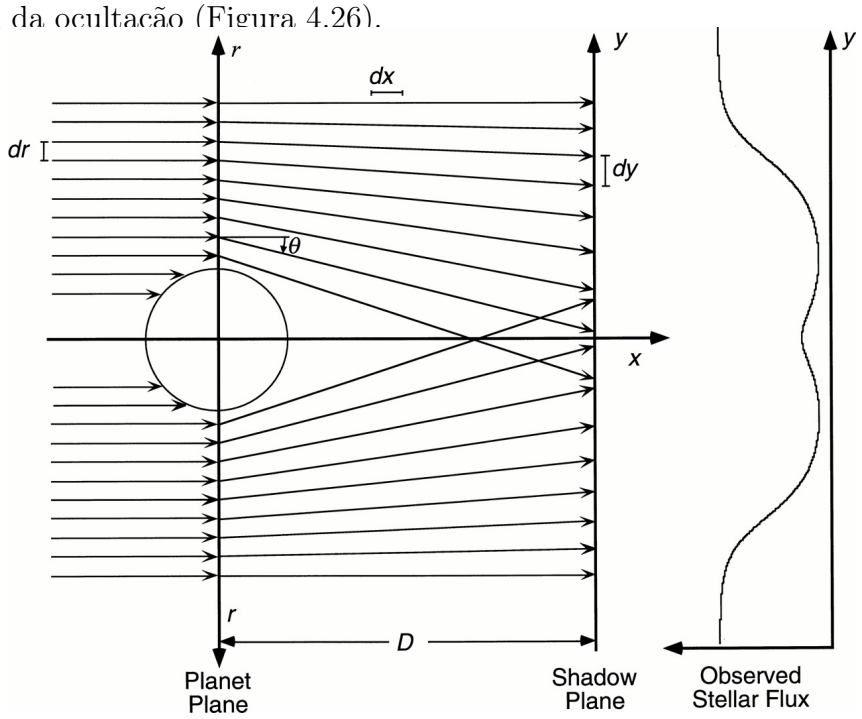


Figura 4.26 Figura esquemática da refração da estrela ao passar pela atmosfera do corpo. A refração dependerá do elemento constituinte da atmosfera, sua temperatura e pressão (Elliot & Olkin 1996).

Usando um código de traçado de raios (*ray tracing*) como descrito em Sicardy et al. (1999) & Sicardy et al. (2006b), modelou-se curvas de luz sintéticas para a atmosfera acima descrita. Estas curvas sintéticas são, então, comparadas com a curva de luz obtida durante a ocultação por Quaoar, pelo telescópio de 0,5 m, em San Pedro de Atacama. Com tempo de integração de 3 s e S/R=52, é a melhor curva de luz positiva do evento.

Como obteve-se diferentes soluções para a forma de Quaoar, a distância do centro do corpo ao limbo sondado não é conhecida. Para contornar este problema, modelou-se as curvas de luz sintéticas para uma gama de raios (de 550 a 560 km). Cada curva sintética foi comparada com a curva observada, e o χ^2 do ajuste para diferentes pressões obtido. Todos os ajustes são então verificados obtendo-se uma curva geral de χ^2 (Figura 4.27). Desta curva pode-se determinar o χ^2_{min}, $\chi^2_{min} + 1$, que corresponde ao erro a 1σ, e o $\chi^2_{min} + 9$, que correspondente ao erro a 3σ.

Obtém-se, então, um limite superior para a pressão superficial da atmosfera de Quaoar de 21 nanobar a 1σ, ou 56 nbar a 3σ. O perfil esperado para uma atmosfera de 21 nbar é mostrado na figura 4.28. O instante de cada exposição é transformado em distância ao

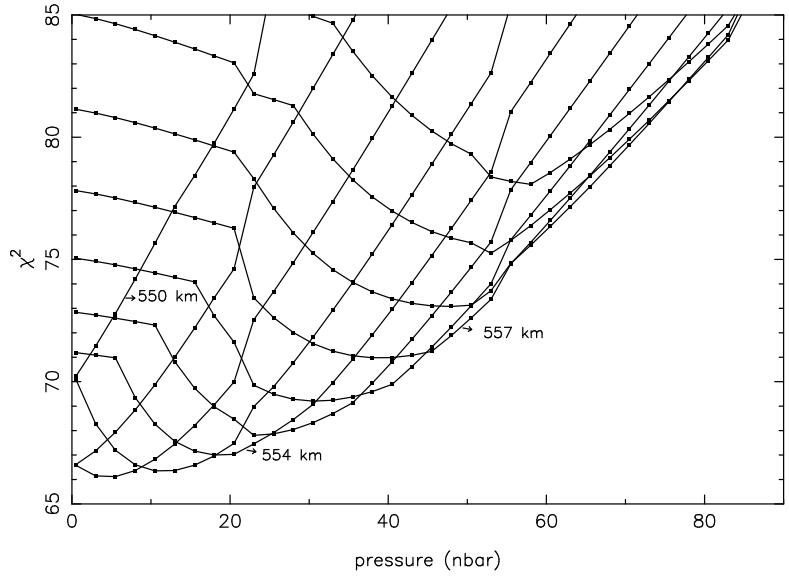


Figura 4.27 Curvas de χ^2 para diferentes distâncias do limbo ao centro do corpo (de 550 a 560 km). Atmosfera de metano como descrita acima (Fig. 4.24). Um limite superior de pressão superficial de 21 nanobar (1σ), ou 56 nanobar (3σ) é obtido.

centro de Quaoar. Os círculos azuis correspondem à curva sintética convolvida ao tempo de exposição de cada ponto observado.

Ajustando apenas os pontos correspondentes à imersão, tem-se uma detecção marginal de uma atmosfera de metano com pressão de 138^{+24}_{-32} nbar (1σ), mas a 3σ tem-se apenas um limite superior a 221 nbar (Figura 4.29(a)). Para o ajuste da emersão isoladamente, obtém-se apenas um limite superior a 145 nbar (Figura 4.29(b)). No entanto pressões como as obtidas deveriam estar globalmente distribuídas.

Mesmo não sendo compatível com "modelo-termostato" do metano, investigou-se o caso de uma atmosfera isotérmica. Neste caso haveria detecção de uma atmosfera a 42 K e pressão superficial de 54 $\pm$ 25 nbar (1σ) (Figura 4.30). As curvas de luz dos outros sítios não apresentam a qualidade suficiente para melhor interpretação dos dados.

Importante notar que o metano tem pressão de vapor $\mathsf{p_v} = 32$ nbar a 42 K, consistente com o limite superior obtido de 56 nanobar a 3σ. O nitrogênio molecular é mais volátil que o metano (Tabela 4.7). À 42 K sua pressão de vapor é de 176 μbar, muito acima dos limites de pressão aqui estabelecidos. Portanto não espera-se que N_2 seja um elemento abundante na atmosfera de Quaoar, mas não é possível descartar a presença de uma tênue atmosfera global de metano.

Como em Quaoar $T_{\mathsf{ss}} = 54$ K, e a pressão de vapor do metano a esta temperatura é de 17 μbar é possível que exista uma atmosfera local no ponto sub-solar que se condensaria ao nível de alguns nanobar no terminador, indetectável em nossos dados.

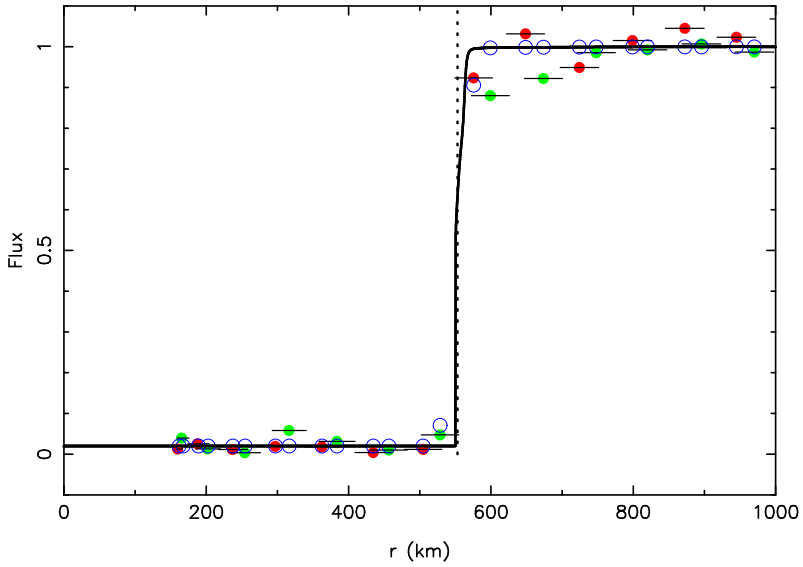
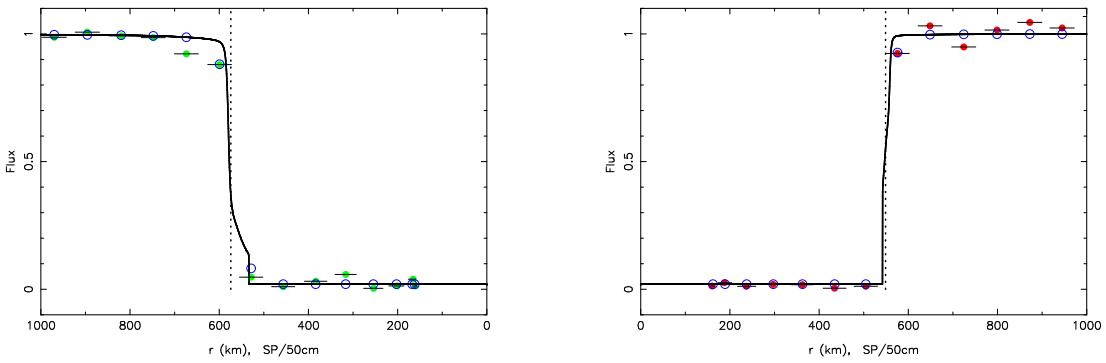


Figura 4.28 Curva de luz sintética de uma atmosfera de metano a 21 nbar de pressão superficial (linhas continuas e pretas). Pontos verdes e vermelhos correspondem aos dados da imersão e emersão respectivamente, os segmentos horizontais representam o tempo de exposição. Círculos azuis são obtidos após a convolução da curva sintética com o tempo de integração.



(a) Imersão

(b) Emersão

Figura 4.29 Curva de luz sintética para a imersão e emersão separadamente, apenas detecção marginal ou limite superior podem ser estabelecidos. Descrição na figura 4.28.

4.13 Outras Ocultações por Quaoar

A primeira ocultação por Quaoar foi observada em 11 de fevereiro de 2011. Detectada pelo grupo de ocultações do MIT (*Massachusetts Institute of Technology*). A ocultação envolveu uma estrela de magnitude R=15,8, prevista no programa caminhos. Foi observada do *Wallace Astrophysical Observatory* (Long.: -7129',1 L; Lat.: 4236',6 N; alt.: 107 m)

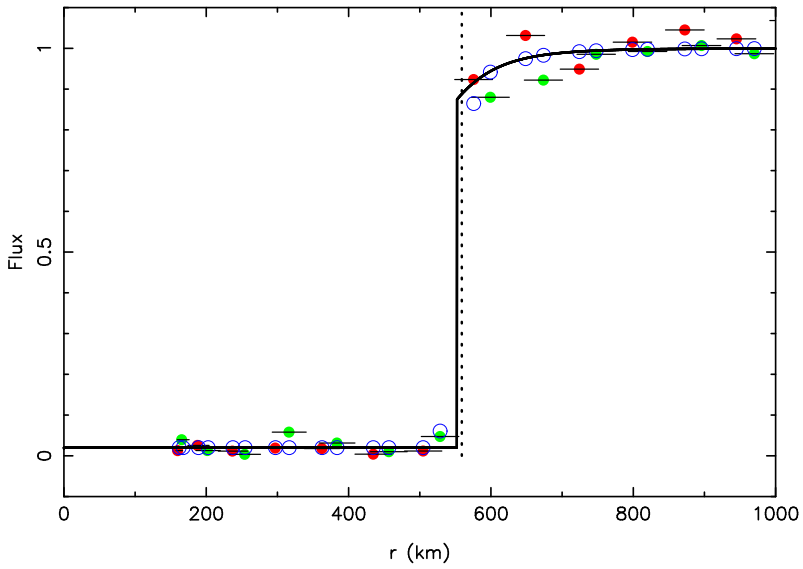


Figura 4.30 Dados de observação comparados à curva de luz sintética para atmosfera isotérmica a 42 K e pressão superficial de 54 ± 25 nbar (1σ). Descrição na figura 4.28.

situado em Westford, Maryland, Estados Unidos da América. O evento teve duração de 37,8 ± 1,9 s, o que corresponde à uma corda de 786 ± 39 km, com instante central às 10:04:02 TU (Person et al. 2011).

Em 2012, outras duas ocultações estelares por Quaoar foram observadas: em 17 de fevereiro, com duas cordas efetivas, e em 15 de outubro, com apenas uma corda. Elas são discutidas a seguir.

4.13.1 17 de Fevereiro de 2012

Em 17 de fevereiro observou-se uma nova ocultação por Quaoar. A predição, também oriunda do programa caminhos, colocava a faixa da ocultação passando, possivelmente, ao norte da Europa. Atualizações nas últimas semanas antes do evento não foram possíveis devido à pequena elongação solar. No entanto, observações astrométricas da estrela, feitas em setembro de 2010, aliadas ao *offset* de Quaoar derivado da ocultação de 04/05/2011, confirmavam que a ocultação poderia ser observável à partir da Europa 4.31.

Uma campanha observacional foi lançada a observadores com a abertura de uma página dedicada na internet (`http://www.lesia.obspm.fr/perso/braga-ribas/campaigns/2012-02-17_Quaoar.html`). A estrela, de magnitude R* = 15,1, poderia ficar ocultada por até 66 s, permitindo exposições longas (<15 s), portanto acessível à telescópios amadores.

Dentre os 19 sítios participantes da campanha, 15 foram impedidos de observar, ou por questões climáticas, ou por problemas técnicos. Dos quatro restantes, todos detectaram

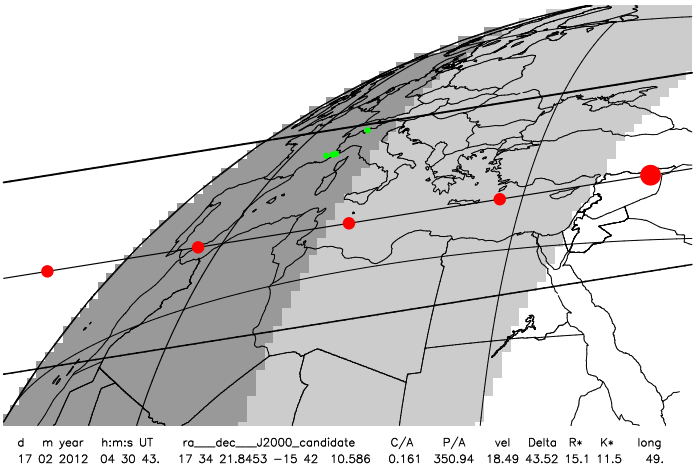


Figura 4.31 Mapa mostrando a posição da faixa de ocultação por Quaoar, observada em 17 de fevereiro de 2012. Os pontos verdes são os sítios que detectaram positivamente o evento. A sombra se desloca da esquerda para a direita. Outros 15 sítios dispostos nas Ilhas Canárias, Marrocos, Espanha, França, Alemanha e Itália, não puderam observar por questões meteorológicas ou problemas técnicos.

o evento (Figura 4.8). No entanto, três deles, todos na França, estavam separados por no máximo 3,1 km, em termos de latitude de Quaoar. O quarto em Gnosca/Suíça, estava a cerca de 136 km mais ao norte. Portanto, na prática, obteve-se duas cordas do perfil de Quaoar (Figura 4.34).

Para a redução fotométrica das imagens de vídeo obtidas, oriundas de Gnosca e Valensole, cada trama do vídeo foi transformada em imagem FITS para posterior fotometria diferencial. Os instantes registrados com dispositivos "incrustadores de tempo em vídeo" (ITV), onde a fonte de tempo oriunda de uma antena GPS é escrita diretamente sobre cada trama do vídeo. A observação de Tourrette-Levens foi feita com câmera CCD. A curva de luz foi obtida por fotometria diferencial e o instante de cada imagem lido diretamente do cabeçalho da imagem. O programa de captura utilizado foi o "MaxImDL", então os instante registrados foram truncados no segundo, e o procedimento descrito na seção 3.4.2 aplicado. Finalmente, no sítio TAROT[3], a observação foi realizada pelo método "traço". Neste método, o motor de acompanhamento sideral do telescópio é desligado, então ao se adquirir imagens com alguns segundos de duração, as estrelas caminham pelo campo da câmera, deixando traços nas imagens. Se a ocultação ocorrer, haverá queda de brilho ao longo do traço da estrela ocultada. Para esta ocultação, por ser uma estrela muito fraca para o instrumento, o acompanhamento foi regulado para 96,4% da velocidade de acompanhamento sideral. Com isto, a estrela ficaria 6 segundos sobre cada pixel, antes

[3]TAROT: Télescopes à Action Rapide pour les Objets Transitoires

Tabela 4.8 Características de observação nos sítios que detectaram o evento. Os instantes de ingresso e egresso são dados em tempo universal, em 17 de fevereiro 2012. † Método "traço", acompanhamento a 96,4%, resolução temporal de 6 s por pixel. ‡ Durante os primeiros minutos da observação o tempo de exposição foi de 10,24 s, e foi reduzido para 5,12 s cerca de 60 s antes do início do evento.

Observatório Cidade Observador	Latitude Longitude Altitude	Abertura Instrumento Escala de Placa	Exposição S/R Fonte de Tempo	Tempo (TU) Imersão Emersão
Gnozca	46° 13' 53",2 N	0,4 m	10,24 s	h:mm:ss,s±ss
Bellinzona/CHE	09° 01' 26",5 L	Watec 120N+	4,4	4:28:37,6±4,5
S. Sposeti	260 m	1",9	ITV IOTA GPS	4:29:09,4±5,5
TAROT	43° 45' 07",3 N	0,25 m	180† s	h:mm:ss,s±ss
Calern/FRA	06° 55' 25",1 L	CCD Andor	4,6	4:28:08,1±3,0 s
A. Klotz & E. Frappa	1270 m	3",3	GPS	4:29:06,5±3,5 s
	43° 51' 52",4 N	0,21 m	5,12‡ s	h:mm:ss,s±ss
Valensole/FRA	06° 00' 23",0 L	Watec 120N+	4,0	4:28:09,2±2,1 s
J. Lecacheux	622 m	0",84	TIM-10 GPS	4:29:03,2±2,2 s
Tourrette-	43° 47' 22",2 N	0,356 m	10 s	h:mm:ss,s±s,s
Levens/FRA	07° 15' 47",2 L	Apogee Alta U1	30	4:28:17,6±2,2 s
P. Tanga	385 m	2",7	NTP BeeperSync	4:29:04,3±0,5 s

de passar para o seguinte, devido à deriva do telescópio. Foram tomadas 3 imagens com 3 minutos de exposição cada e a ocultação foi detectada em uma delas, como pode-se ver na figura 4.32.

Durante a imagem onde detecta-se o início do evento obtida em Tourrette-Levens, ocorreu um problema de acompanhamento que fez com que as estrelas ficassem deformadas. Através de processamento da imagem, pode-se determinar que a ocultação ocorreu durante a exposição, pois ainda foi possível medir um fluxo parcial da estrela ocultada. Como Tourrette-Levens, TAROT e Valensole situam-se sobre a mesma corda (dentro de 3,1 km), corrigindo da paralaxe de suas posições geográficas, espera-se que os instantes de início e fim de Tourrette-Levens seguissem os de TAROT e Valensole de 1,25 e 3,85 s, respectivamente. No entanto, o início do evento em Tourrette-Levens aconteceu 9,5 ± 3 e 8,5 ± 3 s depois de TAROT e Valensole, ou ainda 8,5±3 ou 4,5±3 s, respectivamente, depois do instante esperado. O final do evento foi medido, 2,25 ± 2,5 e 1,1 ± 2,2 s antes de TAROT e Valensole ou 3,5 ± 2,5 e 2,75 ± 2,75 s respectivamente, antes do esperado. Portanto, há, claramente, um problema de datação nas imagens obtidas em Tourrette-Levens.

Para o ajuste da ocultação utilizou-se a corda de Gnosca, e, para a corda mais ao sul, os instantes medidos em Valensole, por apresentar maior precisão e nenhum problema de datação (Tabela 4.8).

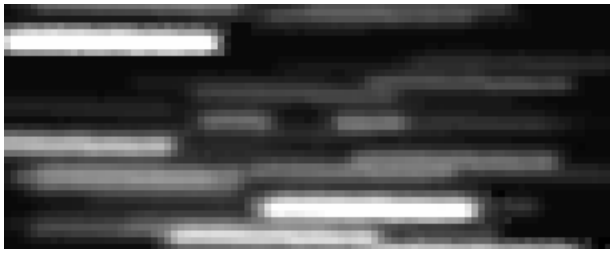


Figura 4.32 Imagem traço obtida pelo telescópio TAROT durante a ocultação. As estrelas se deslocam pelo campo do CCD e a estrela ocultada (no centro da imagem) apresenta uma queda de brilho durante o evento.

Como discutido na seção 4.6, numa elipse há cinco parâmetros a serem ajustados aos pontos das cordas. No entanto, neste evento, apenas quatro pontos estão disponíveis para o ajuste. Pode-se reduzir o número de graus de liberdade fixando a forma da elipse. No caso, limita-se o ajuste a apenas soluções circulares ($\epsilon = 0$) e, neste caso, o ângulo de posição é indiferente, e também fixado a zero (PA=0). Para o ajuste, variou-se o raio do círculo de 400 a 1400 km, com passos de 1 km. Assim os únicos parâmetros livres são f_c e g_c, o centro do círculo. Para cada raio o valor de χ^2 do ajuste é obtido. Assim tem-se uma curva para determinar o melhor ajuste (χ^2_{min}) e a barra de erro (Figura 4.33). A melhor solução é obtida para $R_Q = 688^{+443}_{-154}$ km (1σ). Estabelece-se aqui um raio inferior para Quaoar, caso circular, de $R_{Q,min} = 534$ km. Este resultado está em acordo com as observações da ocultação de maio de 2011. Vale citar que a distância no plano do céu entre as duas cordas negativas OPD e Salto, que cercam as cordas positivas do evento de 2011, é de 1222 km, então, se circular, o raio de Quaoar delimita-se entre 534 e 611 km.

O centro f_c e g_c são calculados usando a efeméride geocêntrica de Quaoar JPL#21. Eles podem ser utilizados para derivar a posição de Quaoar no plano do céu no momento de C/A. Na tabela 4.10, o instante de maior aproximação relativa ao geocentro e a distância são dados para o dia 17 de fevereiro de 2012. Todas as curva de luz deste evento apresentam tempos de exposição muito elevados (>5,12 s) e/ou S/R muito baixo para trazer restrições aos resultados obtidos do evento de 2011.

Tabela 4.9 Solução circular obtida com as cordas de Gnosca e Valensole. Obtém-se um raio mínimo para o caso de um corpo esférico. Barras de erro a 1σ (Figura 4.33).

Raio (km)	f_c (km)	g_c (km)	χ^2_{min}
688^{+443}_{-154}	-3182^{+34}_{-86}	-2895^{+514}_{-232}	0,02

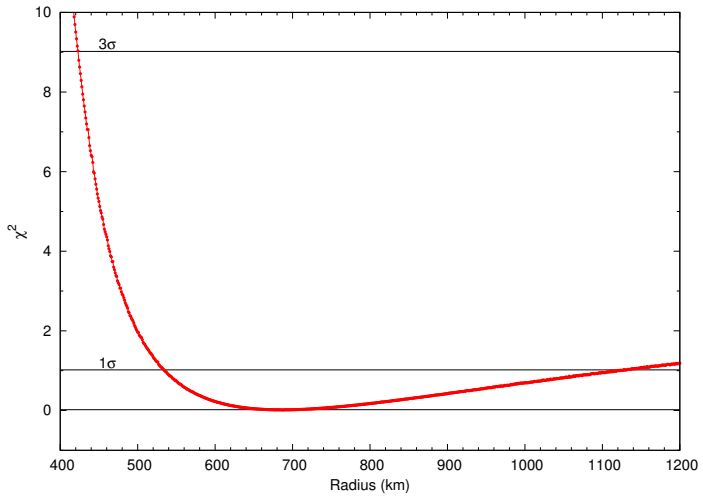


Figura 4.33 Curva de χ^2 do ajuste de um objeto circular às cordas observadas. Pode-se verificar que as soluções são limitadas (1σ) à raios maiores que 534 km, o que representa um limite inferior para o raio de Quaoar, no caso esférico.

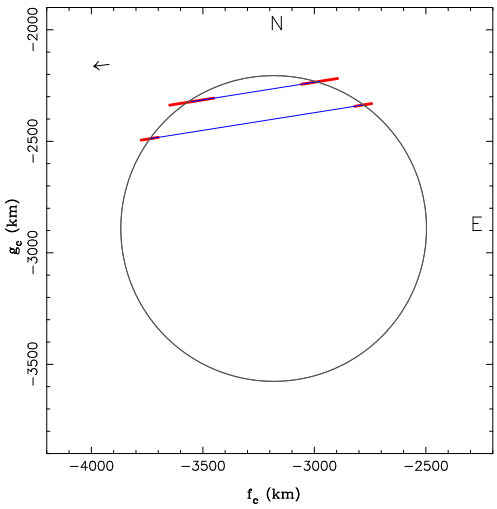


Figura 4.34 O ajuste circular sobre as duas cordas efetivas observadas na ocultação de 17 de fevereiro de 2012. O ajuste tem como solução $R_Q = 688^{+443}_{-154}$ km, estabelecendo um raio mínimo para o caso de Quaoar ser circular. Comparado aos resultados de 2011, conclui-se que o raio de Quaoar delimita-se entre 534 e 557.5 km, caso esférico.

4.13.2 15 de Outubro de 2012

Uma nova ocultação por Quaoar foi detectada no dia 15 de outubro de 2012. Inicialmente o evento era previsto para o norte do Chile, Bolívia e Argentina. A estrela, com magnitudes R*= 17,3 e K*= 14,4 seria um bom alvo para telescópios localizados nesta região. Atualizações apontaram um grande deslocamento da estrela, colocando o evento

Tabela 4.10 Do ajuste da ocultação pode-se obter a posição relativa entre a estrela e Quaoar. O instante de maior aproximação em relação ao geocentro, dado em h:mm:ss,s ± ss, de 17 de fevereiro de 2012. O *offset* corresponde ao centro da solução circular em relação à efeméride JPL#21/DE405, dado em mas, onde 1mas equivale à 31,561 km à distância de 43,5208 ua. A posição de Quaoar em ascensão reta e declinação no instante de C/A e aposição da estrela ocultada também são dadas.

Data	17 de fevereiro 2012
Instante do C/A	04:30:43,73 ± 0,11 s
Distancia do C/A	161 ± 12 mas Norte
offset AR	-101 ± 25 mas
offset DEC	-92 ± 24 mas
Quaoar AR	$17^h\ 34^m\ 21^s$,8438 ± 0^s,0023
Quaoar DEC	-15° 42' 10",4270 ± 0",012
Estrela AR	$17^h\ 34^m\ 21^s$,8453 ± 0^s,022
Estrela DEC	-15° 42' 10",5860 ± 0",0008

no sul do do Chile e Argentina. Tratando-se de uma região desprovida de telescópios adequados e de clima impróprio, não se realizou uma campanha, mas os colaboradores do centro-norte do Chile foram alertados para realizarem a observação, sabendo da baixa probabilidade de detecção.

Quatro sítios, San Pedro de Atacama, La Silla (TRAPPIST[4]) e Cerro Tololo (PROMPT[5]) e Cerro Burek (ASH[6]) participaram da observação. Apenas o PROMPT teve condições técnicas e climáticas para monitorar o evento.

Foram utilizados dois telescópios no mesmo sítio (Long.: 30° 10' 10",78 S, Lat.: 70° 48' 23",49 O, alt.: 2.207 m), sem filtro, com exposição de 8 segundos e tempo-morto de 1 segundo cada, e suas exposições foram iniciadas com 4 segundos de defasagem. Assim, combinando as observações, não houve perda de informação por conta do tempo-morto. A ocultação teve início a 00:45:30 ± 2,5 s e fim a 00:45:50 ± 2,5 s. Análise preliminar das imagens mostra a detecção positiva do evento com duração de 20 s, isto é, uma corda de ∼400 km (Figura 4.35).

Detecções positivas de corda única, envolvendo objetos que já tenham seu tamanho determinado por alguma ocultação multi-cordas (ou outro método), são importantes para determinação da posição astrométrica do corpo, já que neste caso, o erro limita-se à posição da estrela (Tabela 4.11). Esta pode ser determinada com menos de 20 mas em relação ao catálogo UCAC4, por exemplo, o que equivale a cerca de um raio de Quaoar no plano do céu. Vale citar que existem duas soluções para a posição do objeto: uma com a corda sendo medida na parte norte, e outra na parte sul do corpo. Mas, ainda assim, o

[4]TRAnsiting Planets and PlanetesImals Small Telescope (Jehin et al. 2011)
[5]Panchromatic Robotic Optical Monitoring and Polarimetry Telescopes
[6]Astrograph for the Southern Hemisphere

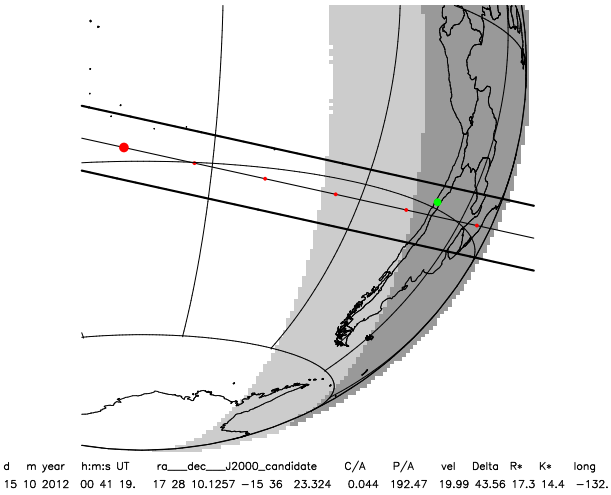


Figura 4.35 Mapa mostrando uma das soluções possíveis para posição da faixa de ocultação por Quaoar em 15 de outubro de 2012. O ponto verde corresponde à localização geográfica de Cerro Tololo, onde se situam os telescópios PROMPT.

erro na determinação da posição do corpo no plano do céu é dominado pela erro sobre a posição da estrela.

Para isto ajusta-se a corda observada à um círculo de tamanho fixo, oriundo de resultados precedentes. Aqui utilizou-se R = 555 km, junto com a efeméride JPL#21/DE405 para se determinar o *offset* de Quaoar. A posição da estrela é oriunda do programa caminhos.

Tabela 4.11 Com a detecção do evento com apenas uma corda, para objetos que ja tenham seu tamanho determinado, pode-se obter sua posição astrométrica com precisão limitada somente à posição da estrela ocultada. O instante de maior aproximação em relação ao geocentro, dado em h:mm:ss,s ± ss, de 15 de outubro de 2012. O *offset* corresponde ao centro da solução circular em relação à efeméride JPL#21/DE405, dado em mas, onde 1mas equivale à 31,59 km à distância de 43,56 ua. A posição de Quaoar em ascensão reta e declinação no instante de C/A e aposição da estrela ocultada também são apresentadas.

Instante do C/A	00:41:19 ± 05 s
Distancia do C/A	44 ± 25 mas Sul
offset AR	-110 ± 3,2 mas
offset DEC	-63 ± 25 mas
AR	$17^h\ 28^m\ 10^s$,1255 ± 0^s,0032
DEC	-15° 36' 23",3610 ± 0",025
Estrela AR	$17^h\ 28^m\ 10^s$,1257 ± 0^s,0013
Estrela DEC	-15° 36' 23",324 ± 0",0013

Capítulo 5

Ocultações estelares positivas por TNOs: resultados já obtidos.

Devido às dificuldades apresentadas nos capítulos 2 e 3, as primeiras detecções positivas de eventos de ocultação estelar por objetos transnetunianos são muito recentes. O sucesso nesta atividade foi possível devido à obtenção de boas predições (Assafin et al. (2010) & Assafin et al. (2012)) e da implementação de uma grande rede de colaboradores. A primeira ocultação por um TNO foi observada em outubro de 2009, envolvendo o objeto 2002 TX_{300}, e as detecções positivas aumentaram em número rapidamente. Em 2010 mais duas ocultações foram detectadas; mais quatro em 2011; mais cinco em 2012; e mais duas só em janeiro de 2013, totalizando 14 eventos que envolvem apenas 8 objetos. Do total de 14 eventos, 12 foram observados pelo grupo no qual me enquadro ou por colaboradores, dois não estavam no catálogo de eventos do programa caminhos e dois foram detectados pela equipe do MIT (seção 5.1).

Vale ressaltar que não estão sendo consideradas aqui as ocultações envolvendo Plutão e Caronte (Figura 5.1), já que suas ocultações vêm sendo observadas desde 1985 (Brosch 1995). Depois de 1985, novas ocultações foram observadas em 1988, 2002 e anualmente, desde então, mostrando a evolução sazonal da atmosfera de Plutão. Caronte foi observado pela primeira vez numa ocultação em 1980 (Walker 1980). Em 2005 realizou-se a primeira observação multi-corda para este corpo, quando foi possivel determinar seu tamanho (Sicardy et al. 2006b). Mais recentemente, novas ocultações por Caronte foram observadas e, por vezes, em eventos duplo com Plutão. Alguns dos resultados mais recentes sobre a atmosfera de Plutão podem ser vistos em (Lellouch et al. (2009) e Zalucha & Gulbis (2012)). A descoberta de Plutão a mais tempo (1930), e o fato de estar transitando uma região do céu que tem como fundo o plano da galáxia, são fatores que contribuíram para que eventos envolvendo este sistema já tenham sido observados em maior número.

Nas seções a seguir cada um dos eventos observados envolvendo os TNOs é brevemente relatado com seu contexto e principais resultados. As ocultações envolvendo Quaoar foram

discutidas em detalhes no capítulo 4. Exceto o evento envolvendo o objeto 2002 TX_{300}, todos os outros foram observados por colaboradores do nosso grupo. Um sumário dos resultados é dado na tabela 5.

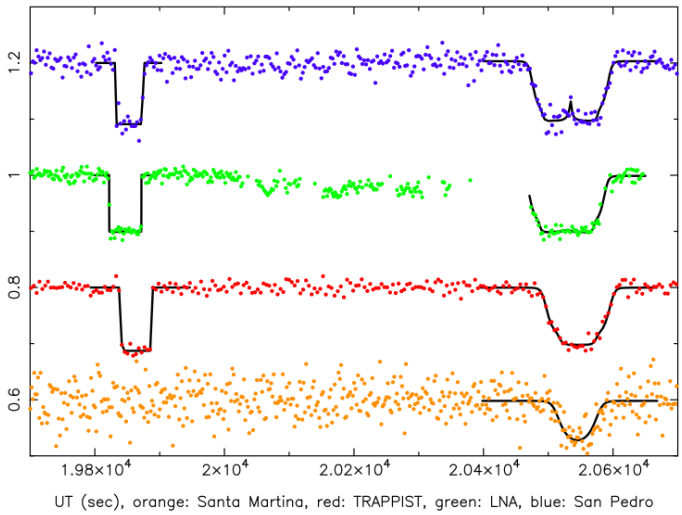


Figura 5.1 Ocultação dupla envolvendo Plutão e Caronte ocorrida em 04 de junho de 2011 e observada de três sítios no Chile e um no Brasil (Sicardy et al. 2012). É clara a diferença entre uma ocultação por um objeto desprovido de atmosfera (Caronte à esquerda) e de um com uma atmosfera de cerca de $12\mu bar$ (Plutão à direita). Ocultações duplas são oportunidades únicas para medir com alta precisão a distância entre um objeto e seu satélite.

5.1 2002 TX_{300} - 09/10/2009

Em 9 de outubro de 2009, detectou-se a primeira ocultação por um objeto transnetuniano (Elliot et al. 2010). Ela envolveu o objeto 2002 TX_{300} e foi observada pela equipe do MIT (*Massachusetts Institute of Technology*).

Descoberto em outubro de 2002 pelo programa de busca de objetos próximos à Terra do Observatório Monte Palomar, 2002 TX_{300} teve seu tamanho estimado em 640 ± 250 km (Stansberry et al. 2008) a partir de observações com o telescópio *Spitzer*. É classificado como objeto quente clássico do cinturão de Kuiper. É um dos maiores membros da família colisional de Haumea (Brown et al. 2007) e possui gelo de água em sua superfície (Peixinho et al. 2012). Possui os seguintes elementos orbitais (Chamberlin 2008):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
43.4	38.1	0.122	25.8	286.03

A ocultação envolveu uma estrela de magnitude R= 13,1 e foi detectada por dois telescópios no Havaí. Outros 19 telescópios participaram da campanha, mas não detectaram

o evento ou não puderam observar. Apesar de ser esperado que o objeto tenha um leve achatamento devido à sua rotação e curva de luz de rotação (P = 8,14 horas, $\Delta m = 0,14$, Thirouin et al. (2010)), uma solução circular com $R_{equiv} = 143 \pm 5$ km foi obtida. Possivelmente por algum problema na datação de um dos telescópios, os instantes centrais das cordas tinham cerca de 32 segundos de separação no plano do céu. Por isto, os autores alinharam as cordas para obter uma solução circular.

Com este raio eles obtiveram um albedo de $0,88^{+0,15}_{-0,06}$. Este alto valor é explicado pelos autores como consequência da presença de gelo de água na superfície do corpo. Como não há elementos leves como N_2 e CH_4 detectados sobre a superfície de 2002 TX300, investigou-se a presença de uma atmosfera de elementos pesados, como o Xe, nas curvas de luz de ocultação obtidas. Obteve-se um limite superior de 2×10^{15} cm^{-3} para uma atmosfera isotérmica com temperatura de 49 K.

5.2 Varuna - 19/02/2010 e 08/01/2013

Descoberto em novembro de 2000 pelo projeto *Spacewatch*, Varuna recebeu designação provisória 2000 WR_{106}. Recebeu a numeração 20000 numa forma de comemoração à sua descoberta, jà que era o maior objeto de sua classe até aquele momento, e também por ter ocorrido 200 anos após a descoberta de Ceres. Recebe a classificação dinâmica de um objeto do cinturão de Kuiper clássico quente, e possui gelo de água em sua superfície (Peixinho et al. 2012). Alguns de seus principais elementos orbitais são (Chamberlin 2008):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
43.2	40.9	0.053	17.1	283.82

Varuna é um candidato a planeta-anão (Tancredi & Favre 2008), uma vez que seu tamanho, curva de luz ($\Delta m = 0,42$) e velocidade de rotação (P= 6,34 h) condizem com um objeto Jacobi em equilíbrio hidrostático (Lacerda & Jewitt 2007). Ajustando uma curva de luz de um objeto Jacobi sobre a curva de luz de rotação de Varuna, e fazendo algumas suposições sobre o albedo, Lacerda & Jewitt (2007) encontram a densidade de $0,992^{+0,086}_{-0,015}$ g/cm^3. Com observações do telescópio *Spitzer*, seu diâmetro equivalente foi estimado em 621 ± 180 km.

Duas ocultações estelares por Varuna já foram observadas, uma em 2010 e outra recentemente em 2013, apresentadas a seguir.

5.2.1 Varuna - 19/02/2010

A ocultação de uma estrela de magnitude R= 12, prevista pelo programa caminhos, foi atualizada com observações astrométricas realizadas poucas semanas antes do evento.

Uma grande campanha foi realizada, com o desdobramento de diversas estações móveis, no nordeste do Brasil e Namíbia (Africa).

Apenas uma estação, localizada em São Luiz do Maranhão, observou com sucesso a ocultação, obtendo uma corda de 1003 km. Outra estação, em Quixadá, cerca de 200 km ao sul, obteve dados mas não detectou o evento. Isto trouxe fortes restrições ao tamanho de Varuna.

No momento da ocultação, Varuna apresentava seu brilho máximo, isto é, maior superfície aparente. Então, considerando fatores como o máximo achatamento esperado para Varuna, limitações sobre seu tamanho obtido por outras técnicas, seu albedo e as cordas positiva e negativa observadas na ocultação, pode-se obter uma solução com limite superior para $a = 860$ km e $b' = 375$ km ($R_{equiv} = 567$ km), para um albedo de 0,04 (Sicardy et al. 2010) (Figura 5.2). Ainda não foi realizado nenhum estudo sobre a assinatura de uma atmosfera na curva de luz, mas a ocultação parece ser bastante abrupta, o que implica num limite na ordem de poucos nanobar à pressão de uma possível atmosfera.

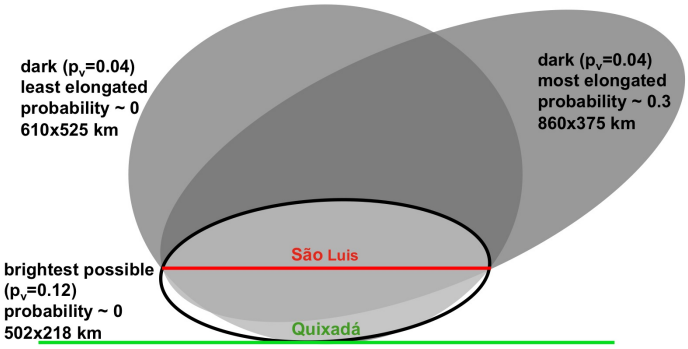


Figura 5.2 Soluções extremas para o tamanho e forma de Varuna, considerando limitações sobre o seu tamanho, o achatamento e o albedo obtidos por outras técnicas, e as cordas observadas na ocultação.

5.2.2 Varuna - 08/01/2013

Uma nova ocultação por Varuna, envolvendo uma estrela de mag. R= 16,5, foi observada no dia 08 de janeiro de 2013. Após observações astrométricas realizadas com os telescópios Calar Alto 1,2 m e Pic du Midi 1 m, as atualizações da predição apontaram uma ocultação sobre China e Japão. Três telescópios no Japão detectaram o evento, um em Hiroshima e dois em Shiga (figura 5.3).

Reduções preliminares apontam para um objeto achatado com $a = 817$ km e $b' = 578$ km, o que equivale à um $R_{equiv} = 687$ km e $\epsilon = 0,29$. Neste caso, seu albedo deve ser ainda

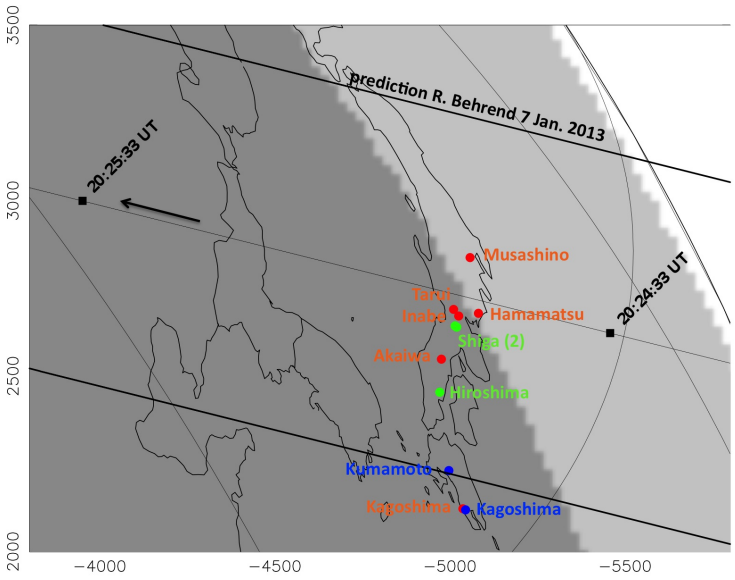


Figura 5.3 Mapa da ocultação observada no Japão em 08 de janeiro de 2013. O pontos verdes indicam locais em que o evento foi detectado, em azul onde os dados não apontam uma ocultação e em vermelho sítios que não puderam adquirir dados.

menor que 0,04. Os dados foram obtidos com tempos de exposição na ordem de 10 s, e, portanto não devem trazer limitações sobre a presença de atmosfera em Varuna.

5.3 Éris - 06/11/2010

O planeta-anão Éris foi descoberto 2005 a partir de imagens tomadas em outubro de 2003 do Monte Palomar, recebendo a designação 2003 UB_{313} (Brown et al. 2005). Seu raio foi estimado em 1.200 ± 100 km e 1.500 ± 200 km, sendo potencialmente maior que Plutão. Sua descoberta foi um dos motivos que levou a IAU a criar a classificação de planetas-anões (Tancredi & Favre 2008).

É atualmente o objeto observável mais distante no Sistema Solar, a 97 ua, e apresenta uma órbita bastante excêntrica, o que faz com que no periélio chegue a 38 uado Sol, tão perto quanto a distância de Plutão. Seus elementos orbitais (Chamberlin 2008):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
67.96	38.2	0.437	43.88	560.23

Encontra-se atualmente numa região do céu bastante desprovida de estrelas (constelação da Baleia). Isto, aliado ao seu lento deslocamento devido à grande distância ao Sol, faz com que ocultações estelares por ele promovidas sejam muito raras. Em 06 de novembro de 2010, uma ocultação de uma estrela de mag. R = 17 foi confirmada depois de uma intensa campanha astrométrica para a atualização da ocultação. Foi apenas nos últimos dias que as predições mostraram que o evento poderia ser observado da América

do Sul.

Uma campanha observacional envolvendo 26 observatórios foi realizada, mas apenas três telescópios no Chile detectaram o evento. Dois dos telescópios encontravam-se no mesmo sítio, San Pedro de Atacama, e o terceiro em La Silla (TRAPPIST). Vários telescópios no Brasil não puderam observar, apesar de preparados para tal, por questões climáticas (Sicardy et al. 2011).

Como até hoje não se detectou nenhuma variação de brilhos em curvas de luz de rotação de Éris, seu período de rotação não é conhecido. Acredita-se que ele esteja em equilíbrio hidrostático devido ao seu tamanho. Então só poderia apresentar a forma de um objeto Maclaurin ou Jacobi. Esta última é descartada, já que exige que seu eixo de rotação esteja orientado exatamente na direção de visada, para não promover curva de luz de rotação.

Como as cordas observadas estão praticamente sobre a mesma mediana, assume-se que a forma de Éris seja próxima à uma esfera. Com o ajuste circular obtém-se R_{equiv} =1,163±6 km (Figura 5.4). Tamanhos maiores são possíveis se ajustes elípticos forem permitidos, mas os ajustes tornam-se rapidamente muito achatados, exigindo rápida rotação o que não é observados em suas curvas de luz. Este tamanho é comparável ao tamanho de Plutão $R = 1,152 \pm 32$ km (Lellouch et al. 2009).

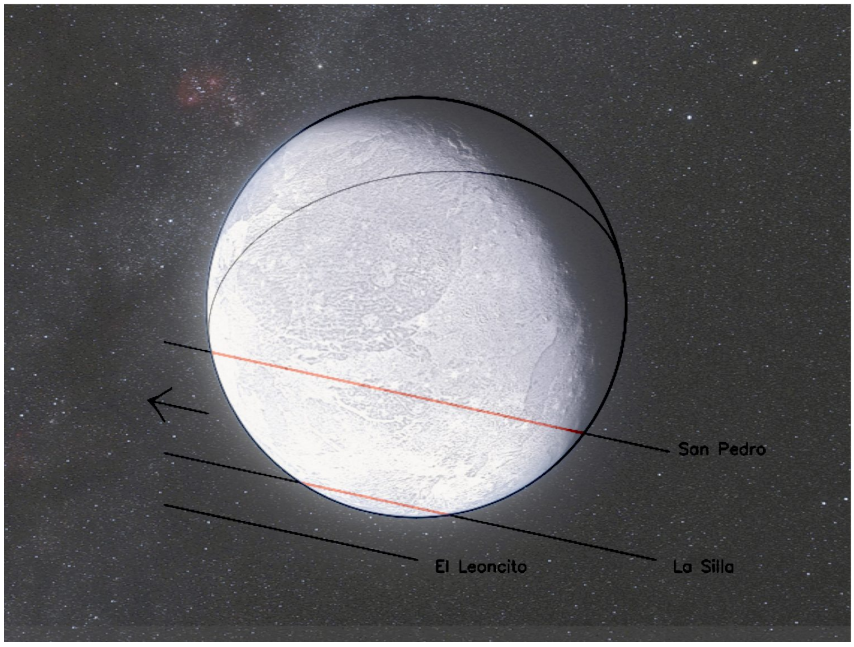


Figura 5.4 Imagem artística de Éris mostra as cordas de ocultação observadas, sobrepostas à um objeto bastante brilhante sem atmosfera. A imagem fui publicada no site *European Southern Observatory*† na ocasião da publicação do artigo (Sicardy et al. 2011).
†`http://www.eso.org/public/news/eso1142/` (acessado em 04/02/2013)

Como Éris possui uma lua, *Dysnomia*, com diâmetro que pode chegar a 650 km, sua

massa é conhecida M $= 1,66 \pm 0,02 \times 10^{22}$ kg (Brown & Schaller 2007). A partir do raio obtido da ocultação, pôde-se então calcular sua densidade $\rho = 2,52 \pm 0,05$g/cm^3. Este valor sugere que Éris seja um corpo rochoso no seu interior, coberto por uma camada de gelo.

Do novo diâmetro obtido para Éris, calculou-se o albedo no visível $p_v = 0,96^{+0,09}_{-0,04}$, colocando Éris como um dos mais brilhantes objetos do Sistema Solar. Investigou-se a presença de uma atmosfera em torno de Éris e determinou-se o limite superior de 1 nanobar para uma atmosfera de N_2 isotérmica a 30 K. Como o espectro de Éris é bastante similar ao de Plutão, com presença de N_2 e CH_4 na superfície (Tegler et al. 2010), imagina-se que ao se aproximar do Sol, elementos de sua superfície sublimem formando uma atmosfera sazonal. Ao se distanciar, essa atmosfera condensa sobre a superfície, o que justifica o alto albedo e a não existência de variação de brilho sobre Éris (Sicardy et al. 2011).

5.4 2003 AZ$_{84}$ - 08/01/2011 e 03/02/2012

Objeto pertencente à classe dos plutinos, em ressonância 2:3 com Netuno, 2003 AZ$_{84}$ foi descoberto em janeiro de 2003 a partir de imagens do Monte Palomar. Possui os seguintes elementos orbitais (Chamberlin 2008):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
39.64	32.73	0.174	13.54	249.60

Com período de rotação P = 6,49 h e amplitude Δm = 0,07 (Thirouin et al. 2010), 2003 AZ$_{84}$ deve estar em equilíbrio hidrostático, assumindo uma forma de equilíbrio Maclaurin (Tancredi & Favre 2008). Em sua superfície observa-se gelo de água e possivelmente metanol (Guilbert et al. 2009).

Estimativas de tamanho para 2003 AZ$_{84}$ variam de D $= 686 \pm 95$ km (Stansberry et al. 2008) a D $= 727 \pm 66$ km (Mommert et al. 2012). Em 2007 uma lua de cerca de 70 km foi descoberta com imagens do *Hubble Space Telescope* (Brown & Suer 2007). Infelizmente novas observações não detectaram novamente a lua. Portanto, ainda não há uma determinação de órbita, de onde poderia se obter a massa do primário.

Dois eventos envolvendo 2003 AZ$_{84}$ já foram observados, sendo o primeiro de apenas uma corda, e o segundo com três cordas. Eles são apresentados a seguir.

5.4.1 2003 AZ$_{84}$ - 08/01/2011

Uma ocultação de uma estrela de mag. R = 18,4 por 2003 AZ$_{84}$ foi prevista para o dia 08 de janeiro 2011. Por se tratar de uma estrela muito fraca, tanto as atualizações

quanto a observação se limitaram a poucos telescópios. O evento foi detectado em San Pedro de Atacama por dois telescópios, mas não foi detectado de La Silla (TRAPPIST), cerca de 610 km, no plano do céu, ao sul.

Com uma única corda, apenas um limite inferior para seu diâmetro pode ser obtido $D_{min} = 573 \pm 21$ km (Braga-Ribas et al. 2011).

5.4.2 2003 AZ_{84} - 03/02/2012

Em 03 de fevereiro de 2012, uma ocultação envolvendo uma estrela mag. R = 15,2 foi objeto de uma campanha envolvendo mais de 20 telescópios na Europa, norte da Africa, Índia e Indonésia. Três telescópios detectaram o evento, dois na Índia (Mount Abu Observatory e IUCAA Girawali Observatory) e um em Israel (Weizmann Istitute). Dois telescópios separados por aproximadamente 1000 km no plano do céu, obtiveram dados mas não detectaram o evento (Alicante e Liverpool), ajudando a limitar o tamanho do TNO.

Devido ao ruído nas imagens oriundas de Mt. Abu, obtida com filtro H, duas soluções são possíveis para o egresso. No entanto, apenas uma das soluções satisfaz uma elipse ao se considerar as outras cordas. No observatório de Girawali, as observações foram feitas com 2 segundos de exposição e 16,5 segundos de ciclo. Uma das imagens apresenta uma queda maior que uma magnitude provocada pela ocultação. Buscou-se então o tamanho máximo de uma corda durante o tempo-morto antes e depois desta imagem ocultada. Com as outras cordas, foi possível determinar que a imagem ocultada corresponde ao egresso de 2003 AZ_{84} naquele sítio.

Com as considerações acima, ajustou-se uma elipse sobre as cordas sobre 2003 AZ_{84}, obtendo um objeto com achatamento $\epsilon = 0,37 \pm 0,1$ e $R_{equiv} = 343 \pm 7$ km (Figura 5.5). Seu albedo no visível também foi calculado em $p_v = 0,147 \pm 0,007$. Como dito, espera-se que 2003 AZ_{84} seja um objeto Maclaurin, então compara-se o achatamento obtido à curva de equilíbrio de objeto Maclaurin com período de rotação P = 6 h, obtendo assim a densidade do objeto $\rho = 0,76^{+0,30}_{-0,17}$ (Braga-Ribas et al. 2012).

5.5 Makemake - 23/04/2011

Com cerca de 2/3 o tamanho de Plutão, Makemake é o terceiro maior objeto transnetuniano. Descoberto em março de 2005 com imagens do Observatório Monte Palomar, recebeu o nome provisório de 2005 FY_9. Ele é um dos poucos entre os grandes objetos transnetunianos a não possuir um satélite, o que dificulta a determinação de sua densidade.

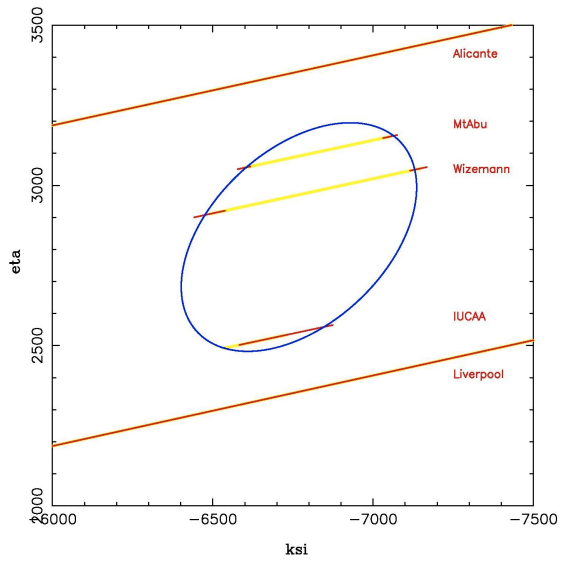


Figura 5.5 Solução elíptica para 2003 AZ_{84} a partir das cordas da ocultação de fevereiro de 2012. Uma elipse de $\epsilon = 0,37 \pm 0,1$ e $R_{equiv} = 343 \pm 7$ km é encontrada. Os traços amarelos são as cordas observadas, os segmentos vermelhos representam a barra de erro de cada ponto. Na corda mais ao sul, o erro do ingresso é grande porque ele ocorreu durante um tempo-morto de 14,5 segundos.(Braga-Ribas et al. 2012)

Makemake pertence à classe dinâmica dos objetos clássicos quentes do cinturão de Kuiper, já que sua órbita é bastante inclinada. Alguns de seus elementos orbitais são (Chamberlin 2008):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
45.55	38.27	0.1599	29.01	307.48

Medidas térmicas feitas pelos telescópios espaciais *Spitzer* e *Hershel* indicam que Makemake deve ter ao menos dois terrenos diferentes, um de maior área com um albedo superior a 0,7 e uma pequena região com albedo próximo a 0,1. Seu tamanho também foi estimado entre $1350 < D < 1510$ km (Lim et al. 2010). Ele possui metano em sua superfície que deve estar em forma de grãos tão grandes quanto 1 cm. Há também evidências de N_2 (Tegler et al. 2008). Ainda, devido ao seu tamanho, baixa amplitude da sua curva de luz (Δm = 0,014) e rotação (P = 7,65 h) (Thirouin et al. 2010), espera-se que ele seja um objeto Maclaurin com pequeno achatamento.

Por sua raridade e valor científico, a ocultação por Makemake prevista para 23 de abril de 2011, envolvendo uma estrela de mag. R = 18,2, foi alvo de intensa campanha astrométrica para realização de atualizações da predição. Um total de 16 telescópios foi utilizado para a observação da ocultação, dos quais 7 detectaram a ocultação em cinco sítios diferentes (4 no Chile e 1 no Brasil) (Figura 5.6).

Das cinco cordas, quatro estavam dispostas a menos de 100 km uma da outra no plano

do céu, próximas à parte central de Makemake. É possível ajustar às cordas uma elipse de $R_{equiv} = 732 \pm 9$ km e pequeno achatamento $\epsilon = 0,05 \pm 0,02$ (Ortiz et al. 2012). Deste raio calcula-se seu albedo no visível $p_v = 0,77 \pm 0,03$ que é intermediário entre o albedo de Plutão ($p_v = 0,52$) e de Éris ($p_v = 0,96$).

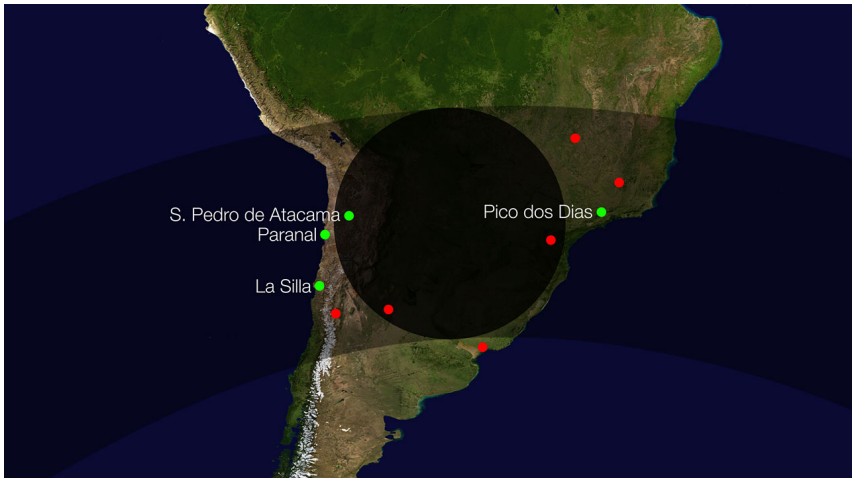


Figura 5.6 Mapa mostrando a disposição dos observatórios ao longo da faixa de ocultação. Os pontos verdes correspondem aos sítios que detectaram o evento. Esta imagem foi publicada no site do *European Southern Observatory*[†] na ocasião da publicação do artigo (Ortiz et al. 2012).
[†] `http://www.eso.org/public/news/eso1246/` (acessado em 04/02/2013)

Por seu tamanho, distância ao Sol e composição, Makemake era tido como um ótimo candidato a ter uma atmosfera global. No entanto, as curvas de luz de ocultação estabelecem um limite superior de 12 nanobar para uma atmosfera de metano. A hipótese de atmosfera local não é descartada, já que poderia provocar o efeito de dois terrenos, um claro e outro escuro, observados com dados dos telescópios espaciais. Neste caso, uma atmosfera numa banda em torno do corpo com pressão menor que 25 μbar não pode ser descartada.

5.6 2002 KX_{14} - 26/04/2012

Descoberto em maio de 2002 em imagens do Monte Palomar, 2002 KX_{14} é um objeto clássico do cinturão de Kuiper frio. Apresenta um espectro plano sem nenhuma linha absorção. Também não tem curva de luz de rotação detectada. Seu diâmetro foi estimado usando o telescópio espacial *Herschel* em D= 455 ± 27 km. Seus elementos orbitais são (Chamberlin 2008):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
38.66	36.92	0.045	0.406	240.39

Como o objeto não está na lista dos 10 objetos do programa caminhos, a predição deste evento foi feita usando estrela NOMAD 0677-041184 de mag. R 17,7, e foi atualizada por telescópios na Espanha. A ocultação foi observadas de La Palma (Ilhas Canárias) e teve duração de 20,9 segundos, que se traduz num limite inferior para seu diâmetro de 415 ± 1 km. Não há nenhum sinal evidente de atmosfera na curva de luz de ocultação.

5.7 2005 TV_{189} - 13/11/2012

Pouco se conhece do objeto 2005 TV_{189}. Descoberto em outubro de 2005, pertence à classe dos objetos do cinturão de Kuiper ressonantes, já que se encontra na ressonância 3:2 de Netuno (a mesma que Plutão). Se destaca por ser o objeto com maior inclinação da classe. Seus elementos orbitais são (Petit et al. 2011):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
39.41	32	0.188	34.4	249.83

Sendo sua magnitude absoluta H = 7,5 (Chamberlin 2008), assumindo um albedo de p = 0,2, pode-se estimar um diâmetro a partir da equação 5.1, onde a magnitude absoluta do Sol é $H_\odot = -26,74$ e a unidade astronômica é dada em quilômetros (au_{km}). Portanto, o diâmetro estimado para 2005 TV_{189} é $\sim$ 95 km.

$$D = 2 \cdot 10^{(H_\odot/5)} \cdot (10^{(-H_v/5)}/\sqrt{p_v}) \cdot 1au_{km} \tag{5.1}$$

A partir de uma rotina automática de predições de ocultações estelares (Preston, S. 2010), uma ocultação por 2005 TV_{189} foi prevista para o dia 13 de novembro de 2012, envolvendo uma estrela de magnitude V = 11,5. O astrônomo amador Tomas Janik observou de Kininice / República Checa, com um telescópio de 0,25 m, visualmente, e com um cronômetro "Chronograph ACH-77". Duas ocultações em sequencia são reportadas, uma de 1,97 s e a segunda de 2,62 s, com um intervalo de 18,15 s entre elas.

Esta observação pode ser interpretada como uma ocultação por um objeto duplo, na qual os dois membros ocultaram a estrela e a distância entre eles é dada pelo intervalo entre os eventos. Neste caso, o diâmetro mínimo para cada componente é $D_1 = 44 \pm 12$ km e $D_2 = 56 \pm 6$ km (Figura 5.7), para o primeiro e segundo evento respectivamente. No instante do evento estavam com uma separação de 440 km (ou 19,4 mas). Vale citar que a soma de seus diâmetros é próxima ao diâmetro estimado.

Outras soluções devem ser investigadas, como por exemplo, um único objeto de $\sim$50 km ocultando uma estrela dupla. Esta hipótese parece pouco provável, já que o objeto 2005 TV_{189} teria que ter um albedo geométrico de aproximadamente 0,7, não esperado para objetos tão pequenos.

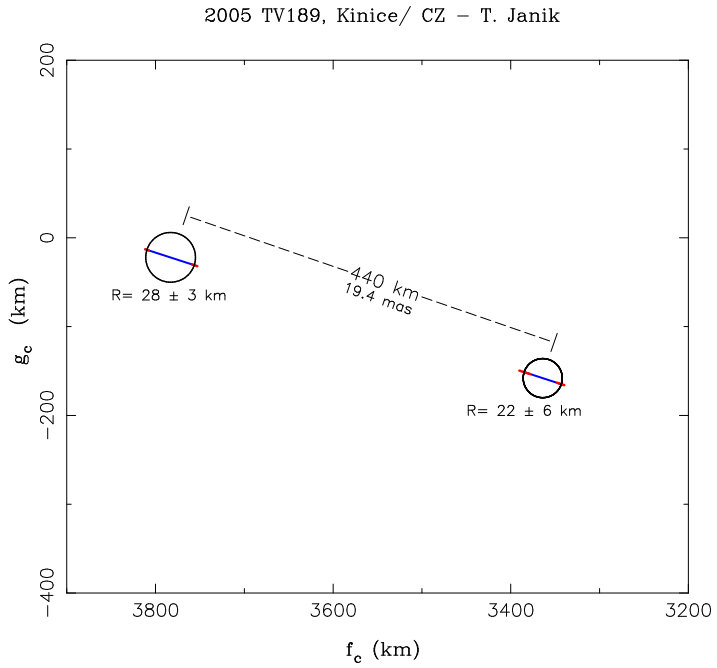


Figura 5.7 Possível solução a partir do relatório de uma observação visual. Um objeto duplo separados por 19 mas.

5.8 Sedna - 13/01/2013

Sedna foi descoberto em novembro de 2003 com observações do Monte Palomar, recebendo a denominação 2003 VB_{12}. É possivelmente um dos TNOs mais interessantes, por conta de sua órbita altamente excêntrica (e = 0,859). Seu periélio se da a q = 76 ua e seu afélio a Q = 1009 ua. Essas características não são facilmente explicáveis e diferentes hipóteses são levantadas para explicar como ele poderia ter sido colocado em tal configuração orbital. Por exemplo: a passagem de estrelas próximas ao Sistema Solar durante a formação do Sol, ou a presença de um grande planeta na nuvem de Oort (Gomes & Soares 2010). Por conta destas características orbitais, pertence à classe de objetos transnetunianos destacados. Alguns de seus elementos orbitais são (Chamberlin 2008):

a_{orb} (ua)	q (ua)	e	i (°)	P (anos)
542.66	76.23	0.859	11.93	12641.67

Como é de se esperar para os maiores e mais distantes TNOs, Sedna apresenta gelo de água, metano e possivelmente nitrogênio molecular em sua superfície. É um dos TNOs mais avermelhados, talvez por sua exposição a radiação galática (Barucci et al. 2010). Sua curva de luz de rotação apresenta uma baixa amplitude $\Delta m = 0,011 \pm 0,001$ com período de revolução P = $10,273 \pm 0,002$ h (Gaudi et al. 2005). Estimativas de tamanho inicialmente o colocavam como um objeto com até 1600 km de diâmetro. Com dados do *Spitzer* e *Herschel* seu diâmetro foi calculado em D= 995 ± 80 km (Pál et al. 2012).

Assim como para Éris, desvido a seu lento deslocamento no céu, ocultações estelares por Sedna são bastante raras (<3/ano). Uma ocultação no dia 13 de janeiro de 2013, envolvendo um a estrela de magnitude R = 18, foi prevista para a região da Austrália. Atualizações foram realizadas apontando uma ocultação para o norte australiano. Dois observadores na cidade de Cairns observaram o evento, com exposições de 60 e 180 segundos. Ambos detectam a ocultação em pelo menos uma imagem. Numa análise preliminar, se for considerado que a ocultação não pode ter durado mais do que 60 segundos, coloca-se um limite inferior para o diâmetro de Sedna de 700 km.

5.9 Sumário dos resultados

Aqui apresenta-se a tabela 5.1, com um sumário dos principais resultados já obtidos a partir de todas as ocultações estelares por objetos transnetunianos já observadas.

Tabela 5.1 Sumário dos principais resultados obtidos a partir de cada ocultação estelar por TNO já observada. A data corresponde à data do evento. O número de telescópios que detectaram cada evento pode ser maior que o número de cordas apresentadas à esquerda na coluna 'N° de cordas', já que por vezes há mais de um telescópio por sítio. O número de telescópios que não detectaram (ou não puderam observar) o evento é dado à direita. O simbolo '>', à frente dos valores na coluna 'R_{equiv} ', representam limites inferiores para o raio do objeto, assim como '<' representam limites superiores para as colunas 'ϵ', 'Albedo' e 'Atmosfera'. A distância do objeto no momento da ocultação é dada na coluna 'r'. Na coluna 'Classe' as letras representam as seguintes classe de objetos transnetunianos: S – espalhados; D – destacados; HC – clássicos quentes; CC – clássicos frios; R – ressonantes. As referências encontram-se na última coluna. Os resultados que ainda estão sob análise são denotados por "preliminar".

Objeto	Data	Local	N° de cordas	R_{equiv} (km)	ϵ	Albedo	Densidade (g cm^{-3})	Atmosfera (nanobar)	r (ua)	Classe	Referência
(136199) Éris	06/11/2010	CL	2+| 24-	1,163±6	0	$0.96^{+0.09}_{-0.04}$	$2,52 \pm 0,05$	< 1 (N_2)	94.74	S	(Sicardy et al. 2011)
(136472) Makemake	23/04/2011	CL,BR	5+|9-	732±9	$0,05 \pm 0,02$	0.77±0.03	1.7±0.3	<12(CH_4)	51.50	HC	(Ortiz et al. 2012)
(20000) Varuna	19/02/2010	BR	1+|>7-	>565	<0.56	-	-	-	42.75	HC	(Sicardy et al. 2010)
(20000) Varuna	08/01/2013	JP	2+| >10-	∼685	0.29	<0.04	-	-	42.67	HC	preliminar
(50000) Quaoar	11/02/2011	US	1+	>380	-	-	-	-	43.62	HC	(Person et al. 2011)
(50000) Quaoar	04/05/2011	CL,UY,BR	5+|11-	555±2.5†	$0.0895^{+0.025}_{-0.016}$	0.109±0.007	1.99±0.42	< 21 (CH_4)	42.35	HC	(Braga-Ribas et al. 2013)
(50000) Quaoar	17/02/2012	FR,CH	2+|<10-	685^{+445}_{-155}	-	-	-	-	43.52	HC	(Braga-Ribas et al. 2013)
(50000) Quaoar	15/10/2012	CH	1+|3-	>200	-	-	-	-	43.56	HC	(Braga-Ribas et al. 2013)
(90377) Sedna	13/01/2013	AU	2+	>700	-	-	-	-	86.20	D	preliminar
(208996) 2003 AZ_{84}	08/01/2011	CL	1+|1-	>286±11	-	-	-	-	44.32	R	(Braga-Ribas et al. 2011)
(208996) 2003 AZ_{84}	03/02/2012	IN,IL	3+|>20-	343±7	0.37±0.1	0.147±0.007	$0.76^{+0,30}_{-0,17}$	-	44.26	R	(Braga-Ribas et al. 2012)
(119951) 2002 KX_{14}	26/04/2012	ES	1+|2-	>207±1	-	-	-	-	38.55	CC	preliminar
(55636) 2002 TX_{300}	09/10/2009	HW	2+ | 19-	143±5	-	$0.88^{+0.15}_{-0.06}$	-	<33μbar (Xe)	40.61	HC	(Elliot et al. 2010)
2005 TV_{189}	13/11/2012	CZ	1+(vis.‡)	>28±3 & >22±6	-	<0.22	-	-	31.26	HC	preliminar

(†Assumindo a solução Maclaurin. ‡Observação visual.)

Capítulo 6

Conclusão

A compreensão do Sistema Solar, da sua formação e de sua evolução, passa pelo estudo dos objetos transnetunianos. Estes corpos guardam informações sobre a evolução dinâmica e são compostos por elementos primordiais da nuvem protoplanetária. Portanto, conhecer seus parâmetros físicos, como distribuição de tamanho, densidade e estrutura interna, é fundamental para remontar a história do nosso sistema estelar.

Empregou-se o método de ocultações estelares para determinar, com precisões sem precedentes, o tamanho, a forma, o albedo, a densidade e a presença de atmosfera sobre alguns TNOS. Como discutido, algumas dificuldades precisaram ser superadas antes de se lograr sucesso em observações de tais ocultações. Os primeiros resultados positivos só começaram a ser obtidos a menos de 3,5 anos, no final de 2009. Desde então, 14 eventos já foram observados, dos quais 12 com participação efetiva de nossa equipe.

A primeira etapa do método consiste em predizer tais eventos. Para isto, utiliza-se as efemérides dos objetos, que são comparadas à catálogos estelares, para determinar eventos de conjunção. No entanto, em consequência de sua grande distância do Sol ($>$30 `ua`), o movimento dos TNOs no céu é bastante lento. Para aumentar o número de eventos possíveis, recorre-se a catálogos estelares menos precisos em termos astrométricos. Suas órbitas também não apresentam as precisões necessárias para uma boa predição. Assim, os erros envolvidos, oriundos das posições das estrelas e do objeto, são superiores a vários diâmetros do objeto, dificultado a detecção do evento.

Para superar este problema, realizou-se um programa de observações astrométricas do caminho que 10 entre os maiores TNOs realizarão no céu até 2015. Com isto, além de melhorar a posição de estrelas relativamente brilhantes, aumentou-se sensivelmente a quantidade de estrelas com posições precisas o suficiente para se realizar predições de ocultações confiáveis. Estrelas de magnitude R $<$18 tiveram suas posições determinadas em relação ao catálogo astrométrico de referência UCAC2. Os TNOs também foram observados para se determinar qual a diferença entre sua efeméride e sua posição real no

céu. Assim, esta diferença é aplicada sobre a efeméride. Com boas posições das estrelas e dos TNOs, gera-se predições de ocultações mais precisas, com anos de antecedência. Este trabalho está sintetizado em duas publicações: Assafin et al. (2010) & Assafin et al. (2012).

Uma vez que uma ocultação favorável é selecionada, inicia-se outra etapa, a de atualizações próximas à data do evento. Devido a possíveis erros em movimento próprio das estrelas, e efemérides que se desviam gradualmente da posição real do objeto, é preciso observar tanto a estrela candidata, quanto o ocultador em épocas próximas ao evento, para corrigir tais erros e melhorar ainda mais as predições. As predições mais precisas são alcançadas dias antes da ocultação, quando a estrela e o corpo podem ser observados sobre a mesma imagem. Assim, erros sistemáticos de catálogos são eliminados, e as predições podem ser tão precisas quanto a um diâmetro do objeto. Sendo a predição favorável, grandes campanhas observacionais são lançadas.

Nos últimos três anos, mais de 40 campanhas foram realizadas. Para isto, páginas na internet dedicadas a cada evento foram criadas com informações e dicas aos observadores. Além disso, milhares de e-mails foram trocados com colaboradores para pedir por observações, e explicar o procedimento observacional. Desde o início de 2010, 12 eventos, de um total de 14, já foram observados positivamente utilizando esta metodologia.

Após a observação e detecção positiva do evento, com as cordas de ocultação obtidas, inicia-se o estudo de tamanho e forma do objeto. Aqui apresentou-se em detalhes a redução da ocultação por Quaoar em 04 de maio de 2011. Como, segundo as cordas obtidas, Quaoar teria uma forma bastante irregular, estudou-se os limites físicos em que estruturas como cratera e montanhas poderiam ser sustentadas sobre a superfície destes objetos. Como as estruturas observadas são fisicamente improváveis, investigou-se o caso de Quaoar apresentar uma figura de equilíbrio hidrostático.

Formas de equilíbrio hidrostático Maclaurin e Jacobi foram apresentadas, bem como suas regiões de transição. Assim determinou-se que Quaoar deve ter a forma de um elipsoide Maclaurin, onde dois de seus semi-eixos são iguais (a=b>c). Neste caso, deslocou-se, em tempo absoluto, 4 das 5 cordas observadas. Usando propriedades dos elipsoides, como o posicionamento de seus instantes centrais em uma linha reta, e associando tais propriedades à curva de equilíbrio (ϵ, ρ) de um corpo Maclaurin, pôde-se determinar achatamento, R_{equiv} , ângulo de posição, densidade e albedo de Quaoar com precisões sem precedentes para este objeto. Investigou-se também a presença de uma atmosfera global, e o limite superior de 21 nbar foi dado. No entanto, a existência de atmosfera local não pode ser descartada. Analisou-se também outras duas ocultações promovidas por Quaoar em fevereiro e outubro de 2012, mas nenhuma nova restrição sobre as propriedades físicas do objeto pode ser obtida (Braga-Ribas et al. 2013).

Outros eventos observados que merecem destaque foram as ocultações por Éris e Makamake. Calculou-se que Éris tem um tamanho similar ao de Plutão (indistinguível qual é o maior), com um albedo e densidades bastante elevados. Ainda determinou-se um limite superior para a pressão superficial de uma atmosfera global de 1 nanobar. Possivelmente, Éris possui uma atmosfera sazonal, que atualmente está condensada sobre a superfície, o que explicaria seu alto brilho e não detecção de curva de luz de rotação (Sicardy et al. 2011). Para Makemake, o terceiro maior objeto da classe, mediu-se seu tamanho com precisão de alguns quilômetros e não detectou-se uma atmosfera global, apesar de ser um dos melhores candidatos a possuir uma. No entanto, uma atmosfera local não pode ser descartada. Seu albedo também é elevado, o que pode ser causado por uma atmosfera parcialmente condensada sobre a superfície (Ortiz et al. 2012).

Ocultações por objetos como Varuna, Sedna, 2003 AZ_{84} já foram detectadas, e análises preliminares já apontam para resultados interessantes como formas achatadas e determinação de densidade para o caso de objetos em equilíbrio hidrostático (Braga-Ribas et al. 2012).

Interessante notar que, para cada evento, diferentes propriedades físicas pré-determinadas a partir de outras técnicas são aplicadas para derivar seus parâmetros físicos, como albedo e densidade. Destacam-se entre estas propriedades a massa, a magnitude absoluta, o período de revolução e a composição superficial.

A metodologia de predição se mostrou muito eficiente, haja visto a quantidade de detecções positivas nos últimos anos. A partir das observações das ocultações, importantes parâmetros físicos e suas implicações puderam ser determinados para alguns dos maiores e mais importantes objetos transnetunianos. O projeto está sendo expandido para mais 39 TNOs (Camargo et al. 2013).

Ao menos duas grandes descobertas ainda podem ser feitas a partir de ocultações estelares por objetos transnetunianos:

1. Detecção de uma atmosfera global sobre algum objeto transnetuniano (além de Plutão). Alguns dos melhores candidatos já foram detectados em ocultações, e apenas limites superiores na ordem de alguns nanobars foram determinados. Estes resultados podem trazer implicações sobre as teorias que explicam os processos de retenção/perda de voláteis por estes objetos.

2. Detecção de um satélite de um TNO. Determinar os parâmetros da lua pode ajudar a determinar sua origem, seja ela capturada ou formada ao mesmo tempo que o corpo principal. Isso pode ajudar a compreender os processos dinâmicos aos quais os TNOs foram submetidos.

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Apêndice A

Classificação dinâmica dos TNOs

O primeiro objeto transnetuniano a ser descoberto foi (15760) 1992 QB_1 por Jewitt & Luu (1993). Desde então, mais de 1260 objetos transnetunianos já foram descobertos. Eles se dividem em alguns grupos dinâmicos, de acordo com sua distância e parâmetros orbitais (Gladman et al. 2008). A seguir apresenta-se os grupos dinâmicos e suas subdivisões:

Objetos Transnetunianos: todos os objetos que orbitam o Sol à uma distância maior que 30 ua:

1. O **cinturão de Kuiper** é uma região que se espalha de 30 a ∼55 ua. Ela se divide em dois subgrupos:
   - **Ressonantes** que se encontram em ressonâncias com Netuno como por exemplo, 1:2 (twotinos) e 2:3 (plutinos);
   - **Clássicos** (ou cubewanos) com órbitas quase circulares não perturbadas por Netuno. Estes ainda se dividem em duas famílias:
     - **Frios**, com baixa inclinações $i < 5,5°$;
     - **Quentes**, com inclinações superiores a $i > 5,5°$.
2. Objetos **espalhados**, sendo aqueles com excentricidade maior que e>0,24, que no seu periélio chegam próximos à influência gravitacional de Netuno (q∼30), mas seu afélio se dá a distâncias muito superiores a isto.
3. Objetos **destacados** são aqueles que no seu periélio não chegam próximos à influência gravitacional de Netuno, nem de nenhum outro planeta. Por isto parecem destacados do Sistema Solar.
4. Objetos da nuvem de **Oort**, são todos aqueles que orbitam o Sol a distâncias superiores a a>2000 ua.

Apêndice B

Résumé Français de la Thèse

Ministério da Ciência, Tecnologia e Inovação
Observatório Nacional

Doutorado em Astronomia

Observatoire de Paris
École Doctorale
Astronomie et Astrophysique D'Ile-de-France

Doctorat
Astronomie et Astrophysique

Felipe Braga Ribas

EXPLORANDO OS OBJETOS TRANSNETUNIANOS PELO MÉTODO DE OCULTAÇÕES ESTELARES :
PREDIÇÃO, OBSERVAÇÃO, QUAOAR E OS PRIMEIROS RESULTADOS.

EXPLORATION DES OBJETS TRANS-NEPTUNIENS PAR LA MÉTHODE DES OCCULTATION STELLAIRES :
PRÉDICTION, OBSERVATION, QUAOAR ET LES PREMIERS RÉSULTATS.

2013

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Doctorat
Spécialité
Astronomie et Astrophysique
École Doctorale
d'Astronomie & Astrophysique d'Île-de-France

RAPPORT DE SOUTENANCE DE THÈSE

Thèse soutenue le mardi 26 Février 2013 à 14h00

par M. Felipe BRAGA-RIBAS

Sujet de la thèse : Etude d'occultations stellaires par de objets transneptuniens

Président du Jury : Rodney GOMES, Chercheur Titulaire/ON/au MCTI au Brésil

Composition du jury :
Mme Antonella BARUCCI, Astronome au LESIA à l'Observatoire de Paris
M. Sylvio FERRAZ-MELLO, Professeur émérite à l'Instituto Astronomico e Geofísico au Brésil
M. Rodney GOMES, Chercheur Titulaire/ON/au MCTI au Brésil
M. Jean-Marc PETIT, Directeur de Recherche à lUTINAM à L'Observatoire de Besançon
M. Bruno SICARDY, Professeur au LESIA à l'Observatoire de Paris
M. Roberto VIEIRA MARTINS, Chercheur Titulaire/ON/au MCTI au Brésil

RAPPORT DE SOUTENANCE :

voir le vers de cette feuille.

MENTION (HONORABLE AU TRÈS HONORABLE) : TRÈS HONORABLE

LE PRESIDENT ET LES MEMBRES DU JURY (SIGNATURES) :

Mme Antonella BARUCCI M. Sylvio FERRAZ-MELLO M. Rodney GOMES

M. Jean-Marc PETIT M. Bruno SICARDY M. Roberto VIEIRA MARTINS

Le jury, dont la liste est donné au recto du présent document, a assisté ce jour (26 février 2013) à la soutenance de thèse de M. Felipe Braga-Ribas, sur le sujet suivant : "Exploration des objets trans-neptuniens par la méthode des occultations stellaires : prédictions, observations, Quaoar et les premiers résultats".

Le jury a apprécié la présentation orale du candidat, qui a répondu de manière tout à fait satisfaisante aux questions du jury.

Le manuscrit est également apprarue comme bien structuré et très complet.

Le jury a souligné la quantité et la qualité du travail effectué. Le candidat a mi en place tout ma chaîne qui permet de prédire les événements, organiser les campagnes d'observation, et finalement analyser les donnés et en déduire les propriétés des objets étudié : taille, forme, densité, albédo, et contraints sur la présence d'atmosphère éventuelles.

Par toutes ces raisons, le jury est très heureux de décerner le titre de docteur à Felipe Braga-Ribas dnas la spécialité Astronome et Astrophysique, aussi bien an titre de l'Observatoire de Paris que de l'Observatório Nacional de Rio de Janeiro, et avec mention très honorable.

Fait à Rio de Janeiro
26 février 2013

Résumé

Les objets trans-neptuniens (OTN) sont des corps primitifs du système solaire qui orbitent au-delà de la planète géante Neptune. Plus de 1200 objets ont été découverts depuis 1992, et on compte dans cette population les planètes naines Pluton, Makemake et Eris. Ce sont des corps primitifs, et à ce titre, leur dynamique orbitale, leur composition interne et surfacique et leur distribution en taille nous renseignent sur la formation et l'évolution du système solaire dans son ensemble.

De nombreuses observations photométriques, spectroscopiques et thermiques (IR lointain) ont fourni des renseignements sur leurs périodes de rotation, leur composition de surface et des estimations de leurs tailles et de leurs albédos.

Cependant, au vu de leurs distances (plus de 30 unités astronomiques) et de leur taille, il est actuellement impossible de résoudre leur diamètre angulaire (en général bien inférieur à 0.1 arcsec) et de détecter directement leur atmosphère.

La méthode d'occultation stellaire consiste à observer le passage d'OTNs devant une étoile. Les durées précises du phénomène, observé en différents endroits, permettent ensuite de retrouver la taille et la forme des objets avec des précisions kilométriques. Ainsi, des valeurs précises de l'albédo peuvent être déduites, et pour les corps dont la masse est connue via le mouvement de satellites, des valeurs précises de la masse volumique peuvent également être calculées. Par ailleurs, des atmosphères éventuelles provoquent des baisses de signal juste avant et après l'occultation, permettant des détections à des niveaux de pression de quelques nanobars.

Cette thèse décrit (1) les méthodes de prédiction des occultations par une cinquantaine de OTNs, avec des précisions de l'ordre de 0.03 arcsec, (2) l'organisation de campagnes mondiales faisant intervenir des dizaines de sites professionnels et amateurs, (3) l'obser-

vation, l'analyse et l'interprétation de plusieurs occultations par Quaoar et d'autres gros OTNs.

Depuis 2009, quatorze occultations par des OTNs ont été observées : douze par le consortium dont fait partie le groupe de Paris/Rio, et deux par une équipe du MIT. Dans cette thèse, outre la discution détaille sur les occultations par Quaoar, un résumé des résultats sur la taille, forme, albédo, densité et sur l'atmosphère, à partir des occultation par d'autres TNOs est présenté. En particulier les occultations par Eris, Makemake, 2003 AZ_{84}, Varuna et Sedna, qui son toutes des premières.

Mots clefs : occultation stellaire, TNO, KBO, 50000 Quaoar, objet de Maclaurin.

Abstract

Transneptunian objects (TNOs) are fossils of the Solar System beyond Neptune. Over 1200 TNOs have been found since 1992, and among them we find dwarf-planets such as Pluto, Makemake and Eris. The dynamics, internal and surface composition, size and mass distribution may have information about the early ages and evolution of the Solar System.

Photometry, spectroscopic, thermal (far infra-red) give information about their rotational period, surface composition, size and albedo estimations.

Nevertheless, due to their distance (greater than 30 Astronomical Units) and their sizes, we do not have the necessary technology to directly measure their size and detect possible atmospheres, as their angular sizes are smaller than a 50 miliarc seconds (mas).

The stellar occultation technique consists of observing the transit of a TNO in front of a star. The precise duration of the event, observed from several different sites, allows to deduce their size and shape with kilometre accuracy. So, precise determination of their albedos can be derived, and for those bodies that have their mass deduced from its satellite orbit, precise determination of their densities is allowed. Besides, tenuous atmospheres at the nanobar level, can be detected as they promote an attenuation of the star light right before and after the occultation.

In this work it is described (1) the prediction method applied to more then fifty TNOs, with accuracy of about 30 mas, (2) the world-wide campaigns with tens of professional and amateurs observatories, (3) the observation, analysis, and interpretation of the observed occultation by Quaoar and other large transneptunian objects.

Since 2009, fourteen occultations by TNOs were already observed, twelve by the Paris/Rio consortium.In addition to the detailed discussion about the Quaoar events, a compilation of all the other results comprising as size, shape, albedo, density and atmosphere, obtained from TNO stellar occultations is presented. In particular occultations by Eris, Makemake, 2003 AZ_{84}, Varuna and Sedna, all of which are new.

We present constraints over size, shape, albedo of these bodies, and also upper limits on possible atmospheres up to the nanobar level.

Keywords : Stellar occultation, TNO, KBO, 50000 Quaoar, Maclaurin object.

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Chapitre 1

Introduction

La compréhension du système solaire passe, entre autres, par l'étude des petits corps, comme par exemple les objets trans-neptuniens (OTN). Ce sont des reliques du système solaire primorial qui orbitent au-delà de Neptune. Après Pluton, découvert en 1930, il a fallu attendre 1992 pour découvrir d'autres corps dans cette région (Jewitt & Luu 1993). Leur existence avait été proposée théoriquement (par exemple par Gerard Kuiper), pour expliquer les comètes à courte période. Depuis lors, plus de 1260 OTNs ont été découverts. Ils ont été divisés en plusieurs groupes, tels que la ceinture classique, les objets résonants et les objets dispersés (Gladman et al. 2008).

Ces groupes portent la signature de l'évolution du système solaire. Ainsi, il se peut qu'au cours d'une migration des planètes géantes, plusieurs OTNs aient été capturés dans la résonance 3/2 avec Neptune, tandis que d'autres étaient dispersés (Gomes 2003). La formation de Neptune a dû se produire en environ 10 millions années, ce qui signifie que les OTNs se sont formés plus rapidement que ce dernier. Cependant, la densité actuelle du disque qui compose ces objets n'est pas compatible avec cette formation rapide (McKinnon et al. 2008). Différentes théories de formation ont été suggérées, et les propriétés de ces objets apportent des informations sur les processus thermiques et chimiques qu'ont subis ces corps, ou permettent au moins de discriminer différentes théories. Par ailleurs, les distributions de taille et de masse sont également liées au processus de formation.

Ces propriétés physiques sont difficiles à déterminer. Des observations photométriques, spectroscopiques et thermiques fournissent des contraintes sur leur périodes de rotation, la taille, l'albédo et leur composition. Par ailleurs, environ 4,5% des OTN doivent avoir des satellites ou être binaires, et de nombreux cas sont déjà connus. Ceci permet entre autres de déterminer la masse de ces systèmes, et leur densité, si les tailles sont connues.

En utilisant la technique des occultations stellaires, qui consiste à observer un OTN lors de son passage devant une étoile, nous cherchons à déterminer les dimensions et formes de ces corps avec des précisions de l'ordre du kilomètre. Nous en déduisons les albédos et les

densités avec une grande précision. Ceci a des implications directes sur la composition de surface et sur la structure interne de ces corps. Aujourd'hui, les occultations stellaires sont la technique la plus puissante pour détecter des atmosphères à des niveaux de pression de l'ordre du nanobar autour de ces corps. Ces atmosphères éventuelles (ou leur absence) ont elles-mêmes des implications pour les processus de sublimation et les propriétés thermiques de ces objets.

Toutefois, de par leurs longues périodes orbitales (> 200 ans), et de leurs faibles éclats dûs à leur grande distance au Soleil (> 30 UA), leurs orbites présentent de barres d'erreur en général dix fois plus grande par rapport à ce qui est nécessaire pour le calcul des occultations. Pour pallier ce problème, nous faisons des mesures astrométriques de ces objets afin de déterminer les différence avec les positions données par ses éphémérides. Ces différences sont ensuite uilisées pour affiner les prédictions d'occultations.

En outre, le lent déplacement des OTNs dans le ciel rend les occultations rares. Cela requiert des catalogues stellaires plus denses, avec des étoiles plus faibles. Cela cause un autre problème, car la précision des positions astrométriques de ces étoiles est également inférieure à celle qui est requise pour une bonne prédiction des occultations. Pour contourner ce problème, nous avons mis sur pied un programme utilisant le télescope Max-Planck de 2.2 m et la caméra WFI (Wide Field Imager) de l'ESO (European Southern Observatory). Nous avons ainsi observé les bandes stellaires balayées dix OTNs parmi les plus gros entre de 2008 à 2015. En utilisant des techniques astrométriques appropriées, nous avons créé des catalogues jusqu'à la magnitude R=18, avec une précision de 40 millisecondes (mas), voir Assafin et al. (2010) & Assafin et al. (2012). Ce travail a permis des prédictions fiables des occultations stellaires par les OTNs.

Des centaines d'événements sont prévus chaque année, et comme il est impossible de les observer tous, une sélection est effectuée selon des critères prenant en compte l'intérêt scientifique du corps, le flux de l'étoile occultée, la localisation de l'ombre sur Terre et la présence de réseaux d'observateurs. Quelques jours avant l'événement, lorsque l'OTN et l'étoile peuvent être observés sur la même image, les mesures astrométriques deviennent plus précises car on évite alors les erreurs systématiques de catalogues. Des précisions de l'ordre de 20 massont alors atteintes et l'endroit où l'ombre de l'occultation passera est bien définie.

Des campagnes internationales sont alors organisées, impliquant des dizaines d'astronomes professionnels et amateurs. Une page internet pour chaque événement est publiée avec les informations nécessaires [1].

Ce travail a déjà permis l'observation de 12 événements, parmi les 14 occultations jamais détectées impliquant des OTNs (excluant Pluton et Charon).

1. http://www.lesia.obspm.fr/perso/braga-ribas/campaigns

La thèse, outre la méthodologie de prédiction indiquée ci-dessus, décrit en détail l'ensemble de la procédure de réduction des données, avec une description plus détaillée des occultations par Quaoar de 2011 et 2012. L'occultation du 4 mai 2011 pu être détectée de 5 sites différents. Chaque site fournit une courbe de lumière qui est modélisée par le masquage de l'étoile par un corps opaque et sans atmosphère, tenant compte de la diffraction, et la convolution par la taille de l'étoile et le temps d'intégration du système. Les instants de début et de fin de l'événement fournissent alors ce que nous appelons une corde d'occultation. Les cordes sont alors utilisées pour ajuster une ellipse qui représente limbe du corps, et fournit donc des contraintes sur sa taille, sa forme, et l'orientation du corps.

Si les temps d'occultation fournis par l'occultation du 4 mai 2011 sont tous correctes, alors Quaoar devrait avoir avoir une irrégulière. Nous examinons deux solutions : la première se caractérise par un crateère qui serait trop profond. L'autre solution suppose la présence d'une montagne, physiquement peu probable car trop haute, compte tenu de la taille, la masse et la composition de Quaoar.

Nous explorons ensuite une troisième solution, dans laquelle on envisage la possibilité de problèmes dans l'enregistrement du temps absolu pour certains des 5 sites. En raison de sa taille, sa rotation et sa composition de surface, on s'attend à ce que Quaoar soit en équilibre hydrostatique, et prenne la forme d'un objet de type Maclaurin. Pour obtenir la forme limbe attendue (une ellipse), les cordes d'occultation sont décalées en temps, c'est-à-dire parallèlement à leur propre direction. Cette procédure conservant l'aire de l'ellipse, nous obtenons une valeur bien contrainte du rayon équivalent de Quaoar (rayon du cercle ayant la même surface que le limbe apparent) : R_{equiv}= 555 ± 2.5 km et de son albédo géométrique visuel : p_V = 0.109±0.007. Par ailleurs chaque solution de limbe fournit le volume de Quaoar et son aplatissement réel, en supposant une valeur donnée de ζ, l'angle d'aspect polaire, c'est-à-dire l'angle entre l'axe polaire de Quaoar et la ligne de visée. Comme la masse de Quaoar est connue via le mouvement de son satellite Weywot, le volume fourni in fine la densité. Par ailleurs, la relation densité/aplatissement étant connue pour objet de Maclaurin, nous obtenons une solution unique rayon équatorial-densité-aplatissement-orientation dans le plan du ciel, une fois ζ fixé. Comme ζ est indéterminé entre 0 et 90° , nous obtenons finalement une densité 1.99±0.46 g cm^{-3} et un aplatissement réel de 0.0897±0.006. De plus, l'orientation de notre solution dans le plan du ciel montre que l'orbite de Weywot ne peut pas être équatorial.

Enfin, nous trouvons une limite supérieure de 21 nanobar pour la présence d'une atmosphère globale de meéthane autour de Quaoar. Cependant, nous ne pouvons pas exclure la présence d'une atmosphère locale.

La première occultation par OTN a été observée à la fin 2009 par un groupe américain

du MIT, impliquant objet 2002 TX_{300} (Elliot et al. 2010). Peu de temps après, en 2010, des occultations par Varuna (19 Février) et Éris (6 Novembre 6) (**?**) ont été détectées par notre groupe. En 2011, nous avons détecté des événements par 2003 AZ_{84} (8 Janvier), Makemake (23 Avril) (Ortiz et al. 2012) et Quaoar (4 mai). En 2012, de nouvelles occultations par 2003 AZ_{84} (3 Février), Quaoar (17 Février et 15 Octobre) ont été enregistrées, ainsi qu'une autre par 2002 KX_{14} (26 Avril) et possiblement par 2005 TV_{189} (13 novembre). Enfin en 2013, nous avons détecté à nouveau Varuna (8 Janvier) et pour la première fois, Sedna (13 Janvier). Ces occultations sont discutées à la fin de la thèse.

Chapitre 2

Ocultations stelaires pour OTNs : résultats obtenus.

À l'exception de Pluton et son satellite Charon, qu'ont eu plusieurs occultations observées, les premières occultations par objets transneptuniens ont commencé a être détecté seulement à la fin de l'année 2009. Dans l'occasion, le groupe américain du MIT a détectée l'occultation par l'objet 2002 TX_{300}. En 2010, les occultations par Varuna et Éris (Sicardy et al. 2012) ont été détectées par notre groupe. En 2011, les nous avons détecté les événements par 2003 AZ_{84} (Braga-Ribas et al. 2011), Makemake (Ortiz et al. 2012) et Quaoar (Braga-Ribas et al. 2013). En 2012, en plus des nouvelles occultations par 2003 AZ_{84} et Quaoar, nous avons également observé un événement par 2002 KX_{14} et 2005 TV_{189}. Que en janvier 2013, deux occultations on été déjà observé, une par Varuna et outre par Sedna.

Dans la table 2.1 ci-dessus un sommaire des principaux résultats est présenté. La liste des campagnes auxquelles j'ai participé est donnée dans les figures 2.1, 2.2.

TABLE 2.1 Résumé des principaux résultats de chaque occultation stellaire par OTN déjà observé. La date correspond à la date de l'événement. Le nombre de télescopes qui ont détecté chaque événement peut être plus grand que le nombre de cordes présentées sur la colonne "Numéro de cordes", parce que parfois il y a plusieurs télescopes par site. Le symbole ">" en face des valeurs dans la colonne "R_{equiv} ", représentent des limites inférieures pour le rayon de l'objet, ainsi que "<" représente des limites supérieures pour les colonnes ϵ, "Albédo" et "Atmosphère". La distance de l'objet au moment de l'occultation est donnée dans la colonne "r". Dans la colonne "Classe" les lettres représentent la classe d'objets transneptuniens suivantes : S - dispersés, D - détaché ; HC - classiques chaud, CC - classiques froid, R - résonnants. Les références sont donnés dans la dernière colonne. Les résultats qui sont en cours d'analyse, sont marqués comme préliminaires.

Objet	Date	Local	N° de cordas	R_{equiv} (km)	ϵ	Albédo	Densité (g cm^{-3})	Atmosphère (nanobar)	r (UA)	Classe	Référence
(136199) Éris	06/11/2010	CL	2+| 24-	1,163±6	0	$0.96^{+0.09}_{-0.04}$	$2,52 \pm 0,05$	< 1 (N_2)	94.74	S	(Sicardy et al. 2011)
(136472) Makemake	23/04/2011	CL,BR	5+|9-	732±9	$0,05 \pm 0,02$	0.77±0.03	1.7±0.3	<12(CH_4)	51.50	HC	(Ortiz et al. 2012)
(20000) Varuna	19/02/2010	BR	1+|>7-	>565	<0.56	-	-	-	42.75	HC	(Sicardy et al. 2010)
(20000) Varuna	08/01/2013	JP	2+| >10-	∼685	0.29	<0.04	-	-	42.67	HC	préliminaire
(50000) Quaoar	11/02/2011	US	1+	>380	-	-	-	-	43.62	HC	(Person et al. 2011)
(50000) Quaoar	04/05/2011	CL,UY,BR	5+|11-	555±2.5†	$0.0895^{+0.025}_{-0.016}$	0.109±0.007	1.99±0.42	< 21 (CH_4)	42.35	HC	(Braga-Ribas et al. 2013)
(50000) Quaoar	17/02/2012	FR,CH	2+|<10-	685^{+445}_{-155}	-	-	-	-	43.52	HC	(Braga-Ribas et al. 2013)
(50000) Quaoar	15/10/2012	CH	1+|3-	>200	-	-	-	-	43.56	HC	(Braga-Ribas et al. 2013)
(90377) Sedna	13/01/2013	AU	2+	>700	-	-	-	-	86.20	D	préliminaire
(208996) 2003 AZ_{84}	08/01/2011	CL	1+|1-	>286±11	-	-	-	-	44.32	R	(Braga-Ribas et al. 2011)
(208996) 2003 AZ_{84}	03/02/2012	IN,IL	3+|>20-	343±7	0.37±0.1	0.147±0.007	$0.76^{+0,30}_{-0,17}$	-	44.26	R	(Braga-Ribas et al. 2012)
(119951) 2002 KX_{14}	26/04/2012	ES	1+|2-	>207±1	-	-	-	-	38.55	CC	préliminaire
(55636) 2002 TX_{300}	09/10/2009	HW	2+ | 19-	143±5	-	$0.88^{+0.15}_{-0.06}$	-	<33μbar (Xe)	40.61	HC	(Elliot et al. 2010)
2005 TV_{189}	13/11/2012	CZ	1+(vis.‡)	>28±3 & >22±6	-	<0.22	-	-	31.26	HC	préliminaire

(†En supposant un corps Maclaurin. ‡Observation visuelle.)

2010		
14 February 2010	Pluto	Europe 🟢
19 February 2010	Varuna	South Africa, Namibia, Brazil 🔴
19 May 2010	Pluto	Run 085.C-0225 (ESO – VTL & NTT) 🟢
04 June 2010	Pluto	Australia & New Zealand 🟢
07 June 2010	Quaoar	Run 085.C-0225 (ESO – VTL & NTT)
22 June 2010	Nix	Australia & New Zealand
04 July 2010	Pluto	Southern Africa, Brazil, Argentina, Chile 🟢
28 August 2010	Hydra	Run 085.C-0225 (ESO – VTL & NTT)
06 November 2010	Eris	Europe, Brazil, Agentina, Chile 🔴
2011		
08 January 2011	2003 AZ84	Chile 🔴
02 February 2011	Quaoar	Europe
06 February 2011	Orcus	Canaries, Venezuela
03 March 2011	2003 VS2	Europe
07 March 2011	Ixion	Chile
02 April 2011	Orcus	Chile
13 April 2011	Quaoar	Chile
23 April 2011	Makemake	Brasil, Argentina, Chile 🔴
01 May 2011	Huya	off Earth
04 May 2011	Quaoar	South Africa, South America 🔴
07 may 2011	Charon	Australia (D. Herald)
14 May 2011	Quaoar	Europe
16 May 2011	Quaoar	Europe (off Earth ?)
17 May 2011	Hydra	North America
18 May 2011	Pluto	New Zealand, Australia 🟢 ?
21 May 2011	Chariklo	Africa (bright)
22 May 2011	Pluto	N. America 🟢
27 May 2011	Nix	S. America
03 June 2011	Pluto	Europe
03 June 2011	Ixion	New Zealand, Australia
04 June 2011	Pluto+Charon	South America 🟢
07 June 2011	Pluto	South America
16 June 2011	Quaoar	South, central America, Canaries
19 June 2011	Ixion	New Zealand, Australia
23 June 2011	Pluto+Charon	Pacific, Mexico 🟢
27 June 2011	Pluton+Hydra	Pacific, Australia, Asia
21 July 2011	Hydra	Europe
05 August 2011	Quaoar	Europe, N. Africa
16 August 2011	2003 OP32	Europe, N. Africa, C. America
30 October 2011	2002 AW197	S. America
01 November 2011	2003 VS2	Indian Ocean ?
08 November 2011	2002 VE95	S. America
15 November 2011	2002 VE95	N. America
28 November 2011	Orcus	S. America
30 November 2011	2002 VE95	Europe
07 December 2011	Haumea	N. and C. America
08 December 2011	2002 WC19	Americas, Europe
2012		
02 January 2012	Varuna	Pacific, Asia
03 February 2012	2003 AZ84	India, Middle East, N. Africa, Spain, Canaries 🔴
17 February 2012	Quaoar	Europe 🔴
19 February 2012	Chariklo	Brazil
13 March 2012	Quaoar	S. America
21 March 2012	Nix	Europe
27 March 2012	Pluto	USA
31 March 2012	Varuna	Europe

FIGURE 2.1 Campagnes auxquelles j'ai participé depuis 2010. En vert, les succès pour Pluton et Charon. En rouge, les succès pour les autres OTNs.

16 April 2012	Ixion	Europe, Spain
17 April 2012	Quaoar	Europe, Africa, Canaries, Cape Verde, Réunion
23 April 2012	2007 JH43	S. America, Chile
25 April 2012	Ixion	Australia, neg. D. Herald
28 April 2012	Ixion	Australia, neg. D. Herald
26 April 2012	2002 KX14	Europe, Canaries, Africa, Middle East 🔴
14 May 2012	Ixion	Africa, neg. G. Dangl Austria
19 May 2012	Quaoar	W. USA, Hawaii
27 May 2012	Quaoar	Europe, N. Africa, Middle East
27 May 2012	Ixion	Antartica, Australia, New Zealand
30 May 2012	Ixion	Australia, Indian Ocean, South Africa
06 June 2012	Quaoar	Africa, Brazil
13 June 2012	Nix	Australia, attempt J. Talbot
14 June 2012	Pluto	SW Europe, N. Africa, Canaries, E. USA 🟢
19 June 2012	Ixion	S. America (including ToO VLT/ISAAC)
22 June 2012	Ixion	S. America
29 June 2012	Pluto	Pacific basin
10 july 2012	Quaoar	Australia, New Zealand, Chile
18 July 2012	Pluto	S. America 🟢 from Chile & Argentina
23 July 2012	Pluto/Charon	Australia, New Zealand
27 July 2012	Pluto	Australia, New Zealand 🟢 Charon from AUS
24 August 2012	Pluto/Charon	S. America, Canaries 🟢 from S. America
26 August 2012	Pluto	Australia 🟢
09 September 2012	Pluto/Charon	NE & Central America 🟢 S. America
27 September 2012	1996 TO66	Europe
02 October 2012	Pluto/Charon	S. America
11 October 2012	Pluto/Charon	S. America
15 October 2012	Quaoar	S. America 🔴 Prompt
7 November 2012	2004 XA192	Europe, Africa
24 December 2012	Sedna	USA (W. Fraser and JL Ortiz) + Chile
24 December 2012	Chaos	USA and SW Europe + Canaries, Morocco
2013		
08 January 2013	Varuna	Asia, N. Europe 🔴 Hiroshima, Shiga
13 January 2013	Sedna	Australia, 🔴 Cairns
27 January 2013	2003 AZ84	S. Africa, La Réunion, Australia

🔴 TNO detections
🟢 Pluto and/or Charon detections

FIGURE 2.2 Suite de la page précédente.

Apêndice C

Assafin et al. 2010

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Astronomy & Astrophysics

Precise predictions of stellar occultations by Pluto, Charon, Nix, and Hydra for 2008–2015⋆,⋆⋆,⋆⋆⋆

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ABSTRACT

Context. We investigate transneptunian objects, including Pluto and its satellites, by stellar occultations.
Aims. Our aim is to derive precise, astrometric predictions for stellar occultations by Pluto and its satellites Charon, Hydra and Nix for 2008−2015. We construct an astrometric star catalog in the UCAC2 system covering Pluto′s sky path.
Methods. We carried out in 2007 an observational program at the ESO2p2/WFI instrument covering the sky path of Pluto from 2008 to 2015. We made the astrometry of 110 GB of images with the Platform for Reduction of Astronomical Images Automatically (PRAIA). By relatively simple astrometric techniques, we treated the overlapping observations and derived a field distortion pattern for the WFI mosaic of CCDs to within 50 mas precision.
Results. Positions were obtained in the UCAC2 frame with errors of 50 mas for stars up to magnitude $R = 19$, and 25 mas up to $R = 17$. New stellar proper motions were also determined with 2MASS and the USNO B1.0 catalog positions as first epoch. We generated 2252 predictions of stellar occultations by Pluto, Charon, Hydra and Nix for 2008−2015. An astrometric catalog with proper motions was produced, containing 2.24 million stars covering Pluto′s sky path with 30′ width. Its magnitude completeness is about $R = 18-19$ with a limit about $R = 21$. Based on the past 2005–2008 occultations successfully predicted, recorded and fitted, a linear drift with time in declination with regard to DE418/plu017 ephemerides was determined for Pluto and used in the current predictions. For *offset (mas)* $= A * (t(\mathrm{yr}) - 2005.0) + B$, we find $A = +30.5 \pm 4.3$ mas yr^{-1} and $B = -31.5 \pm 11.3$ mas, with standard deviation of 14.4 mas for the offsets. For these past occultations, predictions and follow-up observations were made with the 0.6 m and 1.6 m telescopes at the Laboratório Nacional de Astrofísica/Brazil.
Conclusions. Recurrent issues in stellar occultation predictions were addressed and properly overcome: body ephemeris offsets, catalog zero-point position errors and field-of-view size, long-term predictions and stellar proper motions, faint-visual versus bright-infrared stars and star/body astrometric follow-up. In particular, we highlight the usefulness of the obtained astrometric catalog as a reference frame for star/body astrometric follow-up before and after future events involving the Pluto system. Besides, it also furnishes useful photometric information for field stars in the flux calibration of observed light curves. Updates on the ephemeris offsets and candidate star positions (geometric conditions of predictions and finding charts) are made available by the group at www.lesia.obspm.fr/perso/bruno-sicardy/.

Key words. astrometry – occultations – planets and satellites: individual: Pluto – planets and satellites: individual: Charon – planets and satellites: individual: Nix – planets and satellites: individual: Hydra

1. Introduction

Investigating physical properties of Pluto and its satellites is essential for understanding transneptunian objects, keystones in the study of structure, origin and evolution of the solar system.

Lower/upper limits of 1169–1172 km and 1190−1193 km for Pluto′s radius are given by the combination of gaseous CH_4 spectra and stellar occultation observations (Lellouch et al. 2009). The Charon radius ranges from 603.6 ± 1.4 km to 606.0 ± 1.5 km, as estimated by various authors from the 2005 July 11 stellar occultation (Sicardy et al. 2006; Gulbis et al. 2006; Person et al. 2006). Pluto, Charon, Nix, and Hydra have estimated GM values of 870.3 ± 3.7, 101.4 ± 2.8, 0.039 ± 0.034 and 0.021 ± 0.042 km^3 s^{-2} respectively (Tholen et al. 2008). This results in densities of 1.8−2.1 g cm^{-3} and 1.55−1.80 g cm^{-3}

⋆ Tables of predictions for stellar occultations by Pluto, Charon, Nix and Hydra for 2008–2015 and Catalog of star positions for 2008–2015 sky path of Pluto are only available in electronic form at the CDS via anonymous ftp to cdsarc.u-strasbg.fr (130.79.128.5) or via http://cdsweb.u-strasbg.fr/cgi-bin/qcat?J/A+A/515/A32

⋆⋆ Observations made through the ESO run 079.A-9202(A), 075.C-0154, 077.C-0283 and 079.C-0345.

⋆⋆⋆ Also based on observations made at the Laboratório Nacional de Astrofísica (LNA), Itajubá-MG, Brazil.

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for Pluto and Charon, with rock versus ice fractions of 0.65 and of 0.55−0.60, respectively, in agreement with current structural models (McKinnon et al. 1997). Nix and Hydra estimated diameters are 88 km and 72 km, assuming a density of 1.63 gm cm^{-3} like Charon (Tholen et al. 2008).

Stellar occultations have unveiled surprising features in Pluto′s tenuous μbar-level atmosphere. Its upper atmosphere is isothermal with $T \approx 100$ K above about 1215 km from the center. Pressure roughly doubled between 1988–2002 after Pluto′s 1989 perihelion passage and then stabilized over 2002−2007 (Sicardy et al. 2003; Elliot et al. 2003, 2007; Young et al. 2008). Pluto′s atmosphere is thought to be mainly N_2, with a measured CH_4 abundance of 0.5% ± 0.1% and some undetermined amount of CO (Lellouch et al. 2009). The molecular nitrogen is in vapor-pressure equilibrium with the N_2 frost at the surface. Even so, the tiny amount of methane is known to be able to produce a pronounced thermal inversion layer. Indeed, about 10 km below 1215 km, there is a remarkable thermal inversion, which is probably caused by methane heating (see Yelle & Lunine 1989; Lellouch et al. 2009, for details). An additional interesting feature in Pluto's atmosphere are gravity and/or Rossby waves detected via stellar occultation (Hubbard et al. 2009; Person et al. 2008; McCarthy et al. 2008).

The 2005 stellar occultation by Charon brought stringent constraints on the presence of an atmosphere. Considering a surface temperature of 56 K rising up to 100 K above 20 km, a pure N_2 or CH_4 isothermal atmosphere leads to pressure limits of 15 μbar and 110 μbar respectively (Sicardy et al. 2006). These low values are compatible with the expected volatile escape rates for Charon (Yelle & Elliot 1997).

Despite all this knowledge, Pluto′s radius is still dependent on atmospheric models (Stansberry et al. 1994; Lellouch et al. 2009). Also, the thermal inversion could alternatively be explained by the presence of a thin haze layer with opacity >0.15 in vertical viewing. Moreover, because the N_2 vapor pressure is a steep function of temperature, an instantaneous response of the surface to insolation decay of about 3% should have led to a pressure decrease by a factor of 1.4 between 1988 and 2002, instead of the observed increase by a factor of about two. All this points to more complex scenarii at work over the 248-year Pluto′s orbital period. Seasonal effects associated with the recent passage through its equinox (December 1987), also at perihelion epoch (September 1989), may have led to the redistribution of ices on Pluto′s surface. For instance, the recently sunlit southern cap could now sublimate its nitrogen ice, thus feeding the atmosphere with more N_2 despite the decreasing solar energy available. A time lag is now necessary for this nitrogen to condense near the now permanently non-illuminated northern polar region. This kind of scenario was actually predicted in the work by Hansen & Paige (1996). Their best model predicted a pressure maximum in 2005 and a significant decrease only after 2025. Pressure would only be restored back to the 1988 levels in the 2100s. These models are not unique, however, and although they capture the basic physics behind those large pressure variations, the amplitude and duration of the present surge may show a significant discrepancy when compared to models.

Note also that any new measurements of Charon′s radius may furnish even better density figures and thus improve its ice/silicate ratio estimates. Moreover, any stellar occultation by Nix or Hydra will become a benchmark for the Pluto-Charon system. Size and shape could be obtained at kilometer-level precision, finally leading to the determination of density and ice/rock ratio for these small satellites. This in turn would allow for a better selection among plausible scenarios for the collisional history of the Pluto system. The same could be said of serendipitous detection of orbiting material.

All this strongly supports observation of stellar occultations by Pluto and its satellites. In the next years there will be no other observational alternative available to probing at high spatial resolution (km accuracy) Pluto′s atmospheric structure between the surface and about 150 km altitude, at least until 2015 when NASA′s New Horizons spacecraft arrives at Pluto. In this context, efforts toward new and precise predictions for future occultations are important. Note that until the 2000 s, the Pluto-Charon system has only been probed almost exclusively by the 1988 mutual events between Pluto and Charon. Furthermore, a stellar occultation observed in 1985 revealed Pluto′s atmosphere (Brosch 1995), which was observed more extensively during another occultation in 1988 (Millis et al. 1993). Also, besides the first stellar occultation recorded for Charon (Walker 1980), only two others were so far observed, one on 11 July 2005 (Sicardy et al. 2006; Gulbis et al. 2006) and another on 22 June 2008 (Sicardy et al. 2010, in prep.).

The first consistent efforts for the prediction of stellar occultations by Pluto are described in Mink & Klemola (1985) and cover the period 1985−1990. After that, only the work by McDonald & Elliot (2000a,b) is worth noting, now covering the period 1999−2009. Two important common limitations were the astrometric precision of about only 0.″2 and the lack of stellar proper motions leading to uncertainties on the order of the Earth radius for the predicted shadow paths. Also, these earlier predictions were degraded by poorer precision of older ephemerides, an issue which changed with the constant feed of new Pluto positions.

To overcome these and other problems, we carried out an observational program at the ESO2p2/WFI instrument during 2007 and derived precise positions for determining accurate predictions of stellar occultations by Pluto and its satellites Charon, Nix and Hydra for the period 2008−2015.

In Sect. 2, we further develop the astrometric context of predictions and the rationale of the ESO2p2/WFI program. In Sect. 3 we describe the observations. The astrometric treatment is detailed in Sect. 4. As an important part of the work, the derived catalog of star positions along Pluto′s sky path is presented in Sect. 5. Next, in Sect. 6, we describe the determination of ephemeris offsets for Pluto – a necessary refinement for the predictions. The candidate star search procedure is explained in Sect. 7. Predictions of stellar occultations by Pluto and its satellites are finally presented in Sect. 8 and results are discussed in Sect. 9.

2. Stellar occultation predictions: astrometric rationale

Following the release of the ICRS (Arias et al. 1995; Feissel & Mignard 1998) and of the HIPPARCOS catalog (Perryman et al. 1997), denser and astrometrically more precise catalogs became available in the 2000 s, such as the UCAC2 (Zacharias et al. 2004), the 2MASS (Cutri et al. 2003), the USNO B1.0 (Monet et al. 2003) and the GSC2.3 (Lasker et al. 2008). Not by chance a remarkable improvement in the prediction of stellar occultations has taken place since then. Telescopes equipped with CCDs with a relatively small FOV (field-of-view) could now be used. Not only provisional positions of candidate stars could thus be improved, but also better estimates for the Pluto ephemeris offsets could be derived. Another factor was the entering of Pluto in front of the projected Galactic plane, increasing the frequency

of possible events. Successful examples of these new prediction methods are the stellar occultation campaigns of 2002 for Pluto (Sicardy et al. 2003) and of 2005 for Charon (Sicardy et al. 2006).

Since 2004, our group has been engaged on a systematic effort to derive astrometric predictions for stellar occultations by Pluto and its satellites. Using meter-class telescopes and refined astrometric methods, precise positions based in the UCAC2 catalog were obtained since then, not only for candidate stars, but also for Pluto itself. A number of stellar occultations between 2005−2008 were forecast and successfully observed as predicted for stars between $13 < R < 16$. One by-product of those occultation observations has been to provide an accurate Pluto′s offset relative to its ephemeris, revealing a clear linear drift in time for declination, as we show in this paper. This drift can yield to declination offsets larger than 100 milli-arcsec (mas) for 2009 (see Sect. 6). This is comparable to Pluto′s apparent angular diameter. By knowing the ephemeris offset, we could forecast events for Pluto up to 2015. Since the orbits of Charon, Nix and Hydra around Pluto are well known, we could also extend stellar occultation predictions to these satellites.

As time goes by, mostly for magnitudes fainter than about $R = 14$, the estimation of star coordinates for current and future events is severely degraded by increasing errors in proper motion and mean catalog position, amounting to budget uncertainties of more than 70 mas for UCAC2. Position errors can be even worse than 100−200 mas for the 2MASS, USNO B1.0 and GSC2.3 catalogs. In this magnitude regime, predictions based solely on these catalog positions start to become unusable. This is important, as fainter – thus, more numerous – objects are becoming more and more accessible to modern detectors.

Moreover, as Pluto passes in front of regions of denser molecular clouds in the Galactic plane, chances are that relatively faint V or R, but bright infrared-emitting stars might be missed. Another issue is the problem of zero-point reference frame errors inherent to small FOV astrometry.

To overcome these problems, an observational program was carried out at the ESO2p2/WFI instrument during 2007. Precise positions were obtained and accurate predictions derived for stellar occultations by Pluto and its satellites Charon, Nix and Hydra. The astrometry of about 110 GB of acquired/processed images was accomplished with the Platform for Reduction of Astronomical Images Automatically – PRAIA (Assafin 2006). The software provides astrometric solutions suitable for the overlapping WFI CCD moscaics. The covered sky path of Pluto extended from 2008 to 2015-year of the New Horizons flyby. Results for 2008 and 2009 were also included because they might be eventually useful for the adjustment of occultations not yet published and for external checks of the accuracy of our predictions. In the astrometry, we derived a field distortion pattern for the WFI mosaic of CCDs within 50 mas precision. Another feature of our astrometric procedure was the determination of star proper motions using the 2MASS and USNO B1.0 catalogs as first epoch. In this way, we minimized the position error propagation for the 2015 predictions.

From the above procedures, an astrometric catalog of 30′ width was derived encompassing the 2008−2015 sky path of Pluto. It is in the UCAC2 reference frame with magnitude completeness around $R = 18$−19 and limiting magnitude about $R = 21$. Having about 2.24 million stars available in electronic form, the catalog can be very useful in the astrometric calibration of small CCD fields around Pluto and candidate stars, for refining occultation predictions and for star/body astrometric follow up before and after event date. It can be also helpful for deriving the photometric properties of flux calibration stars in the occultation FOV.

Predictions of past 2005−2008 stellar occultations by Pluto and Charon were updated by an astrometric follow-up program carried out in that period at the B&C 0.6 m telescope of the Laboratório Nacional de Astrofísica (LNA), Brazil. From this follow-up program comes an independent set of precise star positions, which is discussed in detail in Sect. 6.

For the future events here predicted, the positions of candidate stars are based on the obtained catalog, with typical errors of 20 mas. This precision is more than enough for a successful record of an occultation by Pluto, as its current apparent radius in the sky is about 50 mas. However, Nix and Hydra are more subject to missings, as their apparent radii are about 7 mas only. The same could be said (to a lesser extent) about Charon (25 mas apparent radius). The probabilities of successful occultation recordings for these satellites are addressed in detail in Sect. 9.

3. Observations at ESO

Observations were made at the 2.2 m Max-Planck ESO (ESO2p2) telescope (IAU code 809) with the Wide Field Imager (WFI) CCD mosaic detector. Each mosaic is composed by eight 4 k × 2 k CCDs of 7.′5 × 15′ (RA, Dec) size, resulting in a total coverage of 30′ × 30′ per mosaic. The pixel scale is 0.″238. A broad-band R filter (ESO#844) was used with $\lambda_c = 651.725$ nm and $\triangle\lambda = 162.184$ nm (full width at half maximum). Exposure time was 30 s. In very few cases, larger exposure times were used to compensate for bad weather conditions. In general, S/N ratios of about 200 were reached for objects with $R = 17$ without saturating bright ($R = 13$−15) stars. The limiting magnitude was about $R = 21$, with completeness about $R = 18.0$−19.0. The seeing varied between 0.″6 and 1.″5, and was typically 1″.

Observations spanned Pluto′s sky path from 2008 to 2015. Runs were carried out in September and October 2007, covering the 2008−2010 and 2011−2015 paths respectively. Mosaic overlapping was optimized for astrometric precision and telescope time consuming, including small shifts so that each star was exposed at least twice in different CCDs. Table 1 lists the WFI mosaic centers for each covered year. A total of 150 WFI mosaics or 1200 individual CCD frames were acquired for science. This resulted in about 40 GB of photometrically calibrated processed images.

Figure 1 illustrates the sky path covered by Pluto.

In all – including the 150 science mosaics for Pluto – 398 observed WFI mosaics (170 in September, 228 in October 2007) or 3184 individual CCD frames were used for determining astrometric field distortions (see Sect. 4.1), resulting in about 108GB of photometrically calibrated processed images. The mosaic centers were distributed along the projected Galactic plane, next to Pluto′s sky paths, where the star-crowded fields particularly favour resolution of distortion maps. The 248 extra mosaics served for another similar astrometric program carried out with the same instrument, covering the 2008−2015 sky path of transneptunian Quaoar, close to Pluto′s own sky path.

4. Astrometry

All CCD images underwent overscan, zeromean, flatfield and bad pixel corrections with IRAF (Tody 1993) via the esowfi (Jones & Valdes 2000) and mscred (Valdes 1998) packages.

Table 1. The (α, δ) ESO2p2/WFI mosaic centers for Pluto sky path from 2008 to 2015.

Year	2008	2009	2010	2011	2012	2013	2014	2015
	h m ° ′	h m ° ′	h m ° ′	h m ° ′	h m ° ′	h m ° ′	h m ° ′	h m ° ′
	18 02 –17 10	18 09 –17 46	18 18 –18 19	18 27 –18 50	18 35 –19 20	18 43 –19 48	18 51 –20 14	18 59 –20 38
	18 03 –17 09	18 10 –17 46	18 19 –18 19	18 28 –18 49	18 36 –19 19	18 44 –19 47	18 52 –20 13	19 00 –20 37
	18 04 –17 07	18 11 –17 45	18 20 –18 18	18 29 –18 49	18 37 –19 18	18 45 –19 46	18 53 –20 12	19 01 –20 36
	18 01 –17 03	18 12 –17 44	18 21 –18 17	18 30 –18 47	18 38 –19 17	18 46 –19 45	18 54 –20 11	19 02 –20 35
	18 00 –17 03	18 13 –17 42	18 22 –18 15	18 31 –18 45	18 39 –19 16	18 47 –19 44	18 55 –20 10	19 03 –20 34
	17 59 –17 04	18 08 –17 41	18 17 –18 17	18 30 –18 45	18 39 –19 15	18 48 –19 43	18 56 –20 09	19 04 –20 33
	17 58 –17 05	18 07 –17 42	18 16 –18 18	18 29 –18 46	18 38 –19 16	18 47 –19 44	18 57 –20 08	19 05 –20 32
	17 57 –17 06	18 06 –17 44	18 15 –18 20	18 28 –18 47	18 37 –19 18	18 46 –19 46	18 56 –20 10	19 04 –20 36
	17 56 –17 08	18 05 –17 46	18 14 –18 21	18 27 –18 48	18 36 –19 19	18 45 –19 48	18 55 –20 12	19 03 –20 38
	17 55 –17 10	18 04 –17 48	18 13 –18 24	18 26 –18 50	18 35 –19 21	18 44 –19 50	18 54 –20 14	19 02 –20 41
	17 54 –17 14	18 03 –17 52	18 12 –18 27	18 25 –18 51	18 34 –19 23	18 43 –19 51	18 53 –20 16	19 01 –20 43
	17 54 –17 29	18 02 –17 59	18 11 –18 37	18 24 –18 53	18 33 –19 24	18 42 –19 54	18 52 –20 18	19 00 –20 45
	17 55 –17 33	18 04 –18 10	18 12 –18 42	18 23 –18 55	18 32 –19 27	18 41 –19 56	18 51 –20 21	18 59 –20 48
	17 56 –17 36	18 05 –18 12	18 13 –18 45	18 22 –18 58	18 31 –19 29	18 40 –19 58	18 50 –20 23	18 58 –20 50
			18 14 –18 47	18 21 –19 01	18 30 –19 32	18 39 –20 01	18 49 –20 25	18 57 –20 53
			18 15 –18 48	18 20 –19 07	18 29 –19 37	18 38 –20 05	18 48 –20 28	18 56 –20 56
				18 21 –19 15	18 29 –19 44	18 38 –20 14	18 47 –20 32	18 55 –21 00
				18 22 –19 18	18 30 –19 47	18 39 –20 16	18 46 –20 38	18 55 –21 06
				18 23 –19 19	18 31 –19 48	18 40 –20 17	18 47 –20 42	18 56 –21 07
				18 24 –19 20	18 32 –19 49	18 41 –20 17	18 48 –20 43	18 57 –21 07
					18 33 –19 50	18 42 –20 17	18 49 –20 43	18 58 –21 07
							18 50 –20 43	
							18 51 –20 43	

Notes. Each overlapping WFI mosaic encompasses a 30′ × 30′ area in the sky. Sky paths 2008−2010 and 2011−2015 were observed in the September and October 2007 runs, respectively.

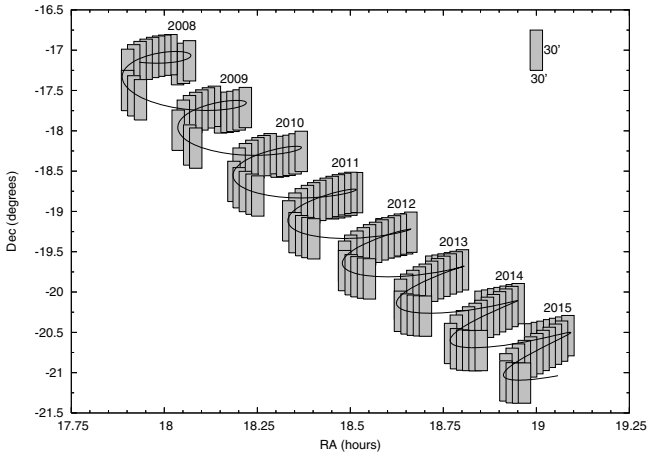


Fig. 1. Sky path covered by the ESO2p2/WFI CCD mosaic observations. Years 2008−2015 follow *from top to bottom*. The continuous line is the sky path of Pluto. Each dashed form represents the 30′ × 30′ area covered by one single WFI mosaic. Note the gaps when, as seen from Earth, the Sun is too close to Pluto′s direction at date (no occultation could then be seen).

Using the PRAIA package, the astrometric treatment consisted of three steps. First, a field distortion pattern was determined for each CCD in the WFI mosaics for each run. Then astrometry was performed over the individual CCDs, with the (x, y) measurements corrected by the pre-determined field distortions. Next all positions of common objects observed over the different CCDs and mosaics were combined in a global solution for each year, when final (RA, Dec) star positions were obtained. Besides positions, proper motions were also computed for each object using the 2MASS and USNO B1.0 catalogs as first epoch. These procedures are described in detail in the following subsections.

4.1. Field distortion pattern

Field Distortion Pattern (FDP) is characterized by the existence of at least two different regions on the CCD field where fixed distances on the sky present different angular distance measurements, even after modeling known astronomical effects (differential refraction, etc.). The ESO2p2/WFI mosaic is affected by FDP due to optical distortions of the third order, which may reach more than twice the size of a pixel (0.″238).

The procedure for mapping the distortions for each CCD of the WFI mosaic started by superposing the observed minus catalog (O−C) position differences of UCAC2 stars computed from the respective individual CCD astrometric solutions of the 398 WFI mosaics observed nearby the projected Galactic plane and along Pluto′s sky path (see Sect. 3). For each CCD, these (O−C) position residuals were averaged over bins of 1.′5 × 1.′5 in (x, y). The bin size was given by the request of a minimum number of stars per bin. With few exceptions, averages counted on more than 15 stars. Most frequently, hundreds of stars were available, furnishing about 500 position residuals per bin. Afterwards, in an iterative process, part of the averages were successively applied as a correction to the distortion. The procedure continued until no significant change occurred in the (O−C) residuals. Independent FDPs were computed for each observation run in September and October 2007. Note that only polynomials of the first order were used to relate the (x, y) measurements with the UCAC2 star coordinates projected in the sky plane. In this way, the third order distortions were consistently mapped onto the FDP. This allowed for the use of first degree polynomials in the individual CCD frame reductions, instead of third order ones, after first applying FDP corrections. The use of simpler polynomials improves the position accuracy, as it increases the ratio between the number of reference stars over the number of coefficients used in the model.

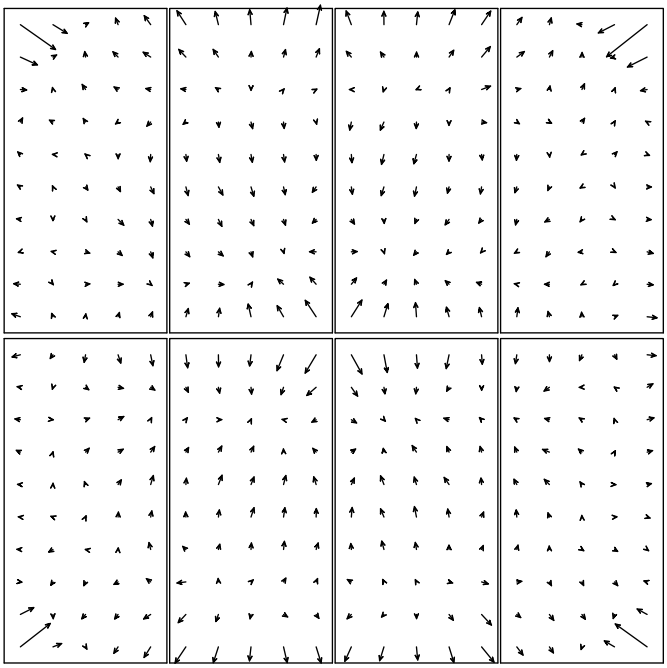


Fig. 2. Field distortion pattern (FDP) for the 8 CCDs of the WFI mosaic for the September 2007 run. North is up, East is left. Arrows point to the FDP-corrected position. The largest one (upper-right corner of plot) is 528 mas. Bins have 1′.5 × 1′.5 sizes. For details see text.

The (Δx, Δy) FDP offsets for each (x, y) bin and CCD from the WFI mosaics of the September and October 2007 runs were computed and stored. The FDP offsets are available by request to the author. Figure 2 illustrates the FDP derived from the September 2007 run for each CCD in the WFI mosaic. North is up, East is left. The largest offset (upper-right corner of plot) is 528 mas. A similar plot is obtained for the October 2007 run. The astrometric procedures used to derive the (RA, Dec)s that feed the FDP computation were the same as those described next in Sect. 4.2.

4.2. Astrometry of individual CCD frames

After obtaining the FDPs, we computed (RA, Dec)s for all stars measured in the CCDs of all the observed mosaics covering the sky paths from 2008 to 2015. The (x, y) measurements were pre-corrected with the FDP of the respective run, according to the respective bin and CCD in the WFI mosaic. Correction values were extrapolated by the inverse square distance to neighbor bin centers.

Positions were obtained with PRAIA (Assafin 2006). This fast astrometric/photometric package automatically identifies objects on the fields. The (x, y) measurements were performed with 2-dimensional circular symmetric Gaussian fits within 1 Full Width Half Maximum ($FWHM$ = seeing). Within 1 $FWHM$, the image profile is well described by a Gaussian profile, free from the wing distortions, which jeopardize the center determination. Theoretical and empirical results support this procedure (Moffat et al. 1969; Stone 1989). PRAIA automatically recognizes catalog stars and determines (RA, Dec) with a number of models relating the (x, y) measured and (X, Y) standard coordinates projected in the sky tangent plane. Positions, (x, y) centers, magnitudes, seeing – among other quantities and respective estimated errors – are computed and archived for all objects.

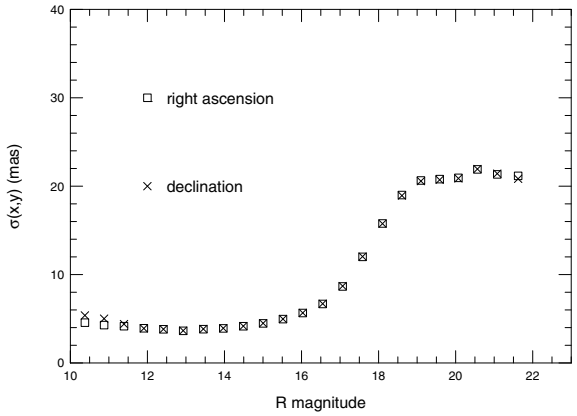


Fig. 3. (x, y) measurement errors as a function of R magnitude from all treated CCDs. Values are averages over 0.5 mag bins.

Magnitudes were obtained from PSF photometry and were calibrated with respect to the UCAC2. Note that the UCAC2 star magnitudes are based on a 579−642 nm filter (between Johnson V and R), and are thus distinct from the filter used in the WFI observations. Thus, the image-to-image magnitude zero point will depend on the mean color of the field stars. However, since the photometric errors of the UCAC2 are somewhat large (about 0.3), we will consider here for all purposes that the derived WFI magnitudes are formally in the UCAC2 system. This issue will be further addressed after future releases of the UCAC catalog, when more refined photometric magnitudes are expected to be available. Furthermore, for simplicity, here we will simply refer to WFI magnitudes as R magnitudes.

A complete and detailed description of the PRAIA package will be published in the future. See further details about performance in Assafin et al. (2007).

We used the UCAC2 as reference frame and the six constants polynomial to model (x, y) measurements to (X, Y) plane coordinates. About 120 UCAC2 stars per frame were used in these Galactic plane star-crowded fields. Reference stars were eliminated in a one-by-one basis until none displayed (O−C) position residuals greater than 120 mas (2−3σ the typical catalog error). The position mean errors from (RA, Dec) solutions were about 60 mas for both coordinates. Estimated (x, y) measurement errors from Gaussian fits were about 20−30 mas between $12 < R < 17$, rising as expected at magnitudes brighter and fainter than this range. Figure 3 shows the distribution of (x, y) errors as a function of R magnitude. Values were averaged over 0.5 mag bins from all measured CCDs.

A summary of the results from the individual CCD astrometric treatment is given in Table 2. For each year, we list the number of CCD frames, mean error of positions from (RA, Dec) solutions and average number of UCAC2 reference stars per frame.

The same procedures described here were applied to individual CCD fields to obtain the FDPs (Sect. 4.1).

4.3. Mosaic global astrometric solution

For each year, global astrometric solution for overlapping mosaics of CCDs was accomplished by an iterative procedure available by PRAIA. Starting from the individual CCD measurements (see Sect. 4.2), all common star positions, magnitudes and other values and errors were averaged. Common stars were recognized among CCD frames by individual CCD positions, which lie within 200 mas from each other. Then a method

Table 2. Astrometry of individual CCD frames of WFI mosaics.

Sky path year	Mean errors $\sigma(\Delta\alpha\cos\delta)$ mas	$\sigma(\Delta\delta)$ mas	Frames per year	No. UCAC2 stars/frame
2008	59	59	112	105
2009	58	56	112	79
2010	58	56	128	164
2011	61	59	160	153
2012	63	62	168	125
2013	63	62	168	107
2014	62	62	184	105
2015	58	57	168	132

Notes. Mean errors come from standard deviations of (O−C) residuals from individual (RA, Dec) solutions with the UCAC2 catalog. The six constants polynomial model was used to relate (x, y) measurements with (X, Y) tangent plane coordinates. The (x, y) centers were pre-corrected by the FDPs. See details in Sect. 4.2.

Table 3. Global astrometric solution for WFI CCD mosaics.

Sky path	Zero-point offset before G.S. $\Delta\alpha\cos\delta$	$\Delta\delta$	(RA, Dec)–UCAC2 before G.S. σ_α	σ_δ	after G.S. σ_α	σ_δ	UCAC2 stars
Year	mas	mas	mas	mas	mas	mas	
2008	−13	−04	112	107	59	58	6001
2009	−09	−01	112	103	58	55	4693
2010	−03	−03	133	126	59	57	11013
2011	−08	−03	134	129	61	59	8988
2012	−06	−05	139	131	62	61	7563
2013	−03	−04	145	141	63	62	6068
2014	−06	+02	135	138	60	60	6439
2015	−06	−03	100	098	57	57	7524

Notes. Tangent plane technique results. Zero point position offsets were computed from averaged observed minus UCAC2 star position differences, before applying the global mosaic solution procedure (G.S.). Due to the tangent plane technique, by definition they are zero after G.S. process. The $(\sigma_\alpha, \sigma_\delta)$s of tangent plane technique solutions refer to observed minus UCAC2 positions computed before and after the G.S. procedure. The last column is the number of used UCAC2 reference stars (see details in Sect. 4.3).

Table 4. Multiplicity flags for WFI global mosaic star positions.

Year	UCAC2 f1 %	f2 %	2MASS f1 %	f2 %	Field f3 %	f4 %	No flag %	Input entries
2008	0.0	0.0	1.4	0.7	1.0	2.6	94.3	296 823
2009	0.1	0.0	5.2	3.1	0.3	1.5	89.8	53 221
2010	0.0	0.0	0.4	0.6	1.3	1.6	96.1	378 236
2011	0.0	0.0	0.7	0.7	2.9	1.9	93.8	502 966
2012	0.0	0.0	0.3	0.8	3.1	2.1	93.7	672 564
2013	0.4	0.0	9.8	3.3	1.1	3.1	82.3	201 541
2014	1.8	0.1	13.0	4.7	2.1	3.5	74.8	159 725
2015	0.0	0.0	1.3	0.6	3.8	3.1	91.2	194 573

Notes. Input entries give the number of mosaic positions before checking for multiplicity. Multiplicity flag percentages refer to the final number of positions, one per object. See text for details.

that we call tangent plane technique, adapted from Assafin et al. (1997), was applied. In this method, all the CCD frame (RA, Dec)s and catalog-extracted UCAC2 reference positions (corrected by proper motions to the mean epoch of observations) are projected in the tangent plane. A complete polynomial model of the third degree is then used to relate these projected coordinates in the same way as in classical photographic field astrometry. After the elimination of UCAC2 stars outliers with (O−C) residuals larger than 120 mas, the tangent plane solution was obtained. Inverse gnomonic projection furnished the (RA, Dec) of all objects in the mosaic. These positions formed an intermediary star catalog in the UCAC2 reference frame. Then, new individual CCD astrometric adjustments were performed, but now using this intermediary catalog as reference frame for all CCD fields. Now, every star in the individual CCD frames participates as reference star. Here, a complete third degree polynomial model was used (instead of the six constant model used in the first step in Sect. 4.2) and new individual CCD positions were obtained. The entire process was then repeated in an iterative fashion, with new averaging of common positions and new application of the tangent plane technique. The procedure stopped after intermediary catalog star positions converged to within 1 mas, which always happened in less than 50 iterations.

Table 3 brings a summary of the results from the global solutions of WFI CCD mosaics for the sky path observed, focusing on the tangent plane technique results. For each year it gives the standard deviations (σ) of observed minus UCAC2 catalog positions before and after the global solution. The listed zero-point position offsets were computed from averaged (RA, Dec) offsets over UCAC2 stars before applying the global mosaic solution. By definition, due to the tangent plane technique, they are zero after global solution. Table 3 also gives the number of UCAC2 reference stars used in the process.

4.4. Mosaic position multiplicity

About 8% of the stars displayed multiple position entries within 1.″5 of each other after global mosaic solutions. This happened whenever individual CCD positions from the same object did not pass the 200 mas criterion for identification of common stars in the global solution procedure. As a result, multiple positions survived the process as if they belonged to distinct objects. In the vast majority of cases, these entries displayed magnitudes fainter than $R = 19$. The cause might be poor faint star deblending in the individual CCD frame (x, y) measurements, as heavily star-crowded sky fields were sampled. Stars with multiple entries were assigned one unique position and flagged. No flag means a star with no multiple entries (good astrometry). Flag cases f1 and f2 apply only for UCAC2 or 2MASS stars. In these cases, multiple entries assigned to one of these catalogs were used, but others wrongly assigned as field stars were rejected. Flag f1 means that more than one entry was used, flag f2 indicates that only one entry was used. Flags f3, f4 and f5 apply only for field stars. Only a single entry was selected according to one of three criteria, in order of priority: a) highest number of used common individual CCD positions (flag f3); b) least (x, y) measurement error (flag f4); c) brightest R magnitude (flag f5).

Table 4 displays multiplicity flag statistics for the derived ESO2p2/WFI global mosaic star positions according to catalog and year. The input number of unflagged entries is furnished, but percentages refer to the final number of WFI stars. Percentages for flag f5 entries were always less than 0.1% and thus are not displayed.

After checking for multiplicity and flagging, the final set of global mosaic star positions is obtained. For the flagged stars, magnitude, mean epoch and other parameters were assigned in

Table 5. Proper motion computations from 2MASS, USNO B1.0 and ESO2p2/WFI global mosaic star positions.

Year	Final mosaic positions	UCAC2 p.m.	2MASS p.m.	USNO B1.0 p.m.	No p.m.
2008	279 127	9826	155 807	47 774	65 720
2009	47 576	7343	32 213	5472	2548
2010	361 830	20 704	121 440	69 433	150 253
2011	466 433	17 740	167 283	83 910	197 500
2012	622 343	15 689	155 970	128 038	322 646
2013	166 283	14 180	109 182	20 722	22 199
2014	122 199	13 634	78 610	14 985	14 970
2015	176 495	11 067	72 908	37 607	54 913
total	2 242 286	110 183	893 413	407 941	830 749

Notes. Number of final global mosaic star positions, number of UCAC2 stars with proper motions (directly extracted from UCAC2), number of stars with proper motions based on the 2MASS and USNO B1.0 and number of stars for which no proper motion could be computed.

the same way as for positions, but no position error could be estimated for them.

4.5. Computation of proper motions

One important step in our astrometric procedure was the derivation of proper motions for stars not belonging to the UCAC2, using the 2MASS and USNO B1.0 catalogs as first epoch. The mean epochs of the 2MASS and USNO B1.0 catalogs are respectively around 2000 and 1980. The 2MASS catalog is based on infrared bandpass observations with modern solid state detectors. The catalog USNO B1.0 was created from astrometric digitalization of photographic Schmidt plates. 2MASS position precision ranges between 100−200 mas, better than the 250−300 mas errors of USNO B1.0 positions. But time span favours USNO B1.0, so that the overall attained error budget of computed proper motions is similar, regardless of the first epoch used.

In the procedure, the first epoch position for brighter stars was chosen from the 2MASS. If the star was fainter – that is, did not belong to that catalog – then the USNO B1.0 position was used, instead. For both catalogs, only matches within 1″ in position were considered. No brightness constraints were applied for matching the USNO B1.0. For the 2MASS case, stars with discrepancies higher than 1 mag were rejected in comparing measured R magnitudes with H band. For multiple matches, the closest magnitude was selected.

In Table 5, we give a summary of the proper motion computations using the WFI global mosaic star positions, 2MASS and USNO B1.0 catalogs. For each year, we list the total number of final global mosaic star positions, the number of UCAC2 stars with proper motions (directly extracted from UCAC2), the number of stars with proper motions based on the 2MASS and USNO B1.0 and the number of non-matched stars, for which no proper motion could be computed. Again, cases where no proper motion could be derived relate almost exclusively to stars fainter than $R = 19$.

5. The catalog of star positions along Pluto's 2008–2015 sky path

The star catalog for the 2008–2015 sky path of Pluto consists of mean (RA, Dec) positions in the ICRS (J2000), proper motions, R magnitudes (also J, H and K for 2MASS stars), mean epoch of observations, position error at mean epoch of observation and magnitude error estimates. It has 2 242 286 stars in the UCAC2 frame. Its mean epoch is approx. 2007.75. The magnitude completeness is about $R = 18-19$. The magnitude limit is about $R = 21$. The position error is about 50 mas for stars up to magnitude $R = 19$, and 25 mas up to $R = 17$. The catalog is freely available in electronic form at the CDS.

The catalog is divided by year. There are small gaps between the years (see Fig. 1). Stars that had multiple entries within 1.″5 in the global mosaic solutions (about 8% of total) are flagged (see Sect. 4.4). The R magnitudes from PSF photometry were calibrated in the UCAC2 system, so magnitude zero-point errors up to 0.3 might be expected for $R > 17$. The position error is estimated from repeatability by the standard deviation (mean error) of contributing individual CCD positions about the final catalog star positions (last iteration in global mosaic solution – see Sect. 4.3). By default, multiple entry flagged stars have no position error estimates. Infrared magnitudes (and errors) were extracted from the 2MASS catalog. Error estimates for R magnitudes come from the standard deviation about the mean from individual CCD frames. Sky coverage of the catalog is detailed in Sect. 3.

Table 6 lists the total number of catalog stars per year, average position errors, R bandpass magnitude limit (including highest values) and completeness. Figure 4 shows the star distribution per R magnitude. Figure 5 plots the position error as a function of R magnitude. Values were averaged over 0.5 mag bins.

6. Pluto's ephemeris offsets

In the recent past, a number of stellar occultations by Pluto and Charon have then been foreseen for 2005−2008. The successful outcome of these complex international observational campaigns were only actually achieved thanks to precise position updates for the candidate stars, which started long enough in advance and continued until the epoch of those events. Beside the efforts by the MIT group and by IOTA (International Occultation Timing Association), among others, an important contribution for these prediction updates came from the astrometric observational program of Pluto carried out at the 0.6 m B&C telescope at the Laboratório Nacional de Astrofísica (LNA), Brazil (IAU code 874). Those observations were made with 1 k^2 and 2 k^2 CCD detectors of 10′ sizes and pixel scales of 0.″3 to 0.″6 (for a detailed description of telescope/instruments, see Assafin et al. 2005). The PRAIA package was employed in the astrometry of these CCD observations. Frames were free from high order optical distortions and were modeled with six constant polynomials. Typical (x, y) measurement errors were about 15 mas for $R < 15$. Position mean errors from astrometric (RA, Dec) solutions ranged between 50−60 mas. Position precision inferred from the repeatability of solutions was about 20 mas. Positions were referred to the UCAC2 catalog. The UCAC2-based candidate star positions were zero-point-corrected toward ICRS, using averaged UCAC2 minus ICRF position offsets of $\Delta\alpha\cos\delta = -12 \pm 8$ mas and $\Delta\delta = -5 \pm 7$ mas. These local offsets were computed from the comparison between optical and VLBI positions for the five nearest ICRF quasars to Pluto's 2006.5 coordinates, distributed within 10 degrees radius (see Assafin et al. 2005). Table 7 summarizes the astrometry of candidate stars with the 0.6 m LNA telescope for 2005−2008 Pluto stellar occultations. Only results regarding the final positions used for deriving ephemeris offsets are displayed. They refer to observations made at least within a month of the event epoch. One exception was the 2006 June 12 event, for which good CCD observations were only

Table 6. Star catalog for the 2008–2015 Pluto sky path.

Year	2008	2009	2010	2011	2012	2013	2014	2015
catalog stars	279 127	47 576	361 830	466 433	622 436	166 283	122 199	176 495
UCAC2 stars	9826	7343	20 704	17 740	15 689	14 180	13 634	11 067
2MASS stars	155 834	32 217	121 489	167 309	155 998	109 201	78 623	72 926
Field stars	113 467	8016	219 637	281 384	450 656	42 902	29 942	92 502
σ_α (mas)	25	27	24	27	25	31	33	30
σ_δ (mas)	24	23	22	24	24	36	39	29
highest R magnitude	22.7	22.3	23.4	22.5	23.7	22.3	21.8	24.0
R magnitude limit	21.0	21.0	21.5	21.0	21.5	21.0	21.0	21.5
R magnitude completeness	9.5–18.5	8.5–18.0	9.5–18.5	10.0–19.0	10.4–19.0	9.1–18.0	8.5–18.0	9.4–18.5

Notes. Number of catalog stars per year, position error (estimated from the standard deviation of contributing individual CCD positions about the final catalog star positions), R bandpass magnitude completeness, magnitude limit and highest value.

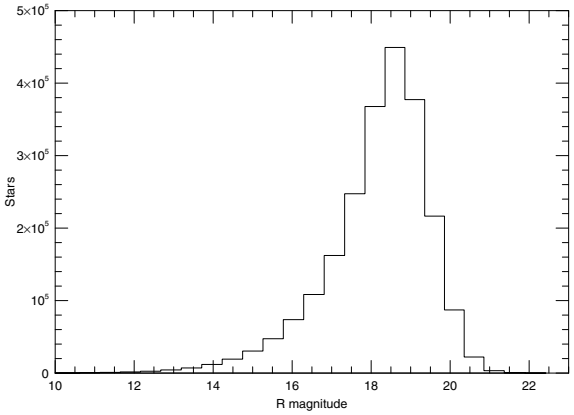


Fig. 4. Star distribution per R magnitude. It illustrates the R magnitude limit and completeness of catalog. Counts were computed over 0.5 mag bins.

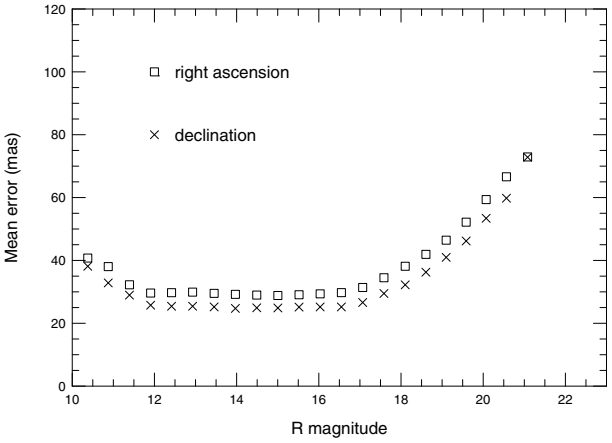


Fig. 5. Catalog position mean errors as a function of R magnitude. Position errors are estimated from the standard deviation of contributing individual CCD positions about the final catalog star positions (last iteration in global mosaic solution – see Sect. 4.3). Values were computed over 0.5 mag bins.

available from an April 2007 run. As it was a UCAC2 star, the derived CCD position could be referred to the epoch of the event by applying UCAC2 proper motions. The 2008 August 25 event star was observed with the 1.6 m P&E LNA telescope (FOV of $5' \times 5'$ and pixel scale of $0\farcs17$).

Observations of Pluto itself were made on a regular basis and also prior to the events with the 0.6 m B&C LNA telescope, for estimating possible ephemeris offsets. More than 1500 Pluto positions where obtained in the time span between 2005−2008, following the same observational and astrometric procedures. These Pluto positions were accurate enough to display the perturbation by Charon, although it was unresolved in the CCD images. We usually solved this problem by looking at the (O−C) ephemeris residuals from observations symmetrically distributed along the 6.4-day orbital period of Charon. Later on, a new procedure for determining position offsets was implemented, which allowed the use of all observations. This new method is based on the modeling of the resulting PSF of unresolved images, in terms of relative apparent distance between components, relative brightness and seeing conditions. Details of these results will be published elsewhere (Vieira Martins et al., in prep).

In practice, we followed a sort of bootstrap method for the 2005−2008 campaigns. In the beginning, occultations were not accurately predicted (sometimes even missed), as no ephemeris drift was applied. Then, based on these LNA observations, significant ephemeris offsets started to be found and applied, which improved the next prediction, until a linear trend in declination eventually appeared. In fact, based on the post-fit of these occultations, we not only found large ephemeris offsets of some tens of mas, but also confirmed this linear ephemeris drift with time in declination for Pluto, as shown in this section. The ephemeris-checking procedure with LNA observations proved to be very important for the successful recording of those past occultations. The predicted Earth locations would be severely misplaced if ephemeris offsets mostly in declination were not properly taken into account in advance during the campaign planning phase.

In a stellar occultation with two or more observed cords, Pluto′s position relative to the occulted star can be derived with mas-level accuracy. If the star position is known, Pluto′s right ascension and declination ephemeris offsets can be determined for that instant. A collection of ephemeris offsets obtained over time thus helps us determining systematic trends, if present.

After fitting synthetic light curves to observations of the events listed in Table 7 (Sicardy et al., in prep.) and taking into account the respective star positions, Pluto′s offsets relative to its ephemeris were computed. Here, the DE418 and plu017 ephemerides were used, as they were specially devised for the New Horizons mission to Pluto (Folkner et al. 2007). They are available through NASAs Navigation and Ancillary Information Facility (NAIF) ftp site (`ftp://naif.jpl.nasa.gov/pub/naif/`) as SPICE kernels DE418 and plu017 (see details about NAIF in Acton 1996). The ephemeris offsets obtained for each event are listed in Table 8. For the 2006 April 10 event, no occultation occurred (this was actually predicted), but the

Table 7. Astrometry of candidate stars observed at the 0.6 m B&C LNA telescope for 2005−2008 Pluto stellar occultations.

Event	(RA, Dec) Epoch years	ICRS (J2000) (RA, Dec) h m s ° ′ ″	Magnitude (R)	Error E_α mas	Error E_δ mas	No. Obs.	Mean error σ_α mas	Mean error σ_δ mas	UCAC2 stars No.
2005 July 11	2005.2560	17 28 55.0167 –15 00 54.726	14.9	11	15	25	50	52	66
2006 April 10	2006.1620	17 46 06.8788 –15 46 10.113	16.0	33	32	255	56	55	107
2006 June 12	2006.4449	17 41 12.0769 –15 41 34.488	15.0	11	15	13	61	59	186
2007 March 18	2007.2165	17 55 05.6948 –16 28 34.369	15.1	08	11	12	59	62	162
2007 May 12	2007.3641	17 53 32.1024 –16 22 47.359	16.5	56	30	8	57	58	176
2007 June 09	2007.4598	17 50 50.6520 –16 22 29.309	17.0	16	03	4	56	60	94
2007 June 14	2007.4588	17 50 20.7402 –16 22 42.207	15.6	48	54	190	58	55	108
2007 July 31	2007.4634	17 45 41.9894 –16 29 31.639	14.0	15	24	68	60	56	72
2008 June 22	2008.4044	17 58 33.0147 –17 02 38.340	13.5	11	09	50	50	48	76
2008 June 24	2008.4044	17 58 22.3941 –17 02 49.346	15.9	13	17	50	50	48	76
2008 August 25	2008.6494	17 53 27.1040 –17 15 27.541	15.6	06	04	30	56	56	26

Notes. Used positions are always within a month of event epoch. Error estimates come from the dispersion (standard deviation) of values from the observations. Mean error of (RA, Dec) solutions and average number of UCAC2 reference stars are given. The UCAC2-based candidate star positions were zero-point-corrected toward ICRS using averaged UCAC2 minus ICRF position offsets ($\Delta\alpha\cos\delta = -12 \pm 8$ mas; $\Delta\delta = -5 \pm 7$ mas) of five ICRF quasars nearby Pluto present coordinates (see Assafin et al. 2005). For the 2006 June 12 event, useful CCD observations were only acquired in April 2007, but the final position is at the event epoch, as proper motions ($\mu_\alpha = -10.6$ mas yr^{-1}, $\mu_\delta = -12.9$ mas yr^{-1}) were applied for this UCAC2 star. The 2008 August 25 event star was observed with the 1.6 m P&E LNA telescope (FOV of 5′ sizes and pixel scale of 0.″17). Actually, 2005 July 11 refers to a stellar occultation by Charon.

offset could be derived (at 20 mas level) due to high resolution adaptative optics observations of Pluto and the star made at the ESO-VLT 8 m Telescope UT4 (Yepun) at Paranal Observatory, Chile, with the NACO instrument. The event of 2005 July 11 involved Charon, while the event of 2008 June 22 involved both Pluto and Charon occulting the same star. In all other cases, the events concern stellar occultations by Pluto alone.

Occultations involving Charon were included in this determination of the ephemeris offsets of Pluto because measuring Charon′s offset with respect to DE418/PLU017 is in practice identical to measuring Pluto′s offset with respect to DE418, with uncertainties dominated by the star position errors. Charon′s plutocentric ephemeris is much better known than Pluto′s barycentric motion in the sky. Hubble Space Telescope observations in 2002−2003 provide an orbital solution with rms residual just around 2 mas, and an orbital solution better than that once $\sqrt{N_{\rm obs}}$ is accounted for (see Tholen et al. 2008). Also, from our fittings to the 2008 June 22 Pluto/Charon occultations of the same star, the agreement between our measurement of Charon and PLU017 is better than 1 mas (Sicardy et al., in prep.).

For the double 2008 June 22 event, only one ephemeris offset was considered for right ascension and declination, as both Pluto/Charon occultations furnish identical values within 1 mas. For the 2008 June 24 event, two ephemeris offsets were used. Although this was a single occultation by Pluto, there was only one cord observed in Hawaii. Fortunately, this was a near-central event. The two possible solutions were only 10 mas apart from each other in the perpendicular direction of motion (mostly in declination). Thus, for practical purposes, we used both values as independent offset measurements in the linear drift solution.

Table 8 also lists four extra offsets measured for four common events, based on an independent set of star positions and occultation observations (Young et al. 2008, 2007; Olkin et al., in prep.; Buie et al. 2009). Averaged offsets from this independent set with our derived values are also furnished for these four common events. The light curve fitting procedure for these cases follow the method described in Young et al. (2008) and references therein. Since only unrealistic differences in the atmospheric modeling/fitting procedures would cause center shifts to be larger than about 1 mas, the common parameter that should be constant across independent analyses of these common occultations is the observed position of the star relative to Pluto at a given time for a given site. These relative distances were obtained from the best light curve fits available and, using the star positions, converted in Pluto positions and then in ephemeris offsets, much in the same way as for all the other offsets.

From Table 8, a clear linear drift with time is seen for the declination offsets. This drift must be taken into account in the predictions, as the occultation shadow path over the Earth is most sensitive to an ephemeris offset in declination. Thus, adopting the empirical relation $offset = A * (t - 2005.0) + B$ with offsets in mas and time in years, and fitting the 12 declination offsets listed in Table 8 (only averaged values were used for the four common events), we find $(A, B) = (+30.5 \pm 4.3$ mas yr^{-1}, -31.5 ± 11.3 mas), with an (O−C) standard deviation of 14.4 mas for the offsets. Figure 6 plots the ($\Delta\alpha\cos\delta$, $\Delta\delta$) ephemerides offsets (DE418 and plu017) of Pluto against time for the 2005−2008 studied occultations. The fitted linear drift with time in declination is illustrated.

From Fig. 6 it can be seen that the ephemeris offsets in right ascension are more dispersed than in declination. One might try to explain it by the appealing scenario of an oscillation pattern related to an error in Pluto′s heliocentric distance (geocentric parallax error). However, contrary to declination, none of the attempted models for this scenario could fit the offsets well below the 50 mas standard deviation, particularly for the events in opposition – even after introducing an empirical linear drift with time in right ascension. This issue will be further addressed in detail in a forthcoming paper (Vieira Martins et al., in prep). In the present work, we have not applied offset corrections of any kind to right ascension. This is here justified by the pragmatic fact that the eventual presence of right ascension ephemeris offsets does not affect the geographic latitude of Pluto′s shadow path over the Earth, except – and only marginally so – far from opposition. Usually it will only cause a slight error in the predicted central instant of the occultation by a few minutes at most, which from the observational point of view is usually easily accommodated by extending the duration of the occultation run.

Table 8. Pluto DE418 and plu017 ephemerides offsets with time.

Event	Central instant years	Observed – Ephem. $\Delta\alpha\cos\delta$ mas	$\Delta\delta$ mas	Note
2005 July 11	2005.5254	+35	−26	p
2006 April 10	2006.2731	−73	+00	p
2006 June 12	2006.4468	+63	+19	p
2006 June 12	2006.4468	+49	+15	1
2006 June 12	2006.4468	+56	+17	a
2007 March 18	2007.2100	−48	+30	p
2007 March 18	2007.2100	−46	+30	2
2007 March 18	2007.2100	−47	+30	a
2007 May 12	2007.3597	−38	+36	p
2007 June 09	2007.4371	−14	+53	p
2007 June 14	2007.4499	−49	+57	p
2007 July 31	2007.5799	+42	+64	p
2007 July 31	2007.5799	+42	+61	3
2007 July 31	2007.5799	+42	+62	a
2008 June 22	2008.4803	+37	+96	p
2008 June 24	2008.4758	+73	+63	p
2008 June 24	2008.4758	+73	+73	p
2008 August 25	2008.6494	+02	+53	p
2008 August 25	2008.6494	+04	+52	4
2008 August 25	2008.6494	+03	+52	a

Notes. DE418 and plu017 ephemerides offsets of Pluto with time. Offsets marked "p" were determined from fittings of past occultations in 2005−2008, taking as reference LNA-based star positions (Table 7) derived for these events. Four extra offsets from four common events were also independently measured ("1" = Young et al. 2008; "2" = Young et al. 2007; "3" = Olkin et al., in prep.; "4" = Buie et al. 2009); for these common occultations, averaged offsets with the corresponding "p" values are marked "a". The offsets are in the sense observed minus ephemeris. For the 2006 April 10 event, no occultation occurred (this was actually predicted), but the offset could be derived (at 20 mas level) due to high resolution adaptative optics observations of Pluto and the star made at the ESO-VLT 8 m Telescope UT4 (Yepun) at Paranal Observatory, Chile, with the NACO instrument. The event of 2005 July 11 involved Charon, while the event of 2008 June 22 involved both Pluto and Charon occulting the same star. In all other cases, the events concern stellar occultations by Pluto alone. For the double 2008 June 22 event, only one ephemeris offset was considered for (α, δ), as both Pluto/Charon occultations furnish identical values within 1 mas. For the 2008 June 24 occultation by Pluto, two ephemeris offsets were used from two possible solutions. There was only one cord observed in Hawaii, but the event was near central with the two possible solutions only 10 mas apart from each other in the perpendicular direction of motion (mostly in declination).

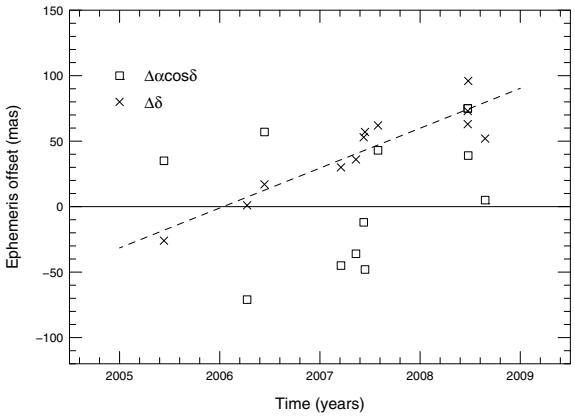


Fig. 6. DE418 and plu017 ephemerides offsets of Pluto in right ascension and in declination against time in the sense observed minus ephemeris. Offsets were determined from fittings of past occultations in 2005−2008, taking as reference LNA-based positions derived for these stars (only averaged values were used for the four common events listed in Table 8). The dotted line is the fitted linear drift in declination. No ephemeris offset correction was attempted for right ascension. (See discussion in the text of Sect. 6.)

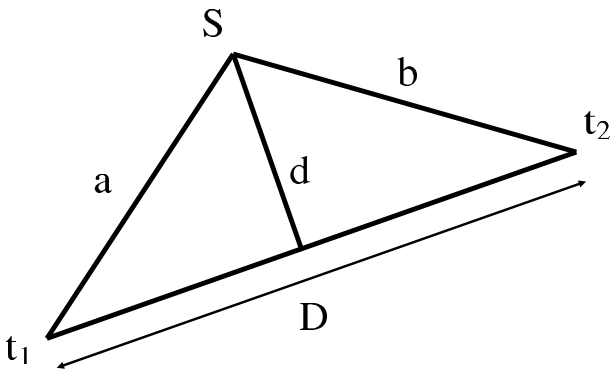


Fig. 7. Geometric configuration of potential close approach. a and b are the apparent geocentric distances in the plane of the sky between the body and the star at arbitrary instants t_1 and t_2 before and after the closest approach. D is the apparent geocentric distance between the body geocentric ephemeris positions at t_1 and t_2 and d is the minimum apparent geocentric distance at closest approach between the body and the star.

7. Search procedure for candidate stars

The search procedure for candidate stars to be occulted by Pluto and its satellites was based on the obtained star catalog described in Sect. 5 and by using the ephemeris drift derived for declination in Sect. 6. All catalog star positions, corrected by proper motions, were crossed against the DE418 and plu017 ephemerides of Pluto, Charon, Nix and Hydra, extracted in a per-minute basis for the whole period between 2008 to 2015. The body's declination ephemeris was offset according to the computed linear drifts for each instant. If the distance between the star position and the (offset-corrected) body ephemeris was less than a given value, a potential occultation was found and all astrometric and geometric data relevant to the possible event were computed and stored.

For each candidate star, besides astrometric and photometric data, the minimum apparent geocentric distance d, the central instant of closest approach t_0, the shadow velocity v across the Earth, the position angle PA of the shadow path and local solar time LST at sub-planet point were computed and stored. These geometric quantities were calculated as follows. Consider the close approach scheme displayed in Fig. 7, where a and b are the apparent geocentric distances in the plane of the sky between the star S and the body at arbitrary instants t_1 and t_2 before and after the closest approach. D is the apparent geocentric distance between the body ephemeris positions at t_1 and t_2 and d is the minimum apparent geocentric distance at closest approach between the body and the star. The minimum apparent geocentric distance d is thus given by

$$d = \sqrt{a^2 - \left(\frac{a^2 - b^2 + D^2}{2D}\right)^2}.$$

If $t_2 > t_1$, the central instant t_0 (UTC) of the occultations is

$$t_0 = t_1 + (t_2 - t_1)\sqrt{\frac{a^2 - d^2}{D^2}}.$$

The velocity v in km s^{-1} of the shadow across the Earth at a distance A(km) from the body is given by

$$v = \frac{A \sin(D)}{(t_2 - t_1)},$$

with t_2 and t_1 expressed in seconds. Positive/negative v means prograde/retrograde velocities respectively, that is, Pluto's geocentric right ascension is increasing/decreasing respectively.

From the (offset-corrected) body's (RA, Dec) ephemeris and from the star position at t_0, one can easily calculate the position angle PA of the shadow path across the Earth surface at central instant t_0. It is defined as the position angle of the body with respect to the star at closest approach. PA is zero when the body is north of the star and is counted clockwise.

The rough local solar time LST at sub-planet point was computed by

$$LST = t_0 + long = t_0 + RA - MSTG,$$

were $long$ is the east longitude of the sub-planet point, $MSTG$ is the Mean Sideral Time in Greenwich at t_0 and RA is the right ascension of the body at closest approach. Note that LST provides a rough indication as to whether the event is mostly observable during night time versus day time at the sub-planet point.

For the search we extracted ephemeris positions using 1 min time intervals. After finding a potential occultation, however, we took ephemeris positions at t_1 and t_2 about 1h apart from each other, around t_0. This precaution allowed for a significant change in the coordinates, thus improving computation precision, particularly far from opposition and close to stationary configurations, when the shadow velocity v is small.

8. Predictions of stellar occultations by Pluto and its satellites

Following the procedure described in Sect. 7, candidate stars for occultations by Pluto, Charon, Nix and Hydra were found. The adopted search radius was 0.″335 – about the apparent radius of Pluto (50 mas) plus the apparent Earth radius (285 mas) as projected in the sky plane at 31 AU (Pluto-Earth distance for 2008–2015). No predictions were discarded due to day light at sub-planet point, as occultations could even so be visible right above the horizon from places still in the dark near Earth terminator. For each body, all relevant astrometric, photometric and geometric information for each potential event found is available in electronic form via anonymous ftp to `cdsarc.u-strasbg.fr`. Table 9 lists a sample of predictions for Pluto. It contains the date and instant of stellar occultation (UTC), the ICRS (J2000) star coordinates at the event date, the closest apparent geocentric distance between star and body, the position angle of the shadow across the Earth (clockwise, zero at North), the velocity in km s^{-1}, the distance to the Earth (AU), longitude of the sub-solar point, local solar time, DE418 and plu017 ephemerides offsets in (RA, Dec) for the central instant, the catalog proper motion and multiplicity flags, the estimated star catalog position errors, the proper motions and the magnitudes R^*, J^*, H^* and K^*. Magnitudes are normalized to a reference shadow velocity of 20 km s^{-1} by

$$M^* = M + 2.5 \log_{10}\left(\frac{v}{20 \text{ km s}^{-1}}\right).$$

The value 20 km s^{-1} is typical of events around the Pluto opposition. Therefore M^* may bring forward faint stars involved in slow events, thus allowing for longer integration time, and consequently reasonably good signal-to-noise ratios (SNRs) without loss of spatial resolution in diameter measurements and in probing atmosphere altitudes in the light curves, in spite of the faintness of the targets. Note however that Pluto's important contribution to the total recorded flux will the be an issue in those situations, so that a case by case estimation of the SNR must be conducted for those candidates.

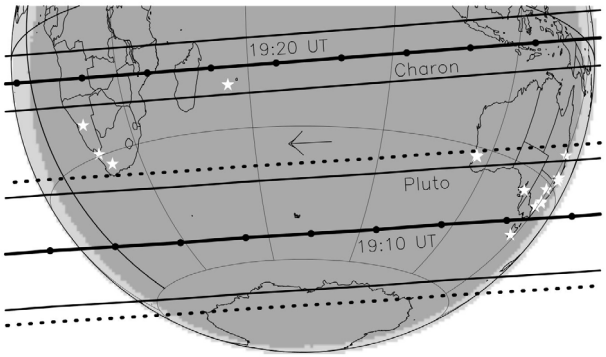


Fig. 8. Geometry of the 2008 June 22 event on Earth, based on reconstructed, post-event paths. See details in the text.

Figure 8 illustrates the geometry of the 2008 June 22 event on Earth, based on reconstructed, post-event paths (see prediction information in Table 9). As predicted, it was actually a double occultation of the same star by Pluto and Charon. The occultations were visible in South Australia, Namibia and La Reunion Island and were eventually recorded from five sites in Australia and one site in La Reunion Island in the Indian Ocean.

Table 10 displays the total number of predicted events for each body for the period 2008–2015.

9. Discussion

We presented predictions for stellar occultations by Pluto, Charon, Nix and Hydra for 2008–2015 based on observations made with the ESO2p2/WFI CCD mosaic. For this purpose, an astrometric catalog of 2.24 M stars with proper motions was derived encompassing the 2008–2015 sky path of Pluto within 30′ width. It is in the UCAC2 reference frame and has a magnitude completeness about R = 18–19 with a limit around R = 21. Its mean epoch is around 2007.75. The position error is about 25 mas for R = 12–17, ranging from 25 mas to 50 mas for R = 17–19 (Fig. 5). The entire astrometric catalog and the complete set of tables with stellar occultation predictions are available in electronic form via anonymous ftp to `cdsarc.u-strasbg.fr`.

The astrometry of about 110GB of processed WFI images, first for the FDP determination, then for the catalog, was made in automatic fashion with speed and precision by PRAIA (Assafin 2006).

One aspect of the work was deriving FDPs for all CCDs in the WFI mosaic. This allowed for the use of a simple linear model to relate measurements and sky-plane projected catalog reference positions, thus granting higher star/parameter ratios and robust astrometric results. Let σ_e be the external standard deviations of $(\Delta x, \Delta y)$ offsets from the same bin, computed over distinct FDPs from different runs, and σ_i the internal standard deviations computed over bins within the same FDP. Figure 9 plots the count distribution of σ_e/σ_i ratios computed for FDPs derived from September and October 2007, as well as from other

Table 9. Sample from prediction tables for stellar occultations by Pluto.

Year m d	h m s	RA (ICRS) h m s	Dec ° ′ ″	C/A mas	P/A °	v km s^{-1}	D AU	$R*$	$J*$	$H*$	$K*$	λ °	LST h:m	Δe_α mas	Δe_δ mas	pm	ct	fg	E_α mas	E_δ mas	μ_α mas	μ_δ mas
2008 06 22	19 10 11	17 58 33.0155	−17 02 38.344	185	176.12	−23.80	30.47	12.9	11.7	11.6	11.4	071	23:53	+00.0	+74.5	ok	uc	0	40	14	−03	−04
2008 06 24	10 36 55	17 58 22.3931	−17 02 49.335	168	355.86	−23.74	30.47	16.3	12.8	11.9	11.7	197	23:46	+00.0	+74.6	ok	2m	0	24	21	+00	+01
2008 08 25	04 34 51	17 53 27.1046	−17 15 27.526	169	328.87	−08.39	31.06	14.9	11.1	10.1	09.8	226	19:38	+00.0	+79.8	ok	2m	0	58	12	−07	−12

Notes. Prediction tables list event date and instant (UTC), the ICRS (J2000) star coordinates at occultation, the closest apparent distance between star and body (C/A), the position angle (P/A) of the shadow across the Earth (counter-clockwise, zero at South), the velocity in km s^{-1}, the distance (D) to the Earth (AU), $R*$, $J*$, $H*$ and $K*$ star magnitudes normalized to a reference shadow velocity (v) of 20 km s^{-1}, longitude (λ) of the sub-solar point, local solar time (LST), (Δe_α, Δe_δ) ephemeris offset correction to the DE418 and plu017 ephemerides in (RA, Dec) for the central instant (see Sect. 6), catalog cross-identification (uc = UCAC2, 2m = 2MASS, fs = field star), proper motion existence and multiplicity flags (see Sect. 4.4), estimated star catalog position errors (E_α, E_δ) and proper motions (μ_α, μ_δ). Positive/negative v means prograde/retrograde velocities respectively, that is, Pluto′s geocentric right ascension is increasing/decreasing respectively. The complete table set of 2008−2015 predictions for Pluto, Charon, Hydra and Nix is available in electronic form at the CDS.

Table 10. Predictions for Pluto and its satellites for 2008–2015.

	2008	2009	2010	2011	2012	2013	2014	2015
Pluto	148	26	184	239	333	90	71	106
Charon	147	27	195	238	325	88	77	104
Hydra	146	24	185	257	335	88	59	121
Nix	156	22	202	276	345	101	78	101

Notes. Number of predicted events per year for each body.

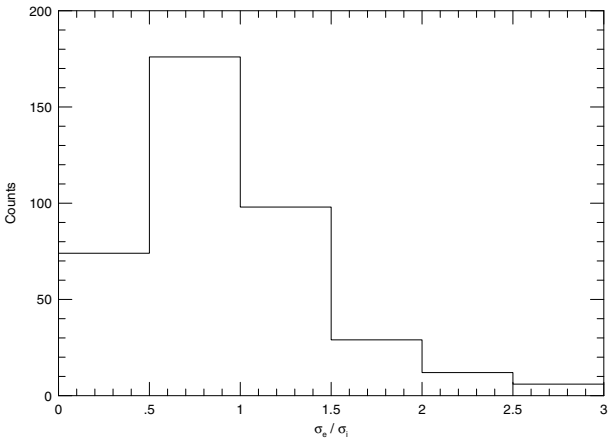


Fig. 9. Count distribution of σ_e/σ_e ratios. σ_e is the standard deviation of (Δx, Δy) offsets of the same bin computed over distinct FDPs from different runs. σ_i is computed over bins within the same FDP. As distribution peaks at 0.75, we conclude that FDPs are stable at least within 50 mas over distinct runs, independent of WFI maintenance.

telescope runs during 2007 and 2008, with detector maintenance in between. Because σ_i is of the order of UCAC2 position mean errors and because the distribution peaks at ratio = 0.75, we conclude that the derived FDPs are representative of the WFI distortions within at least about 50 mas for any run made at the ESO2p2/WFI – even after WFI maintenance. This means that the derived FDPs may be promptly used for WFI astrometry at the 50 mas level. The WFI FDP offsets obtained from the September and October 2007 runs are available by request to the author. But for better astrometric results like in this work dedicated FDP observations for each run are recommended, followed by the relatively simple astrometric procedures described in Sect. 4. In all, our method follows a different approach than that described in Anderson et al. (2006) and references therein for the astrometry of WFI mosaics.

Computation of new stellar proper motions using the 2MASS and USNO B1.0 as first epoch also enhanced the obtained catalog positions. For faint stars in particular, computing of proper motions instead of the direct use of USNO B1.0 own proper motions avoids the zero point issue warned by the authors (Monet et al. 2003). The computed proper motions not only improved the predictions of upcoming events, but also the astrometric prediction and follow-up feasibility of events more distant in the future.

In all, the obtained astrometric catalog represented an improvement over predictions based on GSC2.3, USNO B1.0 or UCAC2 positions. In comparison, stars fainter than about $R = 12$ were better imaged with the ESO2p2/WFI (see Figs. 3 and 5). Also, as the sky path was covered by overlapping 30′ size CCD mosaics, we have overcome the problem of position zero-point errors inherent to predictions based on single catalog positions or originated from CCD observations with small FOV.

Note that we have not applied any UCAC2 to ICRS corrections to the derived catalog of star positions. Contrary to the UCAC2-based star positions derived from the astrometric LNA follow-up program between 2005–2008, which were used to compute ephemeris offsets (see Sect. 6), here we have preserved the original star positions obtained in the catalog. We give freedom to the user to decide what corrections should be applied, if any. Once a correction is established, it can be applied to the positions in the star catalog or can alternatively directly enter as a shift in the occultation shadow paths predicted here. UCAC2 to ICRS corrections can be computed from the comparison of optical versus VLBI positions of selected ICRF quasars nearby the sky path of Pluto along the years. The corrections described in Sect. 6 are only valid for UCAC2-based star positions around about 10 degrees from 2006.5 Pluto′s coordinates, in which case they were $\Delta\alpha \cos\delta = -12 \pm 8$ mas and $\Delta\delta = -5 \pm 7$ mas. Until 2015, Pluto will have moved by more than 15 degrees in the sky, so that new ICRF quasars need to be selected and new corrections evaluated. This problem will be addressed in future releases of the produced star catalog, including the possible use of future improved versions of the UCAC catalog itself (UCAC3, etc.) as a reference frame in the astrometry of the WFI mosaics.

A number of stellar occultations between 2005–2008 were correctly predicted and successfully observed as predicted for stars between $13 < R < 16$ (see Table 7), based on CCD observations made at LNA telescopes in Brazil. From these past occultations successfully recorded and fitted, a linear drift with time in declination for Pluto′s ephemerides (DE418 and plu017) could be determined (see Table 8 and Fig. 6). This drift was taken

into account to correctly describe the sky path of Pluto and its satellites and was an important step in our star candidate search.

On the other hand, no ephemeris correction was applied for right ascension. Although an oscillation pattern related to an error in Pluto′s heliocentric distance (geocentric parallax error) cannot be ruled out, none of the attempted models for this scenario could fit the more dispersed right ascension ephemeris offsets derived from the studied occultations, at least not well below 50 mas (in the case of declination, a standard deviation of only 14.4 mas was achieved after the linear fitting). The phenomenon deserves further investigation, but from a pragmatic point of view it is of secondary importance to predictions, because right ascension ephemeris offsets do not affect the geographic latitude of the occultation shadow path over the Earth and will only cause a marginal error in the predicted central instant of the event by just a few minutes at most, and even so only when far from opposition. Note that if a geocentric parallax error is present, a cyclic declination drift should also be expected, but with amplitudes about five times smaller than those of right ascension in the present orbit configuration. If we take right ascension offset amplitudes of about 70 mas (extreme positive/negative values far from opposition in Table 8), declination amplitudes would be about 15 mas, which is on the order of star position errors. So, although we cannot rule out a small cyclic drift in declination, it could not be (and was not) seen in our data.

No threshold in R magnitude was established in the search for candidates. Pluto is crossing interestellar clouds, so relatively faint R objects may turn out to be bright infrared stars, perfect targets for the SOFIA observatory (Gehrz et al. 2009) and for ground-based instruments well equipped with H, J or K band detectors (H, J and K magnitudes are promptly available in the catalog if the star belongs to the 2MASS). Besides, events may be also favored by slow shadow speeds of less than 20 km s^{-1}. Also, no constraints on a geographic place were applied, as in principle SOFIA observations can be done from any sub-solar point on Earth. Even so, finding charts are also made available for events visible at regions well covered by instruments such as in North and South America, Europe, South Africa, Australia, Japan and Hawaii (see comment on web page reports below). Events in daylight at sub-planet point were not excluded either, as they could yet be observable in the dark, right above the horizon, from places near the Earth terminator.

All through the paper, we did not distinguish between past and future predictions, publishing all found occultations for the sky path covered (or to be covered) by Pluto between 2008–2015. The importance of predictions for occultations still to come is obvious. But the predictions of past occultations are also useful for at least three reasons. First, they can be used by anyone as reference for ongoing fittings of light curves of recent past observed events. Second, they serve to derive ephemeris drifts by comparing expected and observed central instants and C/A values. Finally, they can be used as an external check for the accuracy and precision of our WFI predictions.

In this way we compared star positions WFI-based (Table 9) and LNA-based (Table 7) for three past, common predictions for Pluto stellar occultations occurred in 2008. The star position differences - and thus, the predictions - agree very well within the expected WFI-based star catalog position error estimates. Table 11 displays this comparison.

Table 12 displays a direct comparison between prediction (WFI-based) and actually observed (fit to data) occultation central instants and C/A values for the same three occultations. Here, we note that this comparison results from a somewhat

Table 11. Comparison between WFI-based and LNA-based old predictions: star positions.

	LNA – WFI		WFI stars		
	$\Delta\alpha\cos\delta$	$\Delta\delta$	E_α	E_δ	Mag.
Occultation	mas	mas	mas	mas	R
2008 June 22	−11	+03	40	15	12.9
2008 June 24	+14	−11	24	21	16.3
2008 August 25	−08	−15	58	13	14.9

Notes. Comparison between star positions from WFI-based (Table 9) and LNA-based (Table 7) past predictions for Pluto stellar occultations occurred in 2008. The position differences agree very well within the expected WFI-based star catalog position error estimates.

circular reasoning, as the predictions are based on a linear extrapolation of Pluto′s declination offset with time, which in turn included the occultation data of 22, 24 June and 25 August 2008. The differences in C/A shown in Table 12 are consistent with the standard deviation of the linear fit to Pluto′s declination offset, 14.4 mas. This is expected if our model for Pluto′s ephemeris drift is correct, i.e. a smooth linear trend with time. This 14.4 mas standard deviation should then reflect the typical accuracy of the star positions. Indeed, the announced errors on the 22, 24 June and 25 August 2008 stars (Table 7) are consistent with that value. The same comments apply for Table 8.

In a general sense, assuming a bulk error of 30 mas for C/A from the estimated errors of the WFI catalog star positions and from the errors of the derived ephemeris offsets, we can state that the shadow path uncertainties over Earth are on the order of less than 800 km for the WFI occultation predictions.

Concerning Nix and Hydra stellar occultation predictions, our paper shows that accuracies of 15–20 mas (approx. 300–450 km) can eventually be reached for the star position. But their diameters are approx. 70–100 km only. So, typical probabilities of success of about 15–30% at best can be expected, which is quite good for these small far away bodies. The best reference concerning Pluto's satellites ephemeris accuracy is probably Tholen et al. (2008) (hereafter T08). Table 4 of T08 gives the uncertainties associated with each orbital element, and their Fig. 5 gives the sky-plane positional uncertainties vs. time for Nix and Hydra, from 1980 to 2020. For 2010–2015, they amount to about 10–15 mas or 220–330 km. This is comparable to or a bit smaller than the star position errors we can reach. Thus, this reduces the probability of success quoted before of 15–30%, to about 10–25%. This is not as high as hoped, but not despairingly small, especially if the event occurs above a dense, populated region in terms of astronomers, including amateurs.

For Charon, T08 gives an uncertainty of $e_P = 0.000007$ days for the orbital period $P = 6.4$ days around Pluto. If we propagate this error over the 2008–2015 time span, or approx. $t = 2500$ days, this yields an error in longitude of $e_L = e_P * t / P = 0.15$ degrees, or 50 km along orbital motion, which is then negligible. Thus, for Charon, the important factor in predictions is still the systematic errors (ephemeris offsets) in the orbital motion of Pluto around the Sun.

Continuous observation of Pluto and candidate stars are recommended, and this effort is now facilitated by the produced catalog. Astrometric follow-up is important in predictions due to the need for position refinements, and we hope that this task has been made easier now with the availability of the generated star catalog. Even astrometry with the use of modest FOV observations becomes feasible, as the zero-point error of our catalog is rather small and its magnitude completeness is about $R = 18$–19. Once new occultations are successfully recorded and analyzed, one can further improve the accuracy of Pluto′s ephemeris

Table 12. Comparison between WFI-based and actual observed occultations: central instants and C/As.

	Predictions (Table 9)		Actually observed (fit to data)		Prediction minus observed	
	Central instant	C/A	Central instant	C/A	Central instant	C/A
Occultation	h m s	mas	h m s	mas	s	mas
2008 June 22	19 10 11	185	19 10 58	171	−47	+14
2008 June 24	10 36 55	168	10 37 50	166	−55	+02
2008 August 25	04 34 51	169	04 35 01	154	−10	+15

Notes. Comparison between central instants and C/As from WFI-based (Table 9) and actual observations (fit to data) for the same three past stellar occultations referred to in Table 11.

offsets, allowing for a continuous fine tunning in the predictions. Besides, the photometric information contained in the catalog may be also useful in the observational preparation for the occultation itself.

We remark that updates on the ephemeris offsets or on candidate star positions can be easily taken into account for upgrading the geometric conditions of the predicted events. Updated reports and finding charts are made available in a continuous basis by the group at http://www.lesia.obspm.fr/perso/bruno-sicardy/.

We also emphasize the importance of predictions for stellar occultation by TNOs in general. Astrometry of candidate stars and determination of ephemeris offsets are urgently needed for these objects. Efforts are made right now in this direction by a similar observational program carried out by our group at the ESO2p2/WFI. Among other similar projects conducted by other active groups in the U.S.A., this effort merges with a long term international campaign coordinated by the Observatoire de Paris for this purpose.

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Apêndice D

Assafin et al. 2012

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Candidate stellar occultations by large trans-Neptunian objects up to 2015⋆,⋆⋆,⋆⋆⋆

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ABSTRACT

Context. We study large trans-Neptunian objects (TNOs) using stellar occultations.
Aims. We derive precise astrometric predictions for stellar occultations by Eris, Haumea, Ixion, Makemake, Orcus, Quaoar, Sedna, Varuna, 2002 TX_{300}, and 2003 AZ_{84} for 2011–2015. We construct local astrometric catalogs of stars complete to magnitudes as faint as $R = 18-19$ in the UCAC2 (Second US Naval Observatory CCD Astrograph Catalog) frame covering the sky path of these objects.
Methods. During 2007–2009, we carried out an observational program at the ESO2p2/WFI (2.2 m Max-Planck ESO telescope with the Wide Field Imager) instrument. The observations covered the sky path of the selected targets from 2008 to 2015. We performed the astrometry of 316 GB images using the Platform for Reduction of Astronomical Images Automatically (PRAIA). With the help of field distortion patterns derived for the WFI mosaic of CCDs, we reduced the overlapping mosaics of CCDs.
Results. We derive positions in the UCAC2 frame with 40 mas precision for stars up to the catalog magnitude completeness limit (about $R = 19$). New stellar proper motions are also determined with 2MASS (Two Micron All Sky Survey) and the USNO B1.0 (United States Naval Observatory B 1.0) catalog positions as a first epoch. Astrometric catalogs with proper motions were produced for each TNO, containing more than 5.35 million stars covering the sky paths with 30′ width in declination. The magnitude completeness is about $R = 19$ with a limit of about $R = 21$. We predicted 2717 stellar occultation candidates for all targets. Ephemeris offsets with about from 50 mas to 100 mas precision were applied to each TNO orbit to improve the predictions. They were obtained during 2007–2010 from a parallel observational campaign carried out with telescopes from 0.6 m to 2.2 m in size.
Conclusions. This extends our previous work for the Pluto system to large TNOs, using the same observational and astrometric procedures. The obtained astrometric catalogs are useful for follow-up programs at small to large telescopes used to improve the candidate star positions and TNO ephemeris. They also furnish valuable photometric information for the field stars. For each TNO, updates on the ephemeris offsets and candidate star positions (geometric conditions of predictions and finding charts) are made available by the group at http://www.lesia.obspm.fr/perso/bruno-sicardy/.

Key words. astrometry – occultations – Kuiper belt: general

1. Introduction

The trans-Neptunian objects (TNOs) are located beyond Neptune in a region where only small body differentiation is expected with regard to temperature changes. This makes them 4.5 billion year old interplanetary fossils of the early stages of the formation of the outer solar system. Recent models of planetary migration such as the Nice model (Levison & Morbidelli 2003; Gomes et al. 2004, 2005) also indicate that TNOs are a sensitive laboratory for the study of orbital dynamics. Refining the ephemeris and establishing the fundamental properties of TNOs are thus essential for our understanding of the origin and evolution of the solar system. This in turn contributes to the understanding of exoplanetary systems and helps astrochemical and astrobiological studies (Barucci 2008).

⋆ Tables of predictions for stellar occultations by Eris, Haumea, Ixion, Makemake, Orcus, Quaoar, Sedna, Varuna, 2002 TX_{300}, and 2003 AZ_{84} for 2011–2015 and respective Catalog of star positions for 2008–2015 sky path are only available at the CDS via anonymous ftp to cdsarc.u-strasbg.fr (130.79.128.5) or via http://cdsarc.u-strasbg.fr/viz-bin/qcat?J/A+A/541/A142

⋆⋆ Observations made through the ESO run 079.A-9202(A), 075.C-0154, 077.C-0283 and 079.C-0345.

⋆⋆⋆ Also based on observations made at Laboratório Nacional de Astrofísica (LNA), Itajubá-MG, Brazil.

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The analysis of light curves obtained for a stellar occultation observed from many sites allows the shapes and sizes of TNOs to be measured with a precision of a few kilometers. We can probe their atmospheres or place limits on their existence down to the nanobar level. In addition, occultations can eventually lead to the observation or discovery of close binary companions, satellites, or debris around the central body. For size determinations, occultations are far more suitable than indirect estimations such as those coming from albedo assumptions or from the modelling of optical, infrared (IR), and sub-millimetric observations (Cruikshank et al. 2005; Altenhoff et al. 2004; Stansberry et al. 2008). From these direct size measurements, one can derive more accurate albedos and place tighter constraints on the surface composition of the TNOs. If the mass of the body can be estimated from the orbits of detected satellites or by other indirect means, the density can be estimated more accurately and thus the internal composition and structure of the body can be inferred far more reliably. The detection of atmospheres around TNOs will help to improve our current understanding of their dynamics and relationship with the surface. Detection of Chiron-like jets (Elliot et al. 1995) is also a possibility. A more robust characterization of companions/satellite orbits and rotational periods from stellar occultations and ground-based adaptive optics observations could also improve our models for binary formation and collision (Goldreich et al. 2002; Weidenschilling 2002; Sicardy et al. 2011a).

It is more advisable to select large TNOs as targets for stellar occultation campaigns. They usually have large apparent sizes (30 mas or more) and thus have a greater chance of a positive detection. In addition, theory indicates that larger TNOs are most likely to possess atmospheres (Elliot & Kern 2003). In the case of atmospheres, even negative occultation chords observed nearby locations with positive records are useful, giving valuable upper limits to the atmosphere size and profile. Larger bodies also have smaller relative size errors. The accumulation of positive and negative detections also result in significant improvements in the TNO ephemeris, thus refining the orbit and substantially increasing the accuracy of the subsequent occultation predictions. Improving the chances of success minimizes the money and time spent in the complex international campaigns, which usually involves large telescopes in well-equipped observatories, smaller mobile instruments, and specialized personnel.

Even as one of the most important and larger members of the TNO population, Pluto has only been probed in terms of occultations, prior to 2000, almost exclusively by the 1988 mutual events with its larger satellite Charon. The first observed stellar occultation by Pluto took place only in 1985, when its atmosphere was first discovered (Brosch 1995). The atmosphere could only be more extensively observed in the next occultation in 1988 (Millis et al. 1993). In 2002, a new stellar occultation by Pluto was recorded, but with higher time resolution (Sicardy et al. 2003; Elliot et al. 2003), revealing that there had been a dramatic expansion in the atmosphere pressure. Another high signal-to-noise ratio (S/N) light curve was obtained in the June 2006 stellar occultation (Young et al. 2008), which uncovered evidence of the stabilization of the atmosphere pressure in Pluto for the period 2002–2008. This result is supported by seven other unpublished occultations observed by our group members in Australia, Brazil, Namibia, and Chile with 0.5 m to 4 m size telescopes in 2006–2010 (Sicardy et al., in prep.). Regarding Charon, after its first recorded stellar occultation (Walker 1980), four others have so far been observed, one in 11 July 2005 (Sicardy et al. 2006; Gulbis et al. 2006), one in 22 June 2008 (Sicardy et al. 2011a), and two other unpublished multi-chord events observed in Brazil and Chile (4 June 2011) and in Hawaii (23 June 2011). These last two events, as in 22 June 2008, were double events in which Charon and Pluto occulted the same star. In the 4 June 2011 event in particular, both occultations could be recorded at each site.

The population of TNOs in general, have been far more poorly studied. Up to the time of writing, only seven stellar occultations have been successfully recorded, all with CCD detectors. A double-chord occultation was observed in Hawaii in October 2009 and involved 2002 TX_{300} and a $R = 13.4$ UCAC2 star (Elliot et al. 2009). This was the first time ever that a stellar occultation by a TNO other than Pluto or Charon was recorded. In February 2010, Varuna was observed to occult a $V = 11.1$ star from the north-east of Brazil. One chord was obtained. A solution for the size, albedo, and shape was derived after combining the positive with a near negative chord and astrometric and photometric ground-based data (Ortiz et al., in prep.). Our group subsequently observed a multi-chord occultation of a $R = 16.9$ star by Eris in November 2010 (Maury et al. 2010; Sicardy et al. 2011b), a one-chord occultation of a $R = 18.2$ star by 2003 AZ_{84} in January 2011 (Braga-Ribas et al. 2011a and in prep.), a multi-chord occultation of a $R = 18.4$ star by Makemake in April 2011 (Ortiz et al., in prep.) and a multi-chord occultation of a $R = 15.7$ star by Quaoar in May 2011 (Braga-Ribas et al., in prep.). The MIT group also reported at the 218th meeting of the AAS in 2011 the observation of a one-chord occultation by Quaoar in February of 2011 (Person et al. 2011; Sallum et al. 2011). All seven occultations were predicted in this work. Our group observed them all (except 2002 TX_{300} and the February occultation by Quaoar) with 40 cm to 60 cm size telescopes and also with the NTT and VLT in Chile, Brazil, and Uruguay.

The main difficulty in deriving reliable predictions for the stellar occultation of TNOs is their small apparent diameters, even in the case of larger TNOs. Another cause is the lack of accurate orbital elements, resulting in ephemeris errors as large as a few hundreds of milli-arcseconds (mas). Predictions based solely on published catalog positions such as the USNO B1.0 (Monet et al. 2003) or the 2MASS (Cutri et al. 2003) usually fail because of the poor precision (or lack) of proper motions and the relatively large zonal and/or random errors in their positions (100–200 mas). Even individual positions for fainter stars in the UCAC2 catalog (Zacharias et al. 2004) may need corrections as large as 70 mas.

There are ways to overcome these ephemeris and star position problems. The ephemeris of TNOs must be offset so as to get realistic TNO positions at predicted occultation dates. The straightforward way to do this is to regularly observe TNOs and derive astrometrically the ephemeris offsets as a function of time. As successive positive occultations are collected, their ephemerides can be radically improved down to a few mas. Even a one-chord positive occultation helps to improve the ephemeris, since the apparent diameters of TNOs are smaller than 30 mas. In the case of star positions, one strategy is to select possible occultations based on the positions given in any arbitrary astrometric catalog, then perform follow-up observations to improve the star position and (after applying offsets to the TNO ephemeris) delineate the shadow path.

However, this approach to updating star positions has an intrinsic disadvantage in the case of TNOs. One usually wishes to start searching for occultations from an astrometrically sound catalog in order to select only those truly probable events. The shorter the list of candidates, the greater the chance of

conducting the follow-up observational programme on dedicated telescopes in due course. Since the highest quiality astrometric catalogs, such as the UCAC2 for example, are more complete for the brighter magnitudes, then stars fainter than say $R = 15-16$ will always be left out of the candidate search. This is far from ideal in the case of TNOs. In contrast to Pluto occultations, for TNOs we obtain good contrast in the light curves using very modest instruments for stars as faint as $R = 18.0$, as verified in the recent successfully recorded TNO occultations reported here. This is due to the TNO magnitudes being much fainter than the star brightness and because of the recent developments in CCD detectors. Choosing the USNO B1.0 or other similar catalogs does not help. The simple addition of fainter stars that do not have the minimum required catalog position precision introduces the problem of enlarging the follow-up list with low quality targets. This wastes valuable telescope time in follow-up programs and may lead to the running out of nights to complete the scheduled objects.

A more suitable strategy for TNOs is to derive local astrometric star catalogs with sufficient position precision, of say 50 mas at least, for the time span of the occultations, for stars in the magnitude range $R = 13-19$. In this way, we match the required position precision in the search, preserving faint stars without discarding bright objects. The addition of astrometrically trusted faint stars to the follow-up list, avoiding poor quality targets in turn improves the chances of finding more suitable candidates for TNO occultations, owing to the increase in the star density in the sky path.

The first noteworthy consistent studies of Pluto stellar occultation predictions were those of Mink & Klemola (1985) for the period 1985–1990 and of McDonald & Elliot (2000a,b) for the years 1999–2009. Two common drawbacks of these works were the astrometric precision of only about 0.″2 and the lack of stellar proper motions leading to uncertainties of the order of the Earth radius in the predicted shadow paths. In addition, these earlier predictions were degraded by the poorer precision of the older Pluto ephemerides.

To overcome the problems faced by these first prediction efforts, we used more modern facilities, namely the most up-to-date CCD detectors, reference catalogs, and astrometric procedures, and carried out an observational program at the ESO2p2/WFI instrument. We derived local star catalogs with precise positions for determining accurate predictions of stellar occultations by Pluto and its satellites Charon, Nix, and Hydra for the period 2008–2015. This work was published in Assafin et al. (2010) and is hereafter referred to as P1. Here, we extend that work to the predict the stellar occultations of the large TNOs Eris, Haumea, Ixion, Makemake, Orcus, Quaoar, Sedna, Varuna, 2002 TX_{300}, and 2003 AZ_{84} for 2008–2015, using the same observational and astrometric procedures. The local astrometric catalogs obtained are complete down to about $R = 19$, have positional errors of 40 mas or smaller, have stars with computed proper motions, and are in the UCAC2 frame. The catalogs cover the TNO sky paths with a 30′ width in declination. They can be very useful for the astrometry of small CCD fields for the follow-up of selected candidate stars, and for refining the ephemeris offsets of TNOs for tracking the shadow path across the Earth. The catalogs are also suitable for deriving the photometric properties of calibration and occultation stars.

In Sect. 2, we describe the ESO2p2/WFI program and observations. The astrometric treatment is given in Sect. 3. In Sect. 4, we present the derived catalogs of star positions along the TNO sky paths. In Sect. 5, we describe the determination of ephemeris offsets and the candidate star search procedure. The predictions of the stellar occultations by the large TNOs are finally presented in Sect. 6. An overview is presented in Sect. 7, including discussions about the external comparison with actual occultation results and error budget estimates, and comments about astrometric follow-up programs for TNO ephemeris improvements.

2. The ESO2p2/WFI TNO program and observations

The TNOs of the ESO2p2/WFI program were selected according to their body size, which is usually related to their apparent sizes on the sky. Given the astrometric uncertainties in the star positions and the error in the TNO ephemeris, larger apparent diameters correspond to a higher probability of a successful prediction. Another selection criteria was the star field. We preferred to select TNOs currently orbiting dense star regions close to the Galactic plane at Galactic longitudes of interest. The TNOs of intrinsically high science value were also selected, even if the star density along their sky path was low. In these cases, the advantage of using the large FOV of the WFI was even clearer.

Observations were made at the ESO 2p2 telescope (IAU code 809) using the Wide Field Imager (WFI) CCD mosaic detector. We used the broad-band R filter ESO#844 with $\lambda_c = 651.725$ nm and $\triangle\lambda = 162.184$ nm (full width at half maximum). The exposure time was 30^s. In general, S/N of about 200 were reached for objects of $R = 17$ without saturating bright ($R = 13-15$) stars. The limiting magnitude was about $R = 21$, and the completeness level about $R = 19.0$. The seeing varied between 0.″6 and 1.″5, being typically 1″. The telescope was shifted between exposures in such a way that each star in the path of the TNO was observed at least twice in different CCDs.

The program was completed in seven runs, in April, June, September, and October 2007, May/June 2008, and February and December 2009. Observations spanned each TNO sky path from 2008 to 2015; in six cases, this path started in 2009, since their WFI observations only took place in 2009 and telescope time was limited. Table 1 shows the approximate limits of the sky paths observed for each TNO, hence indicates the period covered. Figure 1 illustrates the sky path covered by Varuna for 2009–2015. A total of 1167 WFI mosaics or 9336 individual CCD frames were acquired for science, resulting in about 316 GB of photometrically calibrated processed images. The two WFI field distortion patterns derived in P1 were used here. The pattern associated with the September 2007 run was used in all earlier runs; the October 2007 pattern was used in all subsequent runs. The used instruments and observation procedures are described in detail in Sect. 3 of P1.

3. Astrometry

We briefly describe the reduction procedures. All CCD images were corrected by overscan, zeromean, flatfield, and bad pixels with IRAF (Tody 1993) using the esowfi (Jones & Valdes 2000) and mscred (Valdes 1998) packages. Using the PRAIA package (Assafin 2006), the astrometric treatment for each TNO consisted of three distinct steps.

An assigned field distortion pattern was first applied to each CCD in the WFI mosaics for each run (see details in Sect. 4.1 of P1).

In the second step, astrometry was performed over the individual CCDs, where the (x, y) measurements were corrected by

Table 1. Approximate (α, δ) limits and sky path periods covered by ESO2p2/WFI mosaics for each TNO.

TNO	(α, δ) (ICRF)		(α, δ) (ICRF)		Time span
	h m	° ′	h m	° ′	yr
Eris	01 38	−04 48	01 42	−02 58	2008–2015
Haumea	13 40	+18 57	14 01	+17 32	2009–2015
Ixion	16 55	−23 58	17 28	−27 15	2009–2015
Makemake	12 30	+28 25	12 52	+25 37	2009–2015
Orcus	09 40	−05 09	10 04	−08 42	2008–2015
Quaoar	17 10	−15 25	17 48	−15 34	2008–2015
Sedna	03 27	+06 28	03 41	+07 17	2009–2015
Varuna	07 30	+25 45	08 03	+26 35	2009–2015
2002 TX_{300}	00 32	+26 59	01 07	+32 43	2008–2015
2003 AZ_{84}	07 36	+12 17	08 02	+09 58	2009–2015

Notes. Approximate geocentric (α, δ) limits and corresponding sky paths periods covered for each TNO. Paths 2008–2010 were observed in the September and October 2007 runs, and 2009–2015 ones were observed in the 2009 runs at ESO2p2. Figure 1 illustrates the actual sky path for Varuna.

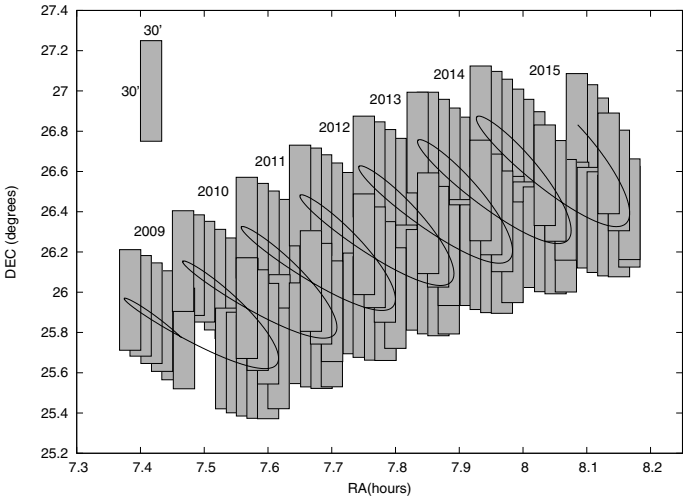


Fig. 1. Sky region covered by the ESO2p2/WFI CCD mosaic observations for Varuna. Years 2009–2015 follow from bottom to top. The continuous line is the geocentric sky path of Varuna. Each dashed form represents the $30' \times 30'$ area covered by one single WFI mosaic.

Table 2. Astrometry of individual CCD frames of WFI mosaics.

TNOs	Mean errors $\sigma(\Delta\alpha \cos\delta)$ mas	$\sigma(\Delta\delta)$ mas	No. of frames	No. ref. stars/frame
Eris	97	95	576	27
Haumea	51	50	784	8
Ixion	54	53	976	108
Makemake	48	51	800	7
Orcus	58	55	1528	14
Quaoar	57	57	904	121
Sedna	54	57	576	11
Varuna	49	53	848	31
2002 TX_{300}	47	50	1184	20
2003 AZ_{84}	50	50	1160	48

Notes. Mean errors are the standard deviations in the (O−C) residuals from individual (α, δ) solutions with the UCAC2 catalog, except for Eris in which case the 2MASS catalogue was used as the reference frame (see text for details). The six constant polynomial model was used to relate the (x, y) measurements to the (X, Y) tangent plane coordinates. The (x, y) centers were pre-corrected by the field distortion patterns (see details of these pre-corrections in Sect. 4.1 of P1).

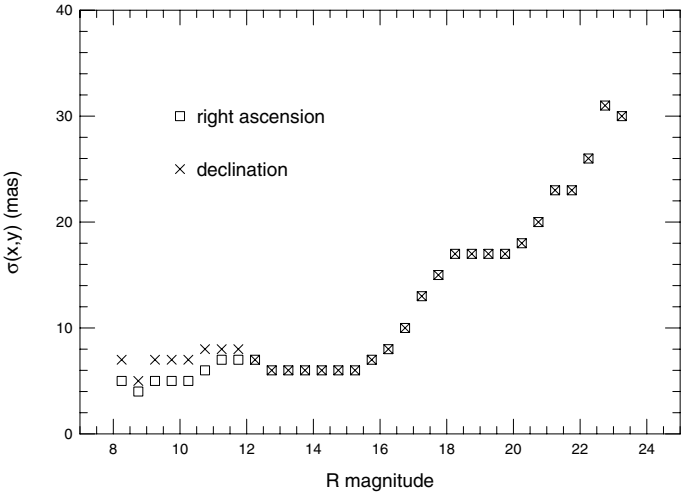


Fig. 2. (x, y) measurement errors as a function of R magnitude from all treated CCDs. Values are averages in 0.5 mag bins.

the pre-determined field distortions. Magnitudes were obtained from PSF photometry and calibrated with respect to the UCAC2. The (x, y) measurements were performed with two-dimensional circularly symmetric Gaussian fits. The estimated (x, y) measurement errors from Gaussian fits were about 5–10 mas for $R < 16$, then increased to 18 mas for $R < 18$, to 20 mas for $18 < R < 20$, and to 30 mas at fainter magnitudes. Figure 2 shows the distribution of (x, y) errors as a function of R magnitude for all measurements of all TNOs. The UCAC2 was the prime reference catalog, except for Eris. In this case, the 2MASS catalogue was used as the reference frame owing to the insufficient number of UCAC2 stars in the FOV. This is not an important change, as in the last step, all frames were consistently corrected to the UCAC2 frame. In steps 1 and 2, we used the six constant polynomial model to relate the (x, y) measurements to the (X, Y) tangent plane coordinates. The position mean errors in the (α, δ) individual CCD solutions are listed in Table 2; the average number of reference stars and number of CCD frames are also given for each TNO set of observations. Detailed information on the procedures in this step are found in Sect. 4.2 of P1.

In the final step, all positions coming from the different CCDs and mosaics were combined to produce a global solution for each sky path year, and final (α, δ) star positions were obtained in the UCAC2 system. In this step, all positions were projected onto the tangent plane and a complete polynomial model of the third degree was used to fix all the coordinates in the UCAC2 frame. Table 3 lists the mean errors of UCAC2 stars before and after the WFI mosaic global solutions. The listed values are averages taken over all sky path years. In addition to eliminating zero-point errors present in the positions in the individual CCD frames, the overall standard deviations with respect to the UCAC2 catalogue were generally smaller, as can be seen in Table 3. Detailed information about the global reduction procedures are given in Sect. 4.3 of P1.

In this final step, multiple entries coming from distinct CCD frames for the same objects are resolved (see Sect. 4.4 of P1 for details). Table 4 displays the statistics for resolved multiple entries. The listed values are taken over the global solutions

Table 3. Global astrometric solution for WFI CCD mosaics.

TNOs	(α,δ)–UCAC2 Before G.S. σ_α mas	σ_δ mas	After G.S. σ_α mas	σ_δ mas	UCAC2 Star No.
Eris	62	63	49	54	232
Haumea	45	44	46	46	617
Ixion	66	65	53	52	6229
Makemake	41	43	46	49	523
Orcus	80	73	56	54	842
Quaoar	96	91	56	55	5996
Sedna	50	53	53	53	396
Varuna	53	57	49	52	1793
2002 TX_{300}	58	64	46	50	1617
2003 AZ_{84}	56	56	49	49	3706

Notes. The $(\sigma_\alpha, \sigma_\delta)$ refer to the observed minus UCAC2 positions computed before and after the global solution (G.S.) procedure. Last column is the number of used UCAC2 reference stars. All values are averages over the results for each sky path year. For more details about the mosaic global solution, see Sect. 4.3 of P1.

Table 4. Multiplicity flags for WFI global mosaic positions.

TNOs	f0 %	f1 %	f2 %	f3 %	f4 %	f5 %
Eris	86.1	4.3	1.3	2.8	5.4	0.1
Haumea	93.1	0.9	0.4	0.7	4.8	0.1
Ixion	93.8	0.3	1.3	2.6	2.0	0.0
Makemake	90.3	1.9	0.6	1.2	5.9	0.1
Orcus	82.3	9.1	3.2	1.9	3.5	0.0
Quaoar	92.9	1.8	1.0	1.7	2.6	0.0
Sedna	92.3	3.0	0.9	1.0	2.8	0.0
Varuna	93.9	1.0	0.5	1.2	3.4	0.0
2002 TX_{300}	94.2	2.0	1.0	0.4	2.3	0.1
2003 AZ_{84}	95.6	0.3	0.2	1.1	2.7	0.1
All TNOs	93.5	0.8	1.1	2.2	2.3	0.1

Notes. Multiplicity flag statistics for each and all TNOs. Percentages refer to the total number of mosaic positions for each TNO and for all TNOs together (last line). No flag (good astrometry) means no multiple entries within 1.″5 for the same object after mosaic solutions. Flag cases f1 and f2 apply only to UCAC2 and 2MASS stars, when only entries assigned to a catalog were used. Flag f1 means that more than one entry was used (more than one position entry was averaged), whereas flag f2 indicates that only one entry was used. Flags f3, f4, and f5 apply only to field stars and mean that only one entry was selected according to one of three criteria, in order of priority: a) highest number of used common individual CCD positions (flag f3); b) least (x, y) measurement error (flag f4); c) brightest R magnitude (flag f5).

from all sky path years. Each final object position is assigned a flag that indicates its multiplicity status (see Table 4 for details). In all, more than 93% of the final positions come with no multiple entries (flag = 0).

Proper motions were extracted directly from the UCAC2 catalog. For the other stars, proper motions were computed for each star using the 2MASS and USNO B1.0 catalogs as a first epoch. Although these two catalogs may have random/systematic errors at the 100–200 mas level, they are still the most suitable choice for this purpose. The proportion of UCAC2, 2MASS-based, and USNOB1-based proper motions of stars are given in Table 5. Details about the computations can be found in Sect. 4.5 of P1. The highest percentage of stars without proper motions was verified for Ixion, namely 62%, which is twice the average fraction (about 30%). Ixion is now crossing the plane of the Milky Way

Table 5. Proper motion computations from 2MASS, USNO B1.0, and ESO2p2/WFI global mosaic star positions.

TNOs	UCAC2 p.m. %	2MASS p.m. %	USNO B1.0 p.m. %	No p.m. %
Eris	8.4	35.5	32.9	23.2
Haumea	6.3	24.2	41.6	27.9
Ixion	1.7	22.3	13.5	62.5
Makemake	6.4	25.2	38.8	29.6
Orcus	15.6	50.3	22.1	12.0
Quaoar	5.9	41.3	21.6	31.2
Sedna	12.2	49.0	22.8	16.0
Varuna	10.7	34.3	34.7	20.3
2002 TX_{300}	18.2	48.0	24.6	09.2
2003 AZ_{84}	10.9	27.8	34.5	26.8

Notes. UCAC2 stars with proper motions (directly extracted from UCAC2), stars with proper motions based on the 2MASS and USNO B1.0, and stars for which no proper motion could be computed. Percentages are relative to the total number of stars for each TNO.

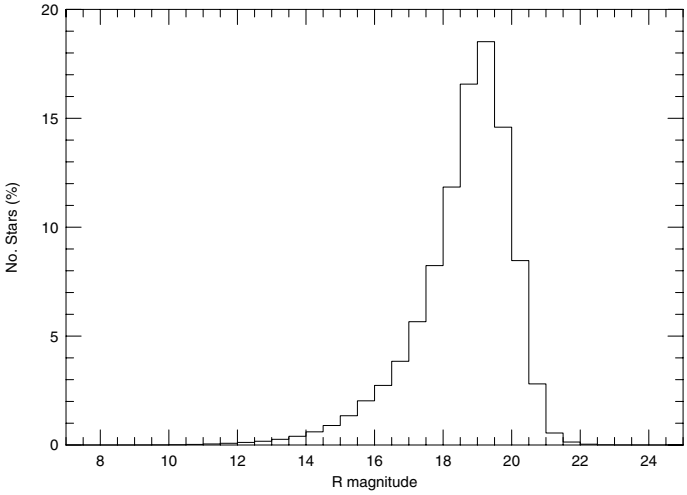


Fig. 3. Star distribution per R magnitude for all catalog stars. It illustrates the R magnitude completeness and limit of the catalogs. Percentages were computed over 0.5 mag bins for all stars in all catalogs.

(see Table 1). One possible explanation is that the WFI magnitude limit is fainter than that of the USNO B1.0 (see star distribution according to magnitude in Fig. 3 next in the text) and many faint stars are present in this region of the sky.

A complete description of the astrometric and photometric procedures used here can be found in Sect. 4 of P1, where data for the same WFI CCD mosaic underwent rigorously the same treatment.

4. The catalogs of star positions of each TNO for 2008–2015 sky paths

For each TNO, we produced star catalogs following their respective sky paths up to 2015. The paths begin in 2008 for Eris, Orcus, Quaoar, and 2002 TX_{300}, and in 2009 for the other TNOs. The star catalogs consist of the mean (α, δ) positions in the ICRS (J2000), proper motions, R magnitudes (also J, H, and K in the case of 2MASS stars), mean epoch of observations, position error at the mean epoch of observation, and the magnitude

Table 6. Star catalogs for each TNO sky path for 2008/2009–2015.

TNO	No. stars	σ_α mas	σ_δ mas	R magnitude completeness	R magnitude limit
Eris	25 685	33	37	19.0	21.0
Haumea	77 860	41	36	19.5	21.5
Ixion	3 452 835	41	29	19.5	21.5
Makemake	64 487	42	39	19.5	21.5
Orcus	56 042	40	39	18.5	20.5
Quaoar	1 150 375	41	37	18.5	20.5
Sedna	27 464	33	31	19.0	21.0
Varuna	140 041	38	39	19.0	21.0
2002 TX_{300}	82 746	33	37	18.0	20.0
2003 AZ_{84}	278 720	33	34	19.0	21.5

Notes. Number of catalog stars per TNO, position error for faint stars with R at completeness magnitude (errors are estimated from the standard deviation of contributing individual CCD positions about the final catalog star positions), R bandpass magnitude completeness, and magnitude limit.

error estimates. In all, 5 356 255 stars were catalogued in the UCAC2 frame. The mean epoch is approximately 2007.75 for the catalogs of TNOs beginning in 2008, and about 2009.5 for the others. Magnitude completeness is about $R = 19$. The magnitude limit is about $R = 21$. Position error is smaller than 40 mas for stars up to magnitude $R = 19$, and 30 mas up to $R = 17$. The catalogs are available in electronic form at the CDS.

The catalogs are divided in terms of year. Stars that had multiple entries within 1.″5 in the global mosaic solutions are flagged (see Table 4). The R magnitudes from PSF photometry were calibrated in the UCAC2 system, thus magnitude zero-point errors up to 0.3 are expected for $R > 17$. The position error is estimated from repeated observations, by evaluating the standard deviation (mean error) in the contributing individual CCD positions about the final catalog star positions (last step of global mosaic solution – see Sect. 3). By default, multiple-entry flagged stars have no position error estimates. Infrared magnitudes (and errors) were extracted from the 2MASS catalog. Error estimates for R magnitudes come from the standard deviation about the mean for individual CCD frames. The sky coverages of the catalogs have widths of at least 30′ in declination and lenghts of a few degrees along the geocentric sky path of each TNO. The proportion of UCAC2, 2MASS, and USNO B1.0 stars in the WFI catalogs may be verified from Table 5.

For each TNO, Table 6 lists the total number of catalog stars per year, the R bandpass magnitude completeness, magnitude limit, and average position errors for faint stars with R at the magnitude completness. Figure 3 shows the star distribution per R magnitude for all catalogs. Figure 4 plots the position error of all catalog stars as a function of R magnitude. Values were averaged over 0.5 mag bins.

5. Ephemeris offsets and search for candidate stellar occultations

To achieve the highest accuracy in predicting stellar occultations, it is necessary to pin down the actual paths of the TNOs in the sky. This can be done by correcting their orbits by offseting their ephemerides in both right ascension and declination. We derived average ephemeris offsets for all TNOs in the ESO2p2/WFI program. Table 7 displays the ephemeris offsets information for each TNO. The TNO observed positions used to derive the offsets were in the UCAC2 system. They came from a parallel astrometric program of CCD observations carried out during 2007–2010 at a variety of 0.6 m to 2.2 m telescopes in Chile (ESO2p2/WFI), Brazil (Laboratório Nacional de Astrofísica – LNA), and France (Pic du Midi; Observatoire Haute de Provence). The obtained offsets are provisional in the sense that the aforementioned program is still ongoing and further refinements and updates are expected. For three TNOs (Varuna, 2002 TX_{300}, 2003 AZ_{84}), we derived the offsets from fits to actual observed stellar occultations (see discussion in next Sect. 7). The offset error estimates come from the standard deviation about the mean of individual nightly averages. They do not depend upon the particular ephemeris errors themselves, which were published by the JPL.

Table 7. TNOs offsets with respect to JPL ephemerides.

	Observed – JPL ephemeris				No.	JPL
	$\Delta\alpha \cos\delta$ mas	$\Delta\delta$ mas	E_α mas	E_δ mas	nights	ephemeris version
Eris	−020	−068	084	055	09	28
Haumea	−090	−098	060	081	21	37
Ixion	+020	+191	092	093	15	09
Makemake	−039	−165	060	058	20	40
Orcus	−087	+001	059	046	19	13
Quaoar	−144	−078	098	043	22	17
Sedna	−043	−160	040	080	05	17
Varuna	−045	+005	015	015	–	21
2002 TX_{300}	−084	−053	040	040	–	14
2003 AZ_{84}	+232	+006	003	015	–	11

Notes. Average "observed minus JPL ephemeris" position offsets. Errors come from the standard deviation about the mean of individual nightly averages. TNO positions in the UCAC2 system come from a parallel astrometric program of CCD observations carried out during 2007–2010 at a variety of 0.6 m to 2.2 m telescopes in Chile, Brazil, and France. Each TNO ephemeris version is displayed. They were extracted using the JPL Horizons service (Giorgini et al. 1996). The listed offsets are provisional in the sense that the parallel program is still ongoing and further updates are expected. For three TNOs (Varuna, 2002 TX_{300}, 2003 AZ_{84}), offsets came from fits to actual observed stellar occultations (see discussion in Sect. 7).

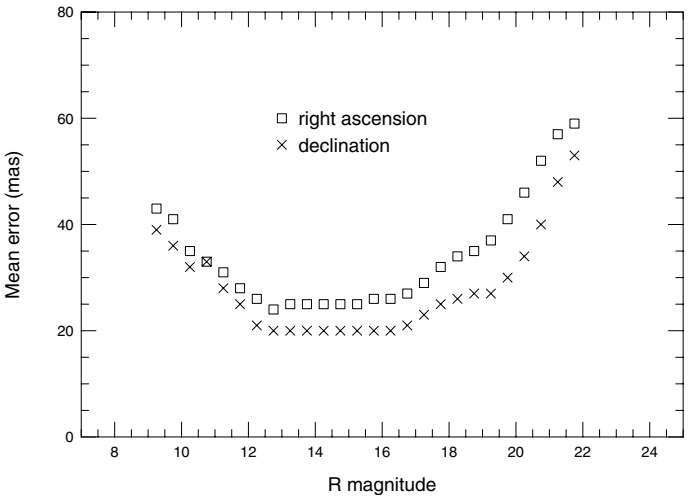


Fig. 4. Position mean errors as a function of R magnitude for all catalog stars. Position errors are estimated from the standard deviation in the contributing individual CCD positions about the final catalog star positions (last step of global mosaic solution – see Sect. 3). Values were computed over 0.5 mag bins.

Table 8. Sample from prediction tables for stellar occultations by large TNOs.

Year m d h m s	RA (ICRS) Dec	C/A P/A v D	R^* J^* H^* K^*	λ LST	Δe_α Δe_δ pm ct fg	E_α E_δ μ_α μ_δ
	h m s ° ′ ″	mas ° km s^{-1} AU		° h:m	mas mas	mas mas mas mas
2009 10 09 10 29 29	00 37 13.6205 +28 22 23.019	034 158.89 −25.29 40.61	13.5 12.6 12.2 12.2	194 23:24	−84.0 −53.0 ok uc 0	11 04 −16 −11
2010 02 19 23 07 40	07 29 22.4672 +26 07 23.191	031 188.28 −18.66 42.75	11.8 10.3 10.0 10.0	336 21:30	−45.0 +05.0 ok uc 0	40 70 −01 −12
2010 11 06 02 08 11	01 39 09.9421 −04 21 12.119	021 347.56 −26.32 95.74	16.8 14.3 13.7 13.5	308 22:38	−20.0 −68.0 ok 2m 0	06 19 +90 −12
2011 01 08 06 31 48	07 43 41.8220 +11 30 23.569	094 187.02 −26.00 44.32	18.5 50.0 50.0 50.0	271 00:33	+232.0 +06.0 ok fs 0	11 21 −03 −03
2011 04 23 01 37 59	12 36 11.3938 +28 11 10.493	174 183.80 −22.09 51.50	18.4 50.0 50.0 50.0	314 22:33	−39.0 −165.0 ok fs 0	18 19 −05 −10
2011 05 04 02 40 45	17 28 50.8021 −15 27 42.788	046 191.77 −18.28 42.35	15.7 13.5 12.8 12.7	000 02:42	−144.0 −78.0 ok uc 0	36 13 −04 −11

Notes. In entry order, the above samples correspond to the predictions for 2002 TX_{300} (October 2009), Varuna (February 2010), Eris (November 2010), 2003 AZ_{84} (January 2011), Makemake (April 2011), and Quaoar (May 2011), respectively. Prediction tables list event date and instant (UTC), the ICRS (J2000) star coordinates at occultation, the closest apparent distance between star and TNO (C/A), the position angle (P/A) of the shadow across the Earth (counter-clockwise, zero at north), the velocity in km s^{-1}, the distance (D) to the Earth (AU), R^*, J^*, H^*, and K^* star magnitudes normalized to a reference shadow velocity (v) of 20 km s^{-1} (50.0 means no magnitude available), longitude (λ) of the sub-solar point, local solar time (LST), ($\Delta e_\alpha, \Delta e_\delta$) JPL ephemeris offsets in (α, δ) for the central instant (see Sect. 5), catalog cross-identification (uc = UCAC2, 2m = 2MASS, fs = field star), proper motion existence and multiplicity flags (see Sect. 3), estimated star-catalog position errors (E_α, E_δ), and proper motions (μ_α, μ_δ). Positive/negative v means, respectively, prograde/retrograde velocities, that is TNO's geocentric right ascension is increasing/decreasing, respectively. The complete table sets of 2008/2009–2015 predictions for Eris, Haumea, Ixion, Makemake, Orcus, Quaoar, Sedna, Varuna, 2002 TX_{300}, and 2003 AZ_{84} are available in electronic form at the CDS. In the electronic version, in addition to the star positions, we provide the TNO estimated coordinates (with ephemeris offsets applied) at occultation instant.

Using the astrometric catalog of stars in the path of the sky through which the TNOs will pass up to 2015, we then searched for stars that could be occulted by the respective TNOs. To achieve this, the whole star catalogs were cross-correlated with the JPL ephemerides of the bodies generated each minute for the whole period between 2008/2009 to 2015. The JPL ephemerides were offset prior to the search using the quantities given in Table 7. If the apparent distance in the sky plane between the star and the body is less than an Earth apparent radius plus 50 mas of safe margin, then a potential occultation was identified and all astrometric and geometric data relevant to the possible event were computed and stored. In addition to astrometric and photometric data, for each candidate star, we computed and stored the minimum apparent geocentric distance d, the central instant of closest approach t_0, the shadow velocity v across the Earth, the position angle P/A of the shadow path, and the local solar time *LST* at the sub-planet point. These geometric quantities are defined and calculated as explained in Sect. 7 of P1.

6. Predictions of stellar occultations by ten large TNOs

Following the procedure described in Sect. 5, we identified potential candidate stars for occultations by Eris, Haumea, Ixion, Makemake, Orcus, Quaoar, Sedna, Varuna, 2002 TX_{300}, and 2003 AZ_{84}. The adopted search radius was 0.″335 – i.e. 285 mas (approximately the apparent Earth radius at 31 AU, i.e., at Pluto for 2008–2015) plus a safe margin of 50 mas. No predictions were discarded due to day light at the sub-planet point, as occultations could, even so, be visible right above the horizon from places still in the dark near the Earth terminator. For each TNO, all relevant astrometric, photometric, and geometric information for each potential event found is available in electronic form[1]. Table 8 lists a sample of predictions for 2002 TX_{300}, Varuna, Eris, 2003 AZ_{84}, Makemake, and Quaoar. These very predictions were used to prepare campaigns actually resulting in successful recordings of stellar occultations by our group. The exception was 2002 TX_{300}, for which no observations could be done at Hawaii (the predicted path), because all telescopes were occupied, must of them having been scheduled to observe the impact of NASA's Lunar Crater Observation and Sensing Satellite (LCROSS; Garner & Dunbar 2009), which occurred about one hour after the predicted central instant of occultation. Prediction tables contain the date and instant of stellar occultation (UTC), both the ICRS (J2000) star and TNO estimated coordinates at the event date, the closest apparent geocentric distance between the star and TNO, the position angle of the shadow across the Earth (clockwise, zero at North), the velocity in km s^{-1}, the distance to the Earth (AU), longitude of the sub-solar point, local solar time, applied JPL ephemerides offsets in (RA, Dec) for the central instant, catalog proper motion and multiplicity flags, estimated star catalog position errors, proper motions, and magnitudes R^*, J^*, H^*, and K^*. Magnitudes were normalized to a reference shadow velocity of 20 km s^{-1} by

$$M^* = M + 2.5 \log_{10}\left(\frac{v}{20 \text{ km s}^{-1}}\right).$$

The reference velocity of 20 km s^{-1} is typical of events around Pluto opposition. In slow events, M^* may become one to two magnitudes brighter than the actual star magnitude, indicating that the observation is feasible after all, because we can use longer integration times, and consequently reach reasonably high S/N, without any losses of spatial resolution in diameter measurements. For the same reasoning, we may also probe the atmosphere altitudes in the light curves, despite the faintnesses of the stars. As mentioned in the introduction, in the case of TNOs, occultations involving stars as faint as $R = 18.0$ have been successfully recorded with 50 cm telescopes.

Table 9 displays the total number of predicted events for each TNO for the period 2008/2009–2015.

7. Discussion

We have presented predictions for stellar occultations by the large TNOs Eris, Haumea, Ixion, Makemake, Orcus, Quaoar, Sedna, Varuna, 2002 TX_{300}, and 2003 AZ_{84} for 2008/2009–2015. We applied the same observational and astrometric procedures as a previous study dedicated to the Pluto system and published in Assafin et al. (2010). Many issues were addressed in that work, such as body ephemeris offset determination, catalog zero-point position errors and field-of-view size, long-term

[1] Anonymous ftp to `cdsarc.u-strasbg.fr`

Table 9. Predictions of stellar occultations by ten large TNOs for 2008–2015.

TNO	2008	2009	2010	2011	2012	2013	2014	2015
Eris	2	1	1	0	0	1	0	0
Haumea	x	3	4	6	4	7	7	6
Ixion	x	102	193	288	352	425	309	147
Makemake	x	4	3	5	3	1	4	0
Orcus	4	3	6	1	3	1	2	0
Quaoar	51	26	27	90	176	53	67	117
Sedna	x	1	0	2	1	2	3	3
Varuna	x	9	11	11	8	14	7	8
2002 TX_{300}	1	4	3	3	2	5	1	8
2003 AZ_{84}	x	18	18	19	16	9	15	11

Notes. Number of predicted events per year for each TNO.

Table 10. TNO diameters.

	TNO diameter		
	km	mas	Reference
Eris	2326 (+/−12)	33	Sicardy et al. (2011b)
Haumea	1324 (+/−167)	36	Lellouch et al. (2010)
Ixion	0650 (+250/−220)	22	Stansberry et al. (2008)
Makemake	1455 (+/−12)	38	Ortiz et al. (2011)
Orcus	0850 (+/−90)	25	Lim, et al. (2010)
Quaoar	1170 (+/−40)	34	Braga-Ribas et al. (2011b)
Sedna	1200–1600	22	Brown (2008)
Varuna	1003 (+/−09)	32	Sicardy et al. (2010)
2002 TX_{300}	0143 (+/−05)	05	Elliot et al. (2009)
2003 AZ_{84}	0910 (+/−60)	28	Mueller et al. (2010)

Notes. TNO diameters in km (uncertainties in parentheses) and as geocentric apparent sizes in mas for 2012.5. Values were obtained from previous stellar occultations and by modeling visible and IR observations (see references).

predictions and stellar proper motions, and faint-visual versus bright infrared stars, which also hold true here.

The local astrometric catalogs that we obtain are complete to about $R = 19$ and in the UCAC2 frame. Up to R magnitude completeness brightness, stars have positional errors of 40 mas or smaller and computed proper motions. The catalogs cover the TNO sky paths with 30′ width in declination and can be very useful in astrometric follow-up programs on telescopes/CCDs with FOVs of any size. The catalogs are also sources of photometric information for calibration and occultation stars. The astrometry was made by the use of the PRAIA package (Assafin 2006). About 316 GB of science-alone WFI images were treated, not to mention calibration observations to derive field distorion patterns, which improved the astrometric results of WFI data to the 40 mas level (see Sect. 9 in P1 for a detailed discussion about the field distortion pattern improvements for WFI astrometry).

As an important by-product, the high-resolution CCD frames obtained (1 pixel = 0.″238) form an image bank useful for the visual inspection of the fields of view. For instance, the analysis of small details in the calibration stars, close-by faint objects, and other elements in the images may prevent problems in the recording and photometry of occultation observations. When it comes to an event, charts extracted from this image bank are usually posted in our web page, but any particular field around some desired position can be obtained in FITS format on request to the authors. Public access to the original images from the observed WFI CCD mosaics will soon be available through the ESO Image Archive Portal.

No threshold in R magnitude was established in the search for occultations, as faint R objects may turn out to be bright IR stars, suitable for observation at adequately equipped IR ground-based facilities or by the SOFIA observatory (Gehrz et al. 2009); H, J, and K magnitudes are readily available in the catalog if the star belongs to the 2MASS. In addition, events that occur at slow shadow speeds of less than 20 km s^{-1} may become observable. No constraints on a geographic place were applied, as in principle SOFIA observations can be done from any sub-solar point on Earth. Events in daylight at sub-planet point were not excluded either, as they remained observable in the dark, right above the horizon, from places near the Earth terminator. Throughout this paper, we do not distinguish between past and future predictions, publishing all found occultations for the sky path covered (or to be covered) by the TNOs over the years 2008/2009−2015. We assume that the WFI predictions of past occultations may be useful as either reference for ongoing fittings of recently obtained light curves, an aid to deriving ephemeris offsets by comparing expected and observed central instants and C/A values, or an external check for the accuracy and precision of our WFI predictions.

The information in the prediction tables allows us to determine the central instant and shadow path over the Earth. For ground-based observations, the selection of events usually favors brighter stars and shadow paths that cross easily accessible places for the use of mobile telescopes. Faint stars may be selected in the case of observation at sites with larger observatories/telescopes. If the star is reddish, observations can be made in daylight with IR detectors. For TNOs with a low rate of occultations per year, such as Eris, opportunities are very rare, so broader nets of observers may be formed over larger areas, even if the event is not as promising in terms of star brightness or location. We leave the choice of potential events among the many predicted occultations listed in the tables to the judgement of the reader. Our group publishes at our web page potential occultation campaigns on a yearly basis.

The errors in our TNO sky path predictions are governed by the internal offset uncertainties listed in Table 7. These uncertainties are based on optical observations made for a period of some years. The budget ephemeris error estimates given by JPL – which increase with each passing year – only express the random error features of the JPL orbit solutions, rather than represent a possible increase in systematic trends. As time goes by and more TNO observations and occultations are accumulated, more accurate JPL ephemeris solutions are being published, although in the new versions no significant shifts in the TNO orbital sky paths are predicted for the next few years. For now, the adoption of a constant ephemeris offset in the predictions has the only side effect of absorbing some remaining systematic errors into the offset uncertainties. These uncertainties, on the other hand, do not increase with time for the period considered and are realistic estimates for the overall sky path prediction uncertainties. In the prediction tables (available in eletronic form), in addition to the star position, we indicate the estimated TNO position (with ephemeris offset applied) at the predicted occultation instant. This should help us to evaluate the events in the future as updates become available at the star position and on the TNO ephemeris by follow-up astrometric programs or by the JPL.

A tentative way to estimate the probability of success in recording an occultation of a particular TNO is to consider the ratio of the apparent TNO diameter in mas to its ephemeris offset errors displayed in Table 7. We list TNO diameter values in Table 10 to support reliability estimates for the predictions.

Table 11. Comparison between WFI-based and actual observed occultations: central instants and C/As.

TNO	Star mag (R)	Prediction minus Occultation Central instant		C/A	
		s	mas	mas	km
2002 TX_{300}	12.9	−15	−13	−09	−265
Varuna	11.0	−20	−12	−63	−1953
Eris	16.9	−604	−229	+39	+2708
2003 AZ_{84}	18.2	+0	+0	+0	+0
Makemake	18.4	+196	+116	+40	+1494
Quaoar	15.7	+130	+78	−44	−1351
average (moduli)		193	90	39	1554
average (Eris out)		90	55	39	1266

Notes. Comparison between central instants and C/As from WFI-based (Table 8) and actual observations (fit to data) for 6 (out of seven) successfully recorded TNO stellar occultations to date for which data is available. Using P/A, distance and the relative speed, central instant differences are expressed in seconds and in mas, and C/A differences in mas and in km. Averages for the modulus of the differences are computed with and without Eris. 2003 AZ_{84} is excluded from the analysis as the fit to the actual occultation already uses the WFI star position, and the current WFI predictions already uses the offsets derived from the occultation fitting.

They are given in km and as geocentric apparent sizes in mas for 2012.5. Values were obtained from previous stellar occultations and by modeling visible and IR observations (see references in Table 10).

The ESO2p2/WFI predictions can be directly compared to actually observed stellar occultations in order to estimate the external accuracy and precision of predictions. Table 11 displays a direct comparison between predictions (ESO2p2/WFI-based) and actually observed occultations, for which P/A, central instant and C/A values were obtained from fits to the light curves of the chords. We used six out of the seven TNO stellar occultations known to date, for which data is available (Elliot et al. 2009; Sicardy et al. 2011b; Braga-Ribas 2011, priv. comm.). Using P/As, distances and relative speeds, we computed both the predicted minus fitted central instant and C/A differences. Differences in central instants are expressed in seconds and in mas (in order to be comparable with C/A results), and C/A differences are given in both mas and kilometers. We give in Table 11 the averages for the modulus of the differences. We note that the WFI prediction for the Eris' central instant was in strong disagreement with observations. This may indicate that much more accurate right ascension ephemeris offsets should be obtained in our follow-up program for Eris. A separate set of averages is provided in Table 11 without taking Eris into account. The TNO 2003 AZ_{84} is excluded from the analysis as the fit to the actual occultation is based on the WFI star position, and the current WFI predictions already uses the offsets derived from the occultation fitting. In all, the differences in C/A and central instant are consistent with the expected errors in the WFI star positions as a function of magnitude and ephemeris offset error (see Table 7). The somewhat larger differences found in Table 11 for the central instant may be associated with ephemeris offset uncertainties in right ascension (the direction of the chords are usually east-west). One possible cause is geocentric parallax error (see discussion about the follow-up program and ephemeris offset determination in the next paragraphs). In all, for $R = 19$ stars (catalog magnitude completeness) and 40 mas errors in the WFI positions, we may assume a bulk error of about 80 mas for C/A, that is dominated by the ephemeris offsets errors of about 70 mas. For about 40 AU, this implies a shadow path uncertainty over the Earth of the order of 2300 km. If the ephemeris offsets can be well-determined to a more accurate precision than 30 mas, then a bulk error of 50 mas in C/A can be achieved, leading to a precision of about 1400 km for the WFI occultation path predictions, which is similar to the obtained values displayed in Table 11.

Long-term predictions are the only way of spotting potential stellar occultations in terms of favourable geographic access, observatories and instrumental facilities, etc. Once an event is chosen, then a complex international campaign is initiated, with astrometric follow-up of the TNO and star being mandatory. The only reliable long-term predictions of stellar occultations that use solely all-sky star catalogs are those based on the UCAC2. This limits the search for candidates to about $R = 16$ stars. The goal of our WFI program was to extend the UCAC2 position precision to stars as faint as about $R = 19$. The availability of data for such faint stars with about 40 mas position precision improves the chances of finding more suitable candidates for TNO occultations, owing to the increase in the number of suitable stars in the sky path and, consequently, the frequency of appulses. Owing to the recent developments of solid state detectors, and because TNOs are usually much fainter than the occulted stars, we can obtain good contrast in the light curves, even using very modest instruments. This has been verified for $R = 17-18$ stars in the occultations by Eris and 2003 AZ_{84} (Maury et al. 2010; Sicardy et al. 2011b; Braga-Ribas et al. 2011a and in prep.), observed with 40 cm and 60 cm telescopes in Chile.

The sky paths covered by the star catalogs were determined by the chosen time span of our predictions (2008–2015) in a compromise between the availability of WFI telescope time and the number of selected TNOs. The upper limit coincides with NASA's New Horizons spacecraft arrival at Pluto in 2015. Given this long time coverage of about eight years, the computation of proper motions for the stars was also of concern. That is why we used as first epoch data the best available astrometric resources at the time of completion of this work. In all, the comparison made with actual events (Table 11) and the successful WFI predictions specifically for the faint stars of Eris, 2003 AZ_{84} and Makemake occultations indicate that the WFI astrometric solution is sactisfatory in the context of long-term predictions.

Owing to the lack of observations of TNOs for ephemerides, to the few positions sometimes derived without the appropriate astrometric care, and to the relatively short orbital arc length coverage, the actual position of TNOs can differ significantly from their ephemerides. To refine the orbits of large TNOs in the short and long runs, by regularly obtaining precise astrometric positions, a parallel program involving many telescopes of 60 cm to 2 m class in Chile, France, and Brazil was started in 2007 (see further details in Sect. 5). Using the first results of this program for the predictions published in this work, we applied average fixed offsets to the current JPL ephemerides of the TNOs in the list (Table 7). In some cases, offsets reached almost 200 mas (about one Earth radius at 30 AU) or more.

Given the time span of 2008–2015, we expect that the fixed corrections applied to the JPL ephemerides suffice for pinpointing the most interesting occultations among all predictions. However, once an event is selected, the shadow path on Earth must be further refined. From the point of view of TNO orbits, improvements can still be made to the accuracy of the WFI predictions. For instance, by observing TNOs at their opposition and conjunctions to obtain the observed ephemeris offsets, one can model the geocentric parallax errors and correctly

extrapolate the observed ephemeris offsets with time for the occultation dates. For a body at 30 AU in the Ecliptic, these errors can amount to 100 mas in right ascension for an ephemeris error of some few tens of thousand kilometers in heliocentric distance (Camargo et al. 2010). As successful occultations are recorded, the ephemeris offsets are more accurately determined. For three TNOs (see Table 7), we could already improve the ephemeris offsets for this work using occultation results. As the new occultation results are consolidated, we will update the ephemeris offsets and release new, higher quality predictions.

An advancement that is beyond the scope of WFI predictions, for selected occultations, is an alternative procedure for improving the shadow path in the short term by direct fitting an orbital arc to the observed TNO positions. This can be done by numerically integrating orbits or, in principle, by using the current ephemeris as a template and then fitting it to the observations following a correction model. In the case of observations when both star and TNO are imaged in the same field of view, this procedure can yield very accurate predictions, as high precision relative positions can be attained for the TNO with respect to the occultation star, once proper care is taken to correct for differential color refraction.

The astrometric follow-up of the selected stars and TNOs on a regular basis is therefore very important, regardless of the approach taken to improve the accuracy of the predictions. In practice, however, the week or two before occultation date is still decisive in pinning down the shadow path on Earth. Updates on the TNO ephemeris offsets, star positions of selected occultations, updated reports, and finding charts are available on a continuous basis from our group[2]. Among other similar projects conducted by other active groups in the USA, our efforts reflect a long-term collaboration with the international campaigns coordinated by the Observatoire de Paris for this purpose.

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Apêndice E

Braga-Ribas et al. 2011

Electronic Telegram No. 2675

Central Bureau for Astronomical Telegrams
INTERNATIONAL ASTRONOMICAL UNION
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STELLAR OCCULTATION BY TRANSNEPTUNIAN OBJECT (208996) 2003 AZ84

F. Braga-Ribas, B. Sicardy, F. Colas, and J. Lecacheux, Observatoire de Paris; A. Maury, San Pedro de Atacama Celestial Explorations Observatory (SPACE); J. L. Ortiz, N. Morales, and I. de la Cueva, Instituto de Astrofisica de Andalucia (IAA), Granada; E. Jehin, J. Manfroid, and M. Gillon, Institut d'Astrophysique de l'Universite de Liege; M. Assafin, Observatorio do Valongo, UFRJ, Rio de Janeiro; and R. Vieira-Martins and J. I. B. Camargo, Observatorio Nacional, Rio de Janeiro, report on the positive detection of an occultation of a faint star (magnitude R about 18) by the transneptunian object (208996) 2003 AZ84 on Jan. 8d06h29m59s UT (mid-time), from the SPACE Observatory. The occultation was recorded by A. Maury with the C. Harlinten 0.5-m Planewave telescope there, and also by J. L. Ortiz with the remotely operated 0.4-m ASH2 telescope of the IAA in Spain. E. Jehin and J. Manfroid report that no drop of the flux was observed with the 0.6-m robotic TRAPPIST telescope of Liege University, which is located 610 km from San Pedro. Using a diffracting model, F. Braga-Ribas determined the times of immersion and emersion of the star. The occultation lasted for 21.7 +/- 0.8 seconds, corresponding to a chord length of 573 +/- 21 km. This result gives a lower limit to the diameter of the TNO, and can be compared to the estimated diameter from Stansberry et al. (2008, *Physical Properties of Kuiper Belt and Centaur Objects*, University of Arizona Press, pp. 161ff; 685.8 +/- 95.5 km, based on Spitzer data) and Mueller et al. (2010, A.Ap. 518, L146; 910 +/- 60 km, based on Herschel data).

NOTE: These 'Central Bureau Electronic Telegrams' are sometimes superseded by text appearing later in the printed IAU Circulars.

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Apêndice F

Sicardy et al. 2011

LETTER

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A Pluto-like radius and a high albedo for the dwarf planet Eris from an occultation

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The dwarf planet Eris is a trans-Neptunian object with an orbital eccentricity of 0.44, an inclination of 44 degrees and a surface composition very similar to that of Pluto[1]. It resides at present at 95.7 astronomical units (1 AU is the Earth-Sun distance) from Earth, near its aphelion and more than three times farther than Pluto. Owing to this great distance, measuring its size or detecting a putative atmosphere is difficult. Here we report the observation of a multi-chord stellar occultation by Eris on 6 November 2010 UT. The event is consistent with a spherical shape for Eris, with radius 1,163 ± 6 kilometres, density 2.52 ± 0.05 grams per cm^3 and a high visible geometric albedo, $p_V = 0.96^{+0.09}_{-0.04}$. No nitrogen, argon or methane atmospheres are detected with surface pressure larger than ~1 nanobar, about 10,000 times more tenuous than Pluto's present atmosphere[2–5]. As Pluto's radius is estimated[3–8] to be between 1,150 and 1,200 kilometres, Eris appears as a Pluto twin, with a bright surface possibly caused by a collapsed atmosphere, owing to its cold environment. We anticipate that this atmosphere may periodically sublimate as Eris approaches its perihelion, at 37.8 astronomical units from the Sun.

The dwarf planet (136199) Eris was discovered[9] in 2005. Its radius has been estimated to be 1,200 ± 100 km on the basis of direct imaging[10], although detection of its thermal flux provided another estimate[11] of 1,500 ± 200 km, potentially making it larger even than Pluto, and the largest known dwarf planet. The motion of Dysnomia (Eris' satellite) provides Eris' mass, $M_E = (1.66 \pm 0.02) \times 10^{22}$ kg, 27% larger than Pluto's mass[12]. No short-term (day-scale) brightness variability has been detected for Eris at the 1% level[13,14], suggesting either a spherical body with no albedo variegation, or—if elongated—a finely-tuned, pole-on viewing geometry. The spectrum of Eris is very similar to that of Pluto and reveals a methane-ice-rich cover, and another dominant ice, presumably nitrogen, but not excluding argon[1].

Stellar occultations by Eris are rare, as it subtends a minuscule angular diameter (~0.03 arcsec) while currently moving in severely depleted stellar fields at an angular rate of ~1.5 arcsec h^{-1} at most. Using the techniques described in ref. 15, we predicted one Eris occultation in 2010, on November 6 UT. We attempted observations from 26 stations, and the occultation was detected from two sites in Chile, with two detections at San Pedro de Atacama (San Pedro for short) with the Harlingten and ASH2 telescopes, 20 m from each other, and one detection at La Silla, with the TRAPPIST telescope (for details, see Fig. 1, Supplementary Figs 1 and 2, and Supplementary Tables 1 and 2). Another station further south at Complejo Astronómico El Leoncito (CASLEO), Argentina, provided a light curve without occultation, but went close to Eris' shadow edge (~200 km; see Fig. 2).

The San Pedro and La Silla observations provide two occultation segments—or 'chords'—whose four extremities are used to constrain Eris' size (red segments in Fig. 2). When deriving the occultation times, it appeared that two equally satisfactory solutions for the star reappearance time at the Harlingten telescope in San Pedro are possible, yielding two different chord lengths. These two solutions are separated by 1.2 s, and are respectively called solution 1 and solution 2, in chronological order. This ambiguity is due to the fact that the star reappearance occurred during a gap between consecutive exposures, corresponding to a net loss of information. The ASH2 data collected next to Harlingten did not provide enough signal-to-noise ratio to discriminate between these two solutions, and are not used in the fit described below (see Supplementary Information). As a dwarf planet, Eris is expected to be in hydrostatic equilibrium under gravity and centrifugal forces. The most general apparent limb shape is then an ellipse with semi-axes $a' > b'$ with effective radius $R_E = \sqrt{a'b'}$, defined as the radius of a disk that has the same apparent surface area as the actual body. This shape stems either from an oblate Maclaurin spheroid

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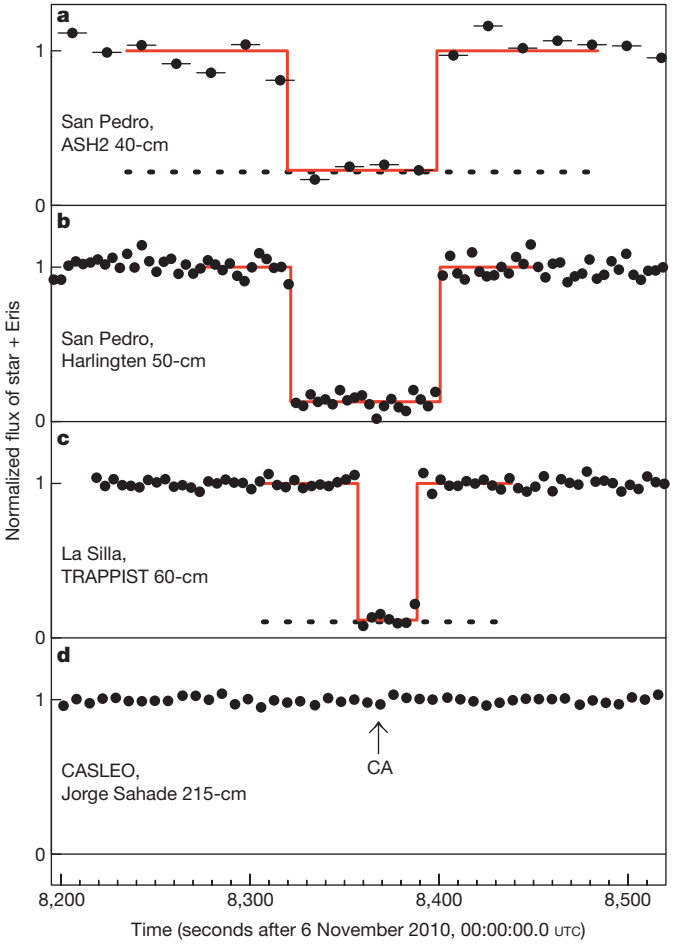


Figure 1 | Eris occultation light curves. The plots (black filled circles) show the flux of the star plus Eris, normalized to unity outside the occultation, versus time. No filter was used at any of the telescopes. **a**, The light curve from the ASH2 40-cm telescope at San Pedro, using a SBIG STL-11000M CCD camera, with 2×2 pixel binning and a sub-frame of 11.24×9.71 arcmin (272×235 pixels). The horizontal error bars indicate the total time intervals associated with each point (15 s, while the cycle time was 18.32 s). Those bars are too small to be visible on the other data sets. **b**, The light curve from the Harlingten 50-cm telescope at San Pedro using an Apogee U42 CCD camera (2×2 pixel binning; sub-frame, 2.67×2.67 arcmin, or 100×100 pixels; integration/cycle times, 3 and 3.88 s). **c**, The light curve from the 60-cm TRAPPIST telescope at La Silla, using a FLI ProLine PL3041-BB CCD camera (2×2 pixel binning; sub-frame, 3.25×3.25 arcmin, or 150×150 pixels; integration/cycle times, 3 and 4.55 s). **d**, The light curve from the 215-cm Jorge Sahade telescope at CASLEO, using a Roper Scientific Versarray 1300B CCD camera (3×3 pixel binning; sub-frame, 2.62×3.50 arcmin, or 77×103 pixels; integration/cycle times, 4 and 7 s). The horizontal dashed lines at the bottom of **a** and **c** represent Eris' contribution to the flux, showing that the star completely disappeared during the event (Supplementary Information, section 2). The red lines are the best square-well models fitted to the events. We show in **b** solution 2 for the light curve (solution 1 being very close at that scale, Supplementary Fig. 3). The vertical arrow in **d** shows the time of closest approach (CA) to the shadow edge at CASLEO, at 8,368 s UT.

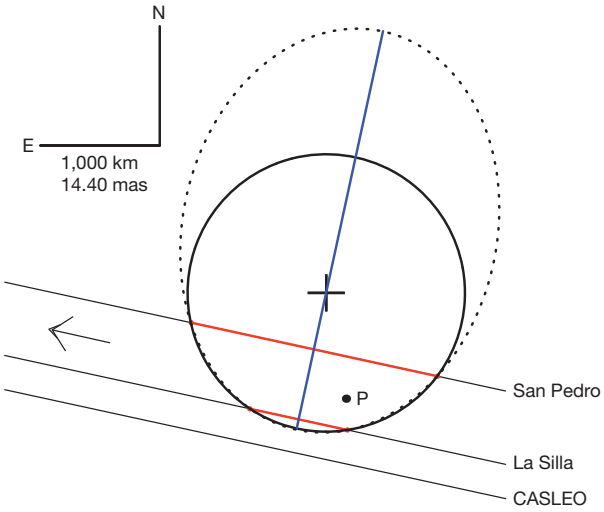


Figure 2 | Measuring Eris' size. The three oblique solid lines show the star trajectories relative to Eris, as seen from San Pedro, La Silla and CASLEO, with the arrow pointing towards the direction of motion. The San Pedro and La Silla timings provide the lengths of the two occultation segments, or 'chords' (in red); see solution 2 in Supplementary Table 3. The median lines of the two red segments are separated by only 5 km and coincide at that scale with the blue line. Celestial north is up and east is left. Scale bars: 1,000 km and 14.40 mas (1 mas corresponds to 69.436 km at Eris). The solid circle has a radius $R_E = 1{,}163$ km, and is our preferred solution for Eris' size and shape, with the cross marking the position of the centre. The dot near 'P' indicates Dysnomia's orbit pole direction[12] projected onto Eris' surface. The dotted curve is an elliptic limb fitted to our occultation chords, with semi-major and -minor axes of $a' = 1{,}708$ and $b' = 1{,}317$ km, respectively, that is, an apparent effective radius of $R_E = 1{,}500$ km, the value of Eris' radius previously derived from thermal measurements[11]. The long axis of the ellipse should be perpendicular to the occultation chords to within $\pm 2^\circ$ in order to match our data points. This has a low probability (2%) of occurring for a random limb orientation between 0 and 180°. Furthermore, the ellipse has an aspect ratio $b/a = 0.771$ that would require a fast rotator (with a period of 4.4 h) observed pole-on to within 18° to suppress the rotational light curve[13,14]. This has also a low probability (5%) of occurring for a randomly distributed pole orientation, making the dotted limb solution unlikely.

(small angular momentum regime) with assumed equator-on viewing, or an elongated triaxial Jacobi ellipsoid (large angular momentum regime) observed pole-on, as implied by the absence of brightness variations.

We have five free parameters to adjust: a', the apparent flattening $(a' - b')/a'$, the ellipse position angle P in the sky plane, and the two coordinates of its centre, f_c, g_c (Supplementary Table 3). With four chord extremities, our observations allow for an infinity of limb solutions. However, as the two chords have almost the same median lines (Fig. 2), this strongly suggests that Eris' shape is indeed close to spherical, unless very special geometries occurred (see below). Using a circular model with three free parameters ($R_E = a', f_c, g_c$) and adopting solution 2, we obtain $R_E = 1{,}163 \pm 6$ km (1σ formal error). The minimum χ^2 per degree of freedom, $\chi^2_{pdf} = 1.38$, indicates a satisfactory fit to the data (Supplementary Table 3). Moreover, the r.m.s. radial residual of 2.1 km is fully consistent with our formal timing errors. We may not exclude, however, the possibility that random topographic features with amplitude approximately ± 3 km exist along the limb, which would result in a slightly larger error bar for Eris' radius, $R_E = 1{,}163 \pm 9$ km (see Supplementary Information). Solution 1 provides $R_E = 1{,}140$ km, but with a high value $\chi^2_{pdf} = 30.7$ (5.5σ level), and radial residuals of $+11$ km and -16 km at the beginning and end of the San Pedro chord, respectively. Topographic features of this size are unlikely on such a large icy body. This indicates that the spherical assumption is not correct for solution 1, and explains why we do not provide a formal error bar for that value.

Allowing for a non-zero flattening of Eris' limb, we find an infinity of possible solutions by fixing the position angle P and semi-major axis a' at various values. If Eris' rotation axis and Dysnomia's orbital pole are aligned, we find values of R_E in the range 1,105–1,155 km, smaller than the value 1,163 km derived above. Relaxing the constraint on Eris' orientation, we find that elliptical limb models can satisfactorily fit the occultation chords in 68.3% of the cases (1σ level) for R_E in the range $1{,}165 \pm 90$ km (Supplementary Fig. 4). However, as R_E departs from 1,165 km, the flattening must rapidly increase, requiring fast rotations which are not supported by observations[13,14]. The extreme case of

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$R_E = 1{,}500$ km previously found[11] can be ruled out, as it requires fine tunings in both Eris' limb and pole orientations (Fig. 2). Thus, the most straightforward interpretation of our observations is that Eris is close to spherical, remembering that larger sizes are possible in a narrow region of the parameter space. Consequently, Eris is close in size to Pluto, whose radius[3–8] is estimated between 1,150 and 1,200 km.

Our radius value implies a density of $\rho = 2.52 \pm 0.05\,\mathrm{g\,cm^{-3}}$, when combined with Eris' mass[12]. This is comparable to Haumea's density[16,17] ($\sim 2.6\,\mathrm{g\,cm^{-3}}$), for which a typical rock/ice ratio of 0.85/0.15 is derived[18]. This suggests that Haumea (and thus also Eris) is a large rocky body with a thin overlying ice shell. Note that the densities of trans-Neptunian objects (TNOs) span a large range, with ρ values of 1.0, 1.6 and $2.0\,\mathrm{g\,cm^{-3}}$ for Varuna[17], Charon[19] and Pluto[19], respectively, pointing to diverse origins and/or evolutions. Our radius value provides a geometric albedo of $p_V = 0.96^{+0.09}_{-0.04}$ in the visible range (Supplementary Information). This makes Eris almost as bright as a perfect isotropic Lambert surface (for which $p_V = 1$ by definition), and one of the intrinsically brightest objects of the Solar System. For comparison, Saturn's satellite Enceladus is even brighter, with a geometric albedo of $p_V \approx 1.4$, associated with its geologically active surface[20]. In contrast, Eris' brightness and lack of light-curve variations may stem from the collapse of a nitrogen atmosphere (see below). We find that Eris is brighter than the TNO 2002 TX_{300}, whose high albedo ($0.88^{+0.15}_{-0.06}$) is probably due to the exposure of fresh water-ice[21].

We now reassess Eris' surface temperature in the light of our new results. Measurements by the Spitzer[22] and IRAM[11] satellites imply disk-averaged brightness temperatures of $T_b = 30 \pm 1.2$ K and $T_b = 38 \pm 7.5$ K at 70 and 1,200 μm, respectively. As a completely absorbing surface at Eris' distance has a temperature $T_0 = 40$ K, the second value is surprisingly high (and consistent with the fact that the previously found radius[11] of 1,500 km is about 30% higher than our value), but we note that a unique brightness temperature $T_b \approx 31$ K matches both the Spitzer (at 70 μm) and IRAM (1,200 μm) measurements at the 1.5σ level (Supplementary Fig. 5). However, Eris' surface temperature is probably not uniform, because an atmosphere (if any) would be too tenuous to isothermalize the surface frosts, as occurs for Triton and Pluto. We therefore consider two extreme standard temperature distribution models, corresponding to (1) a warmer slow rotator (or equivalently, pole-on orientation, or zero thermal inertia, the standard thermal model, STM) with sub-solar temperature T_{ss}, and (2) a cooler fast rotator with equator-on geometry (isothermal with latitude model, ILM), with equatorial temperature T_{eq}.

In the STM, both Spitzer and IRAM fluxes are reproduced satisfactorily with $T_{ss} \approx 35$ K (Supplementary Fig. 5, Supplementary Tables 4 and 5). The thermal equilibrium equation $T_{ss} = T_0[(1 - p_V q)/(\varepsilon\eta)]^{1/4}$ then provides a relationship between the beaming factor η (describing the effects of surface roughness), the phase integral q and the surface emissivity ε, where $A = p_V q$ is the Bond albedo, which measures the fraction of reflected solar energy. Using a standard value[22] $\varepsilon = 0.9$ and a plausible range from $\eta = 1$ (no roughness) to 0.7 (large surface roughness), we obtain $q = 0.49$–0.66, fully consistent with the values for Saturn's brightest icy satellites[20,23]. The ILM in contrast leads to the extreme condition $0 < q < 0.24$, which is an implausible range as bright objects also have large phase integrals[24]. Essentially, the fast rotator model does not provide enough thermal flux given the new, smaller size of Eris. We therefore strongly favour the STM, implying either a pole-on orientation or a very small thermal inertia, as observed in other TNOs[25,26].

The occultation puts an upper limit on a putative atmosphere around Eris. As discussed in Supplementary Information, our preferred model is an isothermal N_2 atmosphere near 30 K, for which we can place an upper limit of about 1 nbar (1σ level) at the surface (Fig. 3). Similar limits are obtained for hypothetical CH_4 or Ar atmospheres. Also discussed in Supplementary Information is the possibility that a Pluto-like atmosphere sublimates as Eris approaches its perihelion, at 37.8 AU from the Sun. In that case, Eris would currently be a dormant Pluto twin, with a bright icy surface created by a collapsed atmosphere. Detailed models are required, however, to confirm this model.

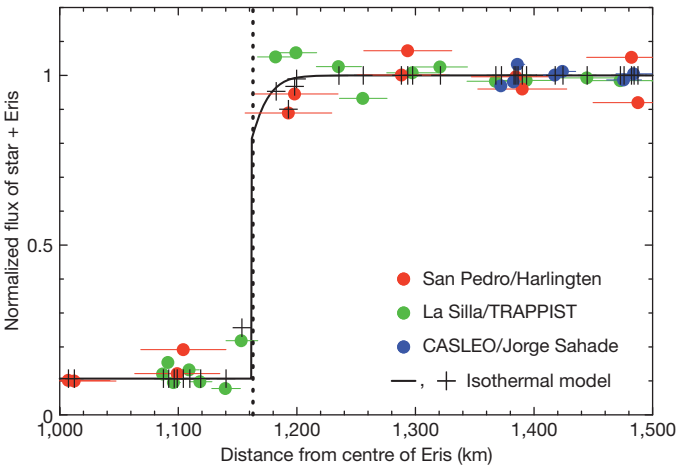


Figure 3 | Upper limit on Eris' atmosphere. Each data point (filled coloured circles) obtained at three of the stations shown in Fig. 2 has been projected onto a radial scale (distance from Eris' centre), using the circular solution 2 displayed in Fig. 2. The horizontal bars indicate the finite radial resolution associated with finite integration intervals; the vertical dotted line shows the adopted Eris radius, $R_E = 1{,}163$ km. The black solid line is a model light curve obtained at 1-km resolution, using an isothermal pure N_2 atmosphere. Black crosses mark the expected flux associated with each data point, once the convolution with the finite integration intervals has been performed. The fit minimizes the differences between the black crosses and the corresponding data points (filled circles). The model shown here is the 3σ-level upper limit of an isothermal N_2 atmospheric profile, with $T = 27.7$ K and a surface pressure of 2.9 nbar. Most of the constraint on the model comes from the two data points obtained at La Silla (the two green filled circles just right of the vertical dotted line), corresponding to the data points obtained just before and just after the event (Fig. 1). The two closest San Pedro data points (red) have only a small contribution to the χ^2 value, while the CASLEO data points (blue) are too far away from Eris' edge ($\sim$200 km) to effectively constrain the atmospheric pressure. Using solution 1 instead of solution 2 for Eris' shape would have a minimal impact on the atmospheric upper limit, as this would slightly displace the San Pedro data points in the plot, leaving the La Silla points where they are shown here.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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Author Contributions B.S. helped plan the campaign, centralized the stellar occultation predictions, participated in the observations, analysed data, wrote and ran the diffraction, limb-fitting and ray-tracing codes, and wrote part of the paper. J.L.O. helped plan the campaign, analysed data for the prediction, participated in the observations, obtained and analysed data, and wrote part of the paper. E.L. analysed the implications of the results for the Eris thermal model, albedo constraints and putative atmospheric structure, and wrote part of the paper. M.A., F.B.-R., A.H.A., J.I.B.C., R.V.M., D.N.d.S.N. and R.B. discovered the star candidate and analysed data for the predictions. E.J. and A.M. obtained and analysed the positive occultation detection at La Silla/TRAPPIST and San Pedro/Harlingten telescopes, respectively. F.B.-R., F.C., M.G. and J.M. analysed data, D.H. calculated Dysnomia's position at the moment of occultation and wrote part of the paper. All other authors participated in the planning of the campaign and/or the observations, and the authors listed in Supplementary Table 2 were responsible for the observations. All authors were given the opportunity to review the results and comment on the manuscript.

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F.1 Suplementos

SUPPLEMENTARY INFORMATION

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1 Prediction and observations

The candidate star for the occultation analysed in this paper was discovered during a systematic search for candidate stars to be occulted by Pluto and other several large TNOs in 2008-15, conducted at the 2.2m telescope of the European Southern Observatory[15]. It revealed the star NOMAD1 0856-0015072 with $V\sim$17.1, $R\sim$16.9 as a potential candidate for an occultation by Eris on the night of 6 November 2010 around 02:20 UT. Initial possible zones of visibility included western Europe and Africa, as well as north and south America (see www.lesia.obspm.fr/perso/bruno-sicardy/predic_occn_10/Eris_2010). This was the unique event predicted for 2010. Actually, stellar occultations by Eris remain rare events as the dwarf planet moves far away from the galactic plane, in poor stellar fields in Cetus constellation (no event involving Eris is expected in our list before 2013).

Measurements made at the 2.2m telescope of the European Southern Observatory (see ref. 15 for the method) provide the following ICRF (loosely speaking J2000) star position as given by the UCAC2 reference frame:

$$\begin{cases} \alpha = 01^{\mathrm{h}}\ 39^{\mathrm{min}}\ 09.9421^{\mathrm{s}} \\ \\ \delta = -04^{\circ}\ 21'\ 12.119'' \end{cases} \qquad (1)$$

with 1σ errors of about 50 milli-arcsec (mas) in both right ascension and declination.

Astrometric updates were conducted from October 18, 2010 onward to pin down the prediction

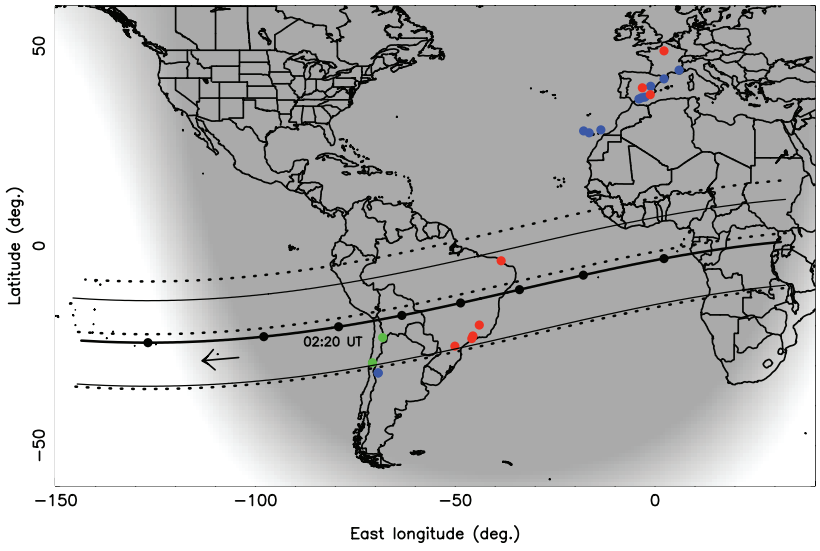


Figure 1S | The Eris observation campaign. The locations listed in Tables 1S-2S are plotted in green for the sites where the occultation was detected, in blue where observations were made but no occultation was observed and in red for stations with clouds. The three parallel dotted lines show the latest prediction (northern, central and southern limits of Eris' shadow) made about 24 hours before the event, assuming a radius of 1,500 km for the dwarf planet. The solid line with dots is the actual path of Eris' shadow centre derived from the solution shown in Fig. 2, with the dots plotted every minute (see the reference dot plotted at 02:20 UT, November 6, 2010) and the arrow shows the direction of motion on Earth. The other two parallel thinner solid lines are the shadow limits, using again the solution of Fig. 2 (radius 1,163 km).

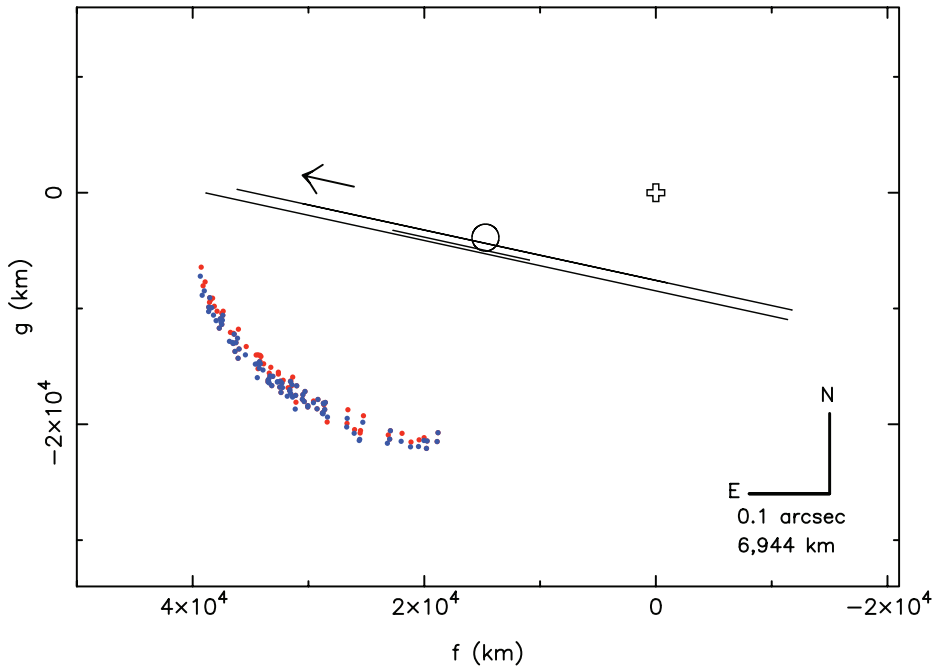


Figure 2S | Star motion with respect to Eris and Dysnomia. Oblique lines: the trajectories of the occulted star relative to Eris, as seen from San Pedro de Atacama, La Silla and CASLEO from top to bottom, respectively. Corresponding start and end times are provided in Table 1S. The arrow shows the direction of the stellar motion. Open cross: the expected Eris' centre position from our reference star position (Eq. 1) and the JPL28 ephemeris. Black circle: circular fit to the occultation chords, see Fig. 2. The shift between the cross and the circle is Eris' offset relative to JPL28, i.e. +211±50 mas and -56±50 mas in right ascension and declination, respectively, see Table 3S. Red (resp. blue) dots: expected position of Dysnomia at 02:20 UT (November 06, 2010) according to the Orbit 1 (resp. Orbit 2) orientation of Dysnomia's orbit[12]. The dots have been generated by using various possible orbital solutions that match Dysnomia observations. The scatter of the points shows the uncertainty on Dysnomia's position at the moment of occultation. Scale and orientation are given at lower right.

and organise the observational campaign, based on images taken at Calar Alto, Cerro Burek, Pico dos Dias and Table Mountain. Only after November 1st, 2010, was the position of Eris relative to the star accurate enough to clearly show that the event would be visible from south America, and possibly northern Africa and southern Europe.

Twenty six stations were eventually involved in our campaign, most of them being clouded out and/or outside the shadow path (Fig. 1S). The occultation was detected from two sites in Chile, one with two telescopes separated by about 20 meters (0.5m-Harlingten and 0.4m-ASH2) at San Pedro de Atacama (San Pedro for short), and the other one at the ESO La Silla site with the new 0.6m TRAnsiting Planets and PlanetesImals Small Telescope[27] (TRAPPIST). A third station in Argentina (Complejo Astronomico El Leoncito, CASLEO, 2.15-m Jorge Sahade telescope) was sufficiently close to the shadow edge (∼200 km, see Fig. 2 and 1S) and with a sufficient SNR to be used for potentially placing an upper limit of Eris' atmosphere. Table 1S provide details on the observations analysed here, while Table 2S lists all our other attempts made during that campaign.

The general geometry of the event is shown in Fig. 2S. Besides the Eris events detected at San Pedro and La Silla (see below), no secondary events were detected in the light curves that we have analysed. Note that Dysnomia predicted position was far away from the stellar tracks that scanned Eris' surrounding (Fig. 2S).

2 Data analysis

All data shown here were obtained with broadband (no filter) CCD images, in order to maximise photon flux. The camera types used at each telescope are given in the caption of Fig. 1. After classical dark subtraction and flat fieldings, aperture photometry was performed to derive the stellar flux vs. time. The aperture used for Harlingten was a square with size 4×4 pixels, while circles with diameters 4, 5 and 6 pixels were used for ASH2, TRAPPIST and Jorge Sahade, respectively. The background flux was estimated near each star and subtracted, so that the zero flux corresponds to the sky level. Images taken at the ASH2 and TRAPPIST telescopes the night after the event allowed us to estimate the fluxes from Eris at the moment of the occultation, see the dotted lines in Fig. 1 and Fig. 3S. This calibration shows that the star completely disappeared during the event, down to the noise level. A nearby reference star (NOMAD1 0856-0015078), roughly two magnitudes brighter than the target, was used to correct the stellar flux for low frequency sky transparency variations at La Silla and CASLEO. Due to the good conditions prevailing at San Pedro (with a target elevation of ∼68 degrees), this correction was not necessary. A further division by the unocculted stellar + Eris flux, averaged over several minutes around the occultation, eventually provides the normalised light curves presented in Figs. 1 and 3S. The flux statistics yields the standard deviation of the signal, which defines the 1σ error bar on each data point, used later for fitting diffraction models to the star disappearances and re-appearances, see below.

Supplementary Table 1S | Circumstances of observations for the data used in this paper

Site	Telescope aperture Integration/cycle times[a] Signal-to-noise ratio[b]	longitude latitude altitude	start end h:min:s	disappearance[c] re-appearance[c] h:min:s±s UT) (6 November 2010)	observers
San Pedro de Atacama	0.5m-Harlingten 3s/3.88s 19.2	68° 10' 47.6" W 22° 57' 12.3" S 2,397 m	02:09:17.7 02:29:20.8	02:18:41.6±0.15 02:19:59.4±0.18[d] 02:20:00.6±0.18[e]	A. Maury S. Saravia
San Pedro de Atacama	0.4m-ASH2 15s/18.32s 15.0	68° 10' 46.9" W 22° 57' 12.2" S 2,397 m	02:02:24.8 02:32:57.3	02:18:39.8±1.3 02:19:58.9±3.1	remotely operated by N. Morales & I. de la Cueva
La Silla	0.6m-TRAPPIST 3s/4.55s 40.2	70° 44' 21.8" W 29° 15' 16.6" S 2,315 m	02:16:58.8 02:24:29.4	02:19:17.1±0.11 02:19:48.4±0.10	remotely operated by E. Jehin
CASLEO	2.15m-Jorge Sahade 4s/7s 56.5	69° 17' 44.9" W 31° 47' 55.6" S 2,492 m	02:02:38.1 02:34:41.4	no occultation	R. Gil-Hutton

[a] All the observations listed here were made with fast broadband visible CCD, see Fig. 1.
[b] The ratio of the star plus Eris flux to the r.m.s. noise per data point.
[c] All error bars are at 1σ level.
[d] and [e] are solution 1 and solution 2 described in the text, respectively, for the star re-appearance at San Pedro, see text and Fig. 3S.

The UT time of each image was taken from the frame headers. However, for the three stations where the event was detected (Harlingten, ASH2 and TRAPPIST), only the integer part of the second is available, due to the truncated format provided by the MaximDL software used at those three stations. To retrieve the correct timing of each image, we fitted a linear model to the set of points $(i, t_{\mathrm{h,i}})$, where i is the image number, and $t_{\mathrm{h,i}}$ is the corresponding time written in the

Supplementary Table 2S | Circumstances of observations at other stations

Site	Telescope aperture	longitude latitude, altitude	Result	observers
Meudon	0.6m	02° 13' 53" E 48° 48' 18" N, 162 m	clouds	F. Braga-Ribas, F. Roques, B. Sicardy
Puimichel	1.0m	06° 01' 15" E 43° 58' 49" N, 700 m	no occ	F. Colas, J. Lecacheux
Esteve Duran	0.6m	02° 15' 32" E 41° 47' 57" N, 720 m	partly cloudy	E. García-Melendo
València	0.6m TROBAR	01° 06' 05" W 39° 56' 42" N, 1,271 m	partly cloudy	V. Peris, J. Fabregat
La Hita	0.77m	03° 41' 10" W 39° 34' 07" N, 675 m	clouds	F. Organero
La Murta	0.4m	01° 12' 10" W 37° 50' 25" N, 404 m	clouds	S. Pastor, J. A. de los Reyes
Calar Alto	0.5m CAB	02° 32' 45" W 37° 13' 25" N, 2,173 m	no occ	remotely operated by L. Cuesta
	1.2m		no occ	remotely operated by S. Mottola
	2.2m		no occ	S. Pedraz
Sierra Nevada	1.5m	03° 23' 05" W 37° 03' 51" N, 2,930 m	no occ	A. Sota
Málaga	0.6m Bootes-2/TELMA	04° 02' 33" W 36° 45' 30"N, 62 m	no occ	A. Castro R. Sánchez-Ramírez
Nazaret	0.3m	13° 33' 22" W 29° 02' 29" N, 366 m	no occ	G. Muller
La Palma	2m Liverpool	17° 52' 45" W 28° 45' 45" N, 2,321 m	no occ	remotely operated by I. A. Steele
Teide	0.8m IAC80	16° 29' 48" W 28° 17' 30" N, 2,112 m	no occ	M. Cebrián, A. Oscoz P. Montañés-Rodríguez
Fortaleza	0.3m	38° 30' 27" W 03° 44' 18" S, 38 m	clouds	D. Weaver
CEAMIG-REA	0.3m	43° 59' 51" W 19° 49' 49" S, 825 m	clouds	C. Jacques, E. Pimentel, B. Giacchini
Pico dos Dias	1.6m	45° 34' 58" W 22° 32' 08" S, 1,840 m	clouds	M. Assafin, W. Corradi, W. Reis Jr., F. P. Santos, A. Ramos Gomes Jr.
São José dos Campos	0.275m	45° 51' 44" W 23° 12' 33" S, 617 m	clouds	A. Milone
Ponta Grossa	0.4m	50° 05' 56" W 25° 05' 22" S, 909 m	clouds	M. Emilio
Cerro Burek	0.45m ASH	69° 18' 25" W 31° 47' 12" S, 2,665 m	no occ	remotely operated by N. Morales
Sevilla	0.3m	06° 19' 35" W 31° 40' 19" N, 471 m	no occ	J. M. Madiedo
San Pedro Mártir	1.5m	115° 27' 48" W 31° 02' 36" N, 2,825 m	no occ	L. Gutiérrez, H. Hernández-Toledo

header. The residuals of those fits, performed over intervals of several minutes around the event, show a regular sawtooth pattern confined between -0.5 and +0.5 seconds. This (1) confirms that the truncation to the integer part of the second is indeed present, and (2) shows that the acquisition cycle was regular. To obtain the mid-exposure time of each frame, we finally added to the fitted

The internal accuracy of those linear fits ensures that the absolute times of individual images can be retrieved to better than 0.05 s. Furthermore, the computers used at San Pedro and La Silla were set up against UT time every 15 minutes at least, using a time server and the 'Dimension 4' synchronisation software, which ensures an absolute time accuracy better than 0.05 s for all computers. In summary, the time associated with each image has an absolute accuracy of about 0.07 s, corresponding to an accuracy of $\pm$ 2km along Eris' shadow track, which is negligible compared to the error associated with the model fitting described below.

3 Occultation timing

The star dis- and re-appearance times are obtained by fitting an abrupt edge shadow model to the light curves, after convolving that shadow by Fresnel diffraction, stellar diameter[28] projected at Eris, finite bandwith of the CCD and finite integration time of the instrument[29]. The final synthetic light curve is largely dominated by finite integration time, which was 3 s or more for all data sets, see Table 1S. At La Silla (resp. San Pedro), this corresponds to a distance of about 30 km (resp. 75 km) perpendicular to Eris' limb, while Fresnel's scale is about 2 km, and the star diameter projected at Eris is about 0.7 km. The free parameter of the fit is the time of star dis- or re-appearance, t_{occ}, i.e. the time at which the occultation would occur in the limit of a point source occulted by a sharp edge in the geometrical optics regime. The fit procedure looks for the minimum value χ^2_{min} of

$$\chi^2 = \sum_{1}^{\mathrm{N}} (\Phi_{\mathrm{i,obs}} - \Phi_{\mathrm{i,cal}})^2 / \sigma_{\mathrm{i}}^2, \qquad (2)$$

where $\Phi_{\mathrm{i,obs}}$ (resp. $\Phi_{\mathrm{i,cal}}$) is the observed (resp. calculated) flux at point i, σ_{i} is the 1σ error bar at that point, estimated as explained earlier, and N is the total number of points considered. The 1σ error bar on t_{occ} is estimated by varying t_{occ} so that to increase χ^2 from the best value χ^2_{min} to $\chi^2_{\mathrm{min}} + 1$.

The best fits are shown in Fig. 3S, and the corresponding occultation times are listed in Tables 1S and 3S. When adjusting the star re-appearance at San Pedro/Harlingten, two equally satisfactory fits (called solution 1 and solution 2, in chronological order) are obtained. This is because it occurred near one of the 0.88 s gaps between consecutive exposures (Table 1S). Consequently, the χ^2 residual exhibits two local minima for two values of t_{occ} separated by 1.2 s, corresponding to a difference of 32 km in the two chord lengths, see Fig. 3S and Table 3S.

4 Occultation geometry and Eris' size

The occultation times given in Table 3S provide the corresponding star positions (f, g) relative to Eris and projected in the plane of the sky, where f is the relative position in right ascension, positive if the star is east of Eris centre, and g is the relative position in declination, positive if the star is north of Eris centre. The quantities f and g are expressed in kilometers. For this calculation, we use the JPL28 geocentric Eris ephemeris (ref. 30 and http://ssd.jpl.nasa.gov/), corrected for parallax effects at each station. Note that due to the larger integration time (15 s) and lower signal

to noise ratio obtained at the San Pedro ASH2 telescope, compared to the Harlingten telescope (3 s integration time), the former data do not bring further constraints to the occultation length. Although they confirm the event (Fig. 3S), they are not included in the analysis below. So, we finally obtain four extremities of the two occultation segments (or 'chords'), derived from the timings at San Pedro/Harlingten and La Silla/TRAPPIST telescopes, see the red segments in Fig. 2.

The most general limb shape considered here is an ellipse projected in the plane of the sky, with

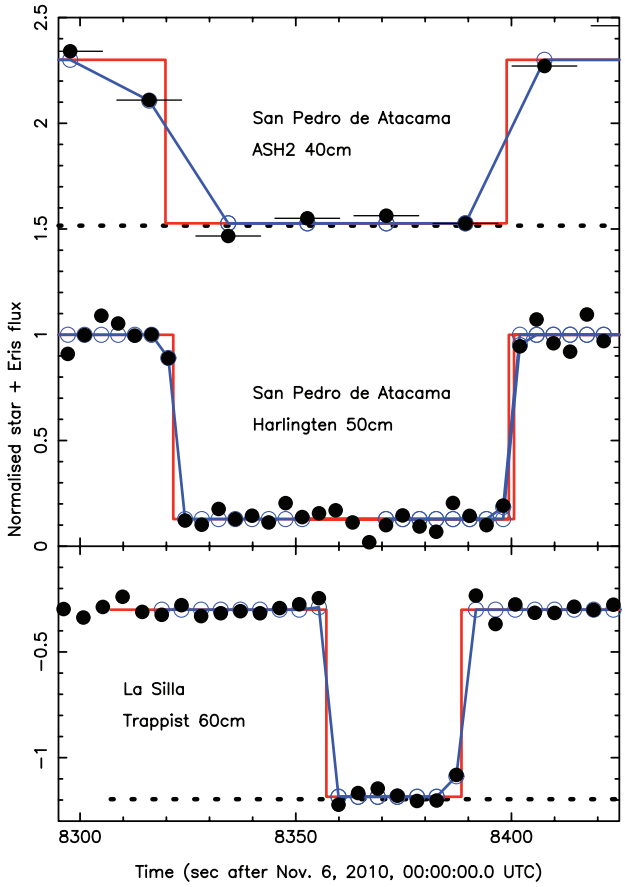


Figure 3S | Fitting diffraction models to the occultation light curves. Black bullets: the occultation data points obtained at the station mentioned in Table 1S (where the star plus Eris fluxes have been normalised to unity). Plots for ASH2 and TRAPPIST data have been shifted vertically by +1.3 and -1.3, respectively, for better viewing. The horizontal bars on the ASH2 data points show the acquisition time intervals corresponding to each point. For the Harlingten and the TRAPPIST data points, the bars have approximately the size of the bullets, and are not visible on the plot. Note that all data sets have 'dead times', i.e. intervals of time when no data are acquired, see Table 1S. The dotted lines under the ASH2 and the TRAPPIST light curves show Eris' contribution to the total flux (0.216 and 0.104, respectively), determined by measuring the occulted star against a nearby reference star the night after the occultation, with the same instrument setup and same airmass. Red curves: the best fit square-well models to the data (corresponding to Eris' shadow edge in the geometrical optics regime). Two equally satisfactory fits are obtained for the star re-appearance at San Pedro (the so-called solutions 1 and 2, from left to right, respectively, see text). Open blue circles: expected flux derived from the red model, and corresponding to each observed data point, once diffraction, star diameter, finite bandwith and finite acquisition time intervals have been accounted for. The open blue circles obtained with both solutions 1 and 2 have been plotted for the Harlingten telescope, but they are almost indistinguishable at the scale used here. The blue curve connects the open blue circles.

Supplementary Table 3S | Fits to the occultation chords[a]

site	6 November 2010 (h:min:s UT)	f[b] (km)	g[b] (km)	shadow velocity[c] (km s^{-1})	radial residual (km, circular fit) solution 1	solution 2
San Pedro, disappearance	02:18:41.6±0.15	+13782.9	-4575.6	26.748	+10.9	+2.4
San Pedro, re-appearance (sol1)	02:19:59.4±0.18	+15816.5	-4133.9	26.748	-16.3	–
San Pedro, re-appearance (sol2)	02:20:00.6±0.18	+15847.9	-4127.1	26.748	–	-3.4
La Silla, disappearance	02:19:17.1±0.11	+14538.7	-5026.2	26.723	-2.0	-0.4
La Silla, re-appearance	02:19:48.4±0.10	+15356.0	-4848.5	26.723	+2.0	+0.4

free parameters of circular fit	Best fit values	
Eris' radius R_E (km)	1,141	1,163±6
Eris' centre[d] f_c (km)	+14,717	+14,716±3
Eris' centre[d] g_c (km)	-3,902	-3,877±10
Goodness of fit, χ^2_{pdf}[e]	30.7	1.38
Radial residual (km)[f]	9.88	2.11
Time of geocentric closest approach (h:min:s UT, 6 November 2010)[g]	02:18:15.7	02:18:15.9±0.1
Distance of closest approach to geocentre (km)[h]	1,246	1,223±10
Offset[i] in right ascension (mas) with respect to DE405/JPL28	+212±50	+212±50
Offset[i] in declination (mas) with respect to DE405/JPL28	-56±50	-56±50

[a] The timings from the San Pedro/0.4m-ASH2 telescope are not included in the fits, as their larger error bars do not bring further constraints on the length of the occultation at that site (Table 1S and Fig. 3S).
[b] The timings of Table 1S provide the star position relative to Eris expected centre, using the DE405/JPL28 Eris' ephemeris (http://ssd.jpl.nasa.gov). This position is projected in the plane of the sky, in km, where f is the relative position in right ascension, positive if the star is east of Eris' centre, and g is the relative position in declination, positive if the star is north of Eris' centre. We use the ICRF/J2000 star position given in Eq. 1.
[c] Velocity of Eris' shadow at the considered site, perpendicular to the observer-Eris line (equivalent to Eris velocity relative to the star, projected in the plane of the sky). The *radial* velocities with respect to the shadow edge (assuming a circular shape) are 9.7 and 24.4 km s^{-1} at La Silla and San Pedro, respectively.
[d] This centre corresponds to the offset to be applied to Eris' position in order to match the occultation chords (see the cross in Fig. 2). This offset is relative to the DE405/JPL28 Eris ephemeris, but also depends on the adopted star position given in Eq. 1.
[e] Value of minimum χ^2 per degree of freedom, see text.
[f] Defined as the r.m.s. of the four radial residuals.
[g] Time at which Eris and the star are the closest in the sky plane, as seen from the geocentre.
[h] Distance between Eris' centre and the star in the sky plane at closest approach, as seen from the geocentre. Conversion to mas can be made using an Eris observer range of 95.738 AU during the occultation, corresponding to a scale of 69.436 km per mas. At closest approach, Eris was at a position angle of 167.57 degrees relative to the star, in the J2000 reference frame, as seen from the geocentre.
[i] These values are derived from the values of f_c and g_c given above. The error bars are dominated by the uncertainties on the star position (±50 mas), not by the error bars of the circular fit to the occultation chords.

semiaxes $a'>b'$ and apparent flattening $\epsilon'=(a'-b')/a'$. In the small angular momentum regime, this shape results from the projection of an oblate Maclaurin spheroid with semiaxes $a=b>c$, where a and c are the equatorial and polar radii, respectively. The true flattening of the body is then $\epsilon=(a-c)/a$, and for small values of ϵ, the latter is related to the apparent flattening through[31] $\epsilon' \sim \epsilon \cdot \sin^2(\zeta)$ where ζ is the angle between the rotation c-axis and the line of sight (ζ=0 corresponding to pole-on geometry). In that case, we assume that the spheroid is observed equator-on (so that its true flattening is $\epsilon=\epsilon'$), remembering that more general viewing geometries will result in true flattenings $\epsilon>\epsilon'$. In the large angular momentum regime, the shape results from an elongated triaxial Jacobi ellipsoid with semiaxes $a>b>c$. In that case, the ellipsoid has to be observed almost pole-on because no brightness variations larger than 1% are observed for the dwarf planet[13,14]. We finally define Eris' effective radius R_E as the radius of a disk that has the same apparent surface area as the actual

body, so that $R_E=\sqrt{a'b'}$.

Our problem has M=5 free parameters: the apparent semi-major axis a', the apparent flattening $\epsilon'=(a'-b')/a'$, the ellipse orientation in the sky (position angle P of the semi-minor axis, counted eastward from the local celestial north), and the two coordinates of its centre f_c, g_c. Note that it is sufficient to consider $0 \leq P < 180$ degrees, as other values of P yield the same solutions by symmetry.

We fit the N=4 chord extremities by finding the minimum value χ^2_{min} of

$$\chi^2 = \sum_1^N \left[(f_{i,obs} - f_c)^2 + (g_{i,obs} - g_c)^2 \right] / \sigma^2_{i,r}, \quad (3)$$

where ($f_{i,obs}$, $g_{i,obs}$) are the coordinates of the chord extremities, and $\sigma_{i,r}$ are the corresponding radial uncertainties obtained by multiplying the 1σ uncertainties on the occultation times by the local star velocity perpendicular to Eris' limb. The 1σ error bar on R_E is obtained by exploring values of R_E (the other parameters being free) so as to change χ^2 from χ^2_{min} to $\chi^2_{min}+1$. The value of χ^2 per degree of freedom for the best fit, χ^2_{pdf} (see Table 3S) is given by $\chi^2_{pdf}=\chi^2_{min}/(N-M)$. With N=4 chord extremities, and M=5 free parameters, there is an infinity of possible solutions and χ^2_{pdf} is not defined. In order to obtain a preferred solution, we have to rely on independent observations and on physical arguments.

4.1 Circular fit

We first note in Fig. 2 that the two chords have almost the same median lines, i.e. the lines going through the middle of the segments, and perpendicular to them, see the blue line in Fig. 2. More precisely, the two median lines are separated by 21 km for solution 1, and 5 km for solution 2. Consequently, a circle can be fitted to the two chords to within our error bars, especially for solution 2.

This strongly suggests that Eris is indeed close to spherical, unless a very special geometry occurred, as discussed later. A circular fit has M=3 free parameters ($R_E=a'=b'$, f_c and g_c), so that $N-M$=1 and $\chi^2_{pdf}=\chi^2_{min}$. Table 3S gives the values of R_E, f_c and g_c corresponding to the best fits using solutions 1 and 2, respectively. Note that due to the high value of χ^2_{pdf} in the case of solution 1, we do not provide error bars, since the assumption of a circular limb is not valid in this case, see main text. For solution 2, we obtain a radius R_E=1,163±6 km for the best circular fit to the chords extremities. The position of Eris' centre, (f_c,g_c) corresponds to the offset to apply to Eris ephemeris in order to fit the occultation chords (Fig. 2). Note that since we actually determine Eris' position relative to the star at a given moment, this offset depends (1) upon the adopted ephemeris (here DE405/JPL28) and (2) upon the adopted star position (Eq. 1). Thus, if one uses an ephemeris different from JPL28 for Eris and/or a star position different from the one mentioned here, the offset should be corrected, taking into account both the differences between the ephemerides and star positions.

Using the star position in Eq. 1 and the ephemeris DE405/JPL28, we obtain offsets of +212±50 and -56±50 mas for Eris, in right ascension and declination, respectively, where the error bar is dominated by the uncertainty in the star position. Alternatively, one may use the time and distance of geocentric closest approach (C/A) in the plane of the sky to calculate those offsets (see Table 3S), once a star position and a reference ephemeris is given, knowing that Eris was at a position angle of 167.57 degrees relative to the star at C/A, as seen from the geocentre.

4.2 Topographic features

Even if globally spherical, a body composed of compact ice with Eris' size may sustain small reliefs with maximum amplitude of typically $\lesssim \pm$ 5 km (ref. 32). They may create randomly distributed local, radial limb distortions, thus altering the occultation chord lengths, and finally, the retrieved Eris' radius. However, $\pm$5km corrugations are probably too high in the case of Eris. First, Eris' interior has a density of $\sim$2.5 g cm^{-3} (see main paper), more than twice the density of ice, resulting in a surface gravity larger than for a pure ice body. Second, the low thermal inertia of TNOs[25,26] suggests low conductivity, porous surfaces that are unlikely to support large reliefs. We also note that other icy bodies like the Uranian satellite Titania (radius 789 km) have local topographic features of $\pm$2 km at most (see the discussion in ref. 29). Another large icy satellite, Triton (radius 1,353 km), has even smaller reliefs, with amplitude of less than $\sim$0.5 km (ref. 33)

We thus consider that an amplitude of $\pm$3 km is a safe maximum for local topographic features on Eris' surface. Generating circular fits to the chords obtained with solution 2, and allowing for residuals up to 3 km, we obtain values of R_{E} in the range 1,163$\pm$6 km. This error bar may be combined quadratically with the formal error bar of $\pm$6 km found above, yielding a final error bar of $\pm$9 km for Eris' radius, in the circular limb assumption with (probably overestimated) reliefs of $\pm$3 km.

4.3 Elliptical fit

As mentioned earlier, we can fit an infinity of elliptical limbs to the two occultation chords. If Dysnomia orbits in Eris' equatorial plane, then the limb orientation is determined and its position angle P can be fixed, see the next subsection. If this is not the case, and since we are primarily interested in Eris' effective radius, we want at least to estimate a plausible range of values for R_{E}.

To do so, we have fitted the two occultation chords with elliptical models where P and a' have been fixed to prescribed values. The free parameters of the fits are then ϵ', f_c and g_c, each best fit returning a value for R_{E} and providing a radial residual which indicates the quality of that fit. We have varied P between 0 and 180 degrees by steps of 1 degree, and a' between 1,000 and 1,800 km by steps of 10 km, first using the chords derived from solution 1, and then, from solution 2. Moreover, we have retained only the fits with radial residuals smaller than 4.2 km, which result from the quadratic addition of possible timing errors (corresponding to about $\pm$3 km in radial direction) and/or topographic features on Eris' surface (with amplitude of about $\pm$3 km, see above).

The results are summarised in Fig. 4S. The histogram of derived R_{E}'s reaches a maximum near 1,165 km, as expected because it is close to the value derived from the circular fit, 1,163 km. The distribution rapidly drops as R_{E} deviates from that value. This is because elongated limb shapes require a fine tuning of P in order to go through the chord extremities, resulting in low probability of realisation, see Fig. 2. The 68.3% probability of occurrence centered around the maximum of the distribution (roughly corresponding to the usual 1σ level error bar for a normal distribution) corresponds to the interval R_{E}=1,165$\pm$90 km.

However, we note in Fig. 4S that the limb apparent flattening ϵ' rapidly increases as R_{E} deviates from 1,165 km, with a strong correlation betwen R_E and ϵ'. At the boundaries of the interval R_{E}=1,165$\pm$90 km, ϵ' reaches about 0.08, implying that the true flattening $\epsilon \sim \epsilon / \sin^2(\zeta)$ is even larger, as mentioned at the beginning of this section.

We can relate ϵ to the body spin frequency ω, assuming hydrostatic equilibrium. In the low angular momentum regime, the corresponding oblate Maclaurin spheroid with equatorial and polar radii a

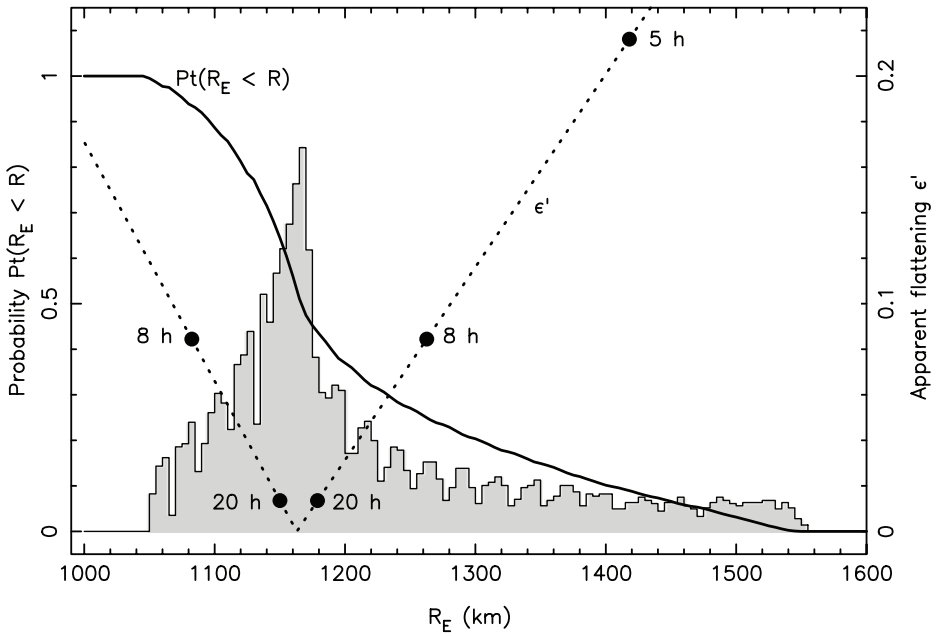


Figure 4S | Distribution of possible Eris' effective radius. Shaded surface: histogram, or probability density (arbitrary units), of Eris' effective radius R_E, derived from a wide variety of fits using elliptic limb models, see text. The integral of that histogram, starting from the right, is the probability $Pt(R_E < R)$ that R_E is larger than a given value R (thick solid line), see scale at left. The dotted line is the approximate correlation relation between R_E and the apparent flattening ϵ' obtained in the fits, see scale at right. The median of the distribution, $Pt(R_E < R)$=0.5, corresponds to R_E~1,165 km, which is close to the circular fit value for solution 2, R_E=1,163 km. Approximate values (see Eq. 5) for the rotation periods corresponding to a few values of ϵ' are shown near the dotted lines, see the black bullets.

and c, respectively, must satisfy[34]:

$$\frac{\omega^2 a^3}{GM} = \frac{3\left\{2\psi\left[2+\cos(2\psi)\right]-3\sin(2\psi)\right\}}{4\sin^3(\psi)}, \qquad (4)$$

where G is the gravitational constant, M is the mass of the body, and $\cos(\psi)=c/a=1-\epsilon$. Note that this equation assumes that the object is incompressible (uniform density). If compressible, complications arise and the body will rotate with slightly different periods. For small flattenings ϵ, Eq. 4 provides the approximation $\omega^2 a^3/GM \sim 4\epsilon/5$, yielding a rotation period of:

$$T = 2\pi/\omega \sim 2.33/\sqrt{\epsilon} \text{ hours}, \qquad (5)$$

taking a=1,163 km and M=(1.66±0.02)×10^{22} kg (ref. 12).

For $\epsilon > \epsilon' \sim 0.08$, we obtain $T < 8.2$ hours. In other words, values of R_E outside the interval 1,165±90 km require a fast rotating Eris. Rotation periods (more precisely, their upper bounds, as $\epsilon > \epsilon'$) corresponding to a few values of ϵ' are shown in Fig. 4S. They show that the required rotation period rapidly decreases as R_E deviates from 1,165 km, i.e. Eris must be a faster and faster rotator.

We may consider more extreme situations. For instance in Fig. 2, we may reconcile our results with IRAM flux[11] (yielding R_E=1,500 km) by considering a very elongated limb with b'/a'=0.771 (and a'=1708 km, ϵ'=0.229). This can be achieved with a fast rotator, where Eris has a triaxial shape (Jacobi ellipsoid) with semiaxes $a > b > c$, observed pole-on to avoid rotational light curve. The ratios

β=b/a=b'/a' and γ=c/a are related to ω and M through:

$$\beta^2 \int_0^\infty \frac{du}{(1+u)(\beta^2+u)\Delta(\beta,\gamma,u)} = \gamma^2 \int_0^\infty \frac{du}{(\gamma^2+u)\Delta(\beta,\gamma,u)} \quad (6)$$

$$\frac{\omega^2 a^3}{GM} = \frac{3}{2} \int_0^\infty \frac{u du}{(1+u)(\beta^2+u)\Delta(\beta,\gamma,u)}, \quad (7)$$

where $\Delta(\beta,\gamma,u)$=$[(1+u)(\beta^2+u)(\gamma^2+u)]^{1/2}$, see e.g. ref. 17, noting the misprint in their Eq. 2, where the term Δ is missing in the denominator. Once β=b/a is given, Eq. 6 yields γ, which in turn allows to calculate ω, using Eq. 7. In the example given in Fig. 2, we have β=0.771, from which we derive γ=0.506, and then a rotation period of 4.4 hours, i.e. a very fast rotator.

In this case, the long axis of the object should be perpendicular to the chords to within $\pm$2 degrees in order to keep the radial residuals of the fit under 4.2 km, as explained at the beginning of this subsection. This has a low probability of $\sim$2% to occur for P randomly distributed between 0 and 180 degrees.

Moreover, Eris must be observed close to pole-on in this case, to avoid a rotational light curve. The light curve amplitude of such an object is, in magnitude[35]:

$$\Delta m = -1.25 \cdot \log_{10} \left[\frac{1+\gamma^2 \tan^2\zeta}{1+(\gamma/\beta)^2 \tan^2\zeta} \right], \quad (8)$$

where we recall that ζ is the angle between the rotation c-axis and the line of sight. With the condition $\Delta m \lesssim 0.01$ imposed by observations[13,14], and taking the case β=0.771, γ=0.506 discussed above (see also Fig. 2), we find that ζ should be smaller than 18 degrees. This has also a low probability of 5% to occur for a randomly oriented rotation axis.

In conclusion, elongated limbs cannot be excluded by our observation, but they require a fast rotating body (period less than $\sim$8 hours) outside the interval R_E=1,165$\pm$90 km (Fig. 4S). Moreover, very elongated bodies, for instance with $R_E\sim$1,500 km, require a simultaneous fine tuning on the limb and the rotation axis orientations, and thus does not appear as plausible. We finally note that if a rotational light curve is eventually detected for Eris, then ϵ would then be determined from Eq. 4 or Eqs. 6-7, and Fig. 4S can be used to pin down the value of R_E.

4.4 Constraints from Dysnomia

The satellite is about 60 times fainter than Eris in K' band[36], and 480 times fainter in V band[12]. The satellite spectrum is thus very different from Eris' (and quite redder), suggesting a significantly darker surface.

We may assess an upper limit for Dysnomia's mass by considering a very dark satellite with a visible albedo of $\sim$0.05. Then its radius would be $R_D \sim$240 km (using an estimation of $R_E\sim$1,150 km for Eris' radius), its projected surface area would be $\sim$4% of Eris', and its mass ratio to the primary would reach $\sim$0.01 (assuming the same densities for the two bodies). Let us first assume that Dysnomia was formed during a giant impact on Eris and tidally migrated in the primary equatorial plane, from the Roche limit ($\sim 2R_E$) to its present orbit (semi-major axis $\sim$37,400 km, eccentricity lower than 0.01 and period 15.77 days[12]) by extracting orbital angular momentum from the primary rotation. With the maximum mass estimated above, Dysnomia could have decreased Eris' spin rate from a primordial rotation period of $\sim$7.5 hours (typical of rotation periods observed among TNOs[37]) to a present period of $\sim$11.5 hours.

At this rate, Eris' flattening would be $\epsilon'\sim$0.041 (Eq. 5). However, as the angle between Dysnomia orbital pole (which coincides with Eris' pole in the present scenario) and the line of sight is close to 50 degrees[12], the apparent flattening of Eris' limb in the plane of the sky would be reduced to about 0.024. Also, Eris' rotation axis position angle would then be $P\sim$10 degrees[12]. Using these values, we can fit the occultation chords with an effective radius of R_{E}=1,143 km, smaller than the value derived from the circular fit (R_{E}=1,163 km).

If we now assume that Dysnomia is much smaller than considered in the extreme case above, but still revolving in Eris' equatorial plane, we still have to fix $P\sim$10 degrees, but we can relax the condition on ϵ'. Using solutions 1 and 2, we then obtain an effective radii R_{E} in the range 1,105–1,155 km, smaller again than the circular fit value.

Finally, if Dysnomia has been captured and migrated inward to its present orbit[38], then no constraints on Eris' rotational period and pole orientation can be derived, and we have to turn back to the discussion developed in the previous subsection.

4.5 Comparison with Pluto's size

Pluto's radius has been estimated using various techniques, ranging from direct imaging to speckle interferometry, mutual Pluto-Charon events (occultations and eclipses), and stellar occultations[8].

Each of those methods is model-dependent and relies on assumptions on physical parameters that must be independently inferred. Pluto's radius cannot be accurately measured using direct imaging, owing to its small angular diameter, about 0.1 arcsec. Pluto's radius was first pinned down using mutual events observed in the late 1980's, with values ranging from 1,151$\pm$4 km (ref. 6) to 1,178$\pm$23 km (ref. 7), depending on assumed Charon's orbital radius and Pluto photometric properties[8].

Pluto stellar occultations cannot reach the usual kilometer accuracy level achieved with airless bodies, as Pluto's atmosphere refracts stellar rays. Actually, during occultations, the stellar image is dimmed but always visible, and in particular, is never blocked by Pluto's opaque limb. However, combining stellar occultation results with Pluto atmospheric models, it is possible to estimate a radius value of 1,152$\pm$32 km (ref. 3), recently improved to $1{,}180^{+20}_{-10}$ km (ref. 5).

Independent results are obtained with high resolution spectroscopy, which provides constraints on the atmospheric CH_4 integrated column density and temperature, yielding an allowed range of 1,159-1,203 km for Pluto's radius, when combined with stellar occultation results[4].

As those values are obtained with different techniques and assumptions, it is not possible to average them in order to reduce error bars. At the present time, we can only say that Pluto's radius safely lies between $\sim$1,150 and 1,200 km. Therefore, it cannot be concluded whether Eris is smaller or bigger than Pluto, until the latter is more precisely measured.

5 Albedo

Eris' geometric albedo p is related to its reflectivity I/F and its radius R_{E} (now assuming a circular limb with R_{E}=1,163$\pm$6 km) through the equations:

$$(I/F)(\alpha) = \Phi(\alpha)\cdot p = \Phi(\alpha)\cdot (AU_{\mathsf{km}}/R_{\mathsf{E}})^2 \cdot 10^{0.4(\mathsf{H}_\odot - \mathsf{H}_{\mathsf{E}})}. \quad (9)$$

Here, $(I/F)(\alpha)$ is the ratio of Eris' surface brightness I observed at phase angle α, to F, the surface brightness of a Lambert surface, where πF is the solar flux density at Eris for the relevant wavelength, $\Phi(\alpha)$ is the phase function (with the convention $\Phi(0)$=1), AU_{km}=1.49598$\times 10^8$ is 1 AU expressed in

km, $H_\odot$ is the Sun apparent magnitude (at 1 AU), and H_E is Eris' absolute magnitude, that would be observed if the object were at 1 AU from the Sun and observed at 1 AU from Earth at exact opposition (α=0). In the visible band, ref. 39 provides $H_{E,V}$=-1.116±0.009. We will use here the more recent values $H_{E,V}$=-1.15$^{+0.05}_{-0.1}$ and $H_{E,R}$=-1.56$^{+0.05}_{-0.1}$ in V and R bands, respectively, where the asymmetric error bars come from the analysis of a possible opposition effect (ref. 40 and I. Belskaya 2011, private communication). Finally, using $H_{\odot,V}$=-26.74 and $H_{\odot,R}$=-27.10, we obtain:

$$\begin{cases} p_V = 0.96^{+0.09}_{-0.04} \\ \\ p_R = 1.01^{+0.10}_{-0.05} \end{cases} \quad (10)$$

Note that the error bars are dominated by the photometric uncertainties attached to $H_{E,V}$ and $H_{E,R}$, not by the uncertainty on R_E. Note also that Eris' phase angle α varies between ∼0.15 and 0.6 degrees as seen from Earth, and that the values of $H_{E,V}$ and $H_{E,R}$ given above assume ordinary opposition effect[40]. If Eris has a strongly peaked opposition surge below 0.15 degrees (which is not known), then we would obtain higher geometric albedos than given in Eqs. 10. More generally, a review of TNO and Centaur photometric properties[41] actually shows that the largest TNOs (Eris, Haumea, Makemake and Pluto) have shallower phase slope when compared with smaller TNOs, possibly associated with active (sublimating atmosphere) surfaces and/or collisional history. However, the lack of data at very small ($\lesssim$0.1 degrees) phase angles prevents conclusions to be drawn about the existence of opposition surges for TNOs. An exception is Varuna, for which a surge has clearly been observed at α∼0.06-0.1 degrees (ref. 41).

The geometric albedos derived here for Eris can be compared to the visual geometric albedos of Enceladus, p_V∼1.4 and Tethys, p_V∼1.2 (ref. 20). However, because those satellites exhibits strong opposition surge of more than 30% for $\alpha\lesssim$0.2 degrees, and because Eris' photometric behavior is not known in this domain, we may instead compare their reflectivity $(I/F)(\alpha)$ in the same range of phase angle. For instance, Eris has I/F∼0.91 near α=0.15 degrees in visual band[40], while Enceladus has I/F∼1.2 in the same conditions[42], about 30% larger than for Eris.

6 Surface temperature

We now use the radius R_E and the geometric albedo p_V derived above, in combination with thermal fluxes at 70 and 1,200 μm (measured by *Spitzer*[22] and *IRAM*[11], respectively), to constrain Eris' surface temperature and global photometric properties. We consider two limiting cases: (1) slow rotation (or equivalently, pole-on orientation, or equivalently, zero thermal inertia), referred to as the 'Standard Thermal Model' (STM) and (2) fast rotation with equator-on geometry and Isothermal Latitude Model (ILM). The first model provides the 'warm' equilibrium situation, where only the illuminated side of Eris is heated by the Sun, while the second model yields the 'cool' situation, where the temperature is constant along a given latitude (on both the lit and unlit sides). Actual models should lie between these extreme cases.

In each model, we use the formulae in Table 4S to calculate the sub-solar temperature T_{ss} for the STM and the equatorial temperature T_{eq} for the ILM. In that table, the angle θ is the local solar zenith angle (STM) and ϕ is the local latitude (ILM). The local temperature T given in the first line is introduced in the integral of the second line in Table 4S. This yields T_{ss} (STM) and T_{eq} (ILM), introducing the emissivity ε (not to be confused with the flattening ϵ defined earlier) and using Planck's law B_ν, once Eris' flux density S_ν at frequency ν and geocentric distance Δ is measured (note that Δ=96.28 UA for the *IRAM* data and Δ=96.41 UA during the *Spitzer* observations). In the calculations, we neglect Dysnomia's contribution to the total thermal flux, as its projected surface area is probably more than 25 times smaller than Eris', see above.

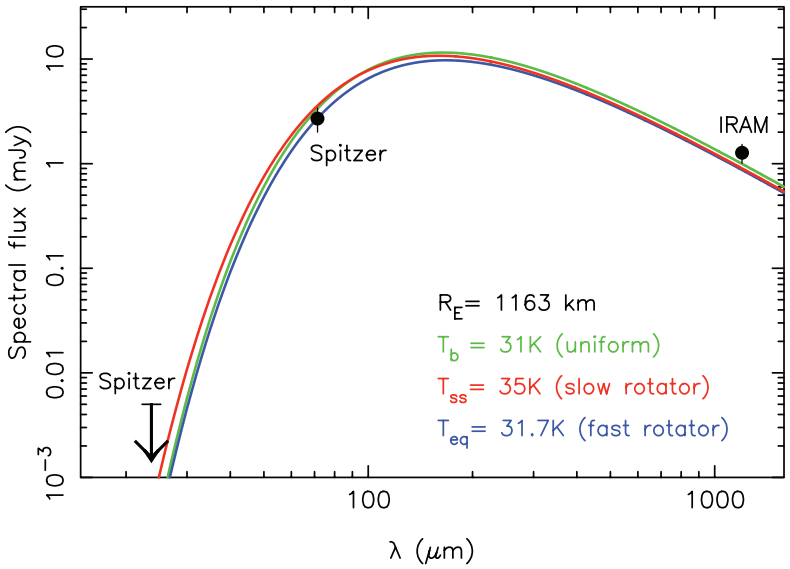


Figure 5S | **Eris thermal models.** The observed Eris' thermal fluxes near 70 and 1,200 μm (see Table 5S and refs. 11,22) in millijansky are compared to various models discussed in the text, assuming a radius R_E=1,163 km and a standard emissivity[22] ε=0.9 for the dwarf planet. Note that only an upper limit is available at 24 μm (arrow). Green: expected flux from a black body with uniform brightness temperature T_b=31 K. Red: expected flux using a Standard Thermal Model (STM), or slow rotator, with a sub-solar temperature T_{ss}=35 K. Blue: expected flux using a Isothermal Latitude Model (ILM), or fast rotator, with an equatorial temperature T_{eq}=31.7 K. All models agree to within 1.5σ or better with the 70 and 1,200 μm measurements, and thus describe satisfactorily the data.

Supplementary Table 4S | **Formulae to calculate temperatures**

slow rotator (STM)	equator-on fast rotator (ILM)
$T = T_{ss} \cdot \cos^{1/4}(\theta)$	$T = T_{eq} \cdot \cos^{1/4}(\phi)$
$S_\nu = \pi\varepsilon \left(\frac{R_E}{\Delta}\right)^2 \cdot \int_0^{\pi/2} \sin(2\theta) B_\nu\left[T(\theta)\right] d\theta$	$S_\nu = 2\varepsilon \left(\frac{R_E}{\Delta}\right)^2 \cdot \int_{-\pi/2}^{\pi/2} \cos(\phi)^2 B_\nu\left[T(\phi)\right] d\phi$
$T_{ss} = T_0 \cdot [(1-p_V q)/(\varepsilon\eta)]^{1/4}$	$T_{eq} = T_0 \cdot [(1-p_V q)/(\pi\varepsilon\eta)]^{1/4}$
$S_\nu = \pi \left(\frac{R_E}{\Delta}\right)^2 \cdot B_\nu\left(T_b\right)$	

The third line of Table 4S provides the thermal equilibrium equations that relate T_{ss} and T_{eq} to Bond's albedo A=$p_V q$, where p_V is the geometric albedo in visible band (where most of the solar flux is emitted), q is the phase integral and η is the beaming factor describing the effects of surface roughness (which enhance the thermal radiation at low phase angles), following the NEATM[43] and Hybrid STM/ILM formalisms[22]. The quantity T_0 is the equilibrium sub-solar temperature of a perfectly diffusing and completely absorbing surface with unit emissivity at Eris distance, T_0=$[F_\odot(r)/\sigma_{SB}]^{1/4}$, where $F_\odot(r)$ is the solar flux integrated over all wavelengths at distance r from the Sun, and σ_{SB}=5.670×10^{-8} W m^{-2} K^{-4} is the Stefan-Boltzmann constant. Using $F_\odot$(1 AU)=1367 W m^{-2}, and r=96.9 AU, we obtain T_0=40.0 K. Solutions in terms of T_{ss} and T_{eq} are found in Table 5S.

Finally, the fourth line of Table 4S relates the flux density S_ν to the disk-averaged brightness temperature T_b of the dwarf planet, i.e. the temperature of a perfect black body with Eris' size that would emit the flux density S_ν toward the observer.

Supplementary Table 5S | Thermal fluxes and temperature constraints for a 1,163 km radius

Wavelength (μm)	Flux (mJy)	Tb (K)[a]	Slow rotator T_{ss} (K)[a]	Fast rotator T_{eq} (K)[a]
23.68	< 0.005 (*Spitzer*)[b]	< 37.2	< 40.2	< 38.6
71.42	2.7±0.7 (*Spitzer*)[b]	30.0±1.2	33.5±1.4	31.7±1.3
1,199	1.27±0.26 (*IRAM*)[c]	38.0±7.5	46.8±9.3	43.6±8.7

[a] The error bars on temperature are dominated by errors on the fluxes, not on our radius determination.
[b] From ref. 22.
[c] From ref. 11.

In the slow rotation model, the 70 μm flux is fit with a sub-solar temperature T_{ss}=33.5±1.4 K, while the 1,200 μm flux gives T_{ss}=46.8±9.3 K. A satisfactory compromise, matching both the *Spitzer* and *IRAM* measurements close to their respective 1σ error bars, is achieved for $T_{ss}\sim$35 K (Fig. 5S). In the fast rotation model, the 70 μm flux implies T_{eq}=31.7±1.3 K, that satisfactorily fits the 70 and 1,200 μm data (Fig. 5S).

Finally, our occultation measurement provides the geometric albedo p_V=$0.96^{+0.09}_{-0.04}$. Adopting a standard value ε=0.9 (ref. 22), the only two free model parameters are η and q in the third line of Table 4S. Using the STM, i.e. $T_{ss}\sim$35 K and a plausible range from η=1 (no roughness) to 0.7 (large surface roughness), leads to q=0.49-0.66, which is fully consistent with Saturn's brightest icy satellites Mimas, Enceladus, Tethys, Dione and Rhea (q = 0.51-0.63), as inferred from their geometric and bolometric albedos[20, 23], see main text.

The ILM in contrast leads to extreme conditions for q. Even at the lower possible value T_{eq}=31.7 K and with 0.9>η>0.7, we obtain 0<q<0.24, an implausible range as bright objects also have large phase integrals[24]. The situation is even worse if we note that the lower bound of 0.7 for η is actually unrealistic in the absence of diurnal temperature variations[44].

7 Limit on atmosphere

Eris' atmosphere, if any, should be composed mainly of nitrogen N_2 (or possibly argon Ar), plus traces of methane CH_4 (ref. 1). However, the atmospheric thermal structure is unknown. If the abundance of methane in Eris' atmosphere is similar to that of Pluto (∼0.5 %, ref. 4), then absorption of near-IR solar radiation by CH_4 may raise the upper atmosphere temperature. In this case, a rough estimate of Eris' upper atmosphere temperature can be simply obtained from the methane 'thermostat-model', in which near-IR heating is balanced through emission in the 7.7 μm methane band[45]. With an incoming solar flux at Eris ten times weaker than at Pluto, this model predicts an Eris' upper atmosphere temperature of 94 K (vs 106 K for Pluto). Then, a 'Pluto-like' thermal profile would start near 30 K at the surface, and ramp up in a few kilometers-thick stratosphere to an isothermal branch at 94 K in the upper atmosphere. Note that the Pluto thermostat-model[45] assumes an optically thin atmosphere. Although more detailed models[46] show this not to be the case, the optically thin approach still provides the correct upper atmosphere temperature, validating the above scaling approach.

However, a 0.5% CH_4 atmospheric abundance is unlikely for Eris. At Pluto, the elevated abundance (compared to expectations based on vapor pressure equilibrium in an ideal N_2-CH_4 mixture) is best interpreted as resulting from the presence of patches of pure, 'warm', methane ice. Those patches are thought to be warmer than nitrogen ice by up to 10-15 K because the N_2 temperature is largely controlled by latent heat flux of a ∼10-μbar atmosphere, large enough to buffer an isothermal N_2 frost

and a spatially uniform atmospheric pressure. This is not the case for Eris, where the atmosphere is presently at least 3 orders of magnitudes more tenuous than on Pluto, so that the local N_2 ice temperature should be driven by local insolation conditions, not by sublimation/condensation processes.

The dominant role of the latent heat flux over thermal flux in establishing the temperature of N_2 can be expressed through the dimensionless parameter $I{=}L\dot{m}/\sigma_{SB}T^4{=}Lf_rP/(c_s\sigma_{SB}T^4)$ where L is the sublimation latent heat, $\dot{m}$ the sublimation mass flux, P the atmospheric pressure, c_s the speed of sound, and f_r is the fractional pressure difference between the saturated and actual atmospheric pressure[47]. For large values of I, sublimation processes dominate the local energy flux from the Sun (insolation) and the ice is forced to be isothermal. With the conditions prevailing at Pluto, one obtains I=5,200, using a typical $f_r{\sim}0.1$ (ref. 47). At Eris, using a maximum pressure of $\sim$5 nbar, and T=30 K, we obtain $I \lesssim 2$, showing that the latent heat flux is at most comparable to the thermal radiation flux from the surface.

Hence, the surface temperature of a N_2 ice is likely to be primarily driven by local insolation conditions. In particular, our predicted sub-solar temperature is at least 32 K (Table 5S), which corresponds to a N_2 equilibrium vapor pressure[48] larger than 260 nbar, much higher than the upper limit derived here. Consequently, a local N_2 atmosphere might exist at Eris' sub-solar point, and that atmosphere could freeze out to undetectable levels (less than a few nbars) at the limb and on the night side, as it is the case for Io[49]. Detailed models will be required to confirm this point.

Note that if there are regions covered by pure methane ice on Eris, they should not be significantly warmer than N_2. The bulk methane abundance on Eris' surface[1], 10%, is similar to the solubility limit of CH_4 in N_2 (ref. 50), suggesting the formation of pure ice grains. Assuming that such pure CH_4 ice grains exist but are at the same temperature as N_2, the atmospheric CH_4/N_2 is given by the ratio of their vapor pressures at $\sim$30 K, i.e. a few times 10^{-5} at most. If pure methane grains are not present and the CH_4/N_2 surface ratio is 10 %, the atmospheric CH_4/N_2 is further decreased by a factor 10.

We conclude that the probable range of methane abundance in Eris' atmosphere is 10^{-5}–10^{-6}. Although detailed models would be needed, this is likely unsufficient to lead to a significant stratosphere, and we favor the case where any Eris' atmosphere is isothermal near $\sim$30 K. More precisely, when considering such atmospheres, we have assumed that the pressure at the surface corresponds to the saturation value at a given temperature[48]. We have generated synthetic light curves (see Fig. 3) caused by a tenuous atmosphere using a ray tracing code, as detailed in ref. 29. Assuming a N_2 isothermal atmosphere near 30 K, we derive an upper limit of about 1 nbar (1σ limit), see Fig. 3. Assuming a Pluto-like atmosphere (30 K near the surface and 94 K in the upper atmosphere), we would obtain upper limits larger by a factor of 3-4 compared to the isothermal model.

We can also place upper limits of about 1 nbar for isothermal pure methane or pure argon atmospheres. In order to sublimate at that pressure[48], the methane ice should be at about 37 K. As our surface temperature estimates are comparable to or smaller than those values, it is normal that a CH_4 atmosphere escaped detection.

Finally, we note that Eris, if indeed covered by N_2 ice, might develop an atmosphere as it approaches perihelion, at 37.8 AU. In that case, the N_2 ice temperature will be buffered by the nitrogen atmosphere, implying a constant temperature on night and day sides. Assuming a maximum Bond albedo of 0.7 (typical of bright objects like Enceladus[23]), we expect a surface equilibrium temperature of at least $\sim$35 K, large enough to maintain a 2 μbar or more Pluto-like atmosphere. This atmosphere might collapse when Eris recedes from perihelion, thus explaining its current bright surface caused by a thin nitrogen-ice coating.

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Apêndice G

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LETTER

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Albedo and atmospheric constraints of dwarf planet Makemake from a stellar occultation

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Pluto and Eris are icy dwarf planets with nearly identical sizes, comparable densities and similar surface compositions as revealed by spectroscopic studies[1,2]. Pluto possesses an atmosphere whereas Eris does not; the difference probably arises from their differing distances from the Sun, and explains their different albedos[3]. Makemake is another icy dwarf planet with a spectrum similar to Eris and Pluto[4], and is currently at a distance to the Sun intermediate between the two. Although Makemake's size (1,420 ± 60 km) and albedo are roughly known[5,6], there has been no constraint on its density and there were expectations that it could have a Pluto-like atmosphere[4,7,8]. Here we report the results from a stellar occultation by Makemake on 2011 April 23. Our preferred solution that fits the occultation chords corresponds to a body with projected axes of 1,430 ± 9 km (1σ) and 1,502 ± 45 km, implying a V-band geometric albedo $p_V = 0.77 \pm 0.03$. This albedo is larger than that of Pluto, but smaller than that of Eris. The disappearances and reappearances of the star were abrupt, showing that Makemake has no global Pluto-like atmosphere at an upper limit of 4–12 nanobar (1σ) for the surface pressure, although a localized atmosphere is possible. A density of 1.7 ± 0.3 g cm^{-3} is inferred from the data.

Stellar occultations allow detection of very tenuous atmospheres and can provide accurate sizes and albedos[9,10,11,3,12], so we embarked on a programme of predicting and observing occultations by (136472) Makemake, also known as 2005 FY_9. The occultation of the faint star NOMAD 1181-0235723 (with magnitude $m_R = 18.22$, where NOMAD is the Naval Observatory Merged Astronomic Dataset) was predicted in 2010 by methods similar to those used to predict occultations by several large bodies[13], but refined as shown in Supplementary Information section 1. We arranged a campaign involving 16 telescopes, listed in Supplementary Table 1. The occultation was successfully recorded from seven telescopes, listed in Table 1, at five sites. From the images obtained, we made photometric measurements as a function of time (light curves).

The light curves of the occultation are shown in Fig. 1. Fitting synthetic square-well models to the light curves yielded the disappearance and reappearance times of the star (Table 1), from which we calculate one chord in the plane of the sky for each site (see Supplementary Information section 3). On the basis of analyses of the light curves, taking into account the cycle time between the images and the dispersion of the data, we deduce that there were no secondary occultations, so we can reject the existence of a satellite larger than about 200 km in diameter in the areas sampled by the chords. The result is consistent with a deep-image survey that did not find any satellites[16]. The chords can be fitted with two shape models (Fig. 2). Our preferred shape, which is compatible with our own and other observations (see Supplementary Information section 8), corresponds to an elliptical object with projected axes of 1,430 ± 9 km and 1,502 ± 45 km. By combining this result with visible photometry at various phase angles[17], we calculated that Makemake has a V-band geometric albedo of $p_V = 0.77 \pm 0.03$ (see Supplementary Information section 4). This is considerably high compared to albedos of other trans-Neptunian objects (TNOs)[5], and is larger than that of Pluto ($p_V = 0.52$)[18] but smaller than that of Eris ($p_V = 0.96$)[3].

The object is large enough to be in hydrostatic equilibrium, so it is possible to use the figures of equilibrium formalism, as done for Haumea[19], to analyse the shape of a body that rotates with Makemake's period of 7.77 h (refs 20, 21). The object could only be a triaxial Jacobi ellipsoid for densities in the range 0.66–0.86 g cm^{-3} (for example, ref. 22). Such low densities are unrealistic for a body as large as Makemake (see Supplementary Fig. 7). Thus, Makemake must be an oblate Maclaurin spheroid for plausible densities between 1.4 and 2.0 g cm^{-3} (see discussion in Supplementary Information section 8).

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Table 1 | Details of the successful observations on 2011 April 23

Site and telescope	Pixel scale (″)	Integration time (s)	Filter name	Dead time (s)	Instrument name/ detector	Immersion time (UT)	Emersion time (UT)	Longitude/Latitude/Height
La Silla, 3.5 m NTT	0.35	0.272	r′	0.0036	ULTRACAM channel r′	1:35:44.59 ± 0.07	1:36:43.51 ± 0.08	289° 15′ 58.5″ E/ 29° 15′ 31.8″ S/2,345.4 m
La Silla, 3.5 m NTT	0.35	0.272	g′	0.0036	ULTRACAM channel g′	1:35:44.64 ± 0.04	1:36:43.66 ± 0.07	289° 15′ 58.5″ E/ 29° 15′ 31.8″ S/2,345.4 m
La Silla, 0.6 m TRAPPIST	0.64	5	Clear	1.435	FLI- PL3041BB	1:35:46.82 ± 1.6	1:36:45.47 ± 1.6	289° 15′ 38.2″ E/ 29°15′ 16.6″ S/2,317.7 m
Paranal, 8 m VLT	0.148	1.521	J	0	ISAAC	1:35:46.00 ± 0.35	1:36:49.60 ± 0.35	289° 35′ 50.1″ E/ 24° 37′ 30.3″ S/2,635 m
Armazones, 0.84 m	0.57	10	Clear	3.5	SBIG-STL6303E	1:35:46.30 ± 1.1	1:36:48.52 ± 3	289° 48′ 13.6″ E/ 24° 35′ 51.9″ S/2,705.7 m
San Pedro de Atacama, 0.5 m Harlingten	1.61	5	Clear	1.048	Apogee U42	1:35:37.86 ± 2.7	1:36:43.56 ± 3.1	291° 49′ 13.0″ E/ 22° 57′ 12.2″ S/2,305 m
San Pedro de Atacama, 0.4 m ASH2	1.21	15	Clear	5.966	SBIG-STL11000	1:35:38.66 ± 4	1:36:41.16 ± 2	291° 49′ 13.0″ E/ 22° 57′ 12.2″ S/2,305 m
Pico Dos Dias, 0.6 m Zeiss	1.98	5	Clear	3.851	SITe SI003AB	1:33:57.27 ± 1.6	1:35:01.08 ± 2.2	314° 25′ 02.5″ E/ 22° 32′ 07.8″ S/1,810 m

Image sequences were obtained with all the telescopes at different image rates and with different dead times, as shown. All the observations were made in the visible, except for the Paranal light curve, which was obtained with ISAAC, a near-infrared instrument[14]. The sequences were started typically 20 min before the nominal occultation time, and finished around 20 min later. The images were bias subtracted and flat-field corrected using calibration frames taken before or after the occultation. From the image sequences, fluxes of the combined light source formed by Makemake and the blended star were obtained. The fluxes were obtained by means of synthetic circular-aperture photometry techniques, and the fluxes of other stars in the field of view were extracted using the DAOPHOT package[15]. The fluxes as a function of time constitute what we call light curves. These were divided by the fluxes of other stars to compensate for transparency fluctuations in the terrestrial atmosphere. The resulting light curves were divided by the average value of the unocculted part of the light curve to obtain a normalized flux. The uncertainties in the fluxes were obtained from the standard deviation of the data outside the occultation drop. The computers that controlled the cameras were all periodically synchronized with UT time servers, except for ULTRACAM at the 3.5-m NTT, the timing of which was directly synchronized by means of a global positioning system that provided a time accuracy better than 1 ms. We tested the accuracy of the timing of the Internet-synchronized computers by checking the error logs. The maximum deviations of the computer clocks were all less than 10 ms. Thus we adopt this value as a conservative estimate of the error in the times of the images.

Thermal measurements indicate that Makemake must have two terrains with very different albedos[5,6,23], and a diameter of 1,420 ± 60 km (ref. 6) if assumed to be spherical. This value is in agreement with, but considerably less precise than, the value of 1,430 ± 9 km determined here under the assumption of spherical shape. One of the terrains in the thermal models must be very dark to explain Makemake's thermal

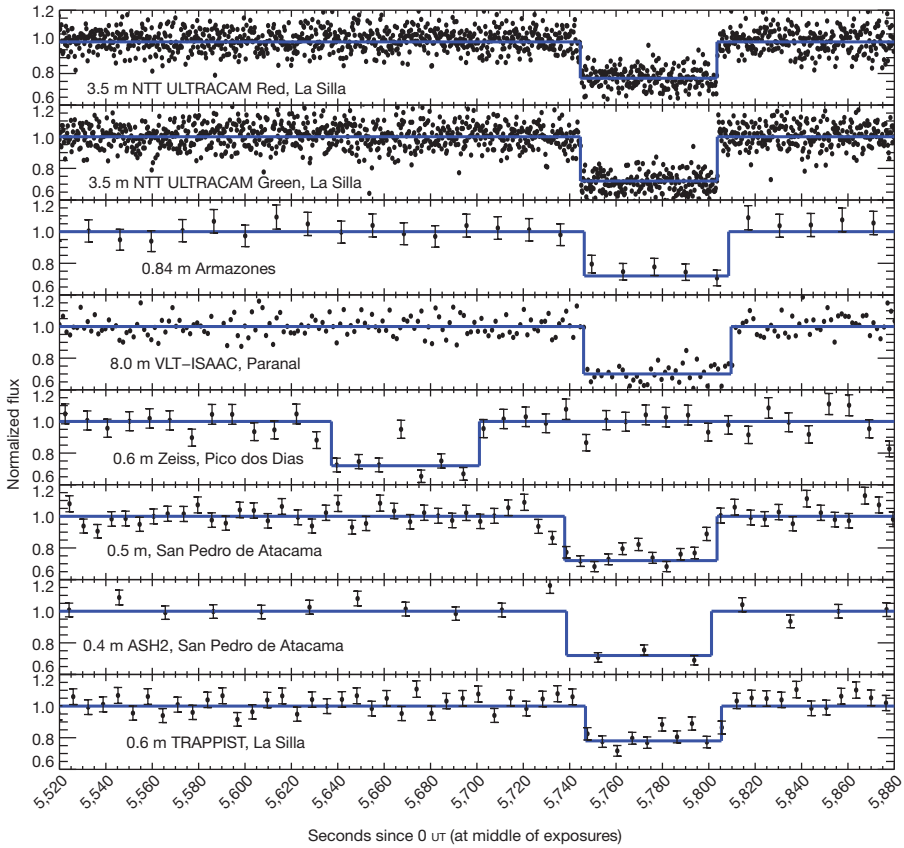


Figure 1 | Light curves of the Makemake event observed from seven telescopes on 2011 April 23. Note that the brightness drop in the Pico dos Dias light curve happens earlier than the rest because the observatory is at a very different longitude (see map in Supplementary Fig. 1). Also note that the ULTRACAM camera provided two channels of useful data (one in the red and the other in the green part of the spectrum). The light curves show the sum of the star and Makemake fluxes, arbitrarily normalized to unity outside the occultation. The R-band star magnitude is about 18.22 according to the NOMAD catalogue, compared with roughly 17.2 for Makemake. Therefore, the expected brightness drop was around 0.35 in normalized flux, as observed. The error bars are 1σ. The NTT and Very Large Telescope Infrared Spectrometer and Array Camera (VLT-ISAAC) light curves are shown without error bars. The blue lines show square-well models that fit the observations, from which the occultation chords of Fig. 2 are obtained. Possible features in the centre of the occultation light curves are analysed in the Supplementary Information. TRAPPIST, Transiting Planets and Planetesimals Small Telescope.

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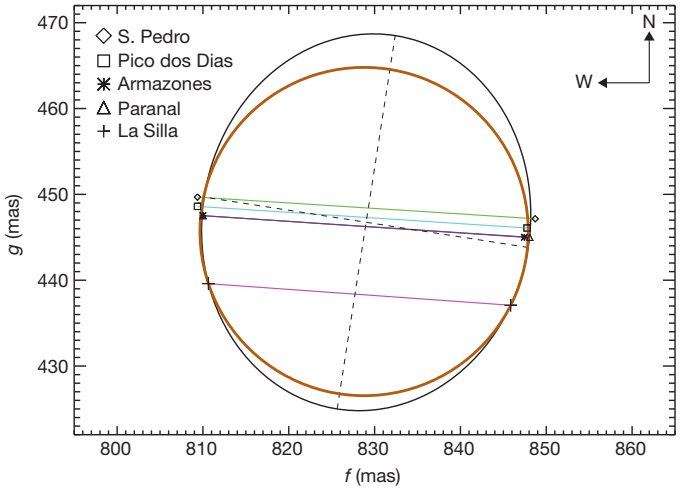


Figure 2 | Occultation chords obtained at five different sites plotted in the projected plane of the sky. The axis marked *g* indicates the north–south direction in the projected plane of the sky; *f* indicates the east–west direction. Units are milliarcseconds (mas). Note that the Paranal and Armazones chords almost overlap. The Paranal, Armazones, Pico dos Dias and San Pedro chords sampled the central part of Makemake. The star disappearance takes place on the left. The chord extremities can be fitted by two different models. The first (pictured in brown) is a circle of diameter 38.28 ± 0.22 mas (1σ level), equivalent to 1,430 ± 9 km, with a reduced χ^2 of 1.032. The second (pictured in black) is an ellipse with a minor axis of 1,428 ± 17 km and an axial ratio of 1.15 ± 0.17, with the long axis tilted by 9 ± 24° (1σ level) with respect to the local celestial north. The reduced χ^2 of the fit is 1.027. The dashed line shows the axes of the best-fitting ellipse. As discussed in the Supplementary Information, the best shape is between the two models. Makemake was 51.5 AU from Earth and 52.21 AU from the Sun at the time of the occultation.

output at 24 μm (ref. 6), which requires a warm terrain on the order of 50 K (see Supplementary Information section 5). The two terrains and Makemake's low rotational variability[20,21,24] can be reconciled if the object is rotating nearly pole-on, if the dark terrain is spread uniformly in longitude (a banded configuration) or by a combination of both conditions.

Makemake is, a priori, a good candidate to have a fully developed atmosphere[4,7,8]. Its albedo and distance from the Sun lie between those of Pluto (which has a global atmosphere) and Eris (which does not, at least currently). Makemake may also have a similar surface composition to Pluto and Eris, on the basis of spectroscopic observations[4]. At the warm temperatures of about 50 K expected from two-terrain thermal models, methane vapour pressure is on the order of a few microbars, whereas nitrogen vapour pressure is around two orders of magnitude higher (as illustrated in Fig. 32 of a work on vapour pressures[25]).

However, a global Pluto-like atmosphere is ruled out by our occultation light curves, which have abrupt ingress and egress profiles (Fig. 1). To get an upper limit on a global atmosphere, it is possible to model its effects on occultation profiles and compare them with observations. The profiles from the New Technology Telescope (NTT) imply an upper limit to the surface pressure of a putative methane atmosphere of only 4–12 nbar at a 1σ confidence level, and 20–100 nbar at the 3σ level (see Fig. 3 and Supplementary Information section 6 for a description of the models, which also consider nitrogen).

One possibility that might explain the lack of a global atmosphere is that Makemake has little or no N_2 ice, because N_2 vapour pressure is well above the microbar level even on the cooler terrain. From an update of the results of the models on retention of volatiles[8], considering new empirical determinations of the vapour pressures of N_2 and CH_4 (ref. 25), Makemake would not have retained N_2 if it were smaller than 1,370 km, which we rule out. With a diameter of 1,430 km, Makemake would have to have a density of less than 1.7 g cm^{-3}, smaller than the adopted nominal value of 1.8 g cm^{-3} (ref. 8), to result in complete N_2 loss. Considering now that CH_4 is abundant on the surface of Makemake, again from the volatile-retention arguments, its density would have to be greater than 1.4 g cm^{-3}. Other constraints on the density based on the observed shape and the figures of equilibrium are discussed in Supplementary Information section 8. Another

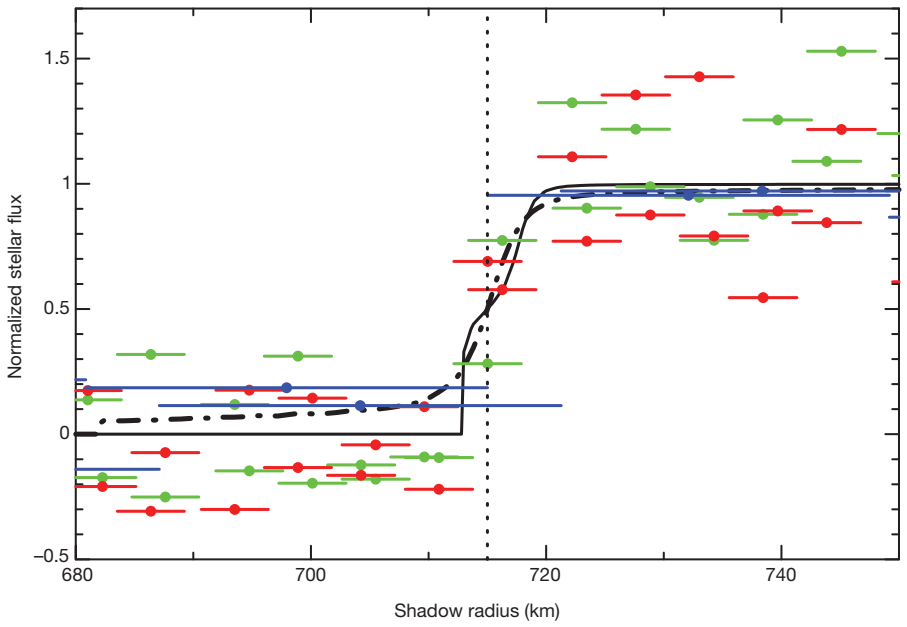


Figure 3 | Observed and synthetic light curves. A comparison of two CH_4 pure atmosphere models with data (ingress and egress profiles: green, NTT g′ points; red, NTT r′ points; blue, VLT J-band) plotted against the distance to Makemake's shadow centre, assuming a circular limb for simplicity. Bars are the radius intervals corresponding to each integration bin. For better reading, and in contrast to Fig. 1, the fluxes have been normalized between zero (average value of the flux during the occultation) and unity (full stellar flux). The models correspond to a CH_4 atmosphere with a surface temperature of 30 K, a near-surface temperature gradient of 17 K km^{-1} followed by an isothermal profile with $T = 100$ K for higher altitudes. Solid line: surface pressure of $P_{surf} = 8$ nanobar, compatible with the data at 1σ. Dash-dotted line: model with $P_{surf} = 100$ nanobar (compatible with the data at 3σ). See Supplementary Information for a full description of the models.

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possibility to explain the lack of a global atmosphere is a nearly pole-on orientation. From a theoretical study[7], TNOs with high obliquity are less likely to have globally distributed atmospheres.

The remarkably high albedo of Eris ($p_V = 0.96$) is thought to be the result of a collapsed atmosphere, which coated Eris with bright, fresh ices[3,26]. A fully condensed atmosphere on Makemake might have resulted in an albedo similar to that of Eris, which is not the case. However, if Makemake had a local rather than a global atmosphere, some parts of the surface could be fully covered with fresh ice from the collapsed part of the atmosphere and be very bright, and others could remain dark. The overall albedo of Makemake could thus be smaller than that of Eris, but larger than that of Pluto. A local atmosphere would also provide a reason for the two-terrain models needed to explain Makemake's thermal data.

Local atmospheres on TNOs are theoretically plausible[7]; they can be confined to a subsolar region or a band at the subsolar latitude. It should be noted that a small drop of only 10 K in surface temperature implies a decrease of three orders of magnitude in the vapour pressure of CH_4 and N_2 at low temperatures.

We can investigate whether the presence of a local atmosphere is consistent with our data. The bottom of the occultation light curves should be flat in an airless body. Flashes in occultations are known to be caused by the focusing effect of an atmosphere when the observer passes near the centre of the shadow[27]. Thus, the noise level of the light curves at their bottoms can put limits on the local atmosphere that can extend to the limbs. Modelling of central flashes for plausible local atmospheres shows that an atmosphere with surface pressure on the order of several microbars can exist and still be consistent with the data, provided that the atmosphere is confined to specific parts of the limb (see Supplementary Information section 7).

From the information gathered on Pluto, Eris and now Makemake using stellar occultations, we hypothesize that the albedos and other surface properties of the largest TNOs are determined by sublimation and condensation processes. In our picture, the largest albedos would result from atmospheres that have fully condensed (collapsed onto the surface), whereas medium-albedo objects would have local atmospheres and the lower-albedo objects would have global atmospheres from sublimation of the volatiles. Future studies will shed light on this possibility and whether sublimation is fully solar driven or is also driven by other mechanisms. The airborne Stratospheric Observatory For Infrared Astronomy, in combination with large aperture telescopes on the ground, would be an excellent tool for this kind of study.

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Supplementary Information is available in the online version of the paper.

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Author Contributions J.L.O. helped to plan the campaign, analysed data for the prediction, made the prediction, participated in the observations, obtained and analysed data, interpreted data and wrote part of the paper. B.S. helped to plan the campaign, participated in the observations, analysed data, interpreted data, wrote and ran the diffraction and ray-tracing codes and wrote part of the paper. F.B.-R. and A.A.-C. helped to plan the campaign, participated in the observations and analysed and interpreted data. E.L. analysed the implications of the results for Makemake's thermal model and putative atmospheric structure and wrote part of the paper. R.D. and V.D.I. helped to plan the observations and analysed data. J.I.B.C., S.P.L., E.U.-S., J.P.C., E.J. and J.M. participated in the observations and analysed data. M.A., F.B.-R., J.I.B.C., R.V.-M., D.N.d.S.N. and R.B. discovered the star candidate and analysed data. P.J.G. and T.M. made thermal models and participated in the interpretation. All other authors participated in the planning of the campaign and/or the observations and/or the interpretations. All authors were given the opportunity to review the results and comment on the manuscript.

Author Information Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Correspondence and requests for materials should be addressed to J.L.O. (ortiz@iaa.es).

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G.1 Suplementos

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1. ASTROMETRIC MEASUREMENTS AND PREDICTION OF THE SHADOW PATH ON EARTH.

For Makemake there are typically 3 potential occultations per year[13], but most of them are too challenging or impossible to observe: some of them involve too faint stars, others take place at too high airmasses or in daylight, or in remote areas of the world where deploying a network of telescopes is too difficult. Besides, the initial shadow path predictions have typical errors of a few thousand km so updates are necessary. Thus, potentially good occultations are rare. This situation may change in the future when high accuracy astrometric data from the GAIA space mission become available, but currently most occultations are difficult to observe and uncertain. The occultation of the faint star NOMAD 1181-0235723 by Makemake was initially predicted in 2010 by following the methods described in a recent work[13]. The star magnitude was 18.24 and 18.55 in the r and g bands respectively, according to the 11th release of the SDSS photometric catalog. The initial occultation prediction was not favorable: it indicated that Makemake's shadow would miss the Earth by about 1000 km on April 23rd at approximately 1:30 UT. Astrometric refinement of the star's and Makemake's positions from CCD images obtained days prior to the occultation showed that the shadow path of the occultation would pass over South America (Fig. S1). A total of 108 astrometric measurements were obtained during 5 nights from images acquired at the f/3 0.77m telescope of La Hita observatory (Spain) in the period April 8th to April 12th 2011. The telescope is equipped with a 4kx4k camera that provided a field of view of 48.1'x 48.1' with a 0.705 arcsec/pixel scale. The observing technique and reductions were identical to those used to obtain high accuracy astrometry of Orcus[28]. The J2000 coordinates of the star determined from the observations were:

Right Ascension (RA): 12h 36m 11.4018s

Declination (Dec): +28° 11' 10.448''

The dispersion in the astrometry of the star was 0.13 arcsec in RA and 0.06 arcsec in Dec. For comparison, Makemake's angular diameter is 0.038 arcsec. From the Makemake astrometry, offsets of Makemake relative to JPL ephemerides JPL#24 were obtained. They were applied to the star coordinates and the shadow path prediction was computed using the Makemake JPL orbital elements for the time of the occultation. The prediction turned out to be off by approximately -3 minutes of time compared to the occultation observations and the shadow path was approximately 300 km south of the final path, shown in Fig. S1.

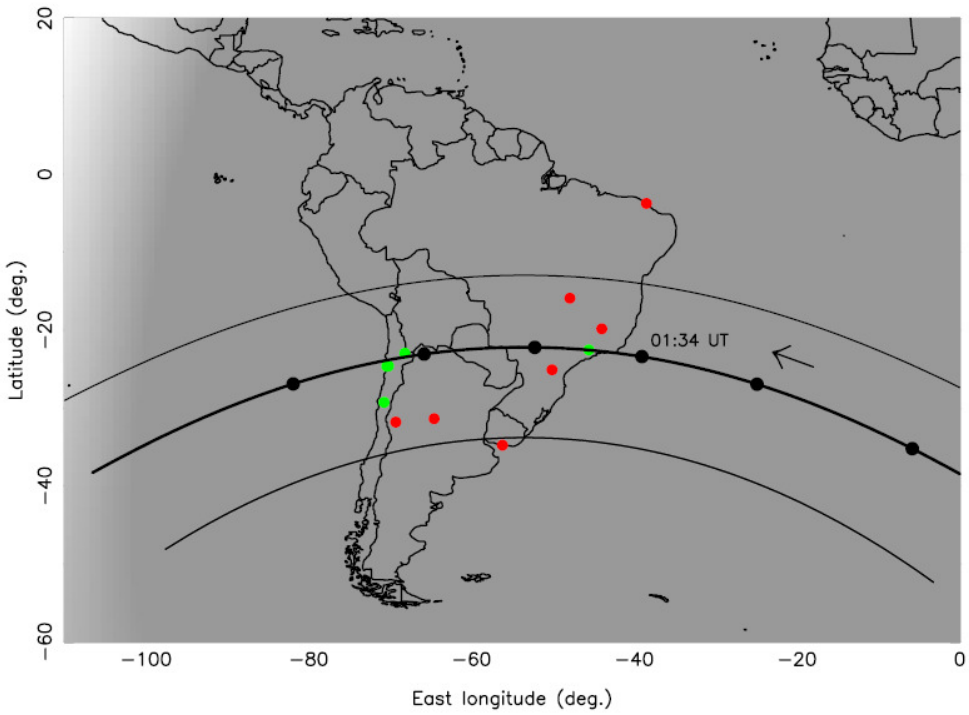


Fig. S1. Makemake's shadow path on Earth based on the occultation results. The width of the shadow path corresponds to an assumed diameter of 1,430 km. The dots on the centreline are plotted every minute. A reference dot is indicated at 01:34 UT, April 23, 2011, and the arrow indicates the direction of motion of the shadow. The locations listed in Tables S1 are plotted in green for the sites where the occultation was detected, and in red for stations with clouds or technical problems. The green dot on the right corresponds to Pico dos Dias observatory where the event was observed about 100 s before the Chile stations.

2. LIST OF ALL THE TELESCOPES THAT OBSERVED OR ATTEMPTED TO OBSERVE THE OCCULTATION.

Table S1. Sites and telescopes that observed the event. In the comments column we include the average signal to noise ratio (SNR) per data point, outside the occultation, for those sites that had a positive result. Note that the integration times were different for each telescope, which explains the various SNR values reached. The synthetic aperture radii used were 1.05, 2.6, 0.6, 2.9, 3.2, 2.4, and 4.0 arcsecs for NTT, TRAPPIST, VLT, Armazones, S. Pedro de Atacama Harlingten, S. Pedro de Atacama ASH2 and Pico dos Dias, respectively. These radii typically corresponded to 2 pixels.

Site	Telescope	Longitude Latitude Height	Results	Comment
La Silla	3.5m NTT+ULTRACAM	70°44'01.5"W, 29°15'31.8"S, 2345.4 m	Positive	SNR=15
La Silla	0.6m TRAPPIST	70°44'21.7"W, 29°15'16.6"S, 2317.7 m	Positive	SNR=22
Paranal	8m VLT+ISAAC	70°24'09.9"W, 24°37'30.3"S, 2635m	Positive	SNR=15
Paranal	8m VLT+FORS2	70°24'09.9"W, 24°37'30.3"S, 2635 m	Negative	Image acquisition started too late
Armazones	0.84m OCA	70°11'46.4"W, 24°35'51.9"S, 2705.7 m	Positive	SNR=25
San Pedro de Atacama	0.5m Harlingten	68°10'47.0"W, 22°57'12.2"S, 2305 m	Positive	SNR=21
San Pedro de Atacama	0.4m ASH2	68°10'47.0"W, 22°57'12.2"S, 2305 m	Positive	SNR=23
Pico dos Dias	0.6m Carl Zeiss	45°34'57.5"W, 22°32'07.8"S, 1810 m	Positive	SNR=15
Pico dos Dias	0.6m Boller & Chivens	45°34'57.5"W, 22°32'07.8"S,	Negative	Instrument problems

		1810 m		
Cerro Burek	0.6m Helen Sawyer Hogg	69°18'25.9"W, 31°47'14.5"S, 2591 m	Negative	Technical problems
Los Molinos	0.45m	56°11'24.6"W, 34°45'19.3S, 80m	Negative	Clouds
Tanti – Córdoba	0.36m	64º35'34.41"W, 31º21'24.58"S, 862 m	Negative	Technical problems
Fortaleza	0.3m	38º30'27"W, 3º44'18"S, 38 m	Negative	Clouds
CEAMIG-REA	0.3m	43º59'51"W, 19º49'49"S, 825 m	Negative	Clouds
Brasilia	0.25m	47º54'40.6"W, 15º53'29.1"S, 1072 m	Negative	Clouds
Ponta Grossa	0.4m	50º05'56"W, 25º05'22"S, 909 m	Negative	Clouds

3. CHORDS AND THEIR ANALYSIS.

Square well models were fit to the occultation light curves by minimizing a χ^2 defined as:

$$\chi^2 = \sum_{1}^{N} \frac{(F_{\mathrm{mod},i} - F_i)^2}{\sigma_i^2} \qquad (1)$$

where $F_{mod,i}$ is the modeled flux at image number i, F_i is the observed flux at image number i and σ_i are the measurement errors, and N is the number of images. The model took into account diffraction, bandwidth, stellar angular diameter and integration time. The effect of time errors as small as 10 ms and 1 ms were entirely negligible compared to the errors in the fluxes (keeping in mind that the ratios of integration times to time uncertainties were always larger than 1,000) so they were not taken into account. The main parameters of the model are the depth of the square well and the disappearance time as well as the reappearance time of the

occultation. The uncertainties in the retrieved parameters were obtained from a grid search in the parameter space. Acceptable values were those that gave a χ^2 within $\chi^2_{\min}$ and $\chi^2_{\min}+1$

From the fitted times at the different sites, one can generate chords in the projected plane of the sky, which represent the motion (in milli-arcsecs) of the star relative to the TNO as seen from the different sites. One can fit a shape to the extremities of the chords. We fitted elliptical and circular shapes by minimizing a χ^2 function defined as follows:

$$\chi^2 = \sum_1^N \frac{(f_{i,obs} - f_c)^2}{a^2\sigma_{i,r}^2} + \frac{(g_{i,obs} - g_c)^2}{b^2\sigma_{i,r}^2} \quad (2)$$

where f_i and g_i are the positions of the star in the plane of the sky relative to Makemake as seen from each observing site at ingress and egress times, *a* and *b* are the semiaxes of the ellipse (in a circular fit both axes are equal). In this case $\sigma_{i,r}$ are the errors of the extremities, which were derived from the errors in the retrieved ingress and egress times and N is the number of extremities. Last, f_c and g_c are the centers of the ellipse in the f,g plane. The best fit was obtained by means of a Marquardt-Levenberg minimization routine. The retrieved a,b parameters in milli-arcsecs (mas) were translated into length in km by using the known distance of Makemake from Earth, according to its ephemerides (which gave 37.35 km/mas). The velocity of Makemake with respect to the star was 22.12 km/s. The 1σ uncertainties in the parameters were obtained from the Marquardt-Levenberg fitting routine.

4. GEOMETRIC ALBEDO COMPUTATIONS.

Once the apparent area (A) of Makemake was obtained from the limb fit to the occultation chords, we used the expression that relates the geometric albedo (p_V), absolute magnitude (H_{obj}), magnitude of the sun (H_{sun}) and the projected area of a solar system body:

$$p_v = \frac{10^{0.4(H_{sun} - H_{obj})}}{(A/\pi)} \quad (3)$$

to derive p_V. We should take into account that the area A in the above equation is expressed in square astronomical units (AU), with 1 AU = 149598000 km. We used an absolute magnitude H_V = 0.091±0.015 from ref. 17 and the usual solar magnitude $H_{sun,V}$ = -26.74. The resulting geometric albedo in the V band is 0.77±0.03 for an object with projected axes a'=1,502±45 km, and b'=1,430±9 km. The main uncertainty in the albedo comes from the error in the projected area

(πa'b'). The geometric albedo in the R band was obtained from the V-R color of Makemake, implying that H_R = -0.395±0.02. Using that result for H_R and a value of $H_{sun,R}$ = -27.08 we obtained the following geometric albedo in the red p_R = 0.90±0.02.

5. THERMAL MODELS OF MAKEMAKE AND EQUILIBRIUM TEMPERATURES.

The equilibrium temperature of an airless spherical body can be computed by using a standard thermal model

$$T_{eq}^4 = \frac{F_{sun}(1 - Alb)}{\sigma \varepsilon \eta \zeta \, r^2} \qquad (4)$$

where *Alb* is the bond albedo, F_{sun} = 1,360 W m^{-2} is the solar constant (Solar Flux at 1 AU), σ = 5.67x10^{-8} W m^{-2} K^{-4} is the Stefan-Boltzmann constant, and the emissivity (ϵ) accounts for the deviation of pure blackbody emission (usually assumed to be 0.9 based on asteroid studies). The beaming parameter (η) takes into account some anisotropic emission effects as a result of surface roughness and thermal inertia and is usually thought to be around 1.2 for TNOs (ref. 5). The variable r is the heliocentric distance of the body in astronomical units and ζ is a parameter that takes the value of π if the object is a fast rotator or 2 if it is a slow rotator. If we want to know the temperature at the subsolar point, this parameter must be set to 1. On the other hand Alb = p_Vq, where p_V is the geometric albedo of the object, and q is the phase integral. This parameter is probably close to 0.75 for Makemake: There is an empirical relation between geometric albedo and phase integral based on TNO observations[29] and also based on observations of the satellites of the outer planets[30]. From those papers almost identical predictions for the phase integral of Makemake are obtained, around 0.75.

Using a value of 0.77 for the geometric albedo, a phase integral of 0.75, an emissivity of 0.9, a beaming parameter of 1.2, and the distance at which the occultation by Makemake took place (52.21 AU) we get an equilibrium temperature for Makemake of only 36 K and 32 K (for the slow rotator and fast rotator cases respectively). However, we know that 0.77 is the average geometric albedo, and in fact we know that Makemake must have a dark terrain to account for the Spitzer and Herschel measurements[5,6,23]. The dark terrain must have a geometric albedo of 0.12 or lower. Using the thermal model with this input for the geometric albedo, we get a temperature of 44 K for the slow rotator case and 40 K for the fast rotator case. The result is 52 K at the subsolar point or even higher if we use a lower than 0.12 geometric albedo (see Fig. S4b). This temperature is high enough for substantial nitrogen and even methane ice sublimation.

6. LIMITS ON A GLOBAL ATMOSPHERE FROM THE NTT AND VLT INGRESS AND EGRESS PROFILES.

Makemake has an icy surface with CH_4 ice (e.g. ref. 4) and perhaps N_2 ice. An atmosphere may then be maintained through vapour pressure equilibrium between the ice and its gaseous counterpart. The ice temperature T_{surf} controls the atmospheric surface pressure P_{surf} through sublimation processes, and for pure ices, we may use laboratory measurements to relate T_{surf} and P_{surf} (ref. 25). More complicated laws apply when the ices are mixed but we use here the approximation that a pure ice (either CH_4 or N_2) is in vapour equilibrium with its gaseous counterpart. The occultation probes Makemake's limb, where the temperature possibly varies from 35 K down to 10 K (see Fig. S4 where we show an example). We examine the possibilities of a pure CH_4 atmosphere, an N_2 atmosphere with a small amount of CH_4, and a pure N_2 atmosphere.

6.1 Pure CH_4 atmosphere

A pure CH_4 atmosphere should start at the surface with a temperature between 10 and 35 K, but near-IR heating should increase this temperature within a few kilometers, to reach an isothermal branch, as observed in Pluto's atmosphere[31,32]. Considering that Pluto's upper atmosphere is at about 106 K, and that the solar insolation at Makemake is reduced by a factor of about 2.8 with respect to Pluto, a rough estimation of a CH_4 atmosphere temperature at altitude is about 100 K. For the thermal gradient near the surface, we adopt a value of 17 K km^{-1}. This high value is typical of what is expected for methane-rich atmospheres[31,32].

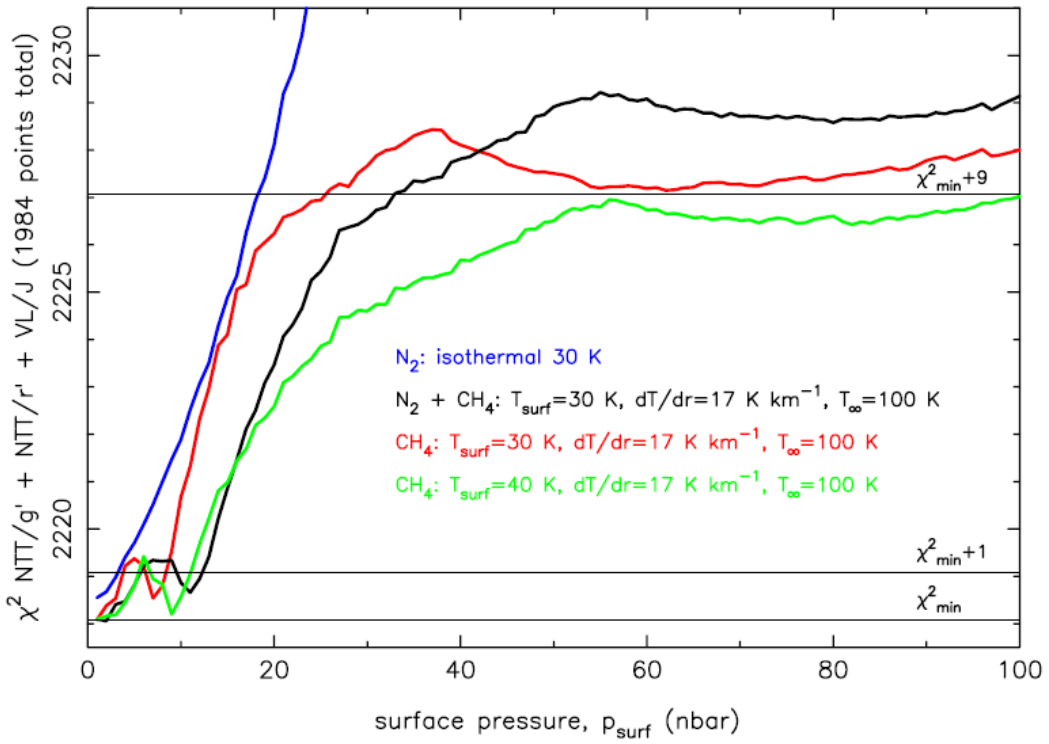


Fig. S2: χ^2 values obtained by varying the surface pressure P_{surf} of the four CH_4 and N_2 atmospheres described in the text. The surface temperature, temperature gradient and the upper level temperature are indicated for each of the three curves. The red line corresponds to a pure CH_4 atmosphere, the blue line is for a pure N_2 atmosphere and the black line corresponds to an N_2 atmosphere with some CH_4 to generate a temperature gradient. The green line is the same as the red line but for a higher surface temperature. Note that most of the constraints come from the NTT data since the VLT data have a lower time resolution.

A ray-tracing program provides the expected ingress and egress occultation light curves caused by such an atmosphere, keeping the surface pressure P_{surf} as a free parameter. We have adjusted this model to the NTT data points (g' and r' channel), as well as to the VLT observations (J-band). Only the NTT and VLT data have been used because they present the highest signal to noise and have good enough time resolution. A standard χ^2 analysis provides the goodness of the fits. We used N = 1984 data points in total, obtaining a minimum χ^2 value of χ^2_{min}= 2218 for P_{surf} = 0 nbar, which corresponds to a value of χ^2_{pdf} = 1.1 per degree of freedom (red curve in Fig. S2). Thus, the data are consistent with a complete absence of atmosphere, at least at the points of the limb probed by the occultation. The increase of χ^2 to χ^2_{min} + 1 actually provides a 1σ upper limit of about 8 nbar for

the surface pressure of the putative CH_4 atmosphere considered here. If one relaxes the 1σ limit to a 3σ level (increase of χ^2 to χ^2_{min} + 9), we obtain a limit of about 25 nbar. Taking a surface temperature of 40 K, instead of 30 K, and everything equal besides, we obtain a 1σ limit of 11 nbar for P_{surf}, and no stringent 3σ limit up to values of 100 nbar or more (Fig. S2). This is because the deep, denser, part of the atmosphere causes a shallow bottom in the occultation light curve that becomes undistinguishable from the noise in the data (see Fig. 3).

In conclusion, a 1σ level upper limit of about 10 nbar can be placed for the surface pressure of a CH_4 atmosphere, but the data could accommodate denser atmospheres (up to 100 nbar or more) if a 3σ level is considered, or if surface temperatures larger than 30 K are considered. This said, 10 nbar corresponds the equilibrium vapour pressure of CH_4 for a temperature of about 40 K (ref. 25). This is larger than the temperature expected along Makemake's limb, and is thus consistent with our non-detection of a global CH_4 atmosphere at the 10 nbar limit.

6.2 N_2 atmosphere

A putative nitrogen atmosphere is likely to be mixed with CH_4, considering the detection of the latter species on Makemake's surface (e.g. ref. 4). Thus, the same general thermal structure is expected as for CH_4, namely an atmosphere starting from a cold surface, and then reaching a warmer isothermal branch after a few kilometers. In that case, an upper limit of P_{surf} = 12 nbar can be derived at the 1σ level, comparable to the pure CH_4 atmosphere case (black curve in Fig. S2), and P_{surf} of about 35 nbar at the 3σ level. The P_{surf} = 12 nbar value would correspond to a vapour pressure equilibrium with a N_2 surface ice at 29 K, consistent with our starting assumption of T_{surf} = 30 K. The possibility still exists, however, that Makemake's atmosphere would be essentially pure N_2, since this gas is much more volatile than CH_4 by about 5 orders of magnitude. In that case, a 1σ upper limit for P_{surf} is about 4 nbar, and the corresponding 3σ limit would be about 20 nbar (blue curve in Fig. S2).

In conclusion, the various atmospheric models considered here place 1σ level upper limits of 4-12 nbar for the surface pressure, while less stringents upper limits in the range 20-100 nbar are obtained at the 3σ level. We note also that we have used a large value of the thermal gradient near the surface (17 K km^{-1}). Lower values of that gradient would provide more stringent limits on P_{surf}. We have actually examined here the less favorable cases, where a thin inversion layer close to the surface remains unnoticed due to the shape of the light curve and the noise in the data (see the dash-dotted curve in Fig. 3)

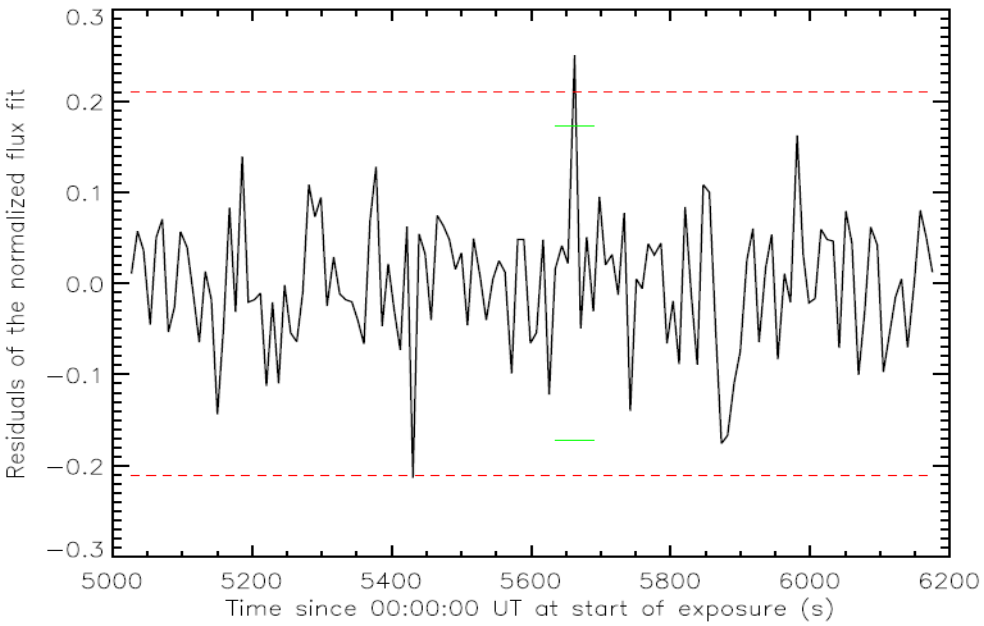
7. DISCUSSION ON A LOCAL ATMOSPHERE FROM FEATURES IN THE LIGHT CURVES.

From the initial inspection of the light curves we realized that there were apparently some signals at the bottom of a few light curves, which might be indicative of refracted light by a local atmosphere somewhere on the limb of Makemake. Because the features were seen in different sites whose chords were nearly central to Makemake, they did not seem a mere coincidence and required closer scrutiny.

We inspected the residuals of a square well fit to the occultation light curve of the S. Pedro de Atacama 0.5m telescope. There is a brightness increase near 5,760s UT, at about the middle of the occultation, when a central flash would be expected. The significance level of this potential spike is 3.1σ, but because the statistics are usually not Gaussian we do not consider this a firm detection.

In Fig. S3 we show the residuals of a square-well fit to the occultation light curve from Pico dos Dias. In this case there is a clear brightening event that also takes place at the middle of the occultation phase and is well above the noise, at a safe 4.8σ significance level. But contrary to the S. Pedro de Atacama 0.5m telescope data, the event takes place in a single image. Visual inspection of the image revealed a bright single pixel slightly away from the core of the point spread function, which raises the suspicion that this might be a cosmic ray hit or a single pixel event of some sort in the detector. If the flux from this pixel is removed from the total flux computation, the light curve spike disappears. Nevertheless we computed the likelihood of such an event by searching for similar single pixel events of the required brightness in the image sequence. It turns out that there were only 24 events with a similar flux level (within 20%) or higher than the alleged single pixel event, in 200 images. We computed the likelihood that such an event took place within a 4x4-pixel box (which would contain Makemake) at the precise image in the middle of the occultation. Given that the images were 50x72 pixels in size, the probability would be only 0.05%. There are alternative explanations to a cosmic ray hit: Because there was image shift of a few pixels in previous images (due to wind, telescope jitter or seeing), this could have also occurred during the acquisition of the image. Hence a very brief central flash while the telescope was shaking might have produced a point source displaced from the core of the point spread function and no smearing of the stars. This seems feasible because the pixel scale was very coarse (1.98 arcsec/pixel), but this would require a very brief and intense central flash (much shorter than the integration time). The seeing in the 5-s images was typically 2 to 2.5 arcsecs (full width at half maximum).

a)



b)

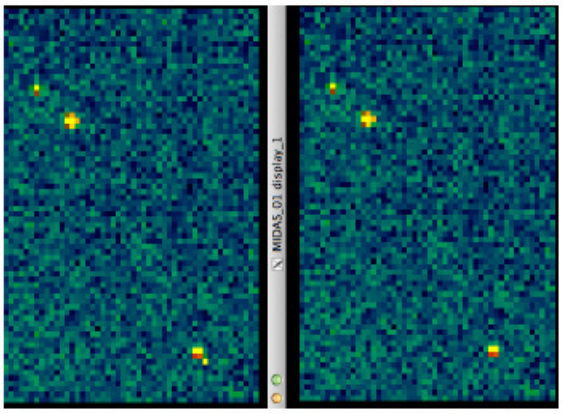


Fig. S3. a) Black solid line: Residuals of a square well fit to the Pico dos Dias occultation light curve. The brightening at around 5,660s after 0 UT is a potential central flash as it takes place in the middle of the occultation time. The brightness increase is well above the noise. By far, there is no comparable brightening in the whole observing sequence. However, this may have been produced by a cosmic ray or an image artifact (see text). The red dashed lines mark the 3σ noise levels of the light curve outside the occultation. The green solid segments represent the expected 3σ noise levels at the occultation. b) Left image: CCD image (in false color) at the time of the brightening event. The plate scale is 1.98 arcsec/pixel. Makemake is in the lower right part of the image. There is an apparent cosmic ray or image artifact close to the core of the point spread function, but slightly

separated from it. The image size is 50x72 pixels. Right panel: Same as left image but with the potential artifact removed by interpolation.

In addition to the above results there is also a very marginal indication that light refracted somewhere on the limb might have reached the detector at Paranal, because there is a slight departure from a flat bottom of the curve. However, this feature in the light curve is not significant enough. It is at just 1.8 σ.

Regardless of whether the features discussed are significant or not, local atmospheres are plausible in TNOs[7], especially in Makemake, given that it has a fraction of its surface covered with a dark terrain[5,6,23] where high enough temperatures are reached and can sustain at least sublimation driven local atmospheres. Hence it makes sense to explore what kind of a local atmosphere would be consistent with the data and to see what limits we can put on such a local atmosphere.

7.1 MODELLING OF THE REFRACTED LIGHT FROM A LOCAL ATMOSPHERE

We considered a pure CH_4 atmosphere localized above patches of warmer methane ice, assuming that the subsolar point temperature reaches 50-55 K as discussed in section 5. A sketch of the location of a possible local atmosphere is show in Fig. S4. Note this is an example of a localized atmosphere, and that other configurations with other orientations and shapes could be examined. The aim of this model is to show that our data are compatible with the existence of a local atmosphere, in spite of the non-detection of a global one.

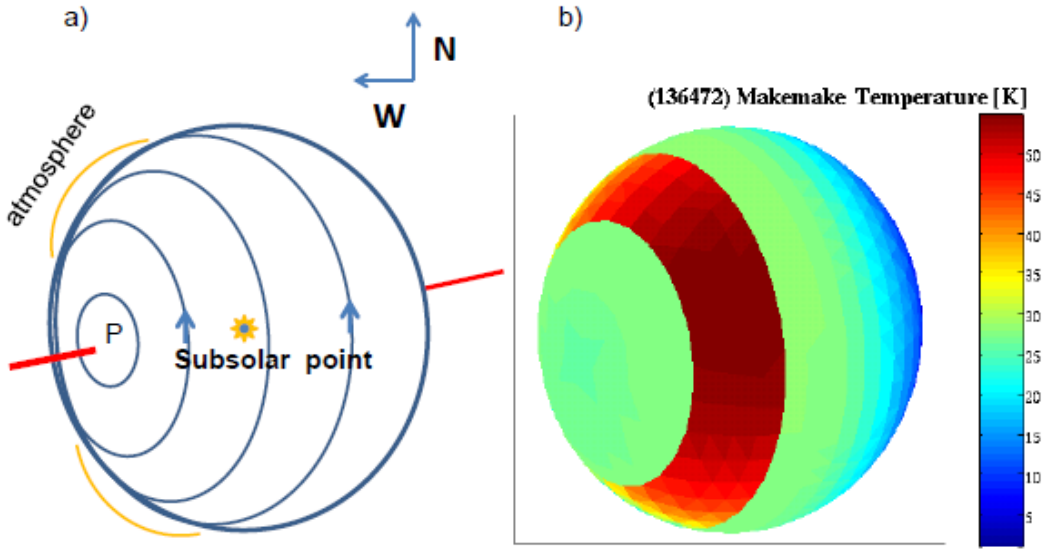


Fig. S4. a) Sketch of where a local atmosphere could roughly reside in Makemake's limb while remaining undetected at the ingresses and egresses of our light curves. A similar configuration with both the local atmosphere and the pole in the East would be valid too. The arrows indicate the assumed direction of rotation of the body and P indicates where the pole might approximately be. In this sketch an atmosphere is formed at the subsolar point and extends to the limb where it can refract light and cause effects in the occultation light curves. The atmosphere would collapse at higher latitudes and might also collapse in the night side where the temperatures are too low to maintain a sublimation driven atmosphere of a few microbar. Such an atmosphere might have a meridional flow to the poles, where the volatiles can condense and form large fresh ice caps of very high albedo. The low albedo terrain is confined to a longitudinal band near the latitude of the subsolar point. b) Thermal map of Makemake based on a thermophysical model[23].

Assuming this model, since the ingresses and egresses in all the light curves sampled areas of the body that were close to its poles, where the atmosphere should be almost collapsed (because of the considerably lower surface temperatures there, which would prevent ice sublimation), no atmosphere would be detected in the ingresses and egresses of our light curves, but a partial atmosphere could exist near the subsolar latitude and extend to the limb away from the extremities of the occultation chords (see Fig. S4). Two slightly different shapes for Makemake were used, one to explain the light curves under the assumption that the spike in the centre of the Pico dos Dias lightcurve is an artifact (and is removed from the light curve), and the other shape can explain the Pico dos Dias original light curve, assuming that the flash is real. The shapes are illustrated in Fig. S5. Both shapes fit the occultation chords. The shape in the right panel provides a slightly lower goodness of fit, but not significant from a statistical point of view.

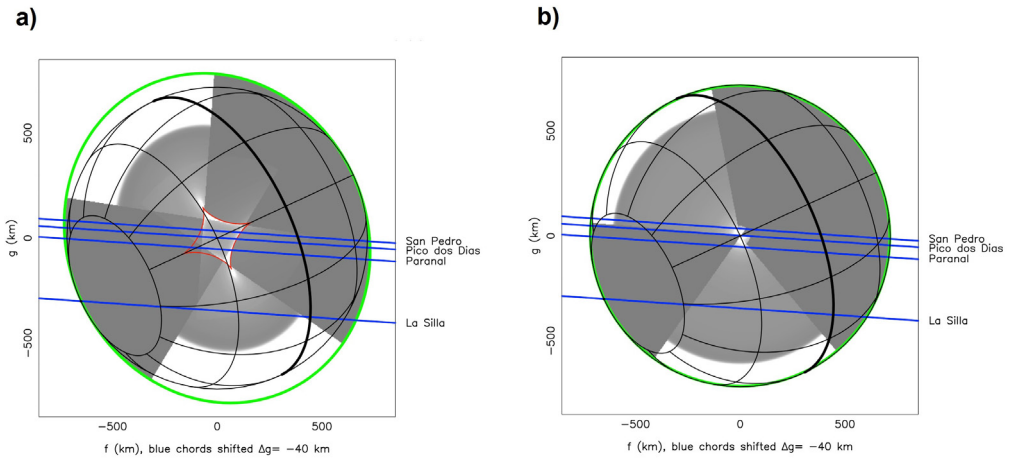


Fig. S5: a) Ray-tracing with a partial CH_4 atmosphere in the limb between 285° and 350° of Position Angle (PA), and between the symmetrical angles with respect to the minor axis (a proxy for the rotation axis). The semi-major and semi-minor axes lengths are 800 and 714 km, respectively. b) Idem, but with PAs between 275° and 350° (plus symmetrical angles), and 717 and 714 km of semi-major and semi-minor axes lengths, respectively. Both models use a temperature profile of 3 K/km until 100K is reached beyond which the atmosphere is isothermal. The surface pressure and temperature are 5 μbar and 50K. The gray scale represents the light intensity. The blue lines mark the chords at the sites indicated with labels, and the green lines indicate the local curvature of the atmosphere. The red line in the left panel marks the centers of curvature of the limb, where caustics are expected due to ray focusing. These models are used only for illustration. The modeled occultation light curves for the left and right cases are shown in Fig. S6 and compared with the observations.

In the first example (Fig. S5a) Makemake's apparent limb has semi-major and semi-minor axes of 800 km and 714 km respectively, with the semi-minor axis at position angle (PA) of 65° (measured from North and in clockwise sense). This particular shape is taken to create a curvature of the isopycnic (iso-density) atmospheric layers that causes the appropriate caustics near the shadow centre. In fact, what is important for the caustics is the curvature of the isopycnic layers. This curvature is likely to be quite different from that of the limb in the presence of a partial atmosphere. This model is not intended to fit the occultation light curves and it is used only for illustration to see whether the features it produces would be at the noise levels of the light curves.

It is also assumed that Makemake's subsolar latitude is ~30°. In that configuration, a local CH_4 atmosphere could be maintained at the subsolar point, with surface pressure P_{surf} = 3-25 μbar, corresponding to the equilibrium vapour pressure of CH_4 ice at 50-55 K and then extend to the limb.

We consider this partial atmosphere located between PA's 285° and 350° along the limb (surrounding the subsolar latitude of 30°), and we used a ray-tracing code

to generate fluxes in the shadow (Fig. S5a). We adopt a temperature profile starting from 50 K at the surface with a gradient of 3 K/km, ramping up to 100 K. The main result is that a P_{surf} = 5 µbar CH_4 atmosphere (corresponding to an ice temperature of 51 K) is able to cause some bumps in the light curves at Paranal, Pico dos Dias and San Pedro de Atacama.

Note that we use a Pico dos Dias light curve corrected for the potential cosmic ray that might have occurred during the occultation. The effect at Paranal is rather modest. One can increase P_{surf} to 25 µbar to get a larger central bump at Paranal (corresponding to a CH_4 ice temperature of 55 K). In this case, the expected peaks at S. Pedro and Pico dos Dias are too big.

We have also investigated the possibility of an N_2 atmosphere at the same location along the limb as the CH_4 atmosphere considered above, with a surface temperature of 38 K, corresponding to P_{surf} = 30 µbar, ramping up like in Pluto to 100 K. It provides a central flash comparable in amplitude with the noise level in the Paranal light curve, but then provides too large flashes at S. Pedro and Pico dos Dias. Conversely, an N_2 atmosphere with P_{surf} <10 µbar would cause bumps small enough to be hidden in the noise level.

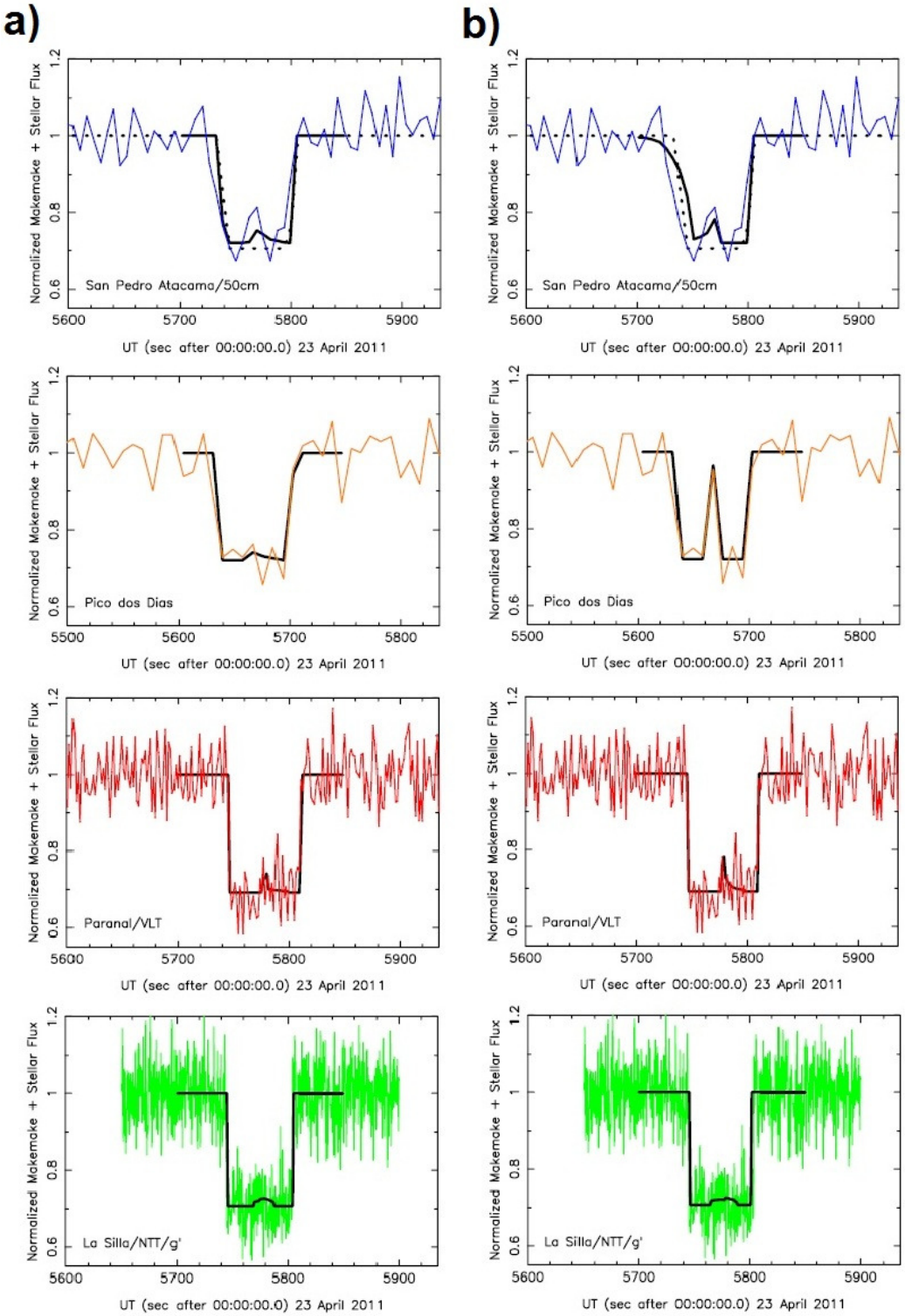

a)
b)
Normalized Makemake + Stellar Flux
San Pedro Atacama/50cm
Pico dos Dias
Paranal/VLT
La Silla/NTT/g'
UT (sec after 00:00:00.0) 23 April 2011

Fig. S6: a) Expected fluxes (black line) at S. Pedro, Pico dos Dias, Paranal and La Silla for a surface pressure P_{surf} ~5μbar using the model in the left panel of Fig. S5. (The Armazones data have not been considered here as the corresponding chord is very close to Paranal, but with a much coarser time resolution, see Fig. 1.) The fluxes from the model have been binned to the time resolution of the observed light curves (blue, brown, red, and green curves). Note that in the Pico dos Dias light curve (brown) the alleged cosmic ray has been removed. The χ^2 values of the fits were 36.2, 13.2, 91.7 and 738.3 for 25, 17, 99, and 540 data points respectively. Thus, the square root of χ^2/N is close to 1 in all cases, indicating that the fits are reasonably good. However, these models are just for illustration. b) Same as panels a, but with the shape model in the right panel of Fig. S5 (note this model can explain a central flash at Pico dos Dias). Note that the Pico dos Dias light curve (brown) is shown as it was initially obtained, with no removal of a potential cosmic ray. The χ^2 values of the fits were 43.3, 22.6, 88.1 and 794.1. The dotted line in the S. Pedro panels is the expected light curve from an airless body.

In the second example, a less elongated Makemake's atmosphere has been used (Fig. S5b). The differences with the other model are the dimensions of the body (1,428 km and 1,434 km for the projected minor and major axes) and the position angles (275° to 350°) for the local atmosphere. The synthetic light curves from the local CH_4 atmosphere are plotted in the right panels of Fig. S6. This model reproduces a flash at Pico dos Dias while preserving the fluxes at S. Pedro and Paranal low. Similar models but with elongations up to 1.06 can give rise to a flash in Pico dos Dias while creating small signals at S. Pedro and Paranal.

We point out that the models presented here depend on several parameters that we do not know. Exploring the possible phase space is clearly out of the scope of this work. This said, the main lesson learnt from our modelling is that a patchy CH_4 or N_2 atmosphere with P_{surf} of a few μbar may be present in the upper and lower left quadrants of the limb and cause structures compatible with the observed light curves. Larger surface pressures would cause too intense features.

8. PREFERRED PROJECTED SHAPE, ASPECT ANGLE AND DENSITY FOR MAKEMAKE.

The elongation direction in the best ellipse fit shown in the main paper is almost perpendicular to the chords direction (Fig. 2). This seems too coincidental. In reality, the elongation in that direction is probably an artificial result because the length of the axis in the direction perpendicular to the chords is not well constrained by the observations. This is usually the case for a nearly spherical body when there are no negative chords sufficiently separated from the positive chords. The same effect was also seen in the occultation by Eris[3], for which a large set of ellipse solutions was possible with considerable elongation in the direction perpendicular to the chords. Thus, we think that the elongation of the nominal best ellipse fit for Makemake in Fig. 2 is too large to be real and is mainly a result of limitations in the observations, similarly to the case of Eris.

We know that the purely circular fit is also unlikely because the general projection of a Maclaurin spheroid is an ellipse and only the pure pole-on orientation can give rise to a circle (but a pure pole-on configuration is very unlikely).

On the other hand, the low rotational variability of Makemake[28,20,21] can be explained if the object has a very homogeneous surface with perhaps a latitudinal band, or if its spin axis orientation is not too far from pole-on. Probably the reality is a mix of the two causes (which we show in Fig. S4). Because the object must have a Maclaurin shape as explained in the main paper, and the projection of a Maclaurin seen far from the equator-on view is closer to a circle than to a very elongated shape, the low rotational variability would favor a fit closer to the circular one than to the very elongated nominal ellipse fit (which has an axial ratio of 1.15).

Besides, the comparison of the Spitzer+Herschel data with the occultation favors the smaller object, which is the nearly circular fit, because the nominal elliptical solution with an axial ratio of 1.15 implies a larger body than is compatible with the Spitzer+Herschel results at almost 2σ. The equivalent diameter of the ellipse with an axial ratio of 1.15 is 1,533 km, away from the maximum diameter allowed for by Spitzer+Herschel (ref. 6).

As mentioned in the main paper, for a spherical Makemake of 1,430 km diameter, a density of ~1.7 g/cm^3 is derived to explain the lack of a global atmosphere, using volatile loss models[8]. In this case the body would not have had enough mass to retain the primordial N_2. The derived 1.7 g/cm^3 density is consistent with the trend observed empirically, as can be seen in Fig. S7. But if Makemake had a projected axial ratio as large as 1.15 (the nominal ellipse fit), the equivalent diameter would be 1,533 km and Makemake would require a smaller density than 1.5 g/cm^3 to be depleted in N_2, according to the volatile loss models[8]. Such a low density for a larger object would be further away from the trend of density versus size than the 1.7 g/cm^3 value (see Fig. S7). Thus, these considerations also indicate that the circular model (the smaller object) is slightly preferred with respect to the nominal ellipse fit. Nevertheless if the lack of a global atmosphere on Makemake is not a result of a surface depletion in N_2 but to a pole-on orientation, this is again in favour that Makemake's projected shape is closer to circular rather than having a 1.15 axial ratio. According to section 6.2 of a theoretical study on TNO atmospheres[7], TNOs with orientation close to pole-on are less likely to develop global atmospheres.

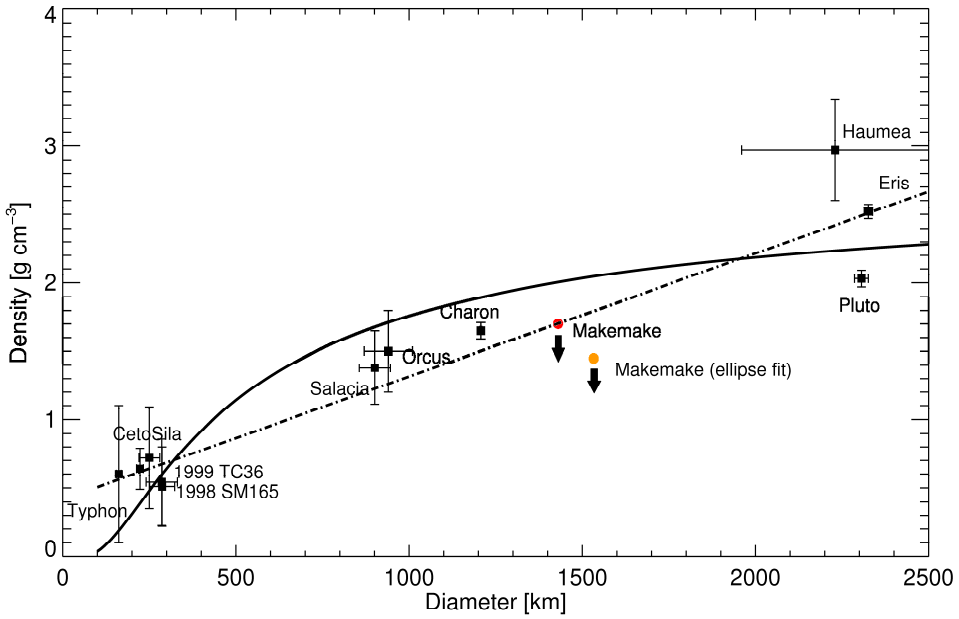


Fig. S7: Plot of density versus diameter for TNOs from a compilation of different sources of data[3,19,33,34,35,36,37,38,39,40]. Two density estimates for Makemake from this paper are shown in red and yellow. The red spot value was obtained by assuming that Makemake is depleted in N_2 ice and using the model in ref. 8 to determine the density below which N_2 would not be retained (using the size determined from the circular fit to the occultation chords). The yellow dot is the same as the red one but using the size derived from the best ellipse fit in Fig. 2 of the main paper, which required an axial ratio of 1.15 and implied a larger body. In reality, the dots indicate upper limits because lower densities also result in no N_2 retention. On the other hand, because methane is retained in Makemake (it is clearly seen in its spectra), the densities must be higher than 1.4 and 1.2 g/cm^3 for the red and yellow dots (using again the model in ref. 8, but this time for methane). This is marked by the arrows. The yellow spot is far from the observed trend and appears less likely. The dashed line is a linear fit to the data, and the continuous line is an exponential fit.

A Maclaurin spheroid of 1.7 g/cm^3 density, spinning at 7.77 hours, would have a true (not projected) axial ratio of 1.17 according to the figures of equilibrium formalism, but when viewed with an aspect angle of 35 degrees (a reasonable angle according to the above discussions) the object would give a projected axial ratio of 1.05. A maximum aspect angle of 45 degrees (corresponding to an orientation exactly in the middle between pole-on and equator-on) would give rise to a projected axial ratio of slightly less than 1.08.

Also, the model of a possible central flash in Pico dos Dias, described in section 7, requires a limb less elongated than 1.06 (in axial ratio) including the atmosphere. Then, if the flash is real, the object would have to be even less elongated than 1.06 in projected axial ratio.

In summary, there are six different indications to favor the solution between the circular model and the nominal ellipse fit: The degeneracy in the direction perpendicular to the chords, the low rotational variability, the fact that for a plausible aspect angle and a plausible density the projected axial ratio would be close to 1.05 (for a body rotating at 7.77h), the lack of a global atmosphere, the comparison with Spitzer+Herschel data, and the possible central flash. These indications point to a projected shape between the circular model and the best ellipse fit, although closer to the circle. The best compromise is for a projected axial ratio of 1.05 ± 0.03. For all the above, the preferred solution from the occultation of Makemake is an ellipse with minor axis 1,430±9 km and major axis 1,502±45 km. Concerning the preferred density, even though the volatile retention arguments point to a density of 1.7 g/cm^3, the models of volatile loss and retention are too simplistic to account for all the processes, and therefore this result is only approximate (but consistent with the value derived from the density vs size trend shown in Fig. S7). A possible indication of the accuracy of this value may be obtained from the dispersion in Fig. S7. Because the dispersion is on the order of 0.3 g/cm^3, we can state that the density of Makemake should probably range from 1.4 to 2.0 g/cm^3.

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Apêndice H

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Size, shape, albedo, density and atmospheric limit of transneptunian object (50000) Quaoar from multi-chord stellar occultations

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ABSTRACT

We present results derived from the first multi-chord stellar occultations by the transneptunian object (50000) Quaoar, observed on 4 May 2011 and 17 February 2012, and from a single-chord occultation observed on 15 October 2012. If the timing of the five chords obtained in 2011 were correct, then Quaoar would possess topographic features (crater or mountain) that would be too large for a body of this mass. An alternative model consists in applying time shifts to some chords, to account for possible timing errors. Satisfactory elliptical fits to the chords are then possible, yielding an equivalent radius $R_{\rm equiv} = 555 \pm 2.5$ km and geometric visual albedo $p_V = 0.109 \pm 0.007$. Assuming that Quaoar is a Maclaurin spheroid with indeterminate polar aspect angle, we derive a true oblateness of $\epsilon = 0.087^{+0.0268}_{-0.0175}$, an equatorial radius of 569^{+24}_{-17} km and a density of 1.99 ± 0.46 g cm^{-3}. The orientation of our preferred solution in the plane of the sky implies that Quaoar's satellite Weywot cannot have an equatorial orbit. Finally, we detect no global atmosphere around Quaoar, considering a pressure upper limit of about 20 nbar for a pure methane atmosphere.

Subject headings: KBO, Maclaurin Object, Quaoar, Stellar occultation, TNO

1. Introduction

Since the discovery of (15760) 1992 QB_1 (Jewitt & Luu (1993)), about 1,260 transneptunian objects (TNOs) have been discovered. Determining their physical and dynamical properties is key to understand the origin and evolution of our Solar System. They were possibly formed closer to the Sun (near the present Uranus's and Neptune's orbits), and displaced to their current location through gravitational perturbations during the migration of the giant planets (Gomes et al. 2005).

Their physical characteristics contain information about the primordial protoplanetary nebula. Moreover, the inferred chemical, thermal and collisional processes that they underwent tell us something about the evolution of the outer Solar System. Due to their large heliocentric distances, our knowledge about their sizes, shapes, albedo, densities and atmospheres remains fragmentary (Stansberry et al. 2008).

Stellar occultations allow accurate determinations of those parameters. They provide sizes and shapes with accuracies which may reach the kilometre level, and detect atmospheres with a sensitivity of a few nanobars in surface pressure. Unfortunately, they are difficult to be predicted. To overcome this obstacle, we have undertaken systematic searches of stars that may be occulted by ten of the largest TNOs, using the 2.2 m telescope of the European Southern Observatory (ESO) at La Silla (Assafin et al. 2010, 2012).

The first successful observation of a stellar occultation by a TNO (besides Pluto and Charon) was the 9 October 2009 occultation by 2002 TX_{300} (Elliot et al. 2010). Since then, the following objects have been measured by stellar occultations: Varuna (19 February 2010, Sicardy et al. (2010)), the dwarf planet Eris (6 November 2010, Sicardy et al. (2011)), 2003 AZ_{84} (8 January 2011, Braga-Ribas et al. (2011)), Quaoar (11 February 2011, Person et al. (2011)), the dwarf planet Makemake (23 April 2011, Ortiz et al. (2012)), Quaoar (4 May 2011, 17 February 2012 and 15 October 2012, this paper), 2003 AZ_{84} (3 February 2012,

Braga-Ribas et al. (2012)) and 2002 KX_{14} (24 April 2012, Alvarez-Candal et al. (2012)). More recently, our group and collaborators successfully predicted and detected occultations by Varuna (8 January 2013) and Sedna (13 January 2013), whose analysis are in progress.

The 11 February 2011 Quaoar occultation provided a single chord and permitted only a probabilistic analysis to constrain Quaoar's size (Person et al. 2011). The first multi-chord stellar occultation by Quaoar was observed on 4 May 2011 (Fig. 1), and is the main topic of this paper. The 17 February 2012 event provided two chords that complement the 4 May 2011 event, but it does not provide further constraints on Quaoar's shape as discussed herein. Finally, the 15 October 2012 event provided a single chord, hence preventing size determination, as briefly commented at the end of this paper.

EDITOR: PLACE FIGURE 1 HERE.

The transneptunian object (50000) Quaoar (also known as 2002 LM_{60}) was discovered by Brown & Trujillo (2004) in June 2002 . Its diameter was estimated to be 1260 ± 190 km from direct Hubble Space telescopes (HST) imagining (Brown & Trujillo 2004). The Spitzer and new HST yielded a smaller diameter of about 850-890 km (Stansberry et al. 2008; Fraser & Brown 2010). With a semi-major axis of 43.51 AU, an orbital period of 287 years, an orbital eccentricity of 0.035 and inclination of 7.98 deg, it is a classical object. Quaoar has a small moon, Weywot, discovered in 2007 (Brown & Suer 2007). Its orbital motion provides Quaoar's mass (Fraser & Brown 2010; Vachier et al. 2012), with the most recent value of $M = (1.40 \pm 0.21) \times 10^{21}$ kg derived by Fraser et al. (2013).

Quaoar has retained water ice in crystalline and amorphous phases, and volatiles such as methane on its surface (Jewitt & Luu 2004; Schaller & Brown 2007b). More tentative detections of ethane and molecular nitrogen are also reported (Guilbert et al. 2009; Dalle Ore et al. 2009). As such, and due to its size and mass, Quaoar might retain a tenuous

atmosphere.

In this paper, we present results derived from the 4 May 2011 stellar occultation by Quaoar. Section 2 describes our prediction method and briefly presents the observations. Data analysis is described in Section 3. Quaoar's size and shape are discussed in Section 4 and their physical implications are presented in Section 5. In Section 6, we give upper limits for the pressure of possible atmospheres. Further constraints (search for satellites and brief comments on the 17 February 2012 and 15 October 2012 occultations) are given in Section 7, before concluding remarks in Section 8.

2. Predictions and Observations

The 4 May 2011 occultation was identified in a systematic search made at the 2.2 m telescope of ESO, using the Wide Field Imager (WFI). The paths of ten large TNOs for the period 2008-2015 were covered by WFI, providing local astrometric catalogs with an accuracy of 40 mas, for stars with magnitudes as faint as R = 18-19. The positions were measured against the UCAC2 (Second US Naval Observatory CCD Astrograph Catalog, Zacharias et al. (2004)) catalog, see Assafin et al. (2010) and Assafin et al. (2012) for details.

Astrometric updates were performed with several telescopes some weeks before the event, at Pic du Midi (1m telescope), La Silla (TRAPPIST 0.6 m telescope), San Pedro de Atacama (ASH2 0.4 m telescope), and Pico dos Dias (0.6 m telescope). On the last few nights prior to the occultation, both Quaoar and the star were observed in the same field of view. This reduced the effects of systematic errors from the reference catalog, thus improving the prediction accuracy. Those updates confirmed that Quaoar's shadow would cross the southern part of South America. A compilation of our measurements provides the

following ICRF/J2000 star position:

$$\begin{cases} \alpha = +17^{\rm h}\ 28^{\rm m}\ 50.801^{\rm s} \pm 0.007" \\ \delta = -15^{\circ}\ 27'\ 42.770" \pm 0.014" \end{cases} \quad (1)$$

An alert was triggered at several potential sites, resulting in data collected with a large diversity of instruments (Table 1). Out of sixteen stations distributed in Chile, Uruguay, Argentina and Brazil, thirteen could acquire data, and six of them obtained a positive detection of the occultation. Moreover two stations, Salto/Uruguay and Pico dos Dias/Brazil, were close enough to the shadow to constrain Quaoar's size, even though they did not detecte the event (Figs. 1 and 2).

EDITOR: PLACE TABLE 1 HERE.

3. Analysis

3.1. Occultation light curves

The various instruments and detectors setup are listed in Table 2, that also provides the ingress and egress times for the sites where the occultation was detected.

All light curves (Fig. 2) were normalized to the unocculted star flux, by applying linear or second degree polynomial fits to the flux. The mid-exposure time of each frame was extracted from the image headers. Only four stations (out of the six that detected the event) have an internal accuracy of a few hundredths of a second: Harlingten and ASH2 (at San Pedro de Atacama), Armazones and UEPG. On the other two sites, only the integer part of the second is available, due to the acquisition software used. In order to retrieve the fraction of a second, a linear fit was performed to the set of points (i, t_i), where i is the

image number and t_i is the corresponding header time. This fit eventually provides the cycle times and mid-exposure times of each frame, once truncation effects are accounted for. This allowed us to retrieve the time of each image with relative accuracies of about 0.05 seconds. All sites excepting one set up their clocks using robust internet servers for time synchronization, and absolute errors larger than one second are not expected. The exception is the Rivera station, where the computer clock was manually set up one hour before the event by looking at a GPS display, due to the lack of internet access. This point is discussed further in Section 4.3.

EDITOR: PLACE TABLE 2 HERE.

3.2. Occultation timing

The start and end times of the occultation were obtained for each light curve by fitting a sharp edge occultation model. This model is convolved by Fresnel diffraction, the CCD bandwidth, the stellar diameter in kilometres, and the finite integration time, see Widemann et al. (2009).

The Fresnel scale ($F = \sqrt{\lambda D/2}$) for the present Quaoar's geocentric distance $D = 42.4$ AU $= 6.34 \times 10^9$ km is 1.4 km for a typical wavelength of $\lambda = 0.65$ μm. The star diameter is estimated using the formulae of van Belle (1999). Its B, V and K apparent magnitudes are respectively 16.5, 15.8 and 12.8 in the NOMAD catalogue (Zacharias et al. 2005). This yields a diameter of about 0.5 km projected at Quaoar's distance. The smallest integration time used in the positive observations was three seconds, which translates to almost 55 km in the plane of the sky. Therefore, the occultation light curves are largely dominated by the integration times, not by Fresnel diffraction nor by the star diameter.

The occultation fits consist in minimizing the χ^2 value for each light curve:

$$\chi^2 = \sum_{1}^{N} (\Phi_{\rm i,obs} - \Phi_{\rm i,cal})^2 / \sigma_{\rm i}^2, \tag{2}$$

where $\Phi_{\rm i,obs}$ and $\Phi_{\rm i,cal}$ are the observed and calculated fluxes at point i, respectively, and $\sigma_{\rm i}$ is the 1σ error bar of point i. The value of σ_i is assumed to be constant for all points, and was derived from the standard deviation of the full stellar flux before and after the occultation. The summation is performed over the N data points around the expected ingress or egress time. The free parameter to adjust is the ingress (disappearance) or egress (re-appearance) time $t_{\rm occ}$, which provides the minimum value of χ^2, noted $\chi^2_{\rm min}$. The 1σ error bar on $t_{\rm occ}$ is estimated by varying $t_{\rm occ}$ so as to increase χ^2 from $\chi^2_{\rm min}$ to $\chi^2_{\rm min} + 1$.

The best fits to the occultation light curves are shown in Fig. 2 and the derived occultation times are listed in Table 2. The error bars are not necessarily symmetric due to the presence of gaps (caused by readout time intervals) between images. No information is gathered during those gaps, causing local plateaux in the χ^2 function, and hence asymmetric error bars.

3.3. Limb fitting

The occultation times provide the chords shown in Fig. 2. The most general shape for Quaoar's limb that we considered here is an ellipse characterized by five adjustable parameters: the coordinates of the body centre, relative to the star in the plane of the sky (f_c, g_c); the apparent semi-major axis a'; the apparent oblateness $\epsilon' = (a' - b')/a'$ (where b' is the apparent semi-minor axis) and the position angle P of the semi-minor axis b'. The quantities f_c and g_c, expressed in kilometers, are positive towards the local celestial east and north, respectively. They are calculated using the JPL#21 Quaoar's ephemeris (Giorgini et al. 1997) at each site, and the star position given in Eq. 1. The position angle P is counted

positively from the direction of local celestial north to celestial east.

EDITOR: PLACE FIGURE 2 HERE.

The six positive observations provide $N = 12$ chord extremities. Among the six chords, two were obtained at San Pedro de Atacama, with the Harlingten and ASH2 telescopes. Although consistent with the Harlingten results, the ASH2 timing has significantly higher error bars (Table 2), so that we kept only $N = 10$ independent points along the limb, and $M = 5$ free parameters to be adjusted. The elliptical fit to the $N = 10$ chord extremities is obtained by minimizing the χ^2 function:

$$\chi^2 = \sum_{1}^{\mathrm{N}} \left[(f_{\mathrm{i,obs}} - f_{\mathrm{i,calc}})^2 + (g_{\mathrm{i,obs}} - g_{\mathrm{i,calc}})^2 \right] / \sigma_{\mathrm{i,r}}^2, \tag{3}$$

where $(f_{i,obs}, g_{i,obs})$ are the coordinates of the chord extremities, and $\sigma_{\mathrm{i,r}}$ are the corresponding radial uncertainties. The latter are obtained by multiplying the 1σ uncertainties on the occultation times by the star velocity perpendicular to the local Quaoar's limb. The value of χ^2 per degree of freedom (or unbiased χ^2) is given by $\chi^2_{\mathrm{pdf}} = \chi^2_{\mathrm{min}}/(N - M)$.

As discussed later, in the hypothesis of hydrostatic equilibrium and in the small angular momentum regime, the apparent oblateness ϵ' is the result of the projection of an oblate Maclaurin spheroid with semi axes $a = b > c$, where a and c are the true equatorial and polar radii, respectively[1]. The true oblateness $\epsilon = 1 - c/a$ is then related to the apparent oblateness through

$$\epsilon' = 1 - \sqrt{\cos^2(\zeta) + (1 - \epsilon)^2 \sin^2(\zeta)}, \tag{4}$$

[1]Note that in this case, the apparent semi-major axis of the limb, a', is equal to the equatorial radius a of the spheroid.

where ζ is the angle between the polar c-axis and the line of sight. We will call this quantity the "polar aspect angle", with $\zeta = 0$ (resp. $\zeta = 90$ deg) corresponding to the pole-on (resp. equator-on) geometry.

To explore the parameter space, we have fitted the chord extremities with ellipses, imposing prescribed values of the position angle P and oblateness ϵ'. We have then varied P between 0 and 180 degrees by steps of 3 degrees, and ϵ' from 0 to 0.5 by steps of 0.004. For each pair of prescribed values (P, ϵ'), we have adjusted and stored the parameters a', f_c, g_c and the resulting χ^2. This allowed us to construct 2D colored maps (see Figs. 4 and 6), where the χ^2 value is shown as a function of oblateness ϵ' and apparent equivalent radius (instead of a'). This quantity is defined by $R_{\rm equiv} = \sqrt{a'b'} = a'\sqrt{1-\epsilon'}$, and is the radius of the disk that has the same area as that enclosed by the apparent limb. The 1σ error bars on $R_{\rm equiv}$ and ϵ are obtained by varying χ^2 from its minimum value $\chi^2_{\rm min}$ to $\chi^2_{\rm min} + 1$.

We note that the colored points are confined inside a V-shaped boundary. This is because the longest occultation chords impose that a' be larger than some value L, which is close to the lengths of those longest chords. Also, those chords are roughly aligned with the longer axis of the fitted ellipse (Fig. 4), which imposes that b' be smaller than $\sim L$. Thus, we must have $L\sqrt{1-\epsilon'} < R_{\rm equiv} < L/\sqrt{1-\epsilon'}$, corresponding to the V-shaped domain seen in Figs. 4 and 6. The value of L depends on the particular model considered to fit the chords, and varies in a range from about 500 to 680 km, see next Section.

4. Quaoar's shape

We expect thousand-kilometer class objects to be in hydrostatic equilibrium, with relaxed to Maclaurin spheroidal or Jacobi ellipsoidal shapes. The critical diameter to reach equilibrium can be determined under some assumptions, as shown in Tancredi & Favre

(2008). Nevertheless, the formation scenario and collisional history of individual objects are not known, so that we cannot discard solutions that diverge somewhat from the equilibrium figure. We examine here possible ranges of sizes and oblateness for Quaoar, making various extreme assumptions.

As seen in Figs. 2 and 3, the chords provide a shape that has a significant departure from the ellipse. We consider here two extreme possibilities: (1) all the timings are correct, implying that Quaoar possesses significant topographic features (craters or mountains), a result which is not expected for such a big object (Section 4.2); or (2) Quaoar is a spheroid in hydrostatic equilibrium (a Maclaurin object), which implies that there were timing problems at some stations (Section 4.3). We now discuss these two possibilities and their implications.

4.1. Nominal solution

EDITOR: PLACE FIGURE 3 HERE.

The *nominal solution* uses the ingress and egress occultation times at face value, and fits an elliptical shape to the chord extremities, accounting for the 1σ error bar of each of them, see Fig. 3. The best fit returns a best value $\chi^2_{\rm pdf}$= 9.1 clearly indicating that an overall elliptical limb shape is not satisfactory. This is obvious from Fig. 3, and at this point it is meaningless to interpret the physical parameters derived from this fit and to discuss their error bars. We thus examine possible extreme solutions in which Quaoar has large topographic features, i.e. a deep crater or a high mountain.

4.2. Large topographics features

EDITOR: PLACE FIGURE 4 HERE.

The first solution we examine is the *crater solution*. It consists in an overall oblate object, from which a part has been excavated, see Fig. 4. An elliptical limb shape is again fitted to the chord extremities, except for the San Pedro de Atacama egress point, so that we now have $N = 9$ data points to fit. This solution provides the most elongated shape of all obtained in this work. It has a position angle $P = 21$ deg, an apparent oblateness $\epsilon' = 0.29$, an equivalent radius of $R_{\rm equiv} = 557$ km and $\chi^2_{\rm min} = 15.1$ (and $\chi^2_{\rm pdf} = 3.78$), better than the nominal solution, but still indicating significant departures from the model. This can be seen by the large radial residual of about -200 km at the San Pedro de Atacama station, see Fig. 5 .

The limit D_c to the diameter of a possible crater on an icy (resp. rocky) body, without shattering it, is $D_{\rm c,icy} \sim 1.2R$ (resp. $D_{\rm c,rocky} \sim 1.6R$), where R is the radius of the body (Leliwa-Kopystyński et al. 2008). So, considering the derived equivalent radius, the largest possible crater has a diameter of about $D_{\rm c,icy} \sim 670$ km ($D_{\rm c,rocky} \sim 890$ km). In the right panel of Fig. 4, the depression related to the San Pedro de Atacama chord has a diameter of about 430 km. However, Leliwa-Kopystyński et al. (2008) give a maximum value of 0.25 for the ratio depth/diameter of a crater in case of a H_2O-ice body, and 0.3 for a CO_2-ice body (and 0.26 for a rocky body). Thus, the 430 km crater yields a maximum depth of about 110-130 km, depending on composition. But, according to Fig. 5, the depression in Fig. 4 has a depth of roughly 200 km, about 45% of the crater diameter. Consequently, such a crater appears to be too deep, unless Quaoar has unexpected cohesion properties.

EDITOR: PLACE FIGURE 5 HERE.

Another solution consists in a less oblate object, but with an important prominence on its eastern limb (Fig. 6). To obtain this solution, that will be referred to as the *mountain solution*, we have discarded the egress extremity of Rivera, the southernmost positive chord. So, again we have $N = 9$ data points to fit. The elliptical fit has a position angle of 7 deg, an oblateness of 0.138, an equivalent radius of 525 km and a $\chi^2_{\rm min} = 15.7$ ($\chi^2_{\rm pdf} = 3.92$), indicating a fit quality comparable to the crater solution.

EDITOR: PLACE FIGURE 6 HERE.

Fig. 7 shows the radial residuals of this fit. The Rivera egress chord extremity displays a residual of more than 140 km, or 24% of the equatorial radius (587 km) obtained here. This can be compared to the much smaller topographic features, ∼20 km at most, expected on an icy body (Johnson & McGetchin 1973). A rocky body could support higher features, about ∼150 km, comparable to the value we obtain here. However, the density for this solution would be around 2 g cm^{-3} (see table 4), implying in a significant fraction of ice. Thus, the mountain solution seems physically implausible, as the mountain is too big to be sustained at Quaoar surface.

EDITOR: PLACE FIGURE 7 HERE.

4.3. Maclaurin solution

Although we do not expect large absolute timing errors at the various stations (Section 3.1), it is difficult to assess timing errors after the fact, as some of the observations are not repeatable. In view of the physically implausible Quaoar's shape obtained in the previous Section, we now allow for time shifts at each station, except for San Pedro de

Atacama. For this site, we have two observations with independent and consistent ingress and egress times (Table 2), so that a systematic timing error is not expected for the two corresponding occultation chords.

Its known that, in an ellipse, the mid point of a set of parallel chords should be aligned, i.e. a straight line must go through all the mid point of the chords. So, as a first exercise, we slide the various chords shown in Fig. 3 along themselves, so as to align the middle of all chords along a line perpendicular to the chords. Thus, a circle is adjusted to the chord extremities with a radius 555 ± 2.5 km and a value $\chi^2_{\rm min} = 1.33$.

To estimate the quality of the fit, however, we have to consider that we implicitly added 4 adjustable parameters to our model (the time shifts of all the chords except San Pedro de Atacama). Besides, we also adjust the radius of the circle R_{circ} and the coordinates of its centre (f_c,g_c). So, we have $M = 7$ parameters to be adjusted to the $N = 10$ chord extremities, implying in a $\chi^2_{\rm pdf} = 0.44$. The fact that the $\chi^2_{\rm pdf}$ is satisfactory is thus an encouraging argument to proceed in our analysis.

Actually, we may now align the middle of the chords, but along a line which makes an angle α relative to the line perpendicular to the chords. Doing so, we transform the original circle (corresponding to $\alpha = 0$) into an infinity of possible ellipses parametrized by α, which can vary in principle between -90 and +90 deg, with the convention that α is increasing from celestial north to celestial east[2]. We will denote β the angle between the long axis of the ellipse and the direction of the chords (with the same convention of sign as for α).

Note that since we slide the chords along their own direction, all the ellipses share the same area. Consequently, the radius of the circle R_{circ} described before actually gives

[2]Note that $\alpha = 0$ does not mean that the semi-minor axis of the ellipse is pointing to the celestial north, as the chords are inclined with respect to the east-west celestial direction.

Quaoar's equivalent radius, $R_{\rm equiv} = 555 \pm 2.5$ km, common to all the ellipses. The next step is to deduce Quaoar's actual shape in space from its apparent elliptical limb.

Quaoar has a small amplitude rotational light curve with $\Delta m = 0.133 \pm 0.028$. Its rotation period is 8.8394 ± 0.0002 hours, if it has a single peaked rotational light curve (Ortiz et al. 2003). The period can be 17.6788 ± 0.0004 hours in case of a double peaked light curve and can be caused by albedo features, shape effects, or a combination of the two. Quaoar's large size and density suggest a body in hydrostatic equilibrium, i.e. a Maclaurin spheroid. We note that a Jacobi-type equilibrium (Lacerda & Jewitt 2007) is not expected here because a 17.7-hour rotation period would require a very low density smaller then 0.6 g cm^{-3} and an oblateness larger than 0.42.

Consequently, we will assume from now on that Quaoar is a Maclaurin spheroid of equatorial radius a, polar radius c, true oblateness $\epsilon = 1 - c/a$, observed with a polar aspect angle ζ and with a rotation period of 8.8394 hours (so that the amplitude of its rotational light curve is entirely due to albedo features). Then, it can be shown from Eq. 4 that Quaoar's apparent surface area is $S = \pi a^2 \sqrt{\cos^2(\zeta) + (c/a)^2 \sin^2(\zeta)}$, where $S = \pi R_{\rm equiv}^2$. Moreover, Quaoar's density is given by $\rho = 3M/(4\pi a^2 c)$. Elementary calculations show that:

$$\begin{cases} \epsilon = 1 - \dfrac{\sqrt{(R_{\rm equiv}/a)^4 - \cos^2(\zeta)}}{\sin(\zeta)} \\ \\ \rho = \dfrac{3M}{4\pi a^3} \cdot \dfrac{1}{1-\epsilon} \\ \\ \tan^2(\beta) = 1 - \epsilon', \text{ with } \mathrm{sgn}(\beta) = -\mathrm{sgn}(\alpha), \end{cases} \tag{5}$$

where the apparent oblateness ϵ' is given as a function of the true oblateness ϵ by Eq. 4. The condition $0 < (1-\epsilon)^2 < 1$ imposes $R_{\rm equiv} < a < R_{\rm equiv}/\sqrt{|\cos(\zeta)|}$. By varying a inside this allowed interval, the first two equations above yield an infinity of solutions (ρ, ϵ) that

fit equally well the occulting chord extremities (after applying the necessary time shift to each chord). These (ρ, ϵ) solutions are shown as blue lines in Fig. 8. They define a V-shape domain inside which the actual solution must reside, with the limit corresponding to the equator-on ($\zeta = 90$ deg) and pole-on ($\zeta = 0$ deg) geometries.

The blue lines are parametrized by $R_{\rm equiv} = 555 \pm 2.5$ km (obtained from our observations) and $M = 1.40 \pm 0.21 \times 10^{21}$ kg, derived from Weywot's orbit (Vachier et al. 2012; Fraser et al. 2013). We note that the errors on the position of the V-shaped domain (the dashed blue lines) are dominated by the uncertainties on M, not on $R_{\rm equiv}$.

The final condition that we impose to the solutions is that they satisfy the Maclaurin equilibrium condition:

$$\frac{\pi G \rho}{\omega^2} = \frac{\sin^2(\psi) \cdot \tan(\psi)}{2\psi\,[2 + \cos(2\psi)] - 3\sin(2\psi)}, \tag{6}$$

where ω is the spin frequency, G is the constant of gravitation and $\cos(\psi) = 1 - \epsilon$ (Plummer 1919; Sicardy et al. 2011). Considering the single peak case (i.e. a rotation period of 8.8394 hours as discussed above), the equation above provides the red curve plotted in Fig. 8.

The intersection between the Maclaurin equilibrium curve and the V-shaped domain finally provides Quaoar's true oblateness $\epsilon = 0.0897 \pm 0.006$ and density $\rho = 1.99 \pm 0.14$ g cm^{-3}, where the error bars come from the indetermination of ζ. Also considering the uncertainty in Quaoar's mass determination (see above), we obtain larger error bars $\epsilon = 0.0897^{+0.0268}_{-0.0175}$ and $\rho = 1.99 \pm 0.46$ g cm^{-3}.

EDITOR: PLACE FIGURE 8 HERE.

From the oblateness, density and equivalent radius given above, the second of Eqs. 5 provides Quaoar's equatorial radius $a = 569^{+24}_{-17}$ km. The third equation indicates that β

must be negative, since α is positive, so $\beta = -44.1 \pm 0.9$ deg (see the position angle of the centers of the chords in Fig. 9). Since the chords are themselves inclined by 12.5 deg with respect to the east-west direction, the position angle of Quaoar's north pole projected in the plane of the sky is $P = -44.1 + 12.5 = -31.8 \pm 0.9$ deg (or $P = 148.4 \pm 0.9$ deg, as there is a indetermination of 180 deg on the pole position), see Fig. 9. This position angle can be compared with that of Weywot's orbital pole projected in the plane of the sky. The various solutions derived by Vachier et al. (2012) and Fraser et al. (2013) provide a position angle between 5 and 15 deg. Thus, our best Maclaurin solution rules out an equatorial orbit for Weywot.

EDITOR: PLACE FIGURE 9 HERE.

Once $R_{\rm equiv}$ is adjusted, Eqs. 5 and 6 impose the values of a, ϵ and P, leaving only the centre position (f_c, g_c) to be adjusted. Thus, the Maclaurin model requires the adjustment of $M = 7$ parameters (4 time shifts, $R_{\rm equiv}$ and (f_c, g_c)). Our preferred fit shown in Fig. 9 has $\chi^2_{\rm min} = 1.33$. Consequently, we have $\chi^2_{\rm pdf} = 0.44$, indicating that we may be slightly overestimating the error bars on the chord extremities.

This satisfactory limb fitting comes at a price, that of applying time shifts of several seconds to the chords, see the values given in Fig. 9. The problem is particularly acute for the Rivera station, where we have to apply a time shift of -9.1 s. However, we note that this is the only site where the computer clock was not systematically corrected by an internet server, due to the lack of internet access at that station. It was set up with respect to a GPS display, half an hour before the event. Such a procedure usually provides an accuracy of a fraction of a second. Its impossible to accesses the errors after the fact, but, although improbable, we cannot exclude a time drift of the clock during the observation or even that the GPS display may not reflect the actual UTC time, due to unknown acquisition problem.

The clocks at the other stations were all connected to an internet time server via NTP. We note that the time shifts applied to the Florianópolis and Ponta Grossa chords are within the 1σ level obtained for the fitted occultation times, see Section 3.2, so the shift could be explained by the errors in the occultation timing. Concerning Armazones, the shift is at about the 2.5σ level of the occultation times, a discrepancy which is more difficult to explain. Finally, we assumed that the San Pedro de Atacama timing is the most reliable because it is confirmed by two experiments. But, although unlikely, an identical systematic error for the two experiments cannot be discarded.

5. Summary of Quaoar's physical properties

We now briefly discuss the various physical parameters derived from our analysis, assuming a Maclaurin shape for Quaoar. Table 3 lists the various parameters of the best fits using each of the models described above, as well as the quality parameter $\chi^2_{\rm pdf}$. Table 4 provides some physical parameters derived from those fits.

EDITOR: PLACE TABLE 3 HERE.

If Quaoar is a Maclaurin spheroid, its density is $\rho = 1.99 \pm 0.14$ (± 0.46) g cm^{-3}, where the first error bar measures our internal accuracy, while the error bar in parentheses takes into account the uncertainty on Quaoar's mass (Fraser et al. 2013). The crater and mountain solutions would imply a range of densities of 1.40 – 2.46 g cm^{-3}.

Similarly, the true oblateness of the Maclaurin solution is $\epsilon = 0.0897 \pm 0.006$ $(^{+0.0268}_{-0.0175})$, with a possible range of 0.0722 – 0.33, taking account the crater and mountain solutions.

The Maclaurin solution has a well defined equatorial radius of $a = 569^{+24}_{-17}$ km and a well defined equivalent radius $R_{\rm equiv} = 555 \pm 2.5$ km at the date of occultation. We can now

derive Quaoar's geometric albedo p through:

$$p = (AU_{\rm km}/R_{\rm equiv})^2 \cdot 10^{0.4(\rm H_\odot - H)}, \quad (7)$$

where $AU_{\rm km} = 1.49598 \times 10^8$ km, $H_\odot$ is the Sun magnitude at 1 AU, and H is Quaoar's absolute magnitude. We find at least two measurements of its absolute magnitude Rabinowitz et al. (2007) gives $H_V = 2.792 \pm 0.025$ and DeMeo et al. (2009) gives $H_V = 2.82 \pm 0.06$. For the results presented below, we have adopted the most recent result given by DeMeo et al. (2009). For the Sun absolute magnitude we have adopted the well established value of $H_{\odot,V}$= -26.74. Adopting the ranges of equivalent radii obtained for each of the presented solution, we calculate Quaoar's geometric albedo in the visible p_V, see Table 4. The error bars represent the range of the albedo obtained for a given solution, combined with the error in absolute magnitude. For the Maclaurin solution, we obtain $p_V = 0.109 \pm 0.007$, and in the case of the crater and moutain solution, $p_V = 0.102$ – 0.139.

EDITOR: PLACE TABLE 4 HERE.

6. Atmosphere

Spectroscopic studies show that the largest TNOs retained water ice and volatiles on their surfaces. Water ice (H_2O), mostly in crystalline phase, was detected on many objects, as well as methane (CH_4), ammonia (NH_3), molecular nitrogen (N_2) and even methanol (CH_3OH) and ethane (C_2H_6) for a few of them (Barucci et al. 2005; Guilbert et al. 2009). In contrast, the smaller objects only reveal water ice bands or mere flat spectra. A simplistic approach is thus to split the TNOs in two groups: large volatile-rich bodies and small volatile-poor objects (Schaller & Brown 2007a).

Consequently, large TNOs may undergo out-gassing from their interiors, or suffer ongoing volatile loss at their surface, leading to the presence of a thin global or local atmosphere. Seasonal sublimation and recondensation can even be expected, causing surface changes, as seems to be the case for Eris (Sicardy et al. 2011).

Only water ice in crystalline and amorphous phase and methane ice are firmly detected on Quaoar's surface (Jewitt & Luu 2004; Schaller & Brown 2007b). More tentative detections were made on ethane and molecular nitrogen (Guilbert et al. 2009; Dalle Ore et al. 2009).

To investigate the stability of volatile on TNOs, Schaller & Brown (2007a) and Brown et al. (2011) developed a simple model of atmospheric loss through Jeans escape, including N_2, CH_4, and CO ices. In the case of Quaoar, given the bulk properties of the object (diameter, mean temperature), they found that N_2 may be too light to have been retained over the age of the Solar System, while CH_4 would be near its limit of stability. These results are consistent with the observed surface composition where only CH_4 is firmly detected, but maybe not N_2 or CO.

However, the Jeans mechanism gives only a lower limit to the escape rates. Since crystalline water is observed in many TNOs including Quaoar, this implies that their surface temperature was larger than $\sim$100 K at some recent epoch (i.e. within the last 10^7 years (Cooper et al. 2003; Jewitt & Luu 2004). This motivated Levi & Podolak (2009) to investigate the transition between the Jeans and hydrodynamic escapes. For a temperature of $\sim$100 K, they found that N_2 is firmly in the Jeans regime, while hydrodynamic escape could be significant for CH_4. However, in a global hydrodynamic regime, the escape rates of the various species depends on their atmospheric mixing ratios, rather than on their individual vapour pressures. Therefore, a partially hydrodynamic regime could potentially enhance the CH_4 escape rates. Nonetheless, the established presence of CH_4 ice on Quaoar'

surface indicates that these effects are not strong enough to have entirely depleted Quaoar's surface of methane ice.

Quaoar's mass (Fraser et al. 2013; Vachier et al. 2012) and our preferred equivalent radius of 555 km imply a surface gravity of about 0.4 m s^{-2}. This is intermediate between the values of 0.64 and 0.30 m s^{-2} obtained for Pluto and Charon, respectively, based on their current mass determinations (Tholen et al. 2008). Also, the mean temperature of Quaoar is similar to that of Pluto, due to their respective albedo and heliocentric distances. Thus, a Pluto-like atmosphere for Quaoar is plausible, considering both its surface composition, temperature and gravity. In this context, ground-based stellar occultations are currently the most powerful technique to detect (or put upper limits on) very thin atmospheres of a few nanobars, see e.g. Sicardy et al. (2011) for the case of Eris.

Quaoar's atmospheric thermal structure is unknown, but we would expect it to be mainly composed of CH_4 (and/or N_2, if present at the surface). The upper atmosphere temperature can be obtained from the methane "thermostat-model" (Yelle & Lunine 1989), where the near-IR heating is balanced through the emission in the 7.7 μm methane band. With an incoming solar flux at Quaoar twice weaker than at Pluto, this model predicts a Quaoar's upper atmosphere temperature of roughly 102 K (vs 106 K for Pluto) at about 10 km above the surface, and a rapid drop below that altitude, connecting the thermal profile to the surface temperature (Fig. 10).

EDITOR: PLACE FIGURE 10 HERE.

We assume an equilibrium surface temperature of t_{surf} = 42 K. This is within the temperature range of a slow or fast rotator body. This crude estimation is sufficient for our purpose. Fig. 10 summarizes the general structure of the adopted temperature profile for a CH4 atmosphere.

The equilibrium surface temperature of an airless spherical body depends on its geometric albedo p_V, phase integral q, emissivity of ε, beaming factor η and heliocentric distance r, see for instance the equations in Table 4S of Sicardy et al. (2011). Using $p_V = 0.120 \pm 0.018$, $\varepsilon = 0.9$, $q = 0.52$, $\eta = 1.7$ (Fornasier et al. 2013) and, $r = 42.35$ AU, we obtain $T_{\rm ss} = 54$ K for the equilibrium sub-solar temperature in the case of a slow rotator, and $T_{\rm eq} = 40$ K for the equilibrium equatorial temperature of a fast rotator.

EDITOR: PLACE FIGURE 11 HERE.

With that prescribed atmosphere, we have modelled synthetic occultation light curves, using a ray tracing code as described in (Sicardy et al. 2006b; Widemann et al. 2009). We compare these synthetic curves with our best light curve obtained with the San Pedro de Atacama 0.5 m telescope.

Because there are various possible solutions for Quaoar's shape, we do not know the exact distance of the sub-occultation points to the body centre. To overcome this problem, we have generated several synthetic light curves for a range of centre-to-limb distances. For each distance, we obtain a χ^2 curve that provides its own detection threshold. The envelope of the various χ^2 curves yields an upper limit of 21 nbar (1σ), or 56 nbar (3σ) for a methane atmosphere, see Fig. 11. The profile expected from a 21 nbar atmosphere is shown in Fig. 12. Fitting only the ingress data points (green dots) in Fig. 12 does provide a detection of an atmosphere with a pressure of 138^{+24}_{-32} nbar (1σ), but at the 3σ level this remains an upper limit of 221 nbar. In conclusion, we can not exclude the presence of a local atmosphere.

EDITOR: PLACE FIGURE 12 HERE.

For completeness, we have also studied the isothermal model, even though it is

unlikely due to the methane heating process, as discussed above. We obtain a marginal detection of a pure isothermal methane atmosphere at 42 K with a pressure of 54 ± 25 nbar. Unfortunately the light curves from the other sites do not have the necessary quality to bring further constrains to this result.

We note that methane has a vapour pressure of 32 nbar at 42 K (Fray & Schmitt 2009), consistent with our absence of detection. Nitrogen is more volatile, with a vapour pressure of 176 μbar at the same temperature. Thus, the atmospheric upper limit tends to rule out the presence of exposed N_2 ice on the surface.

Finally, using the values of the geometric albedo p_V and phase integral q given earlier, we conclude that the surface temperature at the sub-solar point can reach 54 K, which could produce a 17 μbar CH_4 local atmosphere, that would partially condensate to the nanobar level near the limb, thus being undetectable in our data set.

7. Other constraints

7.1. Search for secondary events

Considering its flux ratio to Quaoar and assuming equal albedo, Weywot has a size ratio of about 1/12 with respect to the primary (Fraser & Brown 2010). From the Quaoar size determined here, we can calculate Weywot equivalent diameter. This yelds to a diameter of about 90 km to the secondary. According to the ephemeris[3] calculated by Vachier et al. (2012), the satellite shadow passed at about 800 km north of Quaoar's shadow, and about eleven minutes later. The occultation would have lasted for five seconds at most.

No secondary occultation was seen at Belo Horizonte (CEAMIG), neither at Pico do

[3]available at http://www.imcce.fr/fr/ephemerides/formulaire/form_ephepos.php

Dias (OPD), the two nearest sites to the predicted path. However, the observation strategy was not optimized for such a detection, since the integration time at CEAMIG was 6 seconds, with a readout time of 3 seconds (Table 1). At OPD, the exposure time was 0.95s, with about 1.05s of readout time between frames. So, if an occultation by Weywot had occurred during one exposure in CEAMIG, a conspicuous 2.8 magnitude drop would have been observed. If it had happened during a readout time interval, a magnitude drop of 1.1 would be expected. No such dimming was detected. If a 5 sec occultation had happened at OPD, it would have lasted for at least two complete cycles with a drop of about 3 magnitude, any event like that is seen in the light curve. However, the absence of detection does not bring further constraints on the satellite orbit.

7.2. The 17 February 2012 occultation

On 17 February 2012, another multi-chord stellar occultation by Quaoar was observed from Europe. Also identified in the ESO/WFI program (Assafin et al. 2012), the occulted star has magnitude R= 15.2 and K= 11.6. Astrometric updates could only be made several months in advance, before its conjunction with the Sun. The small solar elongation just prior to the event, prevented last minute predictions. The final ICRF/J2000 star position is given in Table 5.

Fifteen sites reported attempts, but the majority was unable to observe due to poor weather or technical issues. The event was detected from four different sites (Table 6), but three sites were separated by 3 km only cross-track. This means that we have only two effective chords to fit Quaoar's limb.

EDITOR: PLACE TABLE 5 HERE.

The observations were performed with video and CCD cameras. Drift scan mode was used with the TAROT instrument, with a 6 sec/pixel resolution, resulting in a low accuracy chord length. At Tourrette-Levens a sequence of CCD images was acquired, but during ingress a tracking problem occurred. Moreover, the measured chord is too short compared to the other two along the same track, possibly pointing toward a timing problem. As the stations TAROT, Tourrette-Levens and Valensole observed the same chord, we use the occultation times given by Valensole observation, as it presents the best accuracy among them (Table 6).

EDITOR: PLACE TABLE 6 HERE.

With two chords at hand, we can perform a circular fit, with radius R and centre (f_c, g_c) as free parameters. This provides $R = 685^{+445}_{-155}$ km (1σ), and thus a lower limit of $R_{min} = 530$ km. The observed chords are compatible with the Maclaurin solution obtained from the 4 May 2011 event, but does not bring further constraints (Figure 13).

This event can be used, however, to improve Quaoar's ephemeris by giving its time of closest approach to the star, its distance to the star at that moment and its offset relative to the JPL#21/DE405 ephemeris, see Table 5.

EDITOR: PLACE FIGURE 13 HERE.

7.3. The 15 October 2012 occultation

A third stellar occultation by Quaoar was observed on 15 October 2012. The occulted star has magnitudes R= 17.3 and K= 14.4, and was also taken from the Assafin et al. (2012) catalog. Astrometric updates suggested an occultation path crossing the south of Chile.

This unfavourable prediction explains why only four telescopes in Chile were scheduled for that event. The PROMPT[4] instrument was the only one that could eventually monitor the star and detect the event. Two 0.4 m telescopes with identical setups were used with 8 sec integration times and 1 sec readout times. The acquisitions were offset by 4 seconds, so as to overlap the effective observing intervals and thus avoid information loss caused by readout times.

The event lasted from 00:45:30±2.5 to 00:45:50±2.5 UT, which translates into a chord of 400 km in lentgh. Consequently, this observation does not provide new constraints on Quaoar's size, but we may use this result for astrometric purposes. Even not knowing if the chord probed the north or the south part of Quaoar (i.e. we have two possible solutions), this uncertainty is about only 26 mas in the plane of the sky. Table 5 provides the time of Quaoar's closest approach to the star, and its angular distance to the star at that moment.

8. Conclusion

We observed the first multi-chord stellar occultations by (50000) Quaoar, one from South America on 4 May, 2011 with five chords, and another one from Europe on 17 February 2012, with two chords. A third single-chord event was detected from Chile on 15 October 2012. The last two occultations confirm, but do not bring further constraints to the results obtained from the first event. They provide, however, astrometric constraints that may be used for the prediction of future events.

Taking the timings of the 4 May 2011 event at face value, we find topographic features (crater or mountain) that are too large to be maintained on a body as massive as Quaoar.

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[4]Panchromatic Robotic Optical Monitoring and Polarimetry Telescopes, Cerro Tololo (latitude: 30° 10' 10".78 S, longitude: 70° 48' 23".49 W, altitude.: 2,207 m).

An alternative approach is to allow for time shifts at all stations, i.e. accept that there are timing errors. Such errors are admittedly bothersome, since we do not have clear explanations for them. However, as discussed above, the largest time shift has to be applied to the only station, while the shifts at the other stations remain compatible with the occultation time uncertainties at 2.5σ level.

With this approach, we obtain an variety of satisfactory fits by elliptical limbs. All those ellipses have the same apparent area, yielding well constrained values for Quaoar's equivalent radius $R_{equiv} = 555 \pm 2.5$ km and visible geometric albedo $p_V = 0.109 \pm 0.007$. Adding the further constraint that Quaoar is a Maclaurin spheroid with indeterminate polar aspect angle ζ between 0 and 90 deg, we find that Quaoar has an equatorial radius $a = 569^{+24}_{-17}$ km, true oblateness $\epsilon = 0.0897 \pm 0.006$ $(^{+0.0268}_{-0.0175})$ and density $\rho = 1.99 \pm 0.14$ ($\pm$ 0.46) g cm^{-3}. The errors shown within parentheses on ϵ and ρ accounts for the current uncertainty on Quaoar's mass M, and can be improved when M is better known.

The position angle of Quaoar's north pole projected in the plane of the sky is $P = -31.8$ or 148.4 ± 0.9 deg for our preferred solution (Fig. 9), while Weywot's orbit pole has P in the range 5 – 15 deg (Vachier et al. 2012; Fraser et al. 2013). Thus, our preferred solution implies that Weywot is not orbiting into Quaoar's equatorial plane. This is not surprising, considering that Weywot has probably preserved its primordial orbital inclination against tidal damping (Fraser et al. 2013).

The density that we obtain is typical of TNOs, and significantly smaller than the range 2.7 – 5.0 g cm^{-3} given by Fraser et al. (2013). This disagreement fundamentally stems from the fact that our Quaoar's occultation-derived diameter is significantly larger than previous indirect size estimates. In fact, recent measurements using *Herschel* far IR data combined with *Spitzer* provide a preferred equivalent radius of 541 ± 34 km (Fornasier et al. 2013), which agrees with our value at the 0.4σ level. The *Herschel* data also provides a preferred

Quaoar's geometric albedo, 0.13 ± 0.01, which also agrees with our value at the 1.6σ level.

We put an upper limit of about 20 nbar (1σ) to the presence of a global methane atmosphere, but we cannot exclude a local, denser atmosphere. Higher signal to noise ratio occultation light curves will certainly provide better values in the near future. Our limit can be compared to that obtained for Charon: 15 nbar (Sicardy et al. 2006a), Eris: 1 nbar (Sicardy et al. 2011) and Makemake: 12 nbar (Ortiz et al. 2012). The lack of atmosphere around Charon can be explained by the lack of volatiles like CH_4 or N_2 on its surface, which ultimately results from its smaller mass. Concerning Eris, whose surface composition and mass is comparable to that of Pluto, the lack of atmosphere probably stems from its cold environment, which condenses any putative atmosphere on its surface. This is supported by the fact that Eris has a very high albedo, $p_V = 0.96$ (Sicardy et al. 2011), that could be caused by a transient CH_4 or N_2 ice coating, which will sublimate when Eris approaches its perihelion.

The cases of Makemake and Quaoar are more challenging, as they have the right surface composition and heliocentric distances to possess an atmosphere. Both also have dark terrains (locally or globally distributed) and therefore, local surface temperatures over 40 K that can maintain CH_4 or N_2 atmospheres through sublimation. In both cases, the non-detection of an global atmosphere argues against the presence of N2 ice on the surface. Our lack of detections thus constrains the mechanisms that maintain (or prevent) the presence of atmospheres around transneptunian objects.

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This manuscript was prepared with the AAS LaTeX macros v5.2.

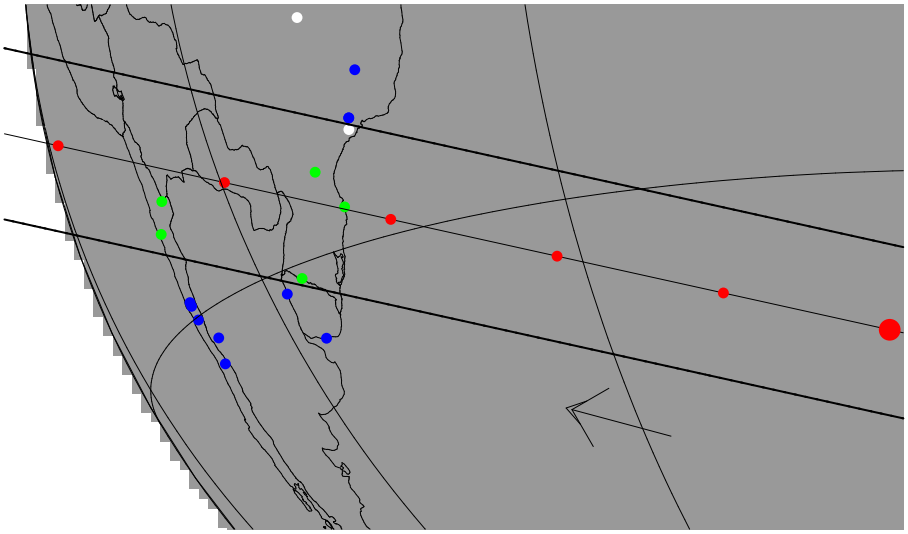


Fig. 1.— The post-occultation reconstruction of Quaoar's shadow path on Earth for the 04 May 2011 event. The shadow moves from right to left, the red dots on the centreline are separated by one minute. The bigger red dot is the closest geocentric approach, at 01:38:37 UT. The s dots are the sites where the occultation was detected. The blue dots are the sites that had obtained data but did not detect the event, and the white ones are the sites that were clouded out (Table 1).

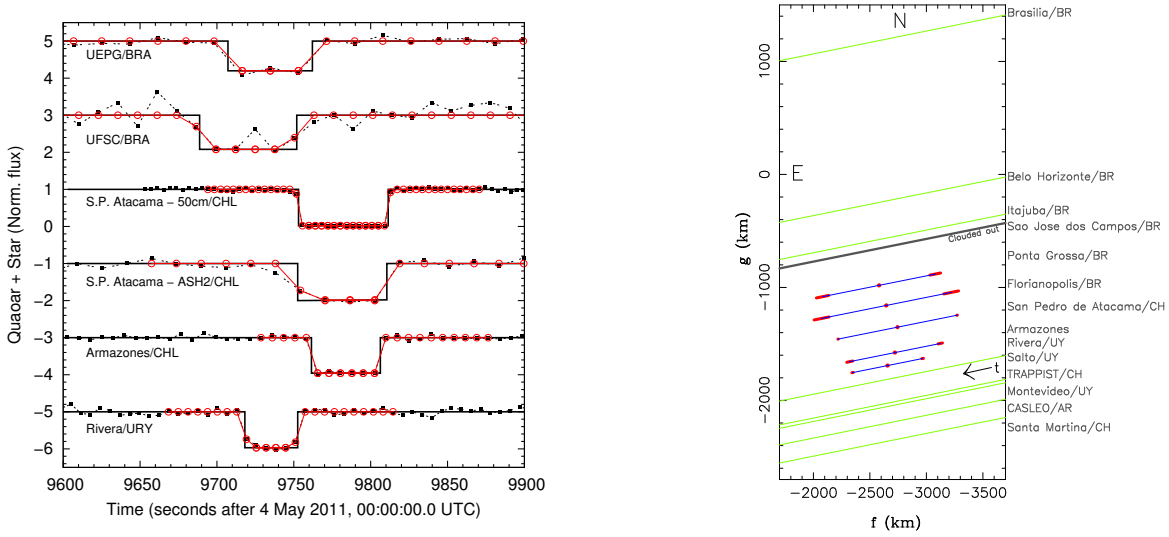


Fig. 2.— **Left panel:** The six occultation light curves, normalized to the unocculted star+Quaoar flux, and vertically shifted by an integer value for better viewing. The solid black lines are the best fit of the square-well model to the data. The red lines and circles are the square-well model convoluted with the Fresnel diffraction, the star diameter and the applied exposure time. The mid-times of the occultations do not coincide due to the propagation delays of the shadow due to the distinct longitudes of the sites (Fig. 1). **Right panel:** All the observed chords showing the region around Quaoar covered by our observations. The red segment are the 1σ error bars on each occultation chord extremity, derived from the corresponding error bars on the occultation times (Table 2). The middle of each chord is marked with a red dot. East is right, North is up, the arrow shows the direction of the body motion, so the ingress at the right and egress at the left. No secondary occultation is observed, as could be caused by a satellite.

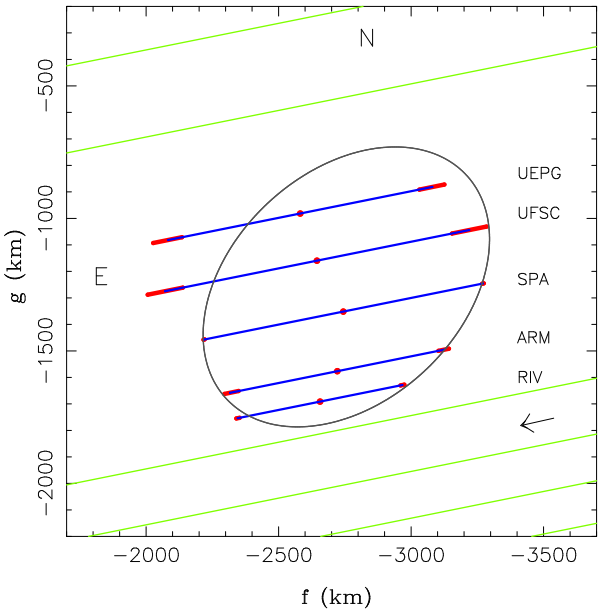


Fig. 3.— **Nominal solution:** The best elliptical fit to the occultation chords, using the timings of Table 2. The elliptical limb model does not satisfactorily fit the observed data. The sites that detected the event are indicated as: UEPG: Ponta Grossa; UFSC: Florianópolis; SPA: San Pedro de Atacama; ARM: Armazones; RIV: Rivera.

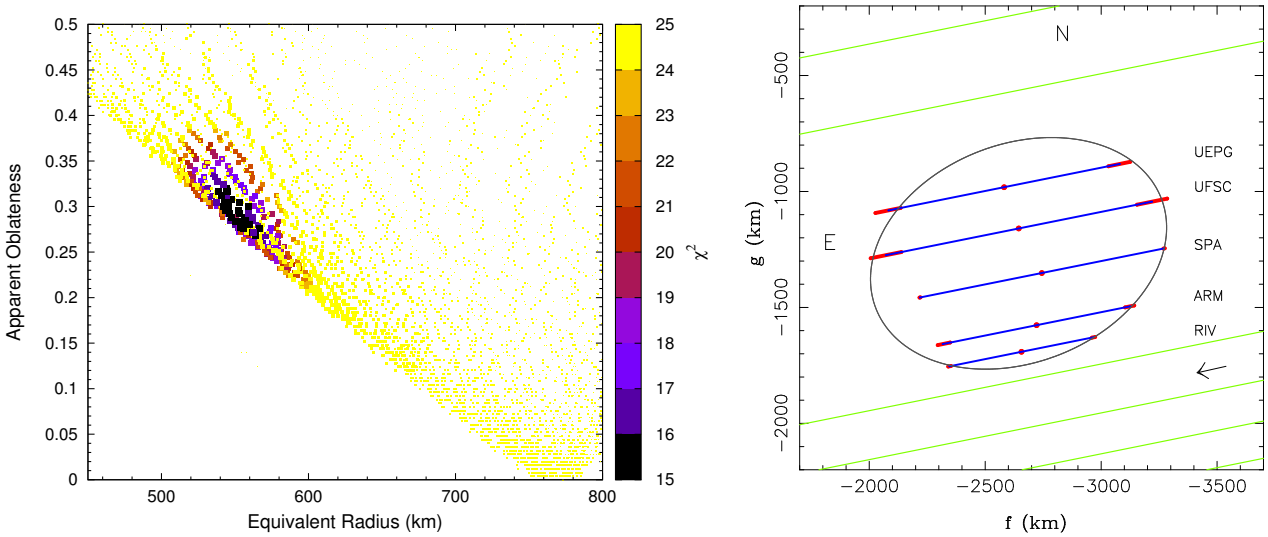


Fig. 4.— **Crater solution.** Left: a 2D-map of χ^2 obtained from various apparent oblateness ϵ' and equivalent radius $R_{\rm equiv}$ in the crater case (see Section 3.3 for details). The darker regions correspond to better solutions (smaller χ^2). The value of χ^2 is coded not only in colors (see the vertical bar at the right of the panel), but also in size: the larger the value of χ^2, the smaller the dot. Right: the best fit for the crater solution, where the San Pedro de Atacama egress point (left) has been discarded from the fit. It has $P = 21$ deg, $\epsilon' = 0.29$ and $R_{\rm equiv} = 557$ km.

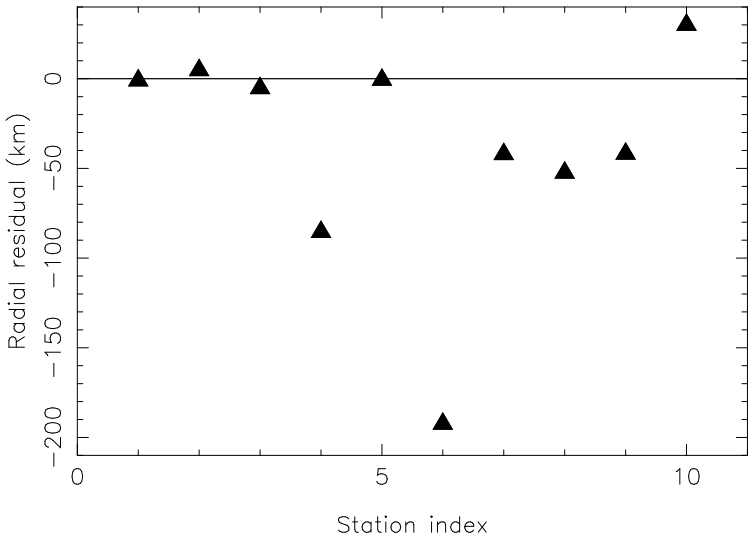


Fig. 5.— **Crater solution:** The radial residuals of the fit shown in Fig. 4. For the San Pedro de Atacama, this residual is about 200 km. The indices refer (from south to north, and from ingress to egress points) to the following stations, Rivera: (1,2), Armazones: (3,4), San Pedro de Atacama: (5,6), Florianópolis: (7,8), and Ponta Grossa: (9,10).

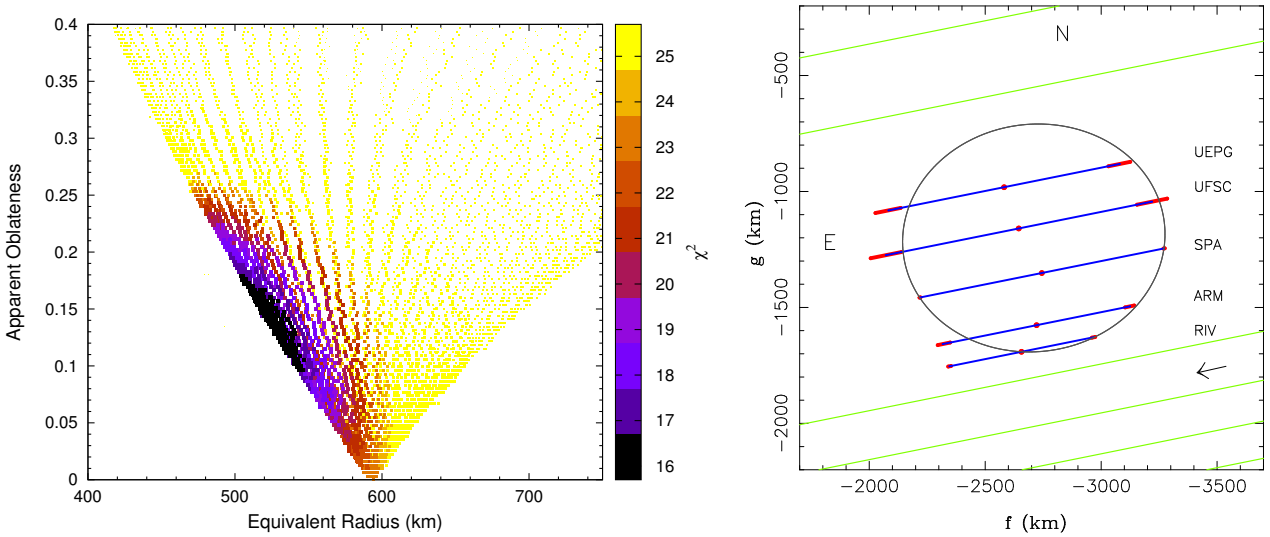


Fig. 6.— **Mountain solution:** The same as in Fig. 4, but for the mountain solution, where the southernmost egress point (from the Rivera station) has been discarded from the fit.

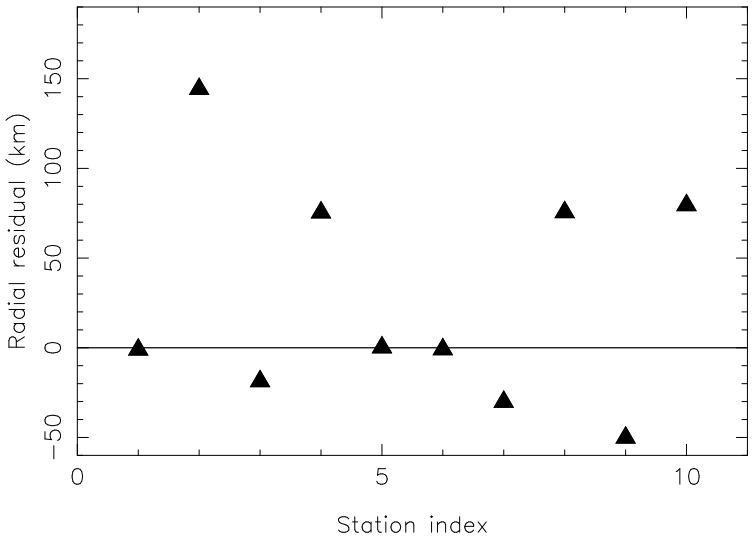


Fig. 7.— **Mountain solution:** The same as in Fig. 5, but for the mountain solution. The Rivera egress chord (i=2) has a positive radial residual of more than 140 km.

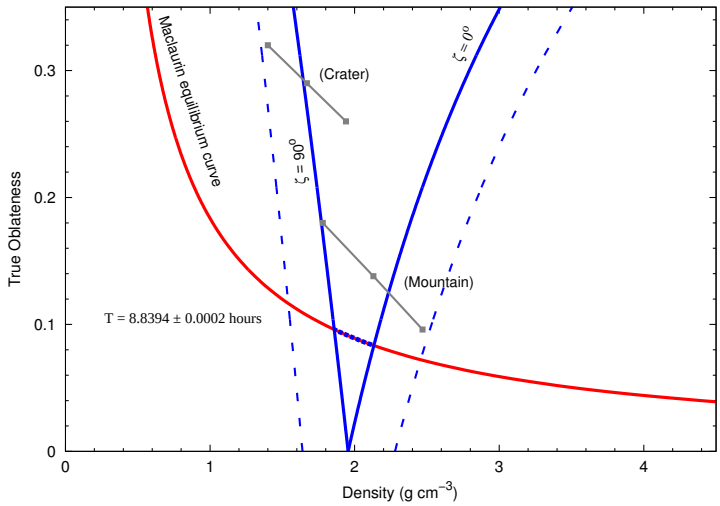


Fig. 8.— Red line: the Maclaurin equilibrium curve relating the density and true oblateness (Eq. 6) for a body rotating with a period of 8.8394 hours. Gray segments: the crater and mountain solutions with their 1σ error bars. In this case, the oblateness is actually the apparent oblateness of the limb. Solid blue lines: the relation between ρ and ϵ derived from Eqs. 5. They correspond to spheroids that all have an apparent equivalent radius $R_{\rm equiv} = 555$ km and a mass $M = 1.40 \pm 0.21 \times 10^{21}$ kg (Fraser et al. 2013). In this figure, we plot the true oblateness, which is derived from the apparent oblateness using Eq. 4. Since the relationship between density and oblateness also depends on the polar aspect angle ζ, we have plotted the extreme equator-on ($\zeta = 90$ deg) and pole-on ($\zeta = 0$ deg) geometries. The dashed lines are obtained by varying Quaoar's mass over its error interval. The spheroid solutions must reside inside the V-shaped blue limits, and the Maclaurin solutions must furthermore be on the intersection with the red line, plotted as the dotted blue segment.

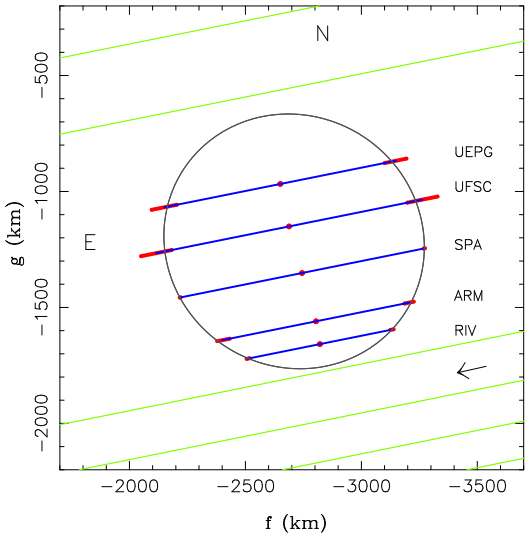


Fig. 9.— The best fit for the Maclaurin solution. It corresponds to the middle point of the blue dotted segment in Fig. 8. It has a position angle $P = 148.4 \pm 0.9$ deg, an equivalent radius $R_{\rm equiv} = 555 \pm 2.5$ km, an equatorial radius $R_{equa} = 569^{+24}_{-17}$ km and apparent oblateness $\epsilon' = 0.0486^{+0.0679}_{-0.0486}$. Once projection effects are accounted for, this corresponds to true oblateness $\epsilon = 0.0897 \pm 0.006$ and density $\rho = 1.99 \pm 0.14$ g cm^{-3}, see text. To obtain this fit, the chords were shifted as follows (leaving the San Pedro de Atacama timing unchanged), Rivera: -9.1 s, Armazones: -4.6 s, Florianópolis: -2.4 s and Ponta Grossa: -3.8 s.

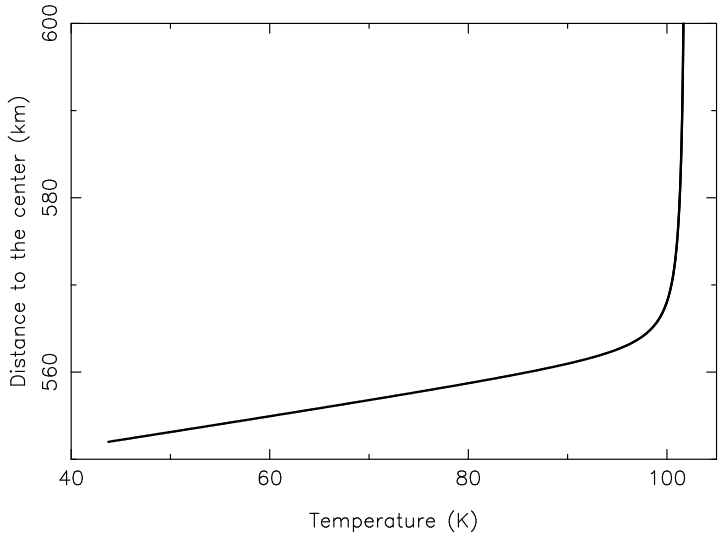


Fig. 10.— The adopted thermal structure of a Quaoar's CH_4 atmosphere. It starts at 42 K near the surface with a gradient of 5.7 K km^{-1} and ramps up to an isothermal profile at 102 K in the upper atmosphere.

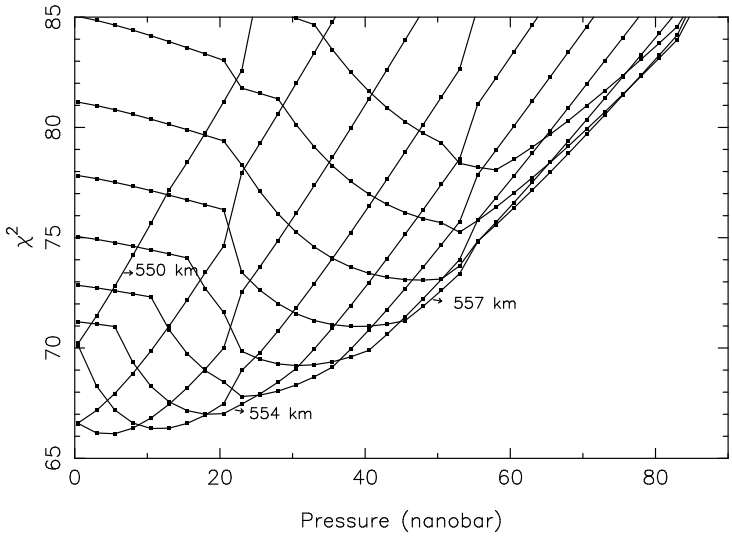


Fig. 11.— Curves of χ^2 for different centre-to-limb radii (from 550 to 560 km), using a pure methane atmosphere with the thermal structure shown in Fig. 10. An upper limit of 21 nbar (1σ), or 56 nbar (3σ) atmosphere is derived. The radius is indicated for some of the curves.

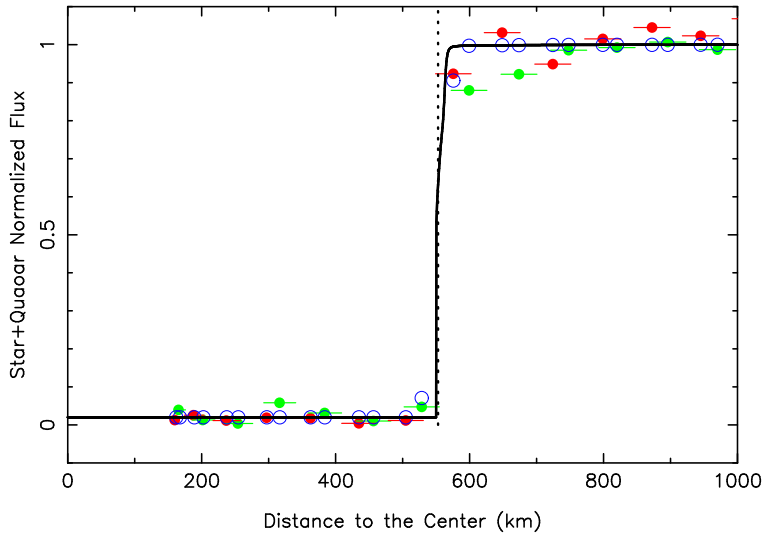


Fig. 12.— Solid black line: a synthetic occultation light curve caused for a 21 nbar methane atmosphere, assuming the thermal profile of Fig. 10 and a centre to limb distance of 553 km. Green and red dots: the ingress and egress data points, respectively, obtained at San Pedro (0.5 m telescope). The horizontal bars represent the finite integration time of each data point. Blue circles: the synthetic points after convolving the synthetic light curve by the finite integration time at each data point.

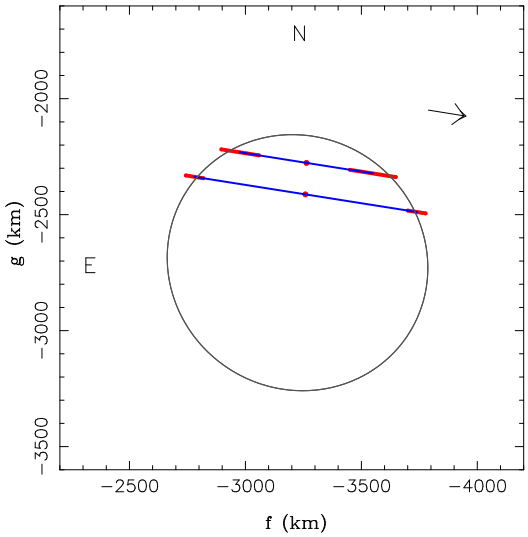


Fig. 13.— The 2012 chords are compatible with the 2011 Maclaurin solution but do not bring further constraints. It corresponds to the middle point of the blue dotted segment in Fig. 8, projected to the 17 February 2012. It has a position angle P= 146.9 deg, an equivalent radius $R_{\rm equiv} = 555$ km, an equatorial radius $R_{equa} = 569$ km, apparent oblateness $\epsilon' = 0.0428$ and a $\chi^2_{pdf} = 0.23$.

Table 1. Circumstances of observation for all observing stations.

Site	Longitude Latitude Altitude	Telescope Aperture	Exposure Readout overhead	Result comment	Observer
Brasília / BRA	47° 54' 41" W 15° 53' 29" S 1,072 m	0.254 m	8.0 sec 2.0 sec	clouds	P. Cacella
Belo Horizonte CEAMIG-REA / BRA	43° 59' 51" W 19° 49' 49" S 825 m	0.30 m	6.0 sec 3.0 sec	No occ. SNR=21	C. Jacques E. Pimentel B. Giacchini
Pico dos Dias - Brasópolis / BRA	45° 43' 58" W 22° 32' 08" S 1,864 m	Carl Zeiss 0.60 m	0.95 sec 1.05 sec	No occ. SNR=85 thin clouds	M. Assafin L. Almeida
Pico dos Dias - Brasópolis / BRA	45° 43' 58" W 22° 32' 08" S 1,864 m	Boller & Chivens 0.60 m	- sec - sec	Instrumet problems	V. Magalhães C. Montaña C. Rodrigues
São José dos Campos - INPE / BRA	45° 51' 44" W 23° 12' 33" S 617m	0.275 m	7.5 sec 2 sec	Humidity	A. Milone
Ponta Grossa - UEPG / BRA	50° 05' 56" W 25° 05' 22" S 909 m	0.40 m	14 sec 4.25 sec	Positive SNR=20	M. Emilio C. Neves
Florianópolis - UFSC / BRA	48° 31' 21" W 27° 36' 12" S 20 m	0.254 m	10 sec 2.75 sec	Positive SNR=12 Humidity	W. Schoenell A. Amorim
San Pedro de Atacama / CHL	68° 10' 48" W 22° 57' 12" S 2,397 m	Harlingten 0.5 m	3.0 sec 1.1 sec	Positive SNR=52	remotely operated by N. Morales
San Pedro de Atacama / CHL	68° 10' 48" W 22° 57' 12" S 2,397 m	ASH2 0.4 m	9.0 sec 7.1 sec	Positive SNR=22	remotely operated by N. Morales
Armazones / CHL	70° 11' 46" W 24° 35' 52" S 2,706 m	OCA 0.84 m	7.0 sec 2.4 sec	Positive SNR=42	E. Unda-Sanzana J.P. Colque
Rivera / URY	55° 36' 00" W 30° 51' 48" S 200 m	0.305 m	4.0 sec 2.3 sec	Positive SNR=26	S. Roland L. Almenares A. Bergengrue
Salto - OLASU / URY	57° 58' 43" W 31° 23' 33" S 37 m	0.305 m	4.0 sec 2.7 sec	No occ. SNR=18	S. Bruzzone
La Silla / CHL	70° 44' 22" W 29° 15' 17" S 2,315 m	TRAPPIST 0.60 m	6.0 sec 1.55 sec	No occ. SNR=50	remotely operated by E. Jehin
Los Molinos OALM / URY	56° 11' 25" W 34° 45' 19" S 80 m	0.35 m	4.0 sec 1.0 sec	No occ. SNR= 10	R. Salvo M. Martinez J. Capeche
CASLEO / ARG	69° 17' 45" W 31° 47' 56" S	J. Sahade 2.15 m	6.0 sec 4.5 sec	No occ. SNR= 250	R. Gil-Hutton

Table 1—Continued

Site	Longitude Latitude Altitude	Telescope Aperture	Exposure Readout overhead	Result comment	Observer
Santa Martina / CHL	2,492 m 70° 32' 04" W 33° 16' 09" S 1,450 m	0.40 m	1.0 sec 0.0 sec	No occ. SNR= 4	R. Leiva L. Vabzi I. Toledo

Table 2. Circumstances of observations for the sites that detected the event (or were very close to it).

Site	Instrument Pixel scale (arcsec)	Telescope Aperture Filter	Exposure Dead Time SNR	Ingress[a] time (UT)	Egress[a] time (UT)
Pico dos Dias - Brasópolis / BRA	Andor/Ixon EMCCD 1.11	Carl Zeiss 0.60 m Clear	0.95 sec 1.05 sec SNR=85	Obs. start: 2:26:10	Obs. end: 2:58:20 No occ.
Ponta Grossa - UEPG / BRA	SBIG STL-6303E 1.74	RCX 400 0.40 m V	14 sec 4.25 sec SNR=20	2:41:47.2±2.7	2:42:42.1±3.1
Florianópolis - UFSC/ BRA	SBIG ST-7 1.48	CM-1100 0.254 m Clear	10 sec 2.75 sec SNR=12	2:41:28.7±3.6	2:42:31.9±3.8
San Pedro de Atacama / CHL	Apogee U42 1.61	Harlingten 0.5 m Clear	3.0 sec 1.1 sec SNR=52	2:42:33.1±0.13	2:43:31.4±0.15
San Pedro de Atacama / CHL	SBIG STL11000 1.21	ASH2 0.4 m Clear	9.0 sec 7.1 sec SNR=22	2:42:32.5±1.0	2:43:30.30±4.3
Armazones / CHL	SBIG STL6303E 0.57	OCA 0.84 m Clear	7.0 sec 2.4 sec SNR=42	2:42:41.5±1.2	2:43:26.3±1.8
Rivera / URY	SBIG ST-7 2.57	LX200 0.305 m Clear	4.0 sec 2.3 sec SNR=26	2:41:58.2±0.4	2:42:32.5±0.4
Salto - OLASU / URY	QSI 516wsg 1.2	LX200 0.305 m Clear	4.0 sec 2.7 sec SNR=18	Obs. start: 2:20:29	Obs. end: 2:54:04 No occ.

[a]Error bars are given at 1σ level

Table 3. Parameters for the best fits shown in Figs. 4, 6 and 9. Error bars are at 1σ level.

Solution:	Crater	Mountain	Maclaurin
Semi-major axis[a] (km)	656 ± 12	568 ± 12	569^{+24}_{-17}
Oblateness[a]	0.29 ± 0.04	0.14 ± 0.05	$0.0897^{+0.0268}_{-0.0175}$
f_c (km)	-2647 ± 19	-2712 ± 16	-2710 ± 4
g_c (km)	-1253 ± 20	-1202 ± 32	-1216 ± 5
Position angle (deg.)	21 ± 3	7.5 ± 11	148.4 ± 0.9
$\chi^2_{\rm min}$	15.1	15.7	1.33
$\chi^2_{\rm pdf}$	1.68[b]	1.74[b]	0.44[b]

[a]For the crater and mountain solutions, these quantities are the apparent semi-major axis and oblateness of the limb. For the Maclaurin solution, they are the true equatorial radius and oblateness, see text.

[b]The calculation of $\chi^2_{\rm pdf}$ from $\chi^2_{\rm min}$ is explicited in the text.

Table 4. Physical parameters of Quaoar obtained from the various solutions of Table 3, see text for details. Error bars are derived from the error bars of each solution, combined with the error bars of the absolute magnitude (H_V= 2.82 ± 0.06, DeMeo et al. (2009)) and of the mass ($M = 1.40 \pm 0.21 \times 10^{21}$ kg, Fraser et al. (2013)). The last line presents the global possible ranges allowed for by all solutions.

Solution	Equivalent radius (km)	Oblateness[a]	Visible geometric albedo	Density (g cm^{-3})
Crater	552 ± 14	0.29 ± 0.04	0.110 ± 0.002	1.67 ± 0.27
Mountain	524 ± 20	0.14 ± 0.05	0.123 ± 0.016	2.12 ± 0.34
Maclaurin	555 ± 2.5	$0.0897^{+0.0268}_{-0.0175}$	0.109 ± 0.007	1.99 ± 0.46
Global range	504 – 566	0.0722 – 0.33	0.102 – 0.139	1.40 – 2.46

[a]For the crater and mountain solutions, this is the apparent oblateness, for the Maclaurin solution, this is the true oblateness, see text.

Table 5: Astrometric constraints derived from the 17 Feb 2012 and 15 Oct 2012 occultations.

Date	17 Feb 2012[a]	15 Oct 2012[b]
Time of closest approach (UT)	04:30:43.76±0.11	00:41:19±5
Distance of closest approach from geocenter [c]	161±12 mas, N	44±25 mas, S
Offset in right ascension[d]	-101±25 mas	-110±26 mas
Offset in declination[d]	-92±24 mas	-63±26 mas
Coordinates of star (ICRF/J2000)	$17^h\ 34^m\ 21^s.8453$±0".022 -15° 42' 10".5860±0".008	$17^h\ 28^m\ 10^s.1257$±0".0013 -15° 36' 23".324±0".0013

[a]Obtained from the circular fit.

[b]Assuming a Quaoar radius of $R = 535$ km, the middle value of the global range of R_{equiv} in Table 4

[c]The label "N" (resp. "S") means that the shadow center passed north (resp. south) of geocenter.

[d]Quaoar's offset relative to the JPL#21/DE405 ephemeris

Table 6. Circumstances of observation for the Quaoar occultation of 17 February 2012.

Site City Observer	Latitude Longitude Altitude	Aperture Instrument Pixel Scale (")	Exposure SNR	Ingress (UT) Egress (UT)
Gnosca/CHE	46° 13' 53".2 N	0.4 m	10.24 sec	4:28:37.6±4.5
S. Sposetti	09° 01' 26".5 E	Watec 120N+	4.4	4:29:09.4±5.5
	260 m	1".9		
TAROT[a]	43° 45' 07".3 N	0.25 m	3 x 180[b] sec	4:28:08.1±3.0
OCA[c]/Calern/FRA	06° 55' 25".1 E	CCD Andor	4.6	4:29:06.5±3.5
A. Klotz &	1270 m	3".3		
E. Frappa				
	43° 51' 52".4 N	0.21 m	5.12[d] sec	4:28:09.2±2.1
Valensole/FRA	06° 00' 23".0 E	Watec 120N+	4.0	4:29:03.2±2.2
J. Lecacheux	622 m	0".84		
Tourrette-	43° 47' 22".2 N	0.356 m	10 sec	4:28:17.6±2.2
Levens/FRA	07° 15' 47".2 E	Apogee Alta U1	30	4:29:04.3±0.5
P. Tanga	385 m	2".7		

[a]Télescopes à Action Rapide pour les Objets Transitoires

[b] Three images obtained with drift scan method, the telescope tracking was set to 96,4% so the time resolution per pixel was 6 sec/pixel.

[c]Observatoire de la Côte d'Azur

[d]The observation started with 10,24 sec of integration time, but about one minute before the beginning of the event it was set to 5.12 sec.